ETAS RTA-RTE v12.7.2

Status: FINAL

Release Notes

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ETAS RTA-RTE 12.7.2 - Release Notes R01 EN – 05.2025

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1 In This Version

Version 12.7.2pr1

Optimize resource locking for one Task simple S/R configs

The behavior of RTE-18341 has been extended to cover scenarios where a participating `RunnableEntity` has both read and write access. Additionally the consistency in resource protection for immaterial configuration changes has been improved.

(RTE-18958)

Set the update flag in resource protection with tranformation

RTA-RTE now sets the update flag(s) in the same concurrency protection as the global data is copied into the RTE managed global buffer in COM/LdCom callbacks where transformation is configured.

(RTE-19675)

2 Known Issues

Limited validation for ISR2s

RTA-RTE has added support for Category 2 ISRs, however, at this time `SWS_Rte_CONSTR_09122` (R23-11) is not validated.

Additionally, due to the similarity of `OsIsr`s and `OsTask`s at this time several of the messages reported by RTA-RTE for ISRs refer to 'Tasks'.

(RTE-19562)

Queued semantics support for Data Conversion and Data Transformation

All release note references to this known issue mention unqueued data (data semantics) being sent or received from the communications layer where the transformation is performed in the relevant callback (ECUC parameter `RteXfrmInvUnqueuedDataCbk` is enabled). However, where possible, code is also generated for data with queued semantics in the relevant callback (command line option `--deviate-cbk-xfrm` is provided). Since queued semantics with Data Conversion and Data Transformation has not been internally tested, it is not officially supported.

(RTE-19511)

Floating point data conversion not supported for data transformation

RTA-RTE does not support floating point data conversion when conversion is to be performed on transformed data.

(RTE-19413)

Limited support for signal fan-in

RTA-RTE supports the reception of multiple `ComSignal`s where transformation is configured for the related `ISignal`s and those signals are mapped to the same `SystemSignal`. Note that:

- RTA-RTE only supports the case where all signals in that inter-ECU communication are mapped to a single partition:
    - RTA-RTE considers first the partition with `EcucPartitionBswModuleExecution` set to `true` (the *bsw partition*) and depending on the configuration may incorrectly identify if IOC is required or not:
        - If all receivers are in the *bsw partition* then RTEGen will generate a global buffer.
        - If all receivers are in the same partition as the `ComSignal`s, but that is not the *bsw partition* then RTEGen will generate IOC calls.
    - When generating the `iocNeeds.arxml` entry for an IOC channel (`OsIocCommunication`) all the sender properties will be based on the partition of the first processed `ComSignal`.
- Several configuration with multiple outer ports lead to duplicated code - use a single outer port where possible.
- Having different `aliveTimeout`s per receiver is not supported.
- In case no data is received via COM the status is updated accordingly

and the `data/transformerError` are also overwritten.

Note: The above configurations are not rejected.

(RTE-18196)

Not all code is moved into partition specific files

RTA-RTE supports the generation of partition specific `C` files for mutli partition ECUs as described by `SWS_Rte_02712` and `SWS_Rte_02713` (R23-11), see RTE-15466.

Note: Not all code that could be moved to partition specific files is currently moved. Anything that is not moved remains in `Rte.c` and may additionally be made visible to the partition files by the generation of an `extern` declaration in `Rte_Partition_Intl.h`. That header file is included by `Rte_Partition_<PartitionName>.c` files only.

(RTE-18160)

No support for passing arguments with trusted functions

RTA-RTE does not support passing arguments via trusted function calls. RTA-RTE doesn't reject configurations where parameters, or where multiple instantiation, is configured. However, the generated call to `CallTrustedFunction()` always passes `NULL` in place of any `FunctionParams` and appends any further arguments, such as the instance, as additional parameters causing the generation of a function that does not match the prototype and will not compile.

(RTE-17080)

Nv writing strategies are invoked during inter-partition NV related callbacks

If any Nv writing strategies are configured for a `RamBlock` involved in inter-partition accesses, they will be triggered during the `Rte_SetMirror` callback and potentially during the `Rte_NvMNotifyInitBlock` callback if a `RomBlock` has been configured.

(RTE-17680)

Inter partition NV access not supported for legacy NvM versions

RTA-RTE does not support inter partition client/server or inter partition reading or writing to RamBlocks when the NvM module is version 4.3.0 or less.

(RTE-17611)

IOC measurement not emitted for inter-partition NvData

RTA-RTE does not emit an `OsIocMeasurementFlatInstanceDescriptorRef` for the IOC channel (`OsIocCommunication`) that implements the `NvData` communication when a `FlatInstanceDescriptor` references that `NvData` in either an `RPortPrototype` or `PPortPrototype` marking it as measurable via `SwDataDefProps.swCalibrationAccess`.

Note that this is not an issue when the `FlatInstanceDescriptor` references the `RamBlock` directly and that `OsIocMeasurementFlatInstanceDescriptorRef` is an ETAS specific extension provided in the OS VSMD file and is not part of the AUTOSAR standard.

(RTE-17518)

Implicit reception with Data Transformation not supported

In the case that there is a ReadAccess to some data that is associated with Data-

Transformationand that data must be inversely transformed, which is not currently supported, then the RTE passes a reference (into the inverse transformation routine(s)) to a non-existent stack allocated variable with symbol `data`.

(RTE-16061)

Incorrect code generation for IIRVs with both read and write access

The change in RTE-9462 fixes inconsistencies in component data structures between contract phase and generation phase when `ImplicitInterRunnableVariables` are accessed from different `OsApplications`. However, RTA-RTE may generate uncompilable or inconsistent code when a configuration contains a runnable that has both read **and** write accesses to an `ImplicitInterRunnableVariable` which is accessed from different `OsApplications` (using the option `--deviate-split-swci-support=1`).

When `--deviate-implicit-modify-for-loopbacks` is **not** used in this case, the component data structure generated for the `ApplicationSwComponentType` will be inconsistent between contract phase and generation phase leading to potential binary incompatibility between components.

When `--deviate-implicit-modify-for-loopbacks` **is** used, component data structures will be consistent but RTE IIRV APIs may attempt to reference data handles which no longer exist leading to uncompilable code.

(RTE-9573)

Inter-OsApplication asynchronous client/server

Inter-`OsApplication` asynchronous client/server has been overhauled to improve freedom from interference. Where previously the generated code made use of data structures shared across partitions RTA-RTE now implements two IOC channels for each client, one is used to transport the call from client to server and the other to pass the result of the server back to the client, additionally RTA-RTE now makes extensive use of `Rte_ReceiverPullCB`s. This note acknowledges the limitations and potential deviations from AUTOSAR and RTA-RTE documentation specifications.

For a particular client, one `ClientServerOperation` in one `RPortPrototype` of one Software Component instance, after `Rte_Call` has been invoked subsequent calls will return `RTE_E_LIMIT` until the corresponding `Rte_Result` has been called. This is true even when the client runnable aggregates an `AsynchronousServerCallResultPoint`. This is a deviation from AUTOSAR R20-11 `SWS_Rte_01105` which allows subsequent calls as soon as the server has finished processing the last request of the client. Note that for intra-`OsApplication` client/server subsequent calls to `Rte_Call` do not return `RTE_E_LIMIT` but are stored in a shared server queue and performed in order of dispatch.

Due to the incompatible implementations of synchronous and asynchronous inter-`OsApplication` client/server in server task bodies, RTA-RTE is not able to interleave synchronous and asynchronous runnable activation. If several `OperationInvokedEvent`s mapped to the same `OsTask` reference `ClientServerOperation`s from both synchronous and asynchronous communication graphs then they must be configured, by use of `RtePositionInTask`, in two contiguous blocks. RTA-RTE will not error in this situation but the task body will not respect the configured server runnable `RtePositionInTask` entries.

WaitPoints are not supported when a runnable is started by an AsynchronousServerCallReturnsEvent, however, RTA-RTE only raises an error when a RunnableEntity which is triggered by an AsynchronousServerCallReturnsEvent aggregates a WaitPoint. RTA-RTE currently does not raise an error if a RunnableEntity is triggered by an RTEEvent of any other type and also aggregates a WaitPoint which references an AsynchronousServerCallReturnsEvent as its trigger despite the configuration being unsupported.

RTA-RTE does not support mapping multiple AsynchronousServerCallReturnsEvents that reference the same client runnable to different OsApplications even when the --deviate-split-swc is provided on the command line. This configuration is not supported but no error is raised.

When configuring an inter-OsApplication asynchronous client/server connection, RTA-RTE will activate any OsTask containing a client/server runnable that is referenced by an AsynchronousServerCallReturnsEvent or OperationInvokedEvent in the Rte_ReceiverPullCB API called when a client request or server response is received via IOC. However, when the respective task is forced to be a basic task via the use of --force-basic-task, the generated code in the respective Rte_ReceiverPullCB should not attempt to activate the relevant task (in accordance with the RTA-RTE Reference Manual to allow the forcing of periodic activations). Currently RTA-RTE will continue to output task activations for tasks containing inter-OsApplication triggered AsynchronousServerCallReturnsEvents or OperationInvokedEvents within the relevant triggering Rte_ReceiverPullCB.

Currently RTA-RTE does not support the use of call serialisation for inter-OsApplication asynchronous client/server communications as per SWS_Rte_04522.

However, RTA-RTE will not raise an error in the case that a configuration contains multiple inter-OsApplication OperationInvokedEvents activating the same server runnable where the server has the attribute canBeInvokedConcurrently set to FALSE and the relevant events are mapped to different OsTasks. This configuration could lead to the relevant server runnable being suspended and executed again concurrently as a result of system dynamics. Currently RTA-RTE does not recognise this scenario and safe behaviour cannot be guaranteed.

RTA-RTE historically uses data types for transformerError parameters passed to server runnables which are a deviation from AUTOSAR R20-11 requirement SWS_Rte_07086. This behaviour is unchanged.

Tasks which only contain a single OperationInvokedEvent or AsynchronousServerCallReturnsEvent triggered by an inter-OsApplication asynchronous client/server connection can be optimised to contain no activation flag as activation of the task implies the triggering of the event. This optimisation is not currently performed by RTA-RTE.

(RTE-9197)

Restart or forcible termination of tasks or partitions is not supported

The *partial restart* of an ECU by forcibly terminating and restarting an OsTask or EcucPartition, is not supported by RTA-RTE.

AUTOSAR does not provide any mechanism to notify the RTE of a forcibly terminated task and the RTE code generates task body code assuming the task runs to completion. Terminating such a task leads to undefined behavior.

RTA-RTE does not support partition restart, and does not provide the `Rte_PartitionTerminated`, `Rte_PartitionRestarting`, or `Rte_RestartPartition` APIs necessary to implement partition restart in an AUTOSAR-conformant way. Terminating a partition leads to undefined behavior.

In this context, *undefined behavior* may include, and is not limited to, sporadic corruption of runtime data or deadlock of parts or the whole of the system.

WARNING

Partial restart of an ECU cannot be used as an error recovery strategy with this or previous versions of RTA-RTE. The design of such a system is beyond the scope of RTA-RTE and remains the responsibility of the system designer or integrator.

(RTE-7286)

Configuration of BSW partitions does not follow AUTOSAR

The configuration of BSW partitions in RTA-RTE does not follow the approach set out by AUTOSAR.

RTA-RTE only allows a single partition where `EcucPartitionBswModuleExecution` is set to `TRUE`. When inter-ECU communication is required, RTA-RTE will assume that this partition is where the COM module is mapped (unless the option `--deviate-event-based-com-mapping` is provided, see RTE-9871). Callbacks generated for the reception of incoming communications will route data via the IOC if the final destination for that data is in a different partition.

The use of the `--deviate-bsw-any-partition` option allows `BswEvent` instances to be mapped to tasks that execute in any partition. To allow the execution of a single BSW module on multiple partitions, the `--deviate-split-swci-support` option must be used in combination with `--deviate-bsw-any-partition`.

Errors will be raised if a configuration contains multiple partitions where `EcucPartitionBswModuleExecution` is set to `TRUE`, or if a configuration attempts to map a `BswEvent` instance to a non-BSW partition without the use of the `--deviate-bsw-any-partition` option.

(RTE-6185)

osNeeds is incomplete when OsApplications are used

Updated by RTE-5611 *and* RTE-6297 *and* RTE-15047 *and* RTE-16646.

Currently, when RTA-RTE generates required OS objects into the `osNeeds.arxml` file they are not configured to allow access from any other `OsApplications` that need it. Additional work is therefore required to integrate the generated objects into a configuration that uses `OsApplications`.

(RTE-1150)

Strict Config Check / OsNeeds is out of step regarding OsResource scheme

Updated by RTE-5611

From RTE-5476, RTA-RTE uses an `OsApplication`-based scheme for resources in the generated code only. Neither OsNeeds generation nor the Strict Config Check feature have been fully updated to match.

This means that if Strict Config Check is enabled, RTA-RTE will not issue errors about missing `OsApplication`-specific resources, allowing the workflow to continue until the problem is detected at link time.

(RTE-5541)

Behavior is undefined when initial or invalid values are used with subElementMappings

Specifying an initial or invalid value on complex data where `subElementMappings` cause a reordering of member elements leads to undefined behavior in RTA-RTE.

(RTE-513)

Not all unmapped runnables are rejected in split-swci configs

When `--deviate-split-swci-support` is enabled, RTA-RTE correctly rejects `OperationInvokedEvents` that are not mapped to `OsTasks`, since it cannot tell what `OsApplication` or `OsApplications` the associated runnable will execute in.

RTA-RTE should reject unmapped RteEvents of any kind that are not mapped, but does not. (*e.g.* `TriggerEvents` or `ModeSwitchEvents`)

(RTE-1312)

Option ——operating-system deprecated

The only valid value for the `--operating-system` is `autosar40`, which must be in lower case. This is its default value, so it is also valid to omit the option altogether.

Support for other values is deprecated and will raise deprecation warning 2977.

(RTE-2456)

Mode Switch Acknowledgement runnables not called when there are no mode users

When there are no Mode Users but a Mode Switch Acknowledgement is configured, RTA-RTE emits misleading warning 1289 about conflicting cores, and does not generate the call to the acknowledgement runnable.

(RTE-2598)

Default SwAddrMethod name for calibration parameters should be CALIB not CALPRM

RTA-RTE uses `CALPRM` as name of the default `SwAddrMethod` for calibration parameters for which no `SwAddrMethod` is specified on the input. This is incorrect by the AUTOSAR `SWS_MemoryMapping` document which specified a section type of `CALPRM` but a name of `CALIB`.This issue affects the memory mapping decoration and `memclass` for calibration parameter allocations and the generated RTE BSWMD output.

(RTE-2688)

Repeated pre-processor code for enumeration literals

When multiple `CompuMethod`s define the same symbolic value, RTA-RTE will emit all the declarations even though it is only necessary to define the preprocessor symbol once. The definitions are protected with a preprocessor guard, so there is no negative effect of the multiple declaration on compilation or runtime.

(RTE-2769)

DataReceivedEvent must be mapped to a task

It is ususally the case that `DataReceivedEvent` must be mapped to a task. However AUTOSAR allows an unmapped `DataReceivedEvent` in the case of a `storeImmediate` NVRAM writing strategy. This is not supported by RTA-RTE.

(RTE-2894)

Incorrect filter code generation

Filters in `UnqueuedReceiverComSpecs` must specify any `MASK`, `X`, `MIN`, or `MAX` attributes within the bounds of the applicable ImplementationDataType or RTA-RTE may generate invalid filtering code.

Filters of type `NEW-IS-OUTSIDE` are not implemented according to the AUTOSAR reference algorithm.

Filters of type `ONE-EVERY-N` are not implemented correctly: the generated code incorrectly sends one message every N+1.

Filters of the following types from AUTOSAR 4.0.2 are deprecated:

- `MASKED-NEW-DIFFERS-MASKED-OLD`
- `NEW-IS-DIFFERENT`
- `NEW-IS-EQUAL`
- `NEW-IS-GREATER`
- `NEW-IS-GREATER-OR-EQUAL`
- `NEW-IS-LESS`
- `NEW-IS-LESS-OR-EQUAL`

(RTE-2908)

Incorrect code when TextValueSpec cannot be found in CompuMethod

When a `TextValueSpecification` is used with an `initialValue`, the only supported interpretation is that RTA-RTE is to generate a numerical literal by looking up the `TextValueSpecification . value` in the `CompuMethod` associated with the associated data type. In the case where the value could not be looked up, either because there was no valid associated `CompuMethod` or the associated `CompuMethod` did not define the symbol in question, RTA-RTE generates the empty string in place of the numerical constant. Usually this leads to a compilation error, but it is also possible to end up with a valid but incorrect comma-expression that causes RTE to compile without errors, despite containing an incorrectinitial value.

(RTE-2940)

COM may be called from any OsApplication

IOC communication is generated for Inter-ECU reception when the receiving software component is in a different `OsApplication` to COM. However the COM

proxy mechanism for Inter-ECU sending ([SWS_Rte_8307] of AUTOSAR 4.2.2) is not implemented so the sending software component must be in an OsApplication from which COM can be called directly.

(RTE-2977)

MinimumStartInterval can expire up to RTE_MAINFUNCTION_PERIOD_MS early

RTA-RTE considers the time between the runnable entering and the next tick of Rte_MainFunction to be RTE_MAINFUNCTION_PERIOD_MS long. If t is the time between the latest tick of Rte_MainFunction and the runnable entering, then the enforced interval will be $(minimumStartInterval - t)$. In saturation conditions, this stabilizes and intervals after the first one will be the correct length. The default value for RTE_MAINFUNCTION_PERIOD_MS is 10, so the backoff interval could be up to, but not including, 10ms too short. This can be mitigated with the --period option, or by increasing the minimum start interval by $(RTE_MAINFUNCTION_PERIOD_MS/1000)$ seconds.

(RTE-2982)

Initial values for transformed signals

When Rte_Read is called for a signal for which a value has never been received, it is usual to return an initial value to the calling SWC. This value is retrieved from a pre-initialized receive buffer. When data transformation is applied to a signal, the receive buffer is stored as transformed data, and the Rte_Read API applies the appropriate inverse transformers to the value in the buffer. Preinitialization of the receive buffer is not possible without detailed knowledge of the transformation process. If an Rte_Read API is called before a signal has been received, it is likely that the buffer contents will not be suitable for transformation, and therefore one or more transformer functions will return an error. In turn this will cause Rte_Read to return RTE_E_HARD_TRANSFORMER_ERROR or RTE_E_SOFT_TRANSFORMER_ERROR.

(RTE-3111)

Inconsistency regarding connected sender-receiver with no writers

RTA-RTE now treats a queued sender-receiver data item as unconnected if there are no configured sender runnables, even if the port is connected in the composition, thereby indicating to the calling software component that no data will ever be received (see RTE-3153).

But the same does not apply in unqueued data: an Rte_Read API generated in the equivalent scenario, where it will always read the initial value, still returns RTE_E_OK.

(RTE-3151)

Repeating and duplicate masks and and bitfield length declarations in types header file

When BITFIELD_TEXTTABLE is used, RTA-RTE may write duplicate #define directives in the types header file. This does not have any effect on the meaning of the compiled code.

(RTE-3349)

COM Signal Invalidation Hook reported but not called

When inter-ECU reception is configured on an invalidatable signal, RTA-RTE can

generate an RTE without a COM Signal Invalidation Hook, yet still report it as a callback in the BSW Module Description report.

(RTE-3456)

No data conversion where a SystemSignal or SystemSignalGroup is mapped to multiple data items of different types

When a single `SystemSignal` or `SystemSignalGroup` is mapped to multiple sender-receiver data items in one or more PPorts, the value written for each data item is passed directly to COM for transmission.

Similarly, when a single `SystemSignal` or `SystemSignalGroup` is mapped to multiple sender-receiver data items in one or more RPorts, the value read for each data item uses the raw data exactly as obtained from COM.

In both cases, RTA-RTE does not perform data conversion or type casting, even when the data items have different data types.

This may lead to the generation of uncompilable code, or to the generation of code that transmits or receives incorrect or invalid data.

You should avoid the use of configurations in which a single `SystemSignal` or `SystemSignalGroup` is mapped to multiple data items.

(RTE-3464)

Spinlock when initial runnable communicates with runnable mapped to task

At the time of writing, it is not supported to use port communcation in runnables started by `InitEvent`, but RTA-RTE does not reject such a configuration.

In the case where an initial runnable writes port data that is read by a runnable mapped to a task, RTA-RTE may generate an unnecessary spinlock.

(RTE-3565)

Rounding of floats

RTA-RTE uses C++ `double`s and `boost::multiprecision::rational`s for storing values from the input XML. Both encodings can show rounding artifacts in decimal numbers beyond 15 significant digits.

This has been observed for example when an initial value of `0.33333333333333333` (17 significant figures) is presented as an `InitialValue`. RTA-RTE renders `0.33333333333333337` in this case.

(RTE-3744)

Identical SHORT-NAME for pre-compile-time Variability

Where Pre-Build-Variability is configured, RTA-RTE does not support elements with the same path, even though at run time the elements will not conflict. This is acceptable when using Configuration-Time-Variability since the strict consistency checking which RTA-RTE performs can ignore the element which is 'disabled'.(Section 7.6.2 of the Generic Structure Template (V3.1.0, R4.0 Rev 2) states "*Strictly speaking, a (not-yet-bound) variant rich M1 model violates AUTOSAR's consistency conditions by having multiple elements with the same ShortName*")

(311298)

Data conversion errors

RTA-RTE does not raise error conditions when (a) the type mappings within a

`ConstantSpecificationMappingSet` conflict, (b) the type mappings across two referenced `ConstantSpecificationMappingSets` within one SWCT conflict, or, (c) an `ApplicationDataType` of category `VALUE` can only be mapped/connected to an `ImplementationDataType` which also has category `VALUE`.

(311301)

Variation in contract phase

RTA-RTE requires all variation points to resolve to a number – however this requires an ECU variation resolver that is not available at contract phase. See also known issue 327067.

(311540)

SwSystemConsts Must Have a Value

RTA-RTE will reject a configuration where a `SwSystemConst` is declared which does not have a value assigned to it (within a `SwSystemConstValueSet`).This means that PreBuildVariability referencing system constants is not supported in Contract phase (since there is no means by which a `SwSystemConstValueSet` can be referenced outside of the ECUC).

(311879)

WaitWithTimeout called with nonexistent event

When generating IOC communication RTA-RTE can invoke the RTE library API `Rte_WaitWithTimeout` with a nonexistent event. This issue will be addressed as part of 312156.

(312157)

Incomplete McSupportData

The content of the McSupportData output from RTA-RTE is incomplete as not all of Table 5.41 (Software Component Template Rev.4.0.3) is implemented - information from Implementation and Application data types is used, but other sources are not.

(312174)

Resource consumption of asynchronous mode switches

The implementation of ModeSwitchEvents for asynchronous mode switches uses two interrupt suspensions to protect state variable manipulations even in cases where there cannot be any interruption.

(322955)

Time-triggered BSW schedulable entities

According to SWS 7574 the `SchM_Init` API function must start the activation of time-triggered BSW schedulable entities. This is not implemented but instead the activation of time-triggered BSW schedulable entities is started by the `Rte_Start` API function, at the same time as the activation of time-triggered application SWC runnable entities is started. Therefore an EcuM implementation that starts the SchM and then delays starting the RTE until a timing event occurs will not work. Similarly the `SchM_Deinit` API function does not stop the activation of time-triggered BSW schedulable entities, the `Rte_Stop` API function having already done that.

(323233)

Optimization of "never written" buffers

If an initialized receive buffer (i.e. a `DataVariablePrototype`) has an initial value but is never written then RTA-RTE generates read accesses that still access a buffer *variable* whereas it would be more efficient to access a constant (ROM) value.

(323703)

Avoid RunToTaskMap for direct function call servers with variable access (ModeSwitchPoints)

RTA-RTE shall support invocation of servers by direct function call where the server makes mode switches.

(324468)

Shared ExclusiveAreas for SchM_Start/SchM_Stop

Shared (synchronized) ExclusiveAreas only work for explicit enter and exit via an RTE API – implicit exclusive area access (RunsInsideExclusiveAreas) is not correctly supported.

(324661)

Call tree analysis

Where such information is provided in the input model RTA-RTE does not perform call tree analysis of BSW modules to optimize protection mechanisms.

(324662)

Rejection of connection from parameter provide port to sender-receiver require port even when compatible

RTA-RTE does not support the connection of a provide port with a parameter interface may be connected to a require port with an unqueued sender-receiver interface provided that the two interfaces have compatible data items. This is valid according to AR4.0 SWCT section 6.4.3 however RTA-RTE rejects such a configuration.

(325960)

Multi-core support

This release of RTA-RTE contains minimal support for multi-core environments. Independent applications (without any interconnections) may be run concurrently on separate cores but the whole RTE for the ECU must be started and stopped at once via the `Rte_Start` and `Rte_Stop` APIs called only from the master core.

(327008)

Not all RTE phases are supported

RTA-RTE does not support the following generation phases:

- PreBuild Data Set Contract phase
- PreBuild Data Set Generation Phase
- PostBuild Data Set Generation Phase

(327067)

Non-present FlatInstanceDescriptor info

RTA-RTE does not detect `FlatInstanceDescriptor` that reference an instance which is not present in the input model.

(327205)

RTEGen does not support flatmaps referencing members of complex data structures

At this time, RTA-RTE does not support flatmap entries which reference members of complex data structures.

(328121)

Redundant RTE queues created for inter-partition client-server calls

When inter-partition client-server calls are implementing using queues, task activation and waiting, an IOC queue is used for each direction. However RTA-RTE also creates but does not use RTE managed queues for the same purpose. This does no harm but wastes RAM and ROM.

(328947)

Unnecessary Status Buffer Allocation for Implicit Sender-Receiver Communication

RTA-RTE allocates status buffer(s) for implicit S/R intra-ECU communications for which the AliveTimeout attribute is set to a value > 0. These status variable(s) is(are) initialized but later never used, although copy is performed in Task body.

(329271)

Multicore: Spinlocks not optimized

When multi-core functionality is required (supported for mode machine instances at present), spinlocks must be used to protect against concurrent access. RTA-RTE currently does not optimize spinlocks based on the concurrency properties of the system. For this reason, spinlocks must be enabled using the command line option `--mcore-spinlocks-always=1`.

(329655)

Multicore: Spinlocks apply globally

This topic only applies to multi-core configurations. When spinlocks are used by RTA-RTE (`--mcore-spinlocks-always` is set, and some Mode Machine Instance is present), the spinlocks configured by RTA-RTE apply to all cores. At this time, no optimization takes place.

(329657)

Optimization of Port Handles

RTA-RTE does not currently optimize the port handle section (e.g. by omitting entries for unused operations).

(329688)

Internal Structures not optimized for ROM/RAM usage

RTA-RTE does not currently order internal structures by size, therefore the generated structures are not optimal for ROM/RAM usage.

(329691)

No Data-Conversion takes place for Initial Values

Where an initial value is calculated for an R-port from a connected P-port (whether inter- or intra-ECU) no data conversion is performed. As a workaround, ensure that initial values are always supplied on R ports.

(329917)

Conditions for invocation of runnables by direct function call do not consider calling task preemption patterns

RTA-RTE now generates a warning message where a runnable-entity is implemented as a direct function call but is not marked as preemptable (i.e. has `CanBeInvokedConcurrently=FALSE` in R4.0, or `IS-REENTRANT` in R3.x).The warning is generated because RTA-RTE does not check the preemption pattern of the calling tasks in this case. Such a configuration is valid if the calling tasks cannot preempt each other (*e.g.* if there is only one calling task) but invalid if they can. It is the user's responsibility to validate the configuration in the case that the warning is generated.

(329925)

VariationPoints do not appear in McSupportData report

RTA-RTE does not export VariationPoints (nor dependent SwSystemconst elements) to McSupportData. McSupportData does reflect the state of Code-GenerationTime variability which has been instantiated by RTA-RTE.

(329953)

Missing Content in McSupportData (Static or Constant Memory)

McSupportData does not include CompuMethods, Units, or DataConstrs for StaticMemory/ConstantMemory containers.

(330319)

SwcBswMapping validation

It is not validated that a `SwcBswMapping` is referenced by exactly one software component instance and exactly one BSW module instance. If multiple instances accidentally reuse the same `SwcBswMapping` the invalid configuration is not detected and the mapping is arbitrarily applied to one of the instances and not the others.

(330391)

Validation of BSW module instance infixes

When a BSW module is instantiated multiple times the instances have unique infixes configured but this restriction is not policed.

(330392)

Potentially unclear error message when Data Conversion contains a division-by-zero

When a conversion formula contains a divisor that is zero, an error message is given but it contains no location information. The same error message is given in the case that a conversion formula with a non-zero divisor is given but for the reverse direction to that which is needed and the formula is not invertable.

(330652)

Generated RTE requires RTE_SPINLOCK

When OsApplications and Partitions are configured the generated RTE code requires `RTE_SPINLOCK` however the generator does not warn if this element is not configured in the OS configuration nor does it create it in `osNeeds.arxml`.

(331949)

Task context for ExclusiveArea

For RunnableEntities and BSE Entities accessing that access ExclusiveAreas RTA-RTE requires information about the executing task context. This is required for ExclusiveAreas realized by `OS_RESOURCE` or `COOPERATIVE_RUNNABLE_PLACEMENT` but is not strictly necessary for ExclusiveAreas realized by `ALL_INTERRUPT_BLOCKING` or `OS_INTERRUPT_BLOCKING`.

(332650)

No integration test exists for the fix to EHI 333876. This relates also to EHI334988

No integration test exists for the fix to EHI 333876. This relates also to EHI334988.

(334550)

Rte ParamDef

This RTA-RTE distribution does not include an ECUC ParamDefs file that is compliant to AUTOSAR AR4.0.3. No vendor specific parameters are required and therefore the standard AUTOSAR file can be used instead.

(334636)

Initialization of calibration values

RTA-RTE supports calibration parameter initialization by values provided as `NumericalValueSpecification`. The initialization of calibration parameter in physical values rather than pure numbers is not supported.

(334958)

Initialization of SwAxisConts and SwValueConts

RTA-RTE does not support ApplicationValueSpecification for parameter initialization with physical values specified via SwAxisConts and via applied SwValueConts.

(335014)

Inter-ECU connections to NvBlock components not rejected

RTA-RTE does not reject invalid configurations containing connections to NvBlock components that span ECUs. It does however correctly reject invalid configurations containg connections to NvBlock components that span partitions (OsApplications) on the same ECU.

(335288)

McSupportData not defined for DATA_REFERENCE

ImplementationDataTypes of category DATA_REFERENCE are not supported in McSupportData. McSupportData output is currently not defined if DATA_REFERENCEs are used.

(335424)

Option -drapm=interrupts still uses RTE library API Rte_GetResource

Option `-drapm=interrupts` still leaves some `Rte_GetResource` (using `RTE_RESOURCE`) in the generated code since it only affects one area of code generation. This is intentional but may be extended in a future revision of RTA-RTE to reduce the reliance on Os resources.

(336856)

Mode evaluations for ModeDisablings per task

As required by AUTOSAR, RTA-RTE performs mode evaluations for ModeDisablings per Runnable Entity. An alternative behaviour (check once per task) will be considered for support in a future version of RTA-RTE.

(336858)

Use of C library

RTA-RTE maps some APIs (e.g. IWrite) to `Rte_memcpy` which in turn can be mapped to memcpy of the C library. To avoid a dependency on the C library ensure that the definition `RTE_LIBC_MEMCPY` is **not** defined when compiling RTE code.

(336860)

Header files generated in contract phase

In Contract Phase the files `Rte_Main.h`, `Rte_Intl.h` and `Rte_Const.h` are generated, but not necesarily used. If not required these output files can be ignored.

(336861)

Compiler abstraction of buffers

The compiler abstraction of buffers does not use the memory class provided in the SwAddrMethod (see examples in chapter `5.2.6.6 Return Values` in RTE SWS).

(336866)

Selectable behaviour for NvBlock buffer access

If a data element of a NvBlockSwComponent is implicitly written/read without possible interruptions (e.g. in the same runnable), then RTEGen accesses the RAM block directly. This is in contradiction to AUTOSAR where the read and write accesses have to go to different buffers and therefore the behaviour of using the same buffer for implicit reads and writes of a Runnable should be selectable per component.

(336872)

Location information missing for data-conversion error

RTA-RTE error message W002688 (data conversion results in division by zero) does not provide any location information. A future release of RTA-RTE will enhance the error report (e.g.by providing file and line information of involved CompuMethod) to enable easier location of the error in the input configuration.

(337599)

Initial modes are not correctly entered on start of SchM / RTE when ASW and BSW mode instances use a shared mode queue

There is a problem with the handling of initial modes in mode queues that are shared between ASW and BSW mode machine instances. In this scenario all mode machine instances begin transition to the initial mode during SchM startup, not

just the BSW mode machine instances. The ASW mode machine instances then fail to complete the transition. Shared mode queues should therefore not be used with a mix of ASW and BSW mode machine instances.

(340407)

Inefficient code generation for external triggers within same task activated by same source

When multiple TriggerOccurredEvents are mapped to the same task RTA-RTE generates an activation flag for each runnable. This is sub-optimal code generation (*e.g.*, if a task contains runnables triggered by multiple sources any that are from the same source could share a flag) and will be addressed in a future release of RTA-RTE.

The RTE-7861 has addressed the case where all runnables in a task are triggered by the same source.

(341063)

Direct global variable access for implicit comms and IOC

RTA-RTE supports implicit inter-partition communication using a task-specific local buffer that is updated/written-back using the IOC APIs during task execution. However the combination of optimization of implicit communication to direct access to global buffers is **not supported** when inter-partition (i.e. IOC) communication is used. If both are enabled then generation will fail.

(343665)

Implicit Sender-Receiver: Optimization of global buffers within a single task fails with multiple runnables per port

This topic relates to issue 341465 concerning the optimization of implicit writers and readers to use the global buffer directly instead of a task-based copy. The current implementation contains a bug determining whether a reader can be pre-empted by a writer. The problem occurs when multiple readers are connected to the same Port and VariableDataPrototype. In this case, if one of the readers is preemptible then all the readers connected to the given Port and VariableDataProtoype are treated as pre-emptible, leading to a failure to use the global buffer and failure to get the latest written value.There are two ways to avoid this problem (you only need to apply one):

- Do not map implicit readers to the same task as their corresponding writer, OR
- If you have multiple readers for a given writer, attach them to unique ports and use multiple AssemblyConnectors to create a 1:n communication

There may be a similar problem when multiple writers are connected to the same Port and VariableDataPrototype.

(347323)

Client-server port interface mappings not honoured for delegation connectors

RTA-RTE fails to find and apply client-server port interface mappings referenced from delegation connectors. When a delegation port on a composition compo-

nentis typed by a different interface than the delegated inner port and the interface elements are not named identically RTA-RTE will consider the interfaces incompatible even if the connector references a valid client-server port interface mapping.

(347355)

Delegation completeness not validated

RTA-RTE no longer validates the completeness of delegation to outer ports of composition components by the AUTOSAR rules. This validation will be restored in a future release.

(347395)

TriggerDirectImplementation does not work together with synchronized Triggers

TriggerDirectImplementation does not work together with synchronized Triggers. RTA-RTE generates flags around the triggered Runnable/BswModuleEntities but these can never be set as the trigger producer activates the task directly. This will be corrected in a future release of RTA-RTE.

(347944)

Multiple ExternalTriggerOccurredEvents mapped to the same task

Multiple ExternalTriggerOccurredEvents mapped to the same task lead to generation of multiple trigger source evaluation flags. This is technically correct but can be more efficiently implemented and will be addressed in a future release of RTA-RTE.

(347950)

Error message 1345 is not very helpful

Limited support has been implemented for expressing complex values for initial value or invalid value of data elements typed by ApplicationRecordDataTypes or ApplicationArrayDataTypes. At the time of writing, it is possible for RTA-RTE to lose track of the type informatio as it descends a complex data type in parallel with a complex value specification in certain invalid configurations. This type information is not needed if the value specification's leaf is specified in the numerical domain, but if it is specified in the physical domain and the type information has been lost in this way then error message 1345 is raised to indicate a faulty configuration. At this point the generator is unable to report much useful information about the original cause of the problem, so the error message is not very helpful. This will improve when the internal modeling of ValueSpecification is complete.

(351344)

Implicit writing of NVRAM data does not work

In the case where a server mapped to the NvMNotifyInitBlock callback writes to the NVDataBlock via implicit API, the Rte_ImplWrite is missed in the callback body, resulting in no written data.

(353801)

ComSignal data type assumed compatible with port data without any validation

RTA-RTE currently generates COM transmission and reception code that assumes that the ComSignal data type is compatible with the port data being transmitted or received, except in the case that linear data conversion is performed. A mismatch could result in failure at runtime without any warning at generation or compile time, for example reception of a ComSignal smaller than the destination will result in partly uninitialised data being received.

(357129)

IncludedDataTypeSets do not cause typedefs to be generated

RTA-RTE now emits enumeration constants for correctly-formed CompuMethods referenced by used ARDataTypes (see TFS ID 346807). Additionally, any prefix specified by means of an IncludedDataTypeSet is honoured. However, [TPS_SWCT 1155] is not implemented. This means that putting a type into an IncludedDataTypeSet does not affect whether the type's corresponding definitions are generated. Only actual use of the type in a DataPrototype of the SwComponentType will cause the generation to occur.

(359012)

Rte_WaitableDatum initializer has different number of elements than its type

In some circumstances, RTA-RTE has been seen to generate an Rte_WaitableDatum structure having 3 members, but being initialized with only a 2-member initializer, causing a compiler error.

(359169)

Incomplete support for SCALE_LINEAR_AND_TEXTTABLE and SCALE_RATIONAL_AND_TEXTTABLE

Although CompuMethod categorys `SCALE_LINEAR_AND_TEXTTABLE` and `SCALE_RATIONAL_AND_TEXTTABLE` are now accepted for the purposes of generating enums, RTA-RTE does not know how to handle any applicable compatibility rules for these categorys of CompuMethod.

(359170)

Incorrect McSupportData generation for DataPrototypes typed by complex ApplicationDataTypes

Except for StaticMemorys (see new feature 390119), McDataInstances exported for DataPrototypes typed by complex ApplicationDataTypes contain sub-element McDataInstances named after the member names of ApplicationRecordDataTypes, not the names of the corresponding record members in the mapped ImplementationDataType. Additionally, resulting properties such as SwBaseTypeRef are not exported from the corresponding elements of the ImplementationDataType; if an explicit mapping of an element's ApplicationDataType to an ImplementationDataType exists then the properties of that ImplementationDataType can be reported, but this type might differ from the corresponding element of the ImplementationDataType that the ApplicationDataType of the DataPrototype is mapped to.

(369794)

Incorrect "keep" behaviour in signal invalidation in intra-ECU case

AUTOSAR 4.0.x rte_sws 7031 states that, in the case of data invalidation with `keep` semantics, RTA-RTE shall behave the same as COM for reception of the invalid

value; that is, any configured DataReceiveErrorEvent should occur, but the invalidvalue should **not** be passed to the ApplicationSw through the Read APIs. In fact, RTA-RTE **does** copy the incoming value to the receive buffer.This has no effect in inter-ECU configurations using AUTOSAR COM, since COM correctly passes the last valid value up to RTA-RTE. But in intra-ECU configurations the copied value is the invalidvalue not the last valid value.

(370788)

Signal size for client-server sequence counter is not validated

For inter-ECU client-server connections a sequence counter is used to associate requests with responses, as required by AUTOSAR. A 16 bit unsigned integer type is used for sequence counters and it is vital that the COM signals mapped to sequence counters are configured as 16 bit for correct operation but RTA-RTE does not currently validate this.

(371173)

Handling of invalidValue for Complex Types

Formerly, the AUTOSAR specification was not clear on how to handle invalidation of complex datatypes. RTA-RTE required that a complex data type had `invalidValue`s specified either for all its leaf members or for none. Invalidation was implemented by copying the invalid values to the signals for all leaf members of the complex data item. From AUTOSAR Revision 4.0.3, the standard specifies how invalidation of complex data shall be performed: by using the `invalidValue` for the first leaf member only. This is currently not implemented in RTA-RTE.

Note, this change will potentially alter the data sent over the bus in the case of invalidatable complex data items compared to previous releases of RTA-RTE. In particular, receivers of invalidatable complex data should be upgraded no later than the senders.

Right now partial invalid values for complex data items are accepted with warning 2025 (see issue 372795), but are not supported in the code generation step.

(372800)

Error 1502 when using SubElementMappings with nested complex implementation datatypes defined inline

The following scenario is not expected to occur in practice, but is possible within AUTOSAR:

- A complex ImplementationDataType contains ImplementationDataElements which are themselves complex (*e.g.* category STRUCT or ARRAY)
- This complex ImplementationDataType is used to characterise port data
- A PortInterfaceMapping attached to the AssemblyConnector references the DataPrototype characterized by the complex ImplementationDataType
- The PortInterfaceMapping contains some SubElementMapping referencing some child of the ImplementationDataElement.

This pattern is not currently supported in RTA-RTE. Current result is that RTA-RTE raises error 1502 stating that an ImplementationDataType was

expected when an ImplementationDataElement was found.

(378234)

TextTableMappings do not work for ImplementationDataTypes in client server

Texttable conversion does not work across client-server connections when the endpoints are characterized by ImplementationDataTypes.

(381683)

Limits are ignored in CompuScales

`Limit`s are ignored in `CompuScale`s except in `CompuMethod`s of category `TEXT_TABLE` where they define enumerations.

(382143)

NetworkRepresentations need review

Network Representations were implemented as a prototype for R4.0.1 and have not been updated to conform with R4.0.2 or R4.0.3. It is not recommended to use Network Representation at this time.

(382146)

Compatibility of ImplementationDataTypes mapped to ApplicationDataTypes not fully validated

RTA-RTE does not fully validate the compatibility of ImplementationDataTypes mapped to ApplicationDataTypes by DataTypeMaps according to all the rules specified by AUTOSAR. For record and array types shape compatibility is validated. For primitive (category VALUE) types and the primitive leaf elements of record and array types it is validated that the range is sufficient. No other checks are made for these types and no validation occurs at all for boolean, string, curve, map and axis types.

(384302)

SubElementMappings not honoured for client-server connections

At present, SubElementMappings on DataPrototypeMappings are not honoured within ClientServerInterfaceMappings, only within VariableAndParameterInterfaceMappings.

(384455)

Data conversion of primitive elements of complex types does not work for client-server connections

At present linear or text-table data conversion on the primitive leaf elements of complex types is supported only for sender-receiver communication (see issue 380241) not for client-server communication.

(384473)

TextTableMappings are applied regardless of referenced CompuMethod

RTA-RTE now implements (for intra-ECU sender-receiver communication) TextTableMappings according to rte_sws 3833. The mapping code is generated whenever a TextTableMapping is aggregated into the DataPrototypeMapping in the VariableAndParameterInterfacemapping referenced by the AssemblySwConnector. However AUTOSAR 4.0.3 TPS_SoftwareComponentTemplate hints that the existence criteria should also depend on the DataPrototypes being enumera-

tiontypes — in Table 4.20: The textTableMapping is only applicable if the referred DataPrototypes are typed by AutosarDataType referring to CompuMethods of category `TEXTTABLE`.

(384505)

Invalid code generated for shared mode queues with SchM-assigned or common mode machine instances using OnExit or OnTransition ModeSwitchEvents

Incorrect code is generated for shared mode queues for tasks with SchM-assigned or common mode machine instances that use OnExit or OnTransition ModeSwitchEvents. For SchM-assigned mode machine instances the generated code fails to compile since it references an undeclared variable. For common mode machine instances the generated code compiles but can fail at runtime.

(392323)

Inter-ECU transmission acknowledgement does not work when an ECU_EXTRACT is used

RTA-RTE only honours a request for transmission acknowledgement for a provided data item when there exists a `Connector` to a require `Port`; when a `System` of `category ECU_EXTRACT` is used so only signal mappings exist for the provided data item, no `Connectors`, the request for transmission acknowledgement is ignored and the `COMCbkTAck` callback is not generated.

(394617)

Example application has wrong PlatformTypes file

The shipped AUTOSAR_PlatformTypes file out of date. This will be fixed in the next release.

(394905)

Runtime range checking not supported

RTA-RTE does not currently support runtime range checking; the `handleOutOfRange` setting is ignored.

(412195)

IncludedDataTypeSets can refer to types with the same name in different packages

It is possible to define two types with the same name and place each type in a different package. Both of these types can then be referenced from an `IncludedDataTypeSet`. RTA-RTE will not detect this invalid configuration.

(413256)

DEST is not checked for references in FlatInstanceDescriptor

The checking of DEST attributes when validating FlatInstanceDescriptors is limited to P and R ports.

(413433)

Validation of FlatInstanceDescriptors does not check the referenced Instance is in the ECU to be generated

Although FlatInstanceDescriptors are now validated if and only if they are in the FlatMap referenced by the RootSwComposition, no check is currently made at validation time that the target object is instantiated in the ECU extract.

(413434)

Initialisation values are ignored for inter OS Application IRVs in split SWCs

When an inter-runnable variable is accessed by runnables of a split SWC that are mapped to different OS applications then the initialisation value assigned to the inter-runnable variable is ignored.

(416795)

Implicit writer of complex data must write whole type

Because of the AUTOSAR rules concerning the visibility and propagation of values in implicit communication, IWriteRef does not return a reference to the global buffer for a data item participating in Implicit Communication, but to an "Implicit Write Buffer" which is not initialized.As a result, when the data item is of a complex data type and IWriteRef is used, it is necessary to write all members of the complex type, or undefined values can propagate across the communication paths.This is not regarded as a bug in RTA-RTE but a shortcoming of the AUTOSAR spec.

(417733)

iocNeeds is wrong for Client-server in split SWC

When `--deviate-split-swc-support` is in use and a client-server communcation crosses an OS protection boundary, `iocNeeds` can incorrectly request an IoC channel where both endpoints are on the same `OsApplication`. This does not apply to servers entered by direct function calls, servers where `--deviate-split-swc-support` is not used, or servers that are in the same `OsApplication` as their client.

(418107)

Corner cases concerning enums, ApplicationDataTypes, Implementation-DataTypes and TYPE_REFERENCEs

Currently, if RTA-RTE is given an `IncludedDataTypeSet` that references an `ApplicationDataType` that is mapped to an `ImplementationDataType`, and both types reference `CompuMethod`s that describe enumerations (i.e. contain point ranges) then RTA-RTE writes all the enumeration literals for both types, using the `IncludedDataTypeSet . literalPrefix`. This is due for review. See AUTOSAR bugzilla issue 72232.

(418862)

QAC warnings/MISRA violations identified

The following QAC/MISRA/Compiler issues were found with RTA-RTE generated code.

- QAC_3313/MISRA Rule 18.1 due to forward declaration of Component Data Structures
- Compiler warning: initialization discards 'volatile' qualifier from pointer target type due to used of `volatile const`
- QAC_605 A declaration must declare a tag or an identifier. (*e.g.* `DeclareResource(RTE_RESOURCE);`)
- QAC_617 'const' qualifier may not appear more than once. (*e.g.* `extern CONST(Rte_SelfType_xx, RTE_DATA) Rte_Self_xx` where `Rte_SelfType`

is already declared `const`)

(426062)

ParameterAccess does not propagate SwInterpolationMethod across connections for calprms of category VALUE

As a side effect of the work for issue 443168, RTA-RTE no longer supports using a `ParameterAccess` to set the `SwInterpolationMethod` across a connection for DataPrototypes of category `VALUE`.Prior to RTA-RTE v5.8.1, the `ParameterAccess` could be used to propagate `SwInterpolationMethod` only, and only for data of `category VALUE`. This is not thought to be a valid use case.

(450600)

FlatInstanceDescriptors do not work when applied to array or record sub-elements

If a `FlatInstanceDescriptor` references a sub-element of an array or record then the `FlatInstanceDescriptor` is silently applied to the array or record rather than the sub-element.

(450613)

DSMQ: Unconnected ModeMachineInstances lead to uncompilable code

When `ModeSwitchEvent`s are mapped to a task participating in DSMQ but there is no mode manager for that ModeMachineInstance, RTA-RTE generates uncompilable code.This scenario is not expected to occur in the field.

(452681)

DSMQ: Possible issue with Common ModeMachineInstances

When a common mode machine instance participating in DSMQ has ModeSwitchEvents, none of which are mapped to the highest-prio task used by the DSMQ, RTA-RTE may fail to check that RTE is initialized (as required in ModeSwitchEvents in common ModeMachineInstances).

(453124)

DSMQ: The limit on the number of ModeMachineInstances is not checked

It is a design limit of DSMQ that only 127 ModeMachineInstances can participate in the DSMQ. RTA-RTE does not reject the configuration if more ModeMachineInstances are present. In the case that more than 127 ModeMachineInstances participate in DSMQ, RTA-RTE may generate code that fails at runtime.This issue has not been seen in the field and has not been verified.

(453127)

DSMQ: Multiple RteDsmq containers will be ignored

The multiplicity of RteDsmq container is 0..1. This is not enforced by RTA-RTE. If more than one RteDsmq container is present, then all are ignored and no Distributed Shared Mode Queue functionality will be generated.

(453135)

Cannot EnableTakeAddress on Implicit Read/Write API

RTA-RTE currently implements AUTOSAR 4.0.3 semantics regarding the `PortAPIOption EnableTakeAddress`, that is, the following APIs are always implemented as macros (`rte_sws_1156`) and cannot be used with `EnableTakeAddress`:

`Rte_Pim`, `Rte_IRead`, `Rte_IWrite`, `Rte_IWriteRef`, `Rte_IStatus`, `Rte_IrvIRead`, `Rte_IrvIWrite`. AUTOSAR 4.2.2 alters the list above to `Rte_Pim`, `Rte_IrvIRead`, `Rte_IrvIWrite` and `Rte_IrvIWriteRef`, but has not yet been implemented.

The change in RTE-13785 has updated the above list. `Rte_IRead`, `Rte_IWrite` and `Rte_IWriteRef` now support `EnableTakeAddress`.

(454627)

swCalibrationAccess on ProvidedModeGroupPrototype in BSW is ignored

Although current-mode variable of Common Mode Machine Instances can be measured by RTA-RTE, only the swCalibrationAccess set on the ASW side is honored. Currently, in a Common Mode Machine Instance, the Rte-prefixed current-mode variable is exported, though it would be desirable to have the Schm-prefixed current-mode variable in this case, since it is correct before the `Rte_Start` lifecycle API is called.

(464026)

DSMQ Callbacks for synchronized mode group sporadic error

When you have synchronized mode groups in DSMQ, RTA-RTE sometimes generates callbacks for the ASW facet of the synchronised runnable, and sometimes for the BSW facet. This can change for the same input model if RTA-RTE is loaded in a different location in memory.

(482808)

Reporting of callbacks, hooks etc to BSWMD

RTA-RTE reports only the shortNames and callTypes of the `BswModuleEntry`s exported to support the reporting of `OutgoingCallbacks` and `ExpectedCallbacks` in Rte_BSWMD.arxml.

Information concerning the arguments and return types of these functions is not reported.

(482812)

DataSendCompletedEvents and DataWriteCompletedEvents do not fire on transmission failure

When a provided data item has transmission acknowledgement configured and is transmiteed via a method other than LdCom, then a failure to transmit results in the `Rte_Feedback` API returning `RTE_E_TIMEOUT` after the timeout period has expired but any configured `DataSendCompletedEvent`s or `DataWriteCompletedEvent`s do not fire. At present RTA-RTE only activates `RunnableEntities` or releases `WaitPoint`s associated with such events when transmission completes successfully before the timeout period expires. In some cases it is possible to work around this limitation by configuring the relevant `Rte_TimeoutAlarm` to perform the necessary action to activate the relevant `RunnableEntity` or set the relevant `OsEvent` upon expiry.

This known issue was updated in RTE-15584

(494429)

Wrong values received in 1:N implicit sender-receiver communication with different receiver settings

In the case that there are multiple implicit receivers in a 1:N sender-receiver

communication whose accessing runnables are mapped to the same task, but that should read different values because of different data types, data conversion, filters, etc. for the different `Connector`s, RTA-RTE currently generates incorrect code. In such a scenario the values written are correctly transformed as necessary and stored in separate global receive buffers for the different receiver representations, but a single implicit shadow buffer is used for all the receivers in the task, filled from an arbitrary choice of global receive buffer. Therefore the receivers all read the same values which will be the wrong values in some cases.
(496920)

3 Known Differences Between OS Versions

The name of the RTA-RTE executable

On Windows the executable is named `RTEGen.exe` on Linux the executable is named `RTEGen`.

(RTE-15023)

Stack size requirements

WARNING

When invoking RTEGen on Linux ensure that `ulimit -s` returns a value of 1024 or greater.

Failure to do this may lead to various unexpected errors being raised by RTEGen such as segmentation faults.

(RTE-14131)

Superficial differences in message wording

The wording of some error messages may vary between Windows and other operating systems. The differences detected are not consequential, such as changing from "`Access is denied`" on Windows to "`Permission is denied`" on Ubuntu20.04.

(RTE-13890)

Superficial differences in messages with ——incremental-build

When the command line option `--incremental-build` is enabled and RTEGen attempts to overwrite an existing file the precise wording of error 1291 varies between Windows and Linux. Additionally, Linux provides more validation when writing to a temporary directory (optionally specified by the environment variable `TMP`). On Linux error 1291 is raised if that directory is not writable, does not exist, or is an ordinary file.

(RTE-13891)

4 Release History

4.1 Version 12.7.1

Version 12.7.1pr1

Accept split BswImplementations

RTA-RTE now accepts input configurations where split `BswImplementation`s are provided. Previously RTA-RTE raised error 3034.

(RTE-19703)

Shared Module Interlink Header files

When there are multiple implemented `BswInternalBehavior`s that result in the same Module Interlink Header file name, RTEGen now generates all relevant prototypes, guards, etc. in that Module Interlink Header file.

(RTE-19558)

Use SenderId in IocWrite APIs

An issue has been resolved where calls to `IocWrite` APIs were output without `SenderId`s for N:1 inter-ECU reception sender-receiver communication graphs that involved `ISignal`s referenced by unused `ISignalGroup`s.

(RTE-19621)

Code formatting with RTE_LOCAL

The format of the configurable storage-class specifier macro `RTE_LOCAL` (see RTE-13167) has been updated to appear on the same line as the related object.

(RTE-13900)

Version number removed from RTEGen.ini

The `RTEGen.ini` file has been updated to remove obsolete version information.

(RTE-7029)

Correct naming of BSW Schedulable Entity Prototypes

An issue has been resolved where when there are multiple implemented `BswInternalBehavior`s containing `BswSchedulableEntity`s referencing different `BswModuleEntry`s, the BSW Module Interlink Header file would use the `BswModuleEntry` referenced within the first alphanumerically sorted `BswInternalBehavior` for all `BswImplementation`s. Note however that it was correctly generated in `Rte.c`.

(RTE-19559)

4.2 Version 12.7.0

Version 12.7.0pr10

Correct SchM API guards

When there are multiple `BswImplementation`s, RTA-RTE now generates the correct pre-compiler API guard definitions (see SWS_Rte_02751 R24-11) for the relevant `BswModuleEntity` in Module Interlink Header Files.

(RTE-19715)

Version 12.7.0pr9

Set the update flag in resource protection

RTA-RTE now sets the update flag(s) in the same concurrency protection as the global data is copied into the RTE managed global buffer in COM/LdCom callbacks where no transformation is taking place.

(RTE-19622)

Version 12.7.0pr8

Improve performance of RTEGen

The performance of RTEGen, in terms of time taken to generate code, had regressed for inputs involving `DataTypeMapping`s and has now been restored to previous levels.

(RTE-19644)

Clarification to previous Issue

RTA-RTE does not currently support naming APIs based on the `BswSchedulerNamePrefix.symbol` and derives the `<bnsp>` from the `BswModuleDescription shortName`. Issue RTE-17948 updated.

(RTE-19602)

Update to the SPDX SBOM and OSS Attribution Document

The SPDX SBOM file and the RTA-RTE OSS Attribution Document have been updated to include the latest license information.

(RTE-19484)

Accept BswImplementations with no BehaviorRef

RTA-RTE now accepts `BswImplementation`s that do not reference a `BswInternalBehavior` provided that they are not used - they are not referenced by the command line option `--contract` in Contract Phase and they not referenced in an `RteBswModuleInstance` in Generation Phase.

(RTE-19591)

RIPS with re-entrant runnable activated in a basic OsTask

RTA-RTE no longer generates RTE managed code related to extended tasks when a RIPS relevant `RunnableEntity` with `canBeInvokedConcurrently` toggled `TRUE` is present in a basic task (see issue RTE-18861).

(RTE-19425)

Updated VSMD formatting

The VSMD file shipped with RTA-RTE has been updated to clean up formatting and inconsistencies with comments.

(RTE-19399)

Replaces tabs with spaces in template files

RTA-RTE templare files have been updated to consistently use spaces (previously both tabs and spaces were used).

(RTE-19324)

Guard typeEmitter header files includes with a command line option

RTA-RTE now only includes the header file defined in the `typeEmitter` of an `ImplementationDataType`, if the value of the attribute typeEmitter ends with `.h` **and**

the command line option `--type-emitter-header-includes` is provided in the input.

This updates the work done in RTE-17266 to require an option, by default no includes are generated.

(RTE-19566)

Version 12.7.0pr7

Support activation of BswSchedulableEntity

RTA-RTE now generates ISR bodies containing the activation of `BswSchedulableEntity`s that are mapped to `BswTimingEvent`s using `RteBswEventToIsrMapping`s.

(RTE-19341)

Added a Known Issue about ISR2 support

See Known Issue RTE-19562.

(RTE-19327)

One OsResourceAccessingApplication Per OsResource

RTA-RTE now generates exactly one `OsResourceAccessingApplication` for each `OsResource`. Previously, all `OsApplication`s would reference each `OsResource`.

(RTE-19339)

Support activation of RunnableEntity by an event mapped to an ISR

RTA-RTE now generates ISR bodies containing the activation of `RunnableEntity`s that are mapped to `TimingEvent`s using `RteEventToIsrMapping`s.

(RTE-15079)

VSA SduLength protection

RTA-RTE now copies the size of the VSA and the payload into local buffers in the Rte_Read API within the same critical section, preventing any changes to data between checking its length and taking the copy of it.

(RTE-19177)

Support Multiple BSW Implementations in Contract Phase

RTA-RTE now generates the necessary header files and APIs in Contract Phase when there are multiple `BswImplementation`s referencing `BswInternalBehavior`s from the same `BswModuleDescription`.

Note: resolves previous issues RTE-19316 and RTE-19364.

(RTE-19347)

Data Conversion for unqueued transformed LdCOM data

In addition to RTE-17059, RTA-RTE can now perform data conversion on unqueued data before transforming and sending it to LdCOM.

See Known Issue RTE-19413. Floating point data conversion is not supported for this use case.

See Known Issue RTE-19511. Queued semantics support.

(RTE-17074)

Data-Conversion for unqueued transformed LdCOM data

RTA-RTE can now perform data conversion on unqueued data received via LdCOM. This is only supported in the LdCOM callback routine which is enabled via the EcuC option `RteXfrmInvUnqueuedDataCbk`.

See Known Issue RTE-19413. Floating point data conversion is not supported for this use case.

See Known Issue RTE-19511. Queued semantics support.

(RTE-17059)

Data Conversion for unqueued transformed COM data

In addition to RTE-17053, RTA-RTE can now perform data conversion on unqueued data before transforming and sending it to COM.

See Known Issue RTE-19413. Floating point data conversion is not supported for this use case.

See Known Issue RTE-19511. Queued semantics support.

(RTE-17051)

Version 12.7.0pr6

Update to the RTA-RTE Reference Manual

The RTA-RTE Reference Manual has been updated to include the new parameter `RteSerialDataE2EInTransformerChain` following the VSMD update in RTE-17477.

(RTE-19400)

Disable cast-suppression

Cast suppression implemented in RTE-14090 is currently disabled.

(RTE-19470)

Reset the update flag in Rte_Read

RTA-RTE now resets the update flag in `Rte_Read` in the same concurrency protection as the global data is copied into a local buffer and before invoking any retransformer functions. This updates the change made in issue RTE-17082.

(RTE-13887)

Do not raise unnecessary error message for disabled server

RTA-RTE no longer raises unnecessary error message E001335 when generating `PreCompileTime` variation code in `Rte_Call` APIs for disabled servers.

(RTE-19490)

Validation of TransformationISignalProps

RTA-RTE now validates the network representation properties defined by the `TransformationISignalProps` for an `ISignal`. In the case that the `TransformationISignalProps` aggregates at least one `DataPrototypeTransformationProps`, RTA-RTE raises error message 100197 if all leaf elements of the `AutosarDataType` defining the port data are not referenced by a `DataPrototypeTransformationProps`. RTA-RTE then validates that for each `DataPrototypeTransformationProps`, if there is at least one `NetworkRepresentationProps`, then all `DataPrototypeTransformationProps` aggregate a `NetworkRepresentationProps`, otherwise error message 100195 is raised. Finally, a `NetworkRepresentationProps` must at least reference a `SwBaseType` which

defines the C-type of the network representation, otherwise error message 100198 is raised.

(RTE-17060)

Reject unmapped BswInterruptEvents

RTA-RTE now raises error 100194 for each `BswInterruptEvent` in the input configuration that is not mapped to an `ISR2` via an `RteBswEventToIsrMapping` violating `[SWS_Rte_08924]@R23-11`.

(RTE-15174)

Update to the RTA-RTE Reference Manual

The RTA-RTE Reference Manual has been updated to include information about *Memory Allocation Keywords* for ISRs.

(RTE-15145)

Version 12.7.0pr5

Add RteSerialDataE2EInTransformerChain to VSMD

RTA-RTE's VSMD has been updated to include the new parameter `RteSerialData/RteSerialDataE2EInTransformerChain`.

(RTE-17477)

Data Conversion for unqueued transformed COM data

RTA-RTE can now perform data conversion on unqueued transformed data received via COM. This is only supported in the COM callback routine which is enabled via the EcuC option `RteXfrmInvUnqueuedDataCbk`.

See Known Issue RTE-19413. Floating point data conversion is not supported for this use case.

See Known Issue RTE-19511. Queued semantics support.

(RTE-17053)

Support activation of BswInterruptEntitys

RTA-RTE now generates ISR bodies containing the activation of `BswInterruptEntity`s that are mapped to `BswInterruptEvent`s using `RteBswEventToIsrMapping`s.

(RTE-15120)

Improvement to the RTA-RTE Reference Manual

The RTA-RTE Reference Manual has been updated to correct the `Type` used in an ARXML example.

(RTE-19439)

Type cast for last retransformer call in reception

RTA-RTE now type casts the third parameter of the last retransformer call in the generated `Com/LdCom` callback when the data element received is an array.

(RTE-19095)

Output activation guards for mixed inter/intra ECU

RTA-RTE now outputs activation flags guarding runnable activation for mixed inter/intra ECU client server communication where client and server runnables are mapped to the same task, even when the command line option

`--deviate-prefer-no-empty-executions` is enabled.

(RTE-14862)

Applying suffixes to VariationPointProxys

RTA-RTE now includes an appropriate suffix for the `#defines` generated in the RTE Configuration Header File for `VariationPointProxy`s with `bindingTime = PreCompileTime` when that `VariationPointProxy` references an `ImplementationDataType` (see R23-11 [SWS_Rte_06513]).

(RTE-5671)

Version 12.7.0pr4

Reject multiple BSW APIs with the same name

RTA-RTE now raises error 100192 when two or more BSW APIs would have the same C symbol.

(RTE-17949)

Support preCompileTime VariationPoints configured on the server port

RTE now supports `PreCompileTime variability` where the `bindingTime` is set to `preCompileTime` on a server port `VariationPoint`.

(RTE-18565)

Support mode user/manager with compatible ModeDeclarationGroups

RTA-RTE now generates the same output when a `mode user` and associated `mode manager` reference different but compatible `ModeDeclarationGroup`s with category `EXPLICIT_ORDER`. The `ModeDeclarationGroup`s may be mapped to the same or compatible `ImplementationDataType`s.

(RTE-19149)

Known issue added

RTA-RTE does not include the `vendorId` and `vendorApiInfix` in BSW APIs and header files when in Contract Phase. See known issue RTE-19364.

(RTE-19371)

Rename SchM_Enter and SchM_Exit APIs

RTA-RTE now generates unique names for `SchM_Enter` and `SchM_Exit` APIs with multiple `InternalBehavior`s by including the `ExclusiveArea shortName` and, when provided, the `vendorId` and `vendorApiInfix`. See Known Issue RTE-19316.

(RTE-18070)

Suppress redundant C-casts

RTA-RTE no longer generates a C-cast when the target type is the same as the cast type.

(RTE-14090)

Rename BswModuleEntitys with multiple InternalBehaviors

RTA-RTE now renames `BswModuleEntity`s in the generated output when the associated `BswModuleDescription` has multiple `InternalBehavior`s. RTA-RTE includes the `<bnsp>` derived from the `BswModuleDescription shortName` in addition to `vendorId` and `vendorApiInfix` if they are configured in the associated `BswImplementation`. See Known Issue RTE-19316.

(RTE-17948)

Reject BswModules with multiple InternalBehaviors and SwcBswMapping

RTA-RTE now raises error 100118 when a configuration contains a `BswModuleDescription` with multiple `InternalBehavior`s and one (or more) of those `InternalBehavior`s is mapped to a `SwcInternalBehavior` via a `SwcBswMapping`.

(RTE-17950)

Version 12.7.0pr3

RIPS with re-entrant runnable can implement basic OsTask

With RIPS enabled, a `RunnableEntity` with `canBeInvokedConcurrently` toggled `TRUE` will not cause the `Task` body to become an extended task.

(RTE-18860)

Improve performance of RTEGen on Linux

The performance of RTEGen on Linux, in terms of time taken to generate code, had regressed with the addition of a combination of recent features and has now been restored to previous levels.

(RTE-18950)

Improved whitespace consistency

RTA-RTE now generates more consistent whitespace, for example, in the initialization of `struct`s.

(RTE-19186)

AliveTimeout variables for LdCom

RTA-RTE now defines the `aliveTimeout` counter variables generated for LdCom with `aliveTimeout` using the storage-class specifier `RTE_LOCAL` (defaults to `static`, see RTE-13167).

In addiction RTA-RTE now always defines the `aliveTimeout` variables in `Rte.c` and no longer generates an external declaration of the same variables in `Rte.h`.

(RTE-18859)

Basic Software Scheduler header file guards

RTA-RTE now generates the `#define`s used to guard against multiple inclusion of Module Interlink Header Files in the format `<UPPERCASE_FILENAME>_H`.

(RTE-17811)

Data type definition omitted for Rte_ or SchM_ symbols

RTA-RTE does not generate the data type definition in the `RTE Header File` for an `ImplementationDataType` if its `Implementation Data Type symbol` starts with `Rte_` or `SchM_`.

(RTE-17268)

RIPS header file inclusion

RTA-RTE now includes `RIPS BSW-SWC instance header file` in `Application Header File` only if that `Software Component` has access to a `Client Server Communication Graph` with enabled `RTE Implementation Plug-In`. Previously that file was included regardless of the access being configured or not.

(RTE-18932)

Reduced whitespace

RTA-RTE now produces fewer blank lines in the generated files. Previously, when the command line option `--disable-comments` was used, the generated output could contains several blank lines in a row (depending on the input configuration).

(RTE-16707)

Version 12.7.0pr2

Optimize transformer calls for COM reception with fanning

RTA-RTE now only calls a transformer chain once on the reception side of inter-ECU communication fanning to multiple receivers with an initial value configured for some but not all of those receivers and transformation taking place in the callback.

Note: RTA-RTE does not support initial values with transformation and generates the code as if the initial value was not present in the input.

(RTE-18169)

Update all status variable(s) with mixed handleNeverReceived

RTA-RTE now updates all relevant status variables in the generated `Rte_Write` APIs associated with an unqueued 1 to N sender receiver communication using IOC where the receivers contain a mixture of `handleNeverReceived` set to `TRUE` and `FALSE`.

(RTE-17411)

Improved Compliance with MISRA Rule 20.7

Template files in RTA-RTE, `Rte_h.template` and `Rte_Intl_h.template`, were modified such that expressions resulting from the expansion of macro parameters are enclosed in parentheses. This addresses violations of MISRA C:2012 Rule 20.7.

(RTE-19022)

RteSerialData callback functions with transformerError

RTA-RTE now passes the argument `transformerError` to the generated `RteCallBackFunction` when the relevant `RteSerialData` configuration has the parameter `RteSerialDataE2EInTransformerChain` set to `TRUE`, if not set or set to `FALSE` then no argument is passed.

(RTE-17475)

Measurement of data in PRPorts

RTA-RTE now outputs appropriate measurement related buffers in the generated `C` code and associated reporting in `Rte_McSupportData.arxml` for `PRPorts` with measurement enabled that are not connected to any other `PortPrototype`.

(RTE-18809)

Include files referenced via the typeEmitter attribute in Rte_Type.h

If the input configuration contains an `ImplementationDataType` with the attribute `typeEmitter` set to a value that ends with `.h` RTA-RTE will now add a `#include` of the provided value in `Rte_Type.h`.

(RTE-17266)

Version 12.7.0pr1

Improved formatting and additional suffixes

RTA-RTE now generates more consistent spacing around `C` operators and adds appropriate suffixes to more constants used in expressions.

(RTE-18641)

Do not error for FUNCTION_REFERENCE ImplementationDataTypes

RTA-RTE now accepts `ImplementationDataType` of category `FUNCTION_REFERENCE` when the attribute `TypeEmitter` ends with `.h`. Previously RTA-RTE rejected the configuration with error 2888.

(RTE-18056)

Set the enableUpdate flag inside the Rte_Read API

RTA-RTE now sets the update flag to `FALSE` in the `Rte_Read` API generated for a data element that has the `enableUpdate` attribute enabled and is read inter-ECU (via COM or LdCom), where transformation is performed in the callback and where IOC is not used.

(RTE-18330)

Deprecated option ——measurement

The command line option `--measurement` has been deprecated, as well as its abbreviation `-ms`. RTA-RTE now raises warning 2976 if `--measurement`, or its abbreviated version, `-ms`, is used. The `--measurement` option should not be used for new projects, and where possible existing projects should be modified to no longer use it.

(RTE-18855)

Type cast for readers in 1:N communication with IOC

RTA-RTE now type casts the data parameter of the generated `IocRead` APIs when called as part of a 1:N inter-`OsApplication` communication graph where the IOC channel and receiver associated with the `IocRead` have different types.

(RTE-18581)

4.3 Version 12.6.0

Version 12.6.0pr6

The RTA-RTE User Guide has been updated

The formatting of the RTA-RTE User Guide has been improved.

(RTE-18964)

Version 12.6.0pr5

Improved Compliance with MISRA Rule 20.7

RTA-RTE now encloses the macro parameters in parentheses when generating `Rte_Write` and `Rte_IrvWrite` as macros in the relevant application header files. This addresses violations of MISRA C:2012 Rule 20.7.

(RTE-16057)

Handle static range checking for scaled CompuMethods

RTA-RTE now correctly handles static range checking for `CompuMethod`s of

category SCLALE_LINEAR_AND_TEXTTABLE. Previously, the upper and lower limits were not correctly honouring the numerical CompuScale of the CompuMethod or RTA-RTE was raising error E002668 due to incompatible internal ranges.

(RTE-18934)

Accept compatible initial values specified in unsigned and signed integers

RTA-RTE now accepts initValues of type Unsigned Integer and Signed Integer and outputs them correctly resolving an issue introduced in an internal refactor related to RTE-14515 where values with the highest order bit set were always output as negative numbers.

(RTE-18895)

Fix issue with optimized resource locking

RTA-RTE now correctly identifies the involvement of a second OsTask involved in a communication, for example, the direct activation in an Rte_Write API. This resolves an issue with the optimization of resource locking for single OsTask configurations added in RTE-18837.

Note: As part of this fix some instance of Rte_GetResource / Rte_ReleaseResource have updated to use Rte_SuspendOSInterrupts / Rte_ResumeOSInterrupts.

(RTE-18873)

Improvements to the RTA-RTE Referenece Manual

The RTA-RTE Referenece Manual has been updated to state that variables explicitly initialized to 0 will go into a VAR_INIT memory section and not, as previously stated, into a VAR_CLEARED memory section.

(RTE-18828)

Accept compatible initial values specified in boolean format

RTA-RTE now accepts initValues of type Boolean when they are provided in more than one place for the same communication with different textual values (for example, 0 vs 0.0). Previously, RTA-RTE detected these values as different and rejected the configuration with error 2009.

(RTE-18882)

RTA-RTE VSMD file updated

The VSMD file shipped with RTA-RTE has been updated and includes the addition of new options for handling the behaviour of ExclusiveAreas, including the addition of RteUserDefinedExclusiveArea and related references (including the USER_DEFINED enum for RteExclusiveAreaImplMechanism) and the option RteBswEnforceExclusiveAreaApisGeneration. Please consult the release notes or the RTA-RTE Reference Manual for descriptions of any new options.

(RTE-18829)

Handle data conversion for scaled CompuMethods

RTA-RTE now performs data conversion for CompuMethods of Category SCALE_LINEAR_AND_TEXTTABLE. Previously, RTA-RTE silently did not honor any data conversion implied by the configured CompuMethod. This also fixes the issue of validating DataTypeMappings, since, previously, RTA-RTE did not derive the correct internal representation of the physical limits when ensuring that the ImplementationDataType / SwBaseType could hold the range of possible values.

(RTE-18167)

Pre-defined OsResources for SchM_Enter/Exit APIs must be STANDARD

Previously in some use cases in which `SchM_Enter` and `SchM_Exit` APIs were being generated RTA-RTE would erroneously demand that an `OsResource` of type `INTERNAL` be used and ignore a pre-defined `OsResource` of type `STANDARD` from the input configuration. In this case RTA-RTE would use and output a new `OsResource` of type `INTERNAL` and raise warning W006800. RTA-RTE now requires any pre-defined `OsResource` used within a `SchM_Enter` and `SchM_Exit` API (referenced via `RteBswExclusiveAreaOsResourceRef`) to be of type `STANDARD` in all cases or error E100189 will be raised.

(RTE-18629)

Fix OsResource handling in Enter and Exit APIs

In some cases, when an `Enter` and `Exit` API were being generated, RTA-RTE would fail to output a new `OsResource` within the `osNeeds.arxml` file or would inappropriately generate one when a pre-defined `OsResource` had been provided in the input configuration. These cases have now been fixed.

(RTE-18635)

Version 12.6.0pr4

Improve performance of RTEGen

The performance, in terms of time taken to generate code, had regressed in **RTA-RTE 12.6.0pr1** and has now been restored to previous levels.

(RTE-18862)

Initialization of ApplicationPrimitiveDataTypes of category VAL_BLK

RTA-RTE now ignores the referenced `SwRecordLayout` when initializing an `ApplicationPrimitiveDataType` of category `VAL_BLK`.

(RTE-18414)

Version 12.6.0pr3

ExclusiveAreas in different BSW modules may share OsResources

RTA-RTE now allows multiple `BswModuleDescriptions` to use the same backing `OsResource` in `SchM_Enter` and `SchM_Exit` APIs when referenced via an `RteBswExclusiveAreaOsResourceRef`.

(RTE-18680)

Optimize resource locking for one Task simple S/R configs

The behavior of RTE-18341 has been extended to cover scenarios where the intra-partition communication is connected to an `outerPort` via a `DelegationSwConnector` but no inter-ECU connections are configured.

(RTE-18837)

ExclusiveAreas in different BSW modules may share OsSpinlocks

RTA-RTE now allows multiple `BswModuleDescriptions` to use the same backing `OsSpinlock` in `SchM_Enter` and `SchM_Exit` APIs when referenced via an `RteBswExclusiveAreaOsSpinlockRef`.

(RTE-18671)

Correct behavior of type compatibility checks and range defines

RTA-RTE type compatibility checking now allows comparisons between `ApplicationDataTypes` and `ImplementationDataTypes`. In addition RTA-RTE now only outputs range defines for `ApplicationDataTypes` when they have a `DataConstr` of type `PhysConstr`.

This improves the behavior previously modified in RTE-16354 and RTE-18163.

(RTE-18534)

Use the BaseTypeEncoding when interpreting upper/lowerLimits

RTA-RTE now determines the type of `upperLimit`/`lowerLimits` to be floating point when the `BaseTypeEncoding` is `IEEE754`. Previously RTA-RTE incorrectly identified values without a decimal point as integral types.

(RTE-15429)

Fix an issue when validating data constraints

RTA-RTE now accepts valid `DataConstrs`, previously error 1413 was raised for input values with high precision and with exponents.

(RTE-18748)

Reject CompuMethod with more than one rational CompuScale

RTA-RTE now raises error 100187 when a `CompuMethod` of `Category` `SCALE-LINEAR-AND-TEXTTABLE` aggregates more than one `CompuScale` that, itself, aggregates a `CompuRationalCoeffs`. This is due to requirement `[TPS_SWCT_01560]`. Also, error messages 100135, 100136 and 100137 have been further clarified.

(RTE-12022)

Accept compatible initial values specified in different formats

RTA-RTE now accepts `initValues` based on the underlying type when they are provided in more than one place for the same communication with different textual values (for example, `1` vs `1.0`). Previously RTA-RTE detected these as different values and rejected the configuration with error 2009.

(RTE-18709)

Further resource locking optimization for same Task S/R

RTA-RTE now optimizes resource locking for 1:1 and 1:N implicit and mixed implicit/explicit unqueued intra-partition sender/receiver communication graphs when the `HandleInvalid` attribute is set to `KEEP` or `REPLACE` for the `VariableDataPrototype` which is being transmitted and the accessing `RunnableEntitys` are triggered by `RteEvents` mapped to the same `OsTask`.

(RTE-18436)

Initialization of ApplicationPrimitiveDataTypes of category VALUE

RTA-RTE now initializes a `DataPrototype` typed by `ApplicationPrimitiveDataType` of category `VALUE` according to the referenced `SwRecordLayout`.

(RTE-18413)

Version 12.6.0pr2

Optimize resource locking for one Task simple S/R configs

RTA-RTE now optimizes resource locking for 1:1 and 1:N explicit unqueued

intra-partition sender/receiver communication graphs when the `HandleInvalid` attribute is set to `KEEP` or `REPLACE` for the `VariableDataPrototype` which is being transmitted and the accessing `RunnableEntity`s are triggered by `RteEvent`s mapped to the same `OsTask`.

(RTE-18341)

Reject configuration when no OsResource referenced for ExclusiveArea

RTA-RTE now raises error 100177 when an `ExclusiveArea` with `RteExclusiveAreaImplMechanism` set to `OS_RESOURCE` is referenced via either a `BswCalledEntity` or `BswInterruptEntity` in the `canEnter` role and no `OsResource` is referenced via the ECUC parameter `RteBswExclusiveAreaOsResourceRef`.

(RTE-17877)

Correctly initialize calibration parameters with decimal values

RTA-RTE now initializes calibration parameters for floating point types with initial values provided via constant specifications including any relevant decimal values. Additionally, RTA-RTE now accepts configurations where the same `COM_AXIS` is referenced by multiple `ApplicationPrimitiveDataType`s.

(RTE-17375)

Round upper/lowerLimits for integers based on ——initial-value-rounding

RTA-RTE now rounds the `upperLimit` and `lowerLimit` defined according to the command line option `--initial-value-rounding` when the `ImplementationDataType` is `integer`. See the RTA-RTE Reference Manual for more details.

(RTE-14514)

Reject configuration if ExclusiveArea not associated with an OsSpinlock

RTA-RTE now raises error message `100175` or `100176` for each `BswSchedulableEntity`, `BswCalledEntity` or `BswInterruptEntity` which references the `ExclusiveArea` via the `canEnter` role if that `ExclusiveArea`s associated `RteBswExclusiveAreaImpl` does not reference an `OsSpinlock` via an `RteBswExclusiveAreaOsSpinlockRef`.

(RTE-17884)

Optimize resource locking for non-preemptive tasks

RTA-RTE now optimizes resource locking for 1:1 and 1:N explicit unqueued intra-partition sender/receiver communication graphs when all `RunnableEntity`s are triggered by `RteEvent`s mapped to a non-preemptive `OsTask`s with no `RteOsSchedulePoint`s configured.

(RTE-18408)

Generate the appropriate suffix for each defined upper/lowerLimit

RTA-RTE now generates the suitable suffix for the `upperLimit` and `lowerLimit` defines as specified by `SWS_Rte_05052` and `SWS_Rte_03619` (R23-11).

(RTE-14515)

Rename OsIocInitValue with ——buffers-use-provider-name

The RTA-RTE option `--buffers-use-provider-name` now includes renaming of the `iocNeeds.arxml` parameter `OsIocInitValue` and its associated variable in `Rte.c`. See the RTA-RTE Reference Manual for more details.

(RTE-13189)

Generate SchM_Enter and SchM_Exit APIs if parameter is enabled

RTA-RTE now generates the `SchM_Enter` and `SchM_Exit` APIs in the case that the RTE ECUC configuration parameter `RteBswEnforceExclusiveAreaApisGeneration` is enabled. Without setting this parameter, RTA-RTE only generates the APIs in the case that a `ExclusiveArea` is ultimately referenced by `CanEnterExclusiveAreaRef` or `canEnter`. The implementation of the generated APIs depends on the value of the associated `RteExclusiveAreaImplMechanism`.

(RTE-17368)

Correct cast of conversion expression for ComSpec.networkRepresentation

RTA-RTE now casts the result of a data conversion expression to the correct destination type when that type is the network representation defined by a `ComSpec.networkRepresentation`. Previously, RTA-RTE would cast the final data conversion expression to an invalid type.

(RTE-18242)

Version 12.6.0pr1

MAKW for Rte_ReceiverPullCB API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition of `Rte_ReceiverPullCB` API, in the format `OS_[START|STOP]_SEC_CODE_<INFIX>`, where `<INFIX>` is the `shortName` of the associated `OsApplication`.

(RTE-17045)

DirtyFlag global variable is now always unique

RTA-RTE now always generates a uniquely named `dirtyFlag` global variable. Previously, RTA-RTE could generate a clashing global variable symbol potentially leading to incorrect dynamic behavior due two distinct `NvBlockDescriptor`s, each with the same `shortName` but in different `NvBlockSwComponentType`s, sharing the same global variable without any generated warning or error.

(RTE-17425)

Support user defined ExclusiveAreas for SchM_Enter and SchM_Exit APIs

When an `RteBswExclusiveAreaImpl` has `RteExclusiveAreaImplMechanism` set to `USER_DEFINED` and its `RteBswUserDefinedExclusiveAreaImplRef` references an `RteUserDefinedExclusiveAreaImpl`, RTA-RTE now generates a call to the function specified by `RteUserDefinedEntryFunction` and `RteUserDefinedExitFunction` in the `SchM_Enter` and `SchM_Exit` APIs respectively, RTA-RTE also generates an include statement specified by `RteUserDefinedHeaderFile`.

(RTE-17369)

Error if [SWS_Rte_CONSTR_09096] is violated

RTA-RTE now raises error 100184 when an input configuration violates AUTOSAR R23-11 constraint `[SWS_Rte_CONSTR_09096]`.

(RTE-18135)

Removed duplication of ActivateTask in Trigger API

RTA-RTE no longer activates multiple times an `OsTask` inside the Trigger API when a configuration with multiple `TriggerOccurredEvent`s mapped to that `OsTask` or to

a successor `OsTask` in the same chain is provided.

(RTE-16930)

Remove composition range defines from SWC header file

RTA-RTE no longer outputs range defines for `DataTypeMappings` to *Application Types Header Files* where the relevant `DataTypeMappingRef` is part of the parent `CompositionSwComponentType` (rather than part of the `SwComponentType` itself).

(RTE-14513)

Use referenced SpinLocks for SchM_Enter and SchM_Exit APIs

The implementation of `SchM_Enter` and `SchM_Exit` APIs now invoke `GetSpinlock` and `ReleaseSpinlock`, passing in the argument defined by the associated `RteBswExclusiveAreaImpl.RteBswExclusiveAreaOsSpinlockRef`, in the case that the `RteBswExclusiveAreaImpl.RteExclusiveAreaImplMechanism` is set to `OS_SPINLOCK`.

(RTE-17879)

Default to full interrupt blocking exclusive area

In the case that RTA-RTE needs to generate `Rte_Enter` and `Rte_Exit` APIs but the associated `ExclusiveArea` is not referenced by a `RteExclusiveAreaImplementation`, RTA-RTE assumes that the implementation mechanism is `ALL_INTERRUPT_BLOCKING` and additionally raises warning 100170. Similarly, when `SchM_Enter` and `SchM_Exit` APIs should be generated and the associated `ExclusiveArea` is not referenced by a `RteBswExclusiveAreaImpl`, `ALL_INTERRUPT_BLOCKING` is assumed and RTA-RTE raises warning 100171.

(RTE-18024)

4.4 Version 12.5.1

Version 12.5.1pr4

Support inter partition NvM service callbacks

RTA-RTE now supports port role-based NvM service callback runnables when the `OperationInvokedEvent` is mapped to an `OsTask`, the triggered `RunnableEntity` is marked as re-entrant and the option `--client-server-global-optimization` is enabled. Even though the `OperationInvokedEvent` is mapped to an `OsTask`, the invocation of the server runnable will be via direct function call. This covers the previous issue in which RTA-RTE could not determine the partition context of the server due to the ambiguity of the component being split across different `EcuCPartitions` which caused it to not fully generate the `iocNeeds` report.

(RTE-18166)

Update the global composite status in all Rte_COMCbkTOut

RTA-RTE now updates the global composite status in all `Rte_COMCbkTOut` callbacks when multiple transformed signals that are mapped to one partition are read from a different partition, transformation is taking place in the callback (`RteXfrmInvUnqueuedDataCbk` is set to `True`), IOC is disabled (by providing the command line option `--inter-ecu-no-ioc`), and there is a `NonqueuedReceiverComSpec` with non-zero `aliveTimeout`.

(RTE-18338)

Physical constraints are correctly interpreted for Data Conversion

The known issue, RTE-17942, has been fixed. Data constraints are now converted into their internal representation via the `CompuMethod` if one is supplied.

(RTE-17985)

Option ——ioc-xml-namespace removed

The command line option `--ioc-xml-namespace` has been removed, as well as its abbreviation `-iocxmlns`. RTE-RTE now exits with error 1103 if either `--ioc-xml-namespace` or `-iocxmlns` options are provided.

(RTE-16584)

Version 12.5.1pr3

MAKW for Com Callback Rte_COMCbkInv API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of `Com` Callback `Rte_COMCbkInv` API, in the format `RTE_<INFIX>_[START|STOP]_SEC_CALLOUT_CODE`, where `<INFIX>` is the `shortName` of the related `ComIPdu's` associated `EcucPartition`.

(RTE-12918)

Initialize status for RteXfrmInvUnqueuedDataCbk and handleNeverReceived

RTA-RTE now initializes the status element of the global composite variable to `RTE_E_NEVER_RECEIVED` when `handleNeverReceived` is `TRUE` and transformation is being performed in the callback (`RteXfrmInvUnqueuedDataCbk` is `TRUE`).

(RTE-18340)

MAKW for Com Callback Rte_COMCbkTOut API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of `Com` Callback `Rte_COMCbkTOut` API, in the format `RTE_<INFIX>_[START|STOP]_SEC_CALLOUT_CODE`, where `<INFIX>` is the `shortName` of the related `ComIPdu's` associated `EcucPartition`.

(RTE-12939)

Option ——have-64bit-int-types removed

The command line option `--have-64bit-int-types` has been removed, as well as its abbreviation `-int64`. RTE-RTE now exits with error 1103 if either `--have-64bit-int-types` or `-int64` options are provided.

(RTE-17479)

Automatically configure OsResourceAccessingApplications

RTA-RTE now automatically configures the `OsResourceAccessingApplication` for any generated `RTE_RESOURCE`s (see RTE-6297).

This updates issue RTE-1150.

(RTE-16646)

Applying suffixes to floats

Improved generation of suffixes [F] for 32-bit floats and consistency of generation by removing superfluous leading + and trailing zeros.

(RTE-18380)

Applying suffixes to integers

Improved generation of suffixes [U, L, UL, LL, ULL] for integers based on their type.

(RTE-18333)

Improved error message for previously raised internal error 900714

Uninformative internal error message, 900714, has been converted into a more informative and new error message, 100162, in the case that a port datum does not have the necessary properties to be mapped to a primitive signal via a `SenderReceiverToSignalMapping`. The port datum must be either a primitive type, associated with DataTransformation or a byte array.

(RTE-14126)

Update receive status for all receivers during COM port fanning

RTA-RTE now updates the status buffers for all receivers in the `Rte_COMCbk` API when fanning to multiple inner ports from a single signal with data transformation and the configuration uses multiple outer ports.

(RTE-17711)

Output OsAppCounterRef and OsAppScheduleTableRef to osNeeds.arxml

RTA-RTE now outputs the `OsAppCounterRef` and `OsAppScheduleTableRef` parameters to the generated `osNeeds.arxml` file for an `OsApplication` when they are provided in the input configuration, and the command line option `--os-output-param=all` is provided.

(RTE-16047)

Improved Compliance with MISRA Rule 20.7

Template file in RTA-RTE, specifically `Rte_h.template`, was modified such that the macros parameters in `RTE_BITSET`, `RTE_BITCLR`, and `RTE_BITTST` are properly enclosed in parentheses. This addresses violations of MISRA C:2012 Rule 20.7.

(RTE-18120)

Reject configuration when NvM service-callback invokes unmapped server

RTA-RTE now generates error message `100178` in the specific case that a server runnable is invoked from an NvM service callback where the associated `OperationInvokedEvent` is not mapped to an `OsTask` and it belongs to an `ApplicationSwComponentType` that is split across multiple `EcuCPartitions`.

(RTE-18221)

Version 12.5.1pr2

MAKW for Com Callback Rte_COMCbk API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of `Com` Callback `Rte_COMCbkTAck` API, in the format `RTE_<INFIX>_[START|STOP]_SEC_CALLOUT_CODE`, where `<INFIX>` is the `shortName` of the related `ComIPdu's` associated `EcucPartition`.

(RTE-12914)

MAKW for Com Callback Rte_COMCbkTAck API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of `Com` Callback `Rte_COMCbkTAck` API, in the format `RTE_<INFIX>_[START|STOP]_SEC_CALLOUT_CODE`, where `<INFIX>` is the `shortName` of

the related `ComIPdu's` associated `EcucPartition`.

(RTE-12920)

Support modern CanEnter ExclusiveAreas references

RTA-RTE will now generate Enter and Exit APIs for `ExclusiveAreas` referenced via the modern attribute `canEnter` (AUTOSAR R21-11) in supported configurations.

(RTE-17367)

Fix DataConstr compatibility checks

RTA-RTE no longer erroneously errors out when a `DataConstr` exists for the left side of a communication and not on the right side. This fixes an issue introduced with RTE-16354

(RTE-18163)

Remove duplicate if conditions in Rte_COMCbkTOut

RTA-RTE no longer outputs duplicate if conditions in the `Rte_COMCbkTOut` API regarding the updating of the status variable.

(RTE-17420)

Version 12.5.1pr1

Automatically create output directories

RTA-RTE now automatically creates the output directories specified by the command line option `--output` if they don't exist. If RTA-RTE isn't able to create a directory it raises error 100173.

(RTE-18025)

Use Rte_memcpy for implicit shared buffer fills in task bodies

RTA-RTE now uses `Rte_memcpy` in the task body to fill shared buffers generated for implicit access when any receiver has a `ComSpec` configured with the attributes that lead to the generation of an `Rte_IStatus` API.

(RTE-17654)

Resolve issue of Rte_memcpy argument order in Rte_Read API

RTA-RTE now invokes `Rte_mempcy` correctly in the `Rte_Read` APIs generated when there is inter-ECU unqueued S/R communication with data transformation performed in the `Rte_COMCbk`.

(RTE-18259)

Generate RTE exclusive area APIs for unmapped RTEEvents

RTA-RTE now generates the `Rte_Enter` and `Rte_Exit` APIs according to the configured `ExclusiveAreaImplMechanism` where the relevant runnable is triggered by an `RTEEvent` that is not mapped to an `OsTask`.

(RTE-15286)

4.5 Version 12.5.0

Version 12.5.0pr8

Allow multiple copies of ARXML containers

RTA-RTE now attempts to merge all input `ARXML` regardless of whether it is `atpSplitable` or not. When considering if a container is identical RTA-RTE disre-

gardscomments, whitespace and UUIDs.

Note: Accepting duplicate non-splitable containers is not AUTOSAR compliant and has a performance overhead. RTA-RTE will raise warning 100172 when this occurs.

Additionally RTA-RTE now correctly identifies CpSoftwareClusters as splitable.

(RTE-18032)

Update the global status when transformation is in the callback

RTA-RTE now always updates the global status when status is included alongside data and transformerError in a composite (managed by RTE or via IOC) and generated for explicit non-queued sender-receiver communication with transformation taking place in the Rte_COMCbk. Previously in the case that the return from COM was not E_OK, RTE did not update the global copy of composite.status.

(RTE-18023)

Added a Known Issue regarding signal fan-in support

RTA-RTE support for signal fan-in via COM with transformation is limited to unqueued sender-receiver communication where all ComSignals are mapped to the same partition. See known issue RTE-18196.

(RTE-17846)

Reset the update flag before retransforming the data in Rte_Read

RTA-RTE now resets the update flag, generated for a data element with the enableUpdate, immediately after the Rte_Read API is called and before invoking any retransformer functions.

(RTE-17082)

Generate initial values using SwRecordLayout

RTA-RTE now generates initial values for ApplicationPrimitiveDataTypes of categorys: CURVE, MAP, CUBOID, CUBE_4, CUBE_5, and COM_AXIS, by iterating through the type's SwRecordLayout based on swRecordLayoutGroupAxis and swRecordLayoutVAxis. Only values of VALUE and COUNT are supported for SwRecordLayoutV.swRecordLayoutVProp.

(RTE-17152)

A known issue has been added regarding splitting Rte into partition files

A known issue has been added regarding splitting the generated Rte files into separate partition files, most but not all code that can be moved to a partition specific file is moved, see RTE-18160.

(RTE-14431)

Ignore Internal Constraints on Application Data Types

RTA-RTE now ignores DataConstr elements of type InternalConstr set on ApplicationDataTypes when checking compatibility between two data types.

(RTE-16354)

Use of memcpy when using a composite global buffer

RTA-RTE now uses Rte_memcpy instead of direct assignment in the Rte_Read and COM callbacks to update the global composite in the case of unqueued, transformed data elements received via COM where fanning to multiple readers.

(RTE-18003)

Known issue added

RTA-RTE does not support passing arguments in trusted function calls. See known issue RTE-17080.

(RTE-16985)

Non-IOC fanning with transformation read from a different partition

RTA-RTE now reads the status from the global composite in the `Rte_Read` API(s) generated when a non-queued data element with transformation is being received inter-ECU, via COM, with the re-transformation in the callback and the command line option `--inter-ecu-no-ioc` enabled, forcing RTEGen to use RTE managed global composite instead of an IOC channel.

(RTE-17113)

Return RTE_E_COM_BUSY from Rte_Read

RTA-RTE now returns `RTE_E_COM_BUSY` from the `Rte_Read` API when the COM receive API returns `COM_BUSY` for an unqueued sender/receiver communication via COM with transformation (performed in the `Rte_Read` API). Previously any non-zero return from COM would result in `Rte_Read` returning `RTE_E_COM_STOPPED`.

Note: In scenarios other than the one described above RTA-RTE still returns `RTE_E_COM_STOPPED`.

(RTE-17761)

Move NV runnable functions to partition specific files

RTA-RTE now generates NV runnable functions in the appropriate split partition file rather than `Rte.c` when `RteSplitPerPartition` is `true` and those functions are mapped to an individual partition.

(RTE-17770)

Version 12.5.0pr7

Accept reception fanning multiple ComSignals with transformation

RTA-RTE now accept explicit unqueued sender-receiver communication with transformation where multiple signals are fanning in. Additionally, the status variable is accessed from IOC in the generated `Rte_Read` APIs.

(RTE-15668)

Allocate a single RTE global buffer for signal fan-In

RTA-RTE now allocates a common RTE buffer typed by the corresponding implementation data type (or the composite type) for the reception of signal `Fan-In` data with transformation.

(RTE-15870)

Include status in the composite type

RTA-RTE now includes the status variable in the composite type (see issue RTE-9593).

(RTE-17104)

Improved static range-check handling

When RTA-RTE generates data conversion code, it now considers any data

constraints that apply to the physical type. Previously, RTA-RTE only considered data constraints that applied to the internal type. This change also handles the case where a SystemSignal.physicalProps.dataConstr exists; previously, RTA-RTE disregarded this property. Additionally, RTA-RTE now raises warning 100168 when the range of possible values of data conversion do not fit into the destination type. The expression that performs the data conversion is now cast to the desintation type. Previously, RTA-RTE would raise no warning in the event that data conversion could potentially result in a value too large for the destination type. In addition, RTA-RTE previously and incorrectly generated an error if the `DataConstr` of a provider and requirer were not compatible, even if a `PortInterfaceMapping` existed. This is no longer the case, as per `[TPS_SWCT_01549]`.

Note: See newly added issue RTE-17942. Note: Removes previous issue 336853. Note: Removes previous issue 312170.

(RTE-17942)

Improve the searchability of the RTA-RTE Error Description

The RTA-RTE Error Description document now contains a new section listing all the variants for each message in the format that they are output by RTEGen.

(RTE-18020)

Set all events inside Rte_IModeInit

RTA-RTE now sets all events inside `Rte_IModeInit` API when a configuration with multiple `ModeDeclarationGroup`s each with a `ModeSwitchEvent` mapped to the same `OsTask` is provided. Previously RTA-RTE could generate unreachable code preventing some calls to `SetEvent` being reachable.

(RTE-17042)

Memory Allocation Keywords for global buffers initialized to zero

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of global buffers initialized to zero with the `INIT_POLICY` '`INIT`'. Previously in some cases the `INIT_POLICY` was '`CLEARED`'.

(RTE-14544)

Non-IOC fanning with transformation read from a different partition

RTA-RTE now uses a single global composite (containing `data`, `transformerError` and `status`) to transmit the retransformed data between the `Rte_ComCbk`s and `Rte_Read`(s) when multiple transformed signals that are mapped to one partition are read from a different partition and IOC is disabled (by providing the command line option `--inter-ecu-no-ioc`).

(RTE-15746)

Signal fan-in via Com with data transformation

RTA-RTE now allows signal fan-in for intra-OsApplication communication via Com with data transformation where all signals and all readers are mapped to the same partition.

(RTE-15756)

IOC for fanning Com with transformation read from a different partition

RTA-RTE now uses IOC to transmit the retransformed data between the

Rte_ComCbks and Rte_Read(s) passing the data, transformerError and status via the same IOC Channel in a composite type. IOC is used when any of the readers are in a different partition to any of the signals.

(RTE-15751)

Version 12.5.0pr6

MAKW for LdCom Callback Rte_LdComCbkTriggerTransmit API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of LdCom Callback Rte_LdComCbkTriggerTransmit API, in the format RTE_<INFIX>_[START|STOP]_SEC_CALLOUT_CODE, where <INFIX> is the shortName of the related LdComIPdu's associated EcucPartition.

(RTE-12921)

MAKW for LdCom Callback Rte_LdComCbkRxIndication API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of LdCom Callback Rte_LdComCbkRxIndication API, in the format RTE_<INFIX>_[START|STOP]_SEC_CALLOUT_CODE, where <INFIX> is the shortName of the related LdComIPdu's associated EcucPartition.

(RTE-12927)

MAKW for LdCom Callback Rte_LdComCbkTxConfirmation API

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition and declaration of LdCom Callback Rte_LdComCbkTxConfirmation API, in the format RTE_<INFIX>_[START|STOP]_SEC_CALLOUT_CODE, where <INFIX> is the ``shortName of the related LdComIPdu's associated EcucPartition.

(RTE-12915)

Update to the ——incremental-build entry in the Reference Manual

The entry for the command line option --incremental-build in the RTA-RTE Reference Manual has been updated to note that when the input changes, resulting in a change to the generated output, RTA-RTE does not remove previously generated files.

(RTE-18010)

RTE trace API parameter types match corresponding RTE API

RTA-RTE now uses the same compiler abstractions for the parameters of RTE Start/Return trace APIs as is used by the corresponding RTE API. Previously, for example, in the case of explicit S/R communication through ports with transformer error handling enabled, transformerError used P2CONST despite it being an OUT parameter.

(RTE-17412)

Version 12.5.0pr5

Accept RIPSified inter-ECU with mapped port interfaces

RTA-RTE now accepts inter-ECU sender-receiver reception of RIPS relevant data where the inner and outer ports require a PortInterfaceMapping. Previously the configuration was rejected with internal error 900712.

(RTE-17888)

Resource locking for reading the global composite buffer

RTA-RTE now guards reading the global composite buffer used when a transformed unqueued data element is received inter-ECU (via Com or LdCom) and retransformed in the callback (`RteXfrmInvUnqueuedDataCbk` is set to True) by calling `Rte_SuspendOSInterrupts()` and `Rte_ResumeOSInterrupts()` APIs (the same as is used in the associated callback).

(RTE-17746)

Resolve issue of (Ld)ComUser callback functions not being split by partition

When *RteSplitPerPartition* is *true*, (Ld)ComUser callback functions are now defined in the appropriate split partition file rather than *Rte.c*.

(RTE-17776)

Exclusive Area access by BSW Interrupt and Called Entities

RTA-RTE now generates the correct `SchM_Enter` and `SchM_Exit` APIs for BSW Interrupt and Called Entities in the role of `CanEnterExclusiveArea` and in the context of the relevant `ExclusiveAreaImplMechanism`. When the implementation mechanism is `OS_RESOURSE`, the corresponding `STANDARD` resource is generated in `osNeeds.arxml`.

(RTE-15287)

Use the SwAddrMethod of the ramBlock with PRPorts

RTA-RTE now uses the `SwAddrMethod` from the `ramBlock` (in preference to the one from the `VariableDataPrototypes` in the `PortPrototypes`) when generating the memory allocation keywords for the RTE managed `ramBlock` global buffer (see [SWS_Rte_07591] R23-11). Previously for `PRPortPrototype` configurations the `ramBlock SwAddrMethod` was ignored.

(RTE-16643)

Version 12.5.0pr4

Resolve issue of elided funtions' definitions not splitting by partition

When *RteSplitPerPartition* is *true* and functions have comments stating that they are elided, the function is now defined in the appropriate split partition file rather than *Rte.c* - the prototypes were already splitting correctly.

(RTE-17777)

Remove unnecessary update flag in Rte_Write

RTA-RTE now no longer sets the update flag inside the `Rte_Write` API when a configuration that contains intra-ecu and inter-partition explicit sender-receiver communication with `enableUpdate` attribute enabled and the command line option `--inter-ecu-no-ioc` is provided.

(RTE-17709)

MAKW for outstanding flag in async c/s for non-split/split SWCI

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition of the outstanding flag, used to determine if an operation call is outstanding, in the format `RTE_<INFIX>_[START|STOP]_SEC_VAR_INIT_8`, where `<INFIX>` is the `shortName` of the client runnable's parent SWC's *associated partition* within a `non-split SWC` or the `shortName` of the *associated partition* of the `RteEvent` which activates the client runnable within a `split SWC`.

(RTE-15505)

Version 12.5.0pr3

Support of partial mapping for multi-partition/multi-core NvM

Following on from issue RTE-15289 RTA-RTE now supports mapping the `NvBlockSwComponentType` and `NvBlock` accessing software components to different partitions/cores in addition to a partial mapping allowing access to specific elements of the `RamBlock` to different readers/writers.

(RTE-15313)

Improve buffer write for PRPorts

RTA-RTE is now consistent when generating the buffer write inside the `Rte_Write` API when a `PRPortPrototype` is connected to an `RPortPrototype` whether that `RPortPrototype` has explicit read access configured or not.

(RTE-17548)

VSMD update for RteRips configuration

The `RteRips` configuration options `RteRipsCfgMachineSupport64Bit` and `RteRipsUseLibcMemcpy` have been updated in the shipped VSMD file to have `SymbolicNameValue` set to `false`.

(RTE-17735)

Fix internal error for VariableAccess with no accessedVariable

In addition to raising the existing warning 2133 (see issue RTE-3169) RTA-RTE now ignores any `VariableAccess` in an input configuration that does not contain an `accessedVariable` and continues with generation. Previously such configurations would lead to internal errors.

(RTE-9095)

Only generate partition files for used partitions

RTA-RTE now only generates a file for each **used** `EcuCPartition` (see issue RTE-15466).

Note: An `EcuCPartition` is determined to be used when there is a relevant `EventToTaskMapping` or the parameter `EcucPartitionBswModuleExecution` is set to true.

(RTE-16286)

Version 12.5.0pr2

Known Issue for the triggering of Nv writing strategies in callbacks added

A Known Issue has been raised with regards to RTA-RTE triggering Nv writing strategies during `Rte_SetMirror` and `Rte_NvMNotifyInitBlock` callbacks.

Note: See issue RTE-17680.

(RTE-17681)

Generate valid IOC function calls to comply with the AUTOSAR schema

RTA-RTE now generates IOC function calls by applying MD5 hash value if the RTA-RTE internal computed name length of the `OsIocCommunication` container `shortName` exceeds 100 characters, thereby ensuring all generated ARXML, including where the `OsIocCommunication.shortName` is concatenated with other data, remain below 128 characters and comply with the AUTOSAR schema.

(RTE-17322)

Improved error handling for multiple RTE event mappings

RTA-RTE now only raises error 1562 when there are multiple RTE event mappings for an event instance. Previously RTA-RTE also raised internal error 904044.

(RTE-16993)

Rte_LdComCbkTxTriggerTransmit for untransformed uint8 arrays

RTA-RTE now generates the `Rte_LdComCbkTxTriggerTransmit` API for data elements typed by a `uint8` array being transmitted via `LdCom` where there is no `TransformationTechnology` associated with that data element.

(RTE-17297)

Error for DataSendCompletedEvents in non-1:1 communication

RTA-RTE now raises error 100160 when a `DataSendCompletedEvent` references a data element that is involved in a 1:N or N:1 communication graph, where N > 1.

(RTE-15192)

Version 12.5.0pr1

Rte Partition Intl header file does not define RTE_CORE

RTA-RTE now does not define `RTE_CORE` inside `Rte Partition Intl` header file when `Rte` is `split per partition`.

(RTE-17285)

Fill implicit write buffers for multi-partition/multi-core access of NvM

RTA-RTE now fills the implicit write buffers, generated in a task body of a runnable with write access to `NvData` in the context of multi-partition/multi-core, by reading the data over IOC.

(RTE-17287)

4.6 Version 12.4.1

Version 12.4.1pr3

Added a Known Issue about lack of NvM inter-partition support

A Known Issue has been raised stating that RTA-RTE does not support inter-partition NvM data access or client-server invocation when the NvM module used is version 4.3.0 or below.

Note: See RTE-17611.

(RTE-17608)

Option ——fixedbanner removed

The command line option `--fixedbanner` has been removed, as well as its abbreviation `-fban`. RTE-RTE now exits with error 1103 if either `--fixedbanner` or `-fban` options are provided.

(RTE-15225)

Remove OS alarms for DSCEs in LdCom transmission

RTA-RTE now no longer outputs the OS alarms for data transmissions via `LdCom` that have a `DataSendCompletedEvent`.

(RTE-17564)

Improve untransformed uint8 array reception via LdCom

RTA-RTE now does not create a global variable for the size of a data element and keeps the `Rte_memcpy` function call consistent inside the `LdComCbkRxIndication` API when that data element is typed by a `uint8` array without an associated `TransformationTechnology` and received via `LdCom`.

(RTE-17596)

Option ——os-fp removed

The command line option `--os-fp` has been removed, as well as the short form `-osfp`. RTA-RTE will now return error 1103 in case either of these options are provided.

(RTE-5843)

Version 12.4.1pr2

Transmit untransformed uint8 array via LdCom

RTA-RTE now supports transmitting data elements typed by a `uint8` array via `LdCom` where there is no `TransformationTechnology` associated with that data element..

(RTE-17246)

Rte_Read API reads uint8 array from LdCom without transformers

RTA-RTE now supports reading of *array* of type *uint8* via *Rte_Read* API, which is received from *LdCom*, without transformers.

(RTE-17247)

Support LdComCbkRxIndication untransformed uint8 array via LdCom

RTA-RTE now supports copying of data elements inside the `LdComCbkRxIndication` API via `LdCom` when the data element is typed by a `uint8` array and there is no associated `TransformationTechnology`.

(RTE-17248)

Remove duplicate buffer write

RTA-RTE now no longer outputs a duplicate buffer write inside the `Rte_Write` API when a `PRPortPrototype` with explicit read and write accesses is the sender in a communication with both inter-`OsApplication` and intra-`OsApplication` receivers.

(RTE-14375)

Support direct server invocation for NvM client ports

RTA-RTE now accepts a configuration where a re-entrant server of the `ApplicationSwComponentType` is not mapped to an `OsTask`, and the `NvBlockSwComponentType` has a client port in the role `NvMNotifyJobFinished` or `NvMNotifyInitBlock`, and the `NvBlockSwComponentType` and `ApplicationSwComponentType` are mapped to different partitions. The server arguments are transmitted using IOC and the server is activated in the context of the `ReceiverPullCB`. Previously RTA-RTE rejected the configuration with error 1563.

(RTE-15328)

Support transformed ComSignal fanout in addition to port fanning

RTA-RTE now supports fanning multiple inner ports in addition to multiple

ComSignals (see issue RTE-15659) leading to the generation of one Rte_Write API for each inner port.

(RTE-15661)

Start timeout alarms for DSCEs in LdCom transmission

RTA-RTE now sets the alarm time in the Rte_Write function for data transmissions via LdCom that have a DataSendCompletedEvent.

(RTE-16280)

Stop optimizing away buffer write for PRPorts

RTA-RTE now no longer optimizes away the buffer write inside the Rte_Write API when a PRPortPrototype with explicit read and write accesses is connected to an RPortPrototype that does not contain a read access.

(RTE-17313)

RamBlock IOC measurement no longer raises internal error

RTA-RTE no longer raises internal error 900596 when a FlatInstanceDescriptor marks NvData in an NvDataInterface as measureable and that NvData is implemented via IOC.

Note: See known issue RTE-17518.

(RTE-17439)

No longer raise an internal error

Internal error 902124 is no longer raised when a configuration contains a RamBlock that is accessed by a different EcucPartition and that RamBlock is partially mapped to NvData via an NvBlockDataMapping. Instead, a new error that states that RTA-RTE does not support this use case is emitted.

(RTE-17468)

Updated Windows runtime libraries and utilities

The Windows runtime libraries and utilities shipped in the RTA-RTE bin directory have been updated.

(RTE-17493)

Version 12.4.1pr1

Updated documentation for DSCEs

The reference manual and user guide have been updated with the support of DataSendCompletedEvents and TransmissionAcknowledgement for data elements transmitted via LdCom.

This updates known issue 494429

(RTE-15584)

NvData with PRPortPrototypes and partial data

RTA-RTE now generates the same code when a PRPortPrototype with an NvDataInterface is aggregated by any of SensorActuatorSwComponentType / ComplexDeviceDriverSwComponentType / EcuAbstractionSwComponentType / ServiceSwComponentType as when that PRPortPrototype is aggregated by an ApplicationSwComponentType. Previously RTA-RTE raised internal error 902155.

(RTE-9500)

Set TransmissionAcknowledgement status in Rte_Write

RTA-RTE now updates the status of a transmitted `DataElement` with `TransmissionAcknowledgement` in the generated `Rte_Write` function. Note: as there is no specified order of precedence for the status, RTA-RTE follows the same order as the return value of `Rte_Write` where possible.

(RTE-16221)

4.7 Version 12.4.0

Version 12.4.0

Fix support for queued RIPS communication with MetaData

RTA-RTE now generates the `Rte_Receive` API with a `MetaData` argument when forwarding to queued RIPS communication with `MetaData`. Previously with this configuration in combination with the command line option `--deviate-cbk-xfrm` the `MetaData` argument was omitted.

(RTE-17337)

Support for multi-partition/multi-core access of NvM

RTA-RTE now supports mapping the `NvBlockSwComponentType` and `NvBlock` accessing software components to different partitions/cores.

Note Support for partial mapping will be implemented in a future release.

(RTE-15289)

Fix behavior when the invalidation policy is set on the sender side

RTA-RTE now generates the same code when different but compatible interfaces are used - and `HandleInvalid` is set to `KEEP` on the sender side - as when a single shared interface is use - and `HandleInvalid` is set to `KEEP` on that single interface. Previously RTA-RTE rejected the configuration with error 1335.

(RTE-15261)

Memory Allocation Keywords for NvM

RTA-RTE now surrounds calls to `Rte_ImplCall`, from `Rte_NvMNotifyJobFinished` and `Rte_NvMNotifyInitBlock`, with partition specific Memory Allocation Keywords.

(RTE-14148)

Direct call of RE for unmapped DRE for Intra-ecu communication

RTA-RTE Supports direct call of *RE* for unmapped *DRE* in case of *implicit / expicit* write with *explicit* read for Intra-ecu communication.

(RTE-17201)

Support transformed ComSignal fanout in Rte_Write

RTA-RTE now supports the generation of an `Rte_Write` API for multiple `ComSignals` whose transforming `ISignals` reference the same `SystemSignal`.

(RTE-15659)

Validate NvBlockDescriptor and NvMBlockDescriptor partitions

RTA-RTE now raises error 100158 when the `NvBlockDescriptor` of an `NvBlockSwComponentType` and the associated `NvMBlockDescriptor` are mapped to different partition.

(RTE-15297)

Version 12.4.0pr5

Fix behavior with RteXfrmInvUnqueuedDataCbk and RIPS

RTA-RTE now generates inter-ECU output for connection graphs that have RIPS enabled the same regardless of the presence or value of the `ECUC` parameter `RteXfrmInvUnqueuedDataCbk`.

(RTE-16332)

OsApplication reference to the Rte_TickCounter

RTA-RTE now maps any generated `OsCounter`(s) of type `Rte_TickCounter` that are not mapped to an `OsApplication` in the input to the `OsApplication` that references the corresponding `OsScheduleTable`. If there is no such `OsApplication`, then they are mapped to the first trusted `OsApplication` on the same core for which the `OsCounter` was generated. If there is no trusted `OsApplication` RTA-RTE will use the first non-trusted `OsApplication` and if there are none there will be no mapping. Note: to determine the *first* `OsApplication` all of the `OsApplication`s on a core are considered alphabetically.

(RTE-16046)

Do not generate <UNDEFINED> in Memory Allocation Keywords

RTA-RTE now correctly generates the Memory Allocation Keywords for LdCom APIs where previously the LdCom signal name resolved as `<UNDEFINED>` when the `LdComIPdu` direction was misconfigured. **Note**: RTA-RTE does not validate the `LdComIPdu` direction in this context.

(RTE-16870)

Fix issue when generating Rte_LdComCbkTriggerTransmit with Metadata

RTA-RTE now directly assigns `struct` data to the local variable when generating `Rte_LdComCbkTriggerTransmit` with metadata for `struct` data types.

(RTE-17254)

OsApplication reference to the Rte_ScheduleTable

RTA-RTE now maps any generated `OsScheduleTable`(s) that are not mapped to an `OsApplication` in the input to the first trusted `OsApplication` on the same core for which the `OsScheduleTable` was generated. If there is no trusted `OsApplication` RTA-RTE will use the first non-trusted `OsApplication` and if there are none there will be no mapping. Note: to determine the *first* `OsApplication` all of the `OsApplication`s on a core are considered alphabetically.

(RTE-16045)

Version 12.4.0pr4

Set feedback variable in Rte_LdComCbkTxConfirmation

RTA-RTE now sets the feedback variable to `RTE_E_TRANSMIT_ACK` or `RTE_E_COM_STOPPED` based on the `result` parameter inside the `Rte_LdComCbkTxConfirmation` API generated for an LdCom data element with `TransmissionAcknowledge` configured.

(RTE-15581)

Metadata support with Rte_LdComCbkTriggerTransmit

RTA-RTE now updates the *metadata element* of the parameter `PduInfoPtr` of the `Rte_LdComCbkTriggerTransmit` API with the last sent data (see issue RTE-16005) when a configuration with inter-ecu queued sender-receiver communication involving `LdCom` and metadata on the sender side is provided and `LdComTxTriggerTransmit` is configured.

(RTE-16008)

Activate LdCom DataSendCompletedEvent on timeout

RTA-RTE now sets the feedback variable to `RTE_E_TIMEOUT` and activates the relevant `RunnableEntity` via the `Rte_MainFunction` when the individual tick counter equals the global tick counter for a particular LdCom data element referenced by a `DataSendCompletedEvent` with `timeout` configured.

(RTE-15137)

RTA-RTE no longer generates RTE_DEFAULT_OS_APP

RTA-RTE no longer generates `RTE_DEFAULT_OS_APP` (see issue RTE-8127). In future improvements RTA-RTE will automatically map generated structures, such as `Rte_ScheduleTable`, to an appropriate `OsApplication`, unless they are already mapped in the input.

(RTE-16044)

VSMD update for RteRips configuration

The VSMD file shipped with RTA-RTE has been updated to include existing `RteRips` configuration options `RteRipsCfgMachineSupport64Bit`, `RteRipsTickIntervalTime` and `RteRipsUseLibcMemcpy`.

(RTE-17160)

Validate MetaDataItemSet consistency

RTA-RTE now raises error 100156 when a `dataElement` referenced by a `MetaDataItemSet` is not owned by the `SenderReceiverInterface` that also owns the `MetaDataItemSet` (see `[constr_10120]` R23-11).

(RTE-16263)

Version 12.4.0pr3

RTA-RTE now initializes the length before calling Com_ReceiveDynSignal

RTA-RTE now initializes the local length variable passed to `Com_ReceiveDynSignal` indicating the maximum length that can be received. Previously, when performing the transformation in the callback (`RteXfrmInvUnqueuedDataCbk` is `true`), the local length variable was not initialized.

(RTE-17168)

Version 12.4.0pr2

Port fanning for explicit S/R reception via COM with data transformation

RTA-RTE now allows port fanning for explicit S/R reception via COM with data transformation, and generates all `Rte_Read` and `Rte_COMCbk` APIs as expected, handling copying to all readers and using IOC where necessary.

(RTE-15230)

Port fanning over IOC for explicit S/R reception via COM with transformation

RTA-RTE now creates appropriate IOC data channels for explicit S/R reception via COM with data transformation when COM resides in a different partition to any of the receiving SWCs.

(RTE-15752)

Generate activation flags for port fanning

RTA-RTE now generates `activation flag`s for receivers with `DataReceivedEvent`s when multiple inner ports are connected to an outer port for explicit S/R reception via COM with data transformation.

(RTE-15754)

Generate ReceiverPullCB APIs for port fanning

RTA-RTE now generates `Rte_ReceiverPullCB` APIs, to set `activation flag`s and activate tasks for receivers with `DataReceivedEvent`s, when multiple inner ports are connected to an outer port for explicit S/R reception via COM with data transformation.

(RTE-15755)

Perform data transformation inside ComCbk function for port fanning

RTA-RTE now execute data transformation inside `Rte_ComCbk` function, when multiple inner ports are connected to an outer port for explicit S/R reception via COM with data transformation and the configuration parameter `RteXfrmInvUnqueuedDataCbk` is set to true.

(RTE-15871)

Generate Rte_Feedback for DataSendCompletedEvents

RTA-RTE now generates an `Rte_Feedback` API for explicitly written data where that data element is also aggregated by a runnable activated by a `DataSendCompletedEvent` and returns the value of the feedback variable from the API.

(RTE-15640)

Stack trace generated for crash

RTA-RTE now generates a stack trace when fatal error 1413 occurs. If you contact support regarding this error then please include the generated stack trace along with the usual information.

(RTE-17038)

Port fanning for explicit S/R transmission via COM with data transformation

RTA-RTE now allows port fanning for explicit S/R unqueued transmission via COM with data transformation, and generates `Rte_Write` APIs as before but with one for each inner port.

(RTE-15656)

SPDX SBOM Improvement

Fixed SPDX SBOM formatting where text field contains quotes.

(RTE-16663)

Optimize buffers for mixed inter and intra partition with data transformation

RTA-RTE now uses IOC for all of the communication in a 1:N communication graph

when data is received via COM and transformation is required and RTEGen detects that any of the receivers require IOC.

(RTE-16872)

Update to the RTA-RTE Reference Manual

RTA-RTE has removed the chapter w.r.t the Lifecycle API functions since their invocations are vendor specific.

(RTE-16768)

Update to the RTA-RTE Reference Manual

The maximum number of tasks supported by RTA-RTE has been updated as per `uint32` max.

(RTE-16767)

AUTOSAR 23-11 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace http://autosar.org/schema/r4.0) against the AUTOSAR 23-11 schema.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to 23-11 will validate against the 23-11 schema.

(RTE-15203)

Improve the documentation for error 2981

The RTA-RTE Error Description document now contains additional information on why error 2981 is raised, and how to change the input configuration to avoid it.

(RTE-15872)

Version 12.4.0pr1

Correct initialization of multidimensional arrays

RTA-RTE now considers the underlying `ImplementationDataType` and not the `ValueGroup` when generating the initialization values for an input described by a `ConstantSpecification`. This primarily fixes an issue whereby the `ValueGroup` description of a multi-dimensional array does not match the `ImplementationDataType`.

(RTE-16271)

Metadata support with LdCom_Transmit in Rte_Send

RTA-RTE now copies the metadata received in the `Rte_Send` function into the `PduInfoType`, which is then passed to the `LdCom_Transmit` function when a configuration with inter-ecu queued sender-receiver communication involving `LdCom` on the sender side is provided.

(RTE-16005)

Validate a data element is referenced by at most one MetaDataItemSet

RTA-RTE now raises error 100155 when a `SenderReceiverInterface.dataElement` is referenced by more than one `MetaDataItemSet` (see `[constr_10121]` R23-11).

(RTE-16262)

Update to the RTA-RTE User Guide

The privacy statement in the RTA-RTE User Guide has been updated.

(RTE-16655)

Fix behavior with RteXfrmInvUnqueuedDataCbk

RTA-RTE now generates intra-ECU (inter- or intra-`OsApplication`) output the same regardless of the presence or value of the `ECUC` parameter `RteXfrmInvUnqueuedDataCbk`.

(RTE-16913)

Fix generation issues caused by RTE-12146

RTA-RTE now correctly generates configurations that contain inter-ECU inter-`OsApplication` connection graphs when the command line option `--no-optimized-ioc-channels` is provided. This corrects a change made in issue RTE-12146

(RTE-16825)

Generate Rte_LdComCbkTxConfirmation for DataSendCompletedEvents

RTA-RTE now generates an `Rte_LdComCbkTxConfirmation` API for explicitly written data where that data element is also aggregated by a runnable activated by a `DataSendCompletedEvent`. Within that API the activation flag guarding the execution of that runnable is set and relevant task is activated.

(RTE-14850)

Metadata support with Rte_LdComCbkRxIndication

RTA-RTE now updates the composite structure metadata received in the `Rte_LdComCbkRxIndication` and can be passed to the connected application Swc when a configuration with inter-ecu queued sender-receiver communication involving `LdCom` on the receiver side is provided.

(RTE-16010)

Correct the return value for Rte_Write with DataReceivedEvents

RTA-RTE now returns `RTE_E_OK` when data is written to the global buffer regardless of the return value of a call to `ActivateTask`, or `SetEvent` or `Rte_ActivateRE` that may be generated for a `DataReceivedEvent`. Previously RTA-RTE based the return value of `Rte_Write` on the return value of the previously mentioned calls.

(RTE-14728)

4.8 Version 12.3.2

Version 12.3.2pr4

Improvements to the RTA-RTE Reference Manual

The RTA-RTE Reference Manual has been updated to contain a chapter, "`RTA-RTE Generation Parameters`", describing the vendor specific ECUC parameters supported by RTA-RTE.

(RTE-16236)

Accept RIPSified inter-ECU with mapped complex port interfaces

RTA-RTE now accepts inter-ECU sender-receiver communication of RIPS relevant complex data where the inner and outer ports require a `PortInterfaceMapping`. Previously the configuration was rejected with internal errors 904120 and 904122.

(RTE-16770)

Accept RIPSified inter-ECU with mapped port interfaces

RTA-RTE now accepts inter-ECU sender-receiver transmission of RIPS relevant data where the inner and outer ports require a `PortInterfaceMapping`. Previously the configuration was rejected with error 7434.

(RTE-16674)

Generate Rte_Feedback for LdCom

RTA-RTE now generates the `Rte_Feedback` API in the context of LdCom. **Note** further work is required to fully support `DataSendCompletedEvent`s with `LdCom`.

(RTE-14848)

Documentation update

The RTA-RTE Reference Manual and RTA-RTE User Guide have been updated so sections referring to COM transmission and reception APIs are more generic to include the recently added support for dynamic signals and signals with meta-data.

(RTE-16011)

VSMD file for RTA-RTE updated to AUTOSAR R23-11

The VSMD file shipped with RTA-RTE has been updated to AUTOSAR R23-11 (4.9.0).

Note that this does not imply the implementation of any new AUTOSAR features. New features will have their own release notes.

(RTE-15204)

Version 12.3.2pr3

MAKWs for Global Status Variables

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition of the global status variables used for the inter-ECU reception of transformed data in the format `RTE_<INFIX>_[START|STOP]_SEC_VAR_INIT_8`, where `<INFIX>` is the concatenation of "`SIG_`" and the `shortName` of the transformer's associated `ComSignal`, `ComSignalGroup` or `LdComIPdu`.

(RTE-15504)

Optimize buffers for mixed inter and intra partition

RTA-RTE now uses IOC for all of the communication in a 1:N communication graph when data is received via COM and RTEGen detects that any of the receivers require IOC.

Note: With RTA-OS 12.3.0 and higher the intra-`OsApplication` IOC communication will be optimized to an `OsApplication` local access.

(RTE-12146)

Correct types for 1:N mixed IOC graphs

In the case of 1:N IOC mixed with non IOC, RTA-RTE now always uses the `ImplementationDataType` of the provider when generating the TDB structures, CDS structures and the iocNeeds.arxml entries for that communication graph.

(RTE-15794)

Merge implicit N:1 communication graph shadow buffers

RTA-RTE now generates at most one write shadow buffer (`wbuf/sbuf`) and at most one read shadow buffer (`rbuf/sbuf`) per implicitly accessing `OsTask` for an N:1 communication graph.

(RTE-14191)

Removed noncompilable redefinition of composite.metaData in Rte_Receive

RTEGen no longer redefines `composite.metaData` inside `Rte_Receive`. Previously this led to noncompilable code.

(RTE-16624)

Update the branding of the RTA-RTE OSS Attribution Document

The branding of the RTA-RTE OSS Attribution Document has been updated.

(RTE-16730)

Omit OsIocReceiverId with ——no-optimized-ioc-channels

RTEGen now does not generate `OsIocReceiverIds` in `iocNeeds.arxml` when the command line option `--no-optimized-ioc-channels` is provided.

(RTE-16698)

Updates to the RTA-RTE MISRA Compliance Statement

The RTA-RTE MISRA Compliance Statement now contains justifications for unavoidable non-conformances in the RTA-RTE implementation. A detailed analysis is provided for each non-conformance stating why the code is considered functionally correct.

(RTE-15168)

Fixed incorrect return code for RTEGen on Linux

RTEGen now returns `0` when execution is successful. Previously on Linux RTEGen generated successfully but failed to shutdown the connection with Etas License Manager correctly and returned `17` leaving a background process running.

(RTE-16700)

MAKWs for Transformer Global Buffers

RTA-RTE now generates the Memory Allocation Keywords surrounding the definition of the global buffer used for the inter-ECU reception of transformed data in the format `RTE_<INFIX>_[START|STOP]_SEC_VAR_CLEARED_8`, where `<INFIX>` is the concatenation of "`SIG_`" and the `shortName` of the transformer's associated `ComSignal`, `ComSignalGroup` or `LdComIPdu`.

(RTE-15503)

Version 12.3.2pr2

IRV macros are generated as function-like-macros

RTA-RTE now generates `Rte_Irv` APIs for Inter-OsApp IRVs as function-like-macros.

(RTE-14988)

Remove duplicates of generated files

RTA-RTE now does not duplicate the generated file `Rte.c`.

(RTE-16555)

Invoke Com_ReceiveSignalGroupArrayWithMetaData to receive meta data

RTA-RTE invokes `Com_ReceiveSignalGroupArrayWithMetaData` instead of `Com_ReceiveSignalGroupArray` when a transformed queued data element is received via COM, with a data transformation protocol of `COMBased`, and `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-15991)

Improvements to the RTA-RTE Reference Manual

The RTA-RTE Reference Manual has been updated to highlight for each option if it is deprecated.

(RTE-16562)

Support for calibration parameter APIs as macros

RTA-RTE now generates the `Rte_CData`, `Rte_Prm` and `SchM_CData` APIs as macros when the **new** `EcuC` Parameter `RteCalibrationAccessApiAsMacro` is set to `true`. **Note** that tracing hooks are not output in the macro even when VFB tracing is enabled.

(RTE-16132)

Version 12.3.2pr1

Invoke Com_ReceiveDynSignalWithMetaData to receive meta data from Com

RTA-RTE invokes `Com_ReceiveDynSignalWithMetaData` instead of `Com_ReceiveDynSignal` when a transformed queued data element is received via COM, with `ComSignal` of type `UINT8_DYN`, and `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-15996)

Support of MetaData in Rte_Receive

RTA-RTE now supports `MetaData` when a transformed queued data element is received via COM and the `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-7686)

Invoke Com_SendSignalGroupArrayWithMetaData to pass meta data to Com

RTA-RTE invokes `Com_SendSignalGroupArrayWithMetaData` instead of `Com_SendSignalGroupArray` when a transformed queued data element is transmitted via COM and `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-15993)

Deprecated option ——have-64bit-int-types

The command line option `--have-64bit-int-types` has been deprecated, as well as its abbreviation `-int64`. RTA-RTE now raises warning 2976 if `--have-64bit-int-types`, or its abbreviated version, `-int64`, is used.

The `--have-64bit-int-types` option should not be used for new projects, and where possible existing projects should be modified to no longer use it.

(RTE-6825)

Invoke Com_ReceiveSignalWithMetaData to receive meta data from Com

RTA-RTE invokes `Com_ReceiveSignalWithMetaData` instead of `Com_ReceiveSignal` when a transformed queued data element is received via COM and the relevant `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-7706)

Composite type with MetaData for queued inter-ECU S/R with transformation

RTA-RTE now declares the relevant composite type (see issue RTE-10621) with `MetaData` when a transformed queued data element is transmitted via COM and the `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-15983)

Replace calls to GetApplicationID with GetCurrentApplicationID

RTA-RTE will now call `GetCurrentApplicationID` instead of `GetApplicationID` within the internal APIs `Rte_GetResource` and `Rte_ReleaseResource`. This fixes a bug in which RTA-RTE may receive an incorrect `OsApplication` ID in scenarios involving `CallTrustedFunction` usage.

(RTE-14497)

4.9 Version 12.3.1

Version 12.3.1pr3

Fix initialization of floating point values

RTA-RTE now correctly initializes floating point values. Previously RTA-RTE was generating a newline before the decimal suffix of the final floating point initial value leading to compilation errors.

(RTE-16338)

Support 64-bit types by default

The command line option `--have-64bit-int-types` is now on by default. RTA-RTE now supports 64-bit types within generated data-transformation functions out-of-the-box.

(RTE-6824)

Version 12.3.1pr2

Transmit untransformed VSA_LINEAR variable size arrays via LdCom

RTA-RTE now supports the transmission of unqueued data via LdCom without data transformation, when the underlying `VariableDataPrototype` is typed by a variable size array having `DynamicArraySizeProfile` set to `VSA_LINEAR`.

(RTE-16079)

Do not generate redundant mapping definitions

RTA-RTE now does not define a macro mapping the AUTOSAR name of an API to the internal function name when those two symbols are identical, as is the case for SchM APIs. Previously RTA-RTE renamed the internal API (see issue RTE-14550). This was not compatible with object code delivery when header files from other vendors were used to compile the object code and has been reverted.

(RTE-16336)

Invoke Com_SendDynSignalWithMetaData to pass meta data to Com

RTA-RTE invokes `Com_SendDynSignalWithMetaData` instead of `Com_SendDynSignal` when a transformed queued data element is transmitted via COM, with `ComSignal` of type `UINT8_DYN`, and `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-15994)

Improve performance of RTEGen

The performance, in terms of time taken to generate code, had regressed with the addition of a combination of recent features and has now been restored to previous levels.

Note: Message 1565 will no longer be raised for SW-C instances outside of the `ECU_EXTRACT` (see issue RTE-12141). Additionally, `WaitPoint`s are now correctly considered when determining if an extended task is required for non-RIPS tasks in RIPS relevant configurations (see issue RTE-14034)

(RTE-16335)

Receive untransformed VSA_LINEAR variable size arrays via LdCom

RTA-RTE now generates an `Rte_LdComCbkRxIndication` API that copies the minimum number of bytes needed when receiving a data element via LdCom without transformation and is typed by a variable size array having `DynamicArraySizeProfile` with value `VSA_LINEAR`.

(RTE-16156)

Update split-swc option in the RTA-RTE Reference Manual

The RTA-RTE Reference Manual's section on the `--deviate-split-swci-support` option has now been updated to reflect recent changes in support of `ExclusiveAreas` for split-swcs. See issue RTE-13449.

(RTE-14891)

Invoke Com_SendSignalWithMetaData to pass meta data to the Com module

RTA-RTE invokes `Com_SendSignalWithMetaData` instead of `Com_SendSignal` when a transformed queued data element is transmitted via COM and the relevant `SenderReceiverInterface` aggregates a `MetaDataItemSet` that references that data element.

(RTE-7705)

Remove additional comments from the generated code

RTA-RTE now suppresses additional comments when the command line option `--disable-comments` is set.

(RTE-8289)

Fix option heading in the RTA-RTE Reference Manual

The heading for the section describing the command line option `--deviate-event-based-com-mapping` in the RTA-RTE Reference Manual was referencing the wrong option and has been corrected.

(RTE-16312)

Version 12.3.1pr1

Generate partition specific C files

RTA-RTE now generates one file per partition, in addition to `Rte.c`, when the input contains more than one active `EcucPartition` and the **new** `EcuC` Parameter `RteSplitPerPartition` is set to `true`.

In addition to this, there is a new header file `Rte_Partition_Intl.h` included in each partition specific file which has the declaration of symbols shared among partition c files.

(RTE-15466)

Read untransformed VSA_LINEAR variable size arrays

RTA-RTE now generates an `Rte_Read` API that copies the minimum number of bytes needed when reading a data element that is received via LdCom without transformation and is typed by a variable size array having `DynamicArraySizeProfile` with value `VSA_LINEAR`.

(RTE-16081)

Support for AXIS initial values provided in the physical domain

RTA-RTE now supports initial values provided in the physical domain for `DataPrototypes` of `category AXIS`, applying the appropriate `CompuMethod` and `Unit` according to the AUTOSAR order of precedence.

(RTE-15253)

Support for CUBOID initial values provided in the physical domain

RTA-RTE now supports initial values provided in the physical domain for `DataPrototypes` of `category CUBOID`, applying the appropriate `CompuMethod` and `Unit` according to the AUTOSAR order of precedence.

(RTE-13848)

Remove ——sws

The `--sws` option and its short form `-sws` have been removed.

If either the `--sws` or `-sws` options are provided, RTE will exit with error 1103.

(RTE-10058)

Parse SystemSignal.physicalProps.compuMethod

RTA-RTE now parses and prioritises `SystemSignal.physicalProps.compuMethod` over `ISignal.networkRepresentationProps.compuMethod` when choosing the resulting `CompuMethod` for an `ISignal`. Warnings 100148 and 100149 are raised in the event that the `networkRepresentationProps.compuMethod` is still used.

(RTE-15570)

Deprecated option ——memory-sections

The command line option `--memory-sections` has been deprecated, as well as its abbreviation `-mem`. RTA-RTE now raises warning 2976 if `--memory-sections`, or its abbreviated version, `-mem`, is used.

The `--memory-sections` option should not be used for new projects, and where possible existing projects should be modified to no longer use it.

(RTE-11277)

Deprecated option ——makedep

The command line option `--makedep` has been deprecated, as well as its abbreviation `-m`. RTA-RTE now raises warning 2976 if `--makedep`, or its abbreviated version, `-m`, is used.

The `--makedep` option should not be used for new projects, and where possible existing projects should be modified to no longer use it.

(RTE-11274)

4.10 Version 12.3.0

Version 12.3.0pr8

Calls to trusted functions now match the generated function name

RTA-RTE now generates matching names for trusted functions and calls to trusted functions where the trusted function name is sufficiently long that it results in hashing (see issue RTE-14701).

Note: For all shortened names the maximum length before any prefix (such as, "`TRUSTED_`") has been reduced.

(RTE-16322)

Version 12.3.0pr7

Correct raising of errors 100147 and 100150

RTA-RTE no longer incorrectly raises errors 100147 and 100150 for variable sized arrays that are transmitted inter-ECU when they are configured with a transformer or they are RIPS activated.

(RTE-16210)

Version 12.3.0pr6

New Software Bill Of Materials

A Software Bill Of Materials (SBOM) has been added to the product documents.

(RTE-15040)

New CERT C Compliance Statement

A CERT C compliance statement has been added to the product documents (see "`RTA-RTE CERT C Compliance Statement.pdf`").

(RTE-15855)

Support for CURVE initial values provided in the physical domain

RTA-RTE now supports initial values provided in the physical domain for `DataPrototypes` of `category CURVE`, applying the appropriate `CompuMethod` according to the AUTOSAR order of precedence.

(RTE-13846)

Support for MAP initial values provided in the physical domain

RTA-RTE now supports initial values provided in the physical domain for `DataPrototypes` of `category MAP`, applying the appropriate `CompuMethod` according to the AUTOSAR order of precedence.

(RTE-13847)

Remove ——com-symbolic-sigs

The `--com-symbolic-sigs` option and its short form `-cy` have been removed.

If either the `--com-symbolic-sigs` or `-cy` options are provided, RTE will exit with error 1103.

(RTE-14196)

Error for nested variable size arrays

RTA-RTE now raises error 100150 when a variable size array contains another variable size array.

(RTE-10709)

Handle transmitting VSA with transformer

RTA-RTE now generate `Rte_Write` as it would be normally when data element is transmitted via LdCom with transformation and typed by a variable size array.

(RTE-15958)

Error for non VSA_LINEAR variable size arrays

RTA-RTE now raises error 100147 when a non RIPS-relevant sender receiver data element without transformation is typed by a variable size array having a `DynamicArraySizeProfile` with a value other than `VSA_LINEAR`.

(RTE-10670)

No Rte_SpinLocks definitions if not used

RTA-RTE now does not define `Rte_SpinlockCount`, `Rte_ReleaseSpinlock`, and `Rte_GetSpinlock` when `Rte_ReleaseSpinlock` and `Rte_GetSpinlock` are not called in the generated RTE files.

(RTE-14366)

Removed unnecessary white spaces when comments are removed

The command line option `--disable-comments` parameter no longer creates extra, unnecessary white spaces.

(RTE-8430)

Version 12.3.0pr5

Change SchM API mappings to improve compliance with MISRA Rule 5.5

RTA-RTE now generates the SchM API functions with identifies distinct from the macro name that maps to those APIs. This addresses violations of MISRA C:2012 Rule 5.5.

(RTE-14550)

Call RteSerialData.RteCallBackFunction with transformerError for LdCom

RTA-RTE now calls the `RteSerialData.RteCallBackFunction` in the `Rte_LdComCbkRxIndication` API passing the argument `transformerError` when `DataTransformation` is configured for the relevant `ISignal`.

(RTE-15689)

Call RteSerialData.RteCallBackFunction in Rte_LdComCbkRxIndication API

RTA-RTE now calls the function specified by the Ecuc parameter `RteSerialData.RteCallBackFunction` inside the associated `Rte_LdComCbkRxIndication` API for the `ISignal` referenced by an Ecuc container `RteSerialData` which is in turn referenced by the Ecuc container

RteSerialDataConfiguration.

(RTE-15688)

Rte_LdComCbkRxIndication retransformation for unqueued inter-ECU S/R

RTA-RTE now performs retransformation in the generated Rte_LdComCbkRxIndication for unqueued explicit inter-ECU sender/receiver communication with data transformation when the Ecuc parameter RteXfrmInvUnqueuedDataCbk is enabled.

(RTE-15676)

Version 12.3.0pr4

Do not generate RTE data structures for RIPS communication

RTA-RTE no longer generates the code that would be used in an RTE-managed communication when that communication is handled by RIPS. This change considers Event and Timeout code in the context of inter and/or intra OsApplication RIPSifed client/server communication graph for non-reentrant servers. Additionally this leads to the generation of a basic task if the cause of the extended task is the runnable calling that non-reentrant server.

(RTE-14364, RTE-14363)

Reduce the length of the OsIocReceiverPullCB parameter value

RTA-RTE reduces the length of the OsIocReceiverPullCB parameter value of an OsIocCommunication container in the generated iocNeeds.arxml by applying MD5 hash value if the RTA-RTE internal computed name exceeds 128 characters, thereby complying to the AUTOSAR schema.

(RTE-15294)

Support for -INF/INF in CompuScales

RTA-RTE now accepts -INF/INF for LowerLimit/UpperLimit defined within CompuScales.

(RTE-15624)

Improved compliance with MISRA Rule 2.7

RTA-RTE now casts all the parameters of an unconnected Rte_Call API to void. This addresses violations of MISRA C:2012 Rule 2.7.

(RTE-15388)

Version 12.3.0pr3

Reduce the length of the OsApplicationTrustedFunction shortName

RTA-RTE reduces the length of the OsApplicationTrustedFunction container shortName in the generated trustedFunctionNeeds.arxml by applying MD5 hash value if the RTA-RTE internal computed name exceeds 128 characters, thereby complying to the AUTOSAR schema.

(RTE-14701)

Use the correct destination type when referencing an OsEventMask

RTA-RTE now uses EcucIntegerParamDef as the DEST attribute of the DestinationRef when referring to an OsEventMask in the generated osNeeds.arxml.

(RTE-13951)

Version 12.3.0pr2

Call RteSerialData.RteCallBackFunction with transformerError

RTA-RTE now calls the `RteSerialData.RteCallBackFunction` RTE-15602 in the Rte_COMCbk API by passing `transformerError` when `DataTransformation` is configured for the relevant `ISignal`.

(RTE-15603)

Rte_ComCbk for unqueued inter-ECU S/R comms with transformation

RTA-RTE now performs retransformation in the generated `Rte_ComCbk` for unqueued explicit inter-ECU sender/receiver communication with data transformation when the Ecuc parameter `RteXfrmInvUnqueuedDataCbk` is enabled.

(RTE-15675)

Improved handling of VariableAndParameterInterfaceMapping when RIPS is enabled

RTA-RTE now supports mapping of primitive `ImplementationDataType` of `Category TYPE_REFERENCE` to a specific sub-element of `ImplementationDataType` of `Category STRUCTURE` via `VariableAndParameterInterfaceMapping` for RIPS enabled inter-ecu `SenderReceiver` communication graph.

(RTE-15789)

Include RteSerialData.RteSerialDataHeaderInclude

RTA-RTE now includes the file specified by the Ecuc parameter `RteSerialData.RteSerialDataHeaderInclude` inside the generated `Rte.c` file when a function call is generated for the Ecuc container `RteSerialData` inside an `Rte_COMCbk` API.

(RTE-15704)

Version 12.3.0pr1

Generate modern Rte_LdComCbkRxIndication APIs

RTA-RTE now generates modern `Rte_LdComRxIndication` callback of the form Rte_<Partition>_LdComCbkRxIndication when the input configuration contains an `RteLdComUser` aggregating an `LdComUserCallback` with `LdComUserCallbackType LDCOM_RX_INDICATION`, where <Partition> is the `EcucPartition` referenced by the Ecuc parameter `RteLdComUserEcucPartitionRef` of that `RteLdComUser`

(RTE-13819)

Call RteSerialData.RteCallBackFunction in Rte_COMCbk API

RTA-RTE now calls the function specified by the Ecuc parameter `RteSerialData.RteCallBackFunction` inside the associated `Rte_COMCbk` API for the `ISignal/ISignalGroup` referenced by an Ecuc container `RteSerialData` which is in turn referenced by the Ecuc container `RteSerialDataConfiguration`.

(RTE-15602)

4.11 Version 12.2.1

Version 12.2.1pr12

Initialization of floats with scientific notation

RTA-RTE now correctly appends "F" to 32-bit floating-point data configured in scientific notation. This fixes an issue with the RTE-14806 where ".0" was also appended to the exponent.

(RTE-15738)

Allow PRPorts to read data which is sent inter-ECU S/R

In addition to the support added in issues RTE-14594, RTE-14595 and RTE-15196, RTA-RTE now allows PRPorts to read the data which is sent inter-ECU if a DataReadAccess or DataReceivePointByArgument or DataReceivePointByValue is configured which references that PRPort and the corresponding DataElement.

(RTE-15802)

Server Runnable Hooks for Vendor Mode

RTA-RTE now generates server runnable hook function declarations when in a client server communication RteVfbTraceEnabled is enabled, the command line option --optimize-task-stack-size is used, and the parameter RteGenerationMode is set to VENDOR_MODE. This works similarly to the case where the same configuration is used but the parameter RteGenerationMode is set to COMPATIBILITY_MODE

(RTE-15747)

Version 12.2.1pr11

Allow inner PRPorts in LdCom in explicit inter-ECU S/R

RTA-RTE now allows PRPorts to be used as the inner port of an inter-ECU explicit sender-receiver communication graph that uses LdCom when connected to a delegation PPort. In this case, the inner PRPort will be treated as a PPort for that communication graph.

(RTE-15196)

Rte_memcpy is now defined by a user provided memcpy

The Rte_memcpy function is now defined by the Rte_UserMemCpy function rather than the C standard memcpy function.

(RTE-15612)

Configure the main function period via BSWTimingEvent

RTA-RTE no longer raises error 100132 (see issue RTE-14678). The generated output when a configuration contains a MinimumStartInterval, a BSWTimingEvent, and the command line option --period is not provided is the same as if the --period options is set to match the period configured by the BSWTimingEvent.

(RTE-15452)

Updated VSMD for RteSerialData and RteXfrmInvUnqueuedDataCbk

The VSMD file shipped with RTA-RTE has been updated with a new container to configure references to the RteSerialData configuration. This is achieved by referencing the RteSerialDatas in an EcucContainerValue with shortName RteSerialDataConfiguration. A new EcucDestinationUriDefSet RteUriDefSet is added to define the structure of RteSerialData container configuration.

Additionally, a new EcucParameter RteXfrmInvUnqueuedDataCbk is added and when set to true, RTEGen calls the transformer APIs to re-transform the data

inside Com/LdCom reception callbacks for unqueued SenderReceiverCommunication.

Note: This change only introduces the new parameters, the functionality will be added in further tickets.

(RTE-15652)

MAKWs for cross core LdCom aliveTimeout partB flag

RTA-RTE now defines the start and stop Memory Allocation Keywords for cross core LdCom aliveTimeout DataReceivedErrorEvent partB flag in the format RTE_<INFIX>_START_SEC_VAR_INIT_BOOLEAN and RTE_<INFIX>_STOP_SEC_VAR_INIT_BOOLEAN respectively, where <INFIX> is the shortName of the EcucPartition that accesses the OsTask mapped to the respective DataReceivedErrorEvent.

(RTE-15233)

Version 12.2.1pr10

Allow inner PRPorts in implicit inter-ECU S/R

RTA-RTE now allows PRPorts to be used as the inner port of an inter-ECU implicit sender-receiver communication graph when connected to a delegation PPort. In this case, the inner PRPort will be treated as a PPort for that communication graph.

(RTE-14595)

Version 12.2.1pr9

Correct types for 1:N IOC

In the case of 1:N IOC, RTA-RTE now always uses the ImplementationDataType of the provider when generating the TDB structures, CDS structures and the iocNeeds.arxml entries for that communication graph.

(RTE-14454)

Make cross-core DREE flags consistent with other MAKWs

RTE-RTE now surrounds the dree_PartA_Flag variables used with LdCom alive timeouts with an MAKW that has MAIN as the infix, VAR_INIT as the swaddr method and BOOLEAN as the alignment.

(RTE-15231)

Allow inner PRPorts in explicit inter-ECU S/R

RTA-RTE now allows PRPorts to be used as the inner port of an inter-ECU explicit sender-receiver communication graph when connected to a delegation PPort. In this case, the inner PRPort will be treated as a PPort for that communication graph.

(RTE-14594)

Generate modern Rte_LdComCbkTriggerTransmit APIs

RTA-RTE now generates modern Rte_LdComCbkTriggerTransmit callback of the form Rte_<Partition>_LdComCbkTriggerTransmit when the input configuration contains an RteLdComUser aggregating an LdComUserCallback with LdComUserCallbackType LDCOM_TX_TRIGGER_TRANSMITACK, where <Partition> is the EcucPartition referenced by the Ecuc parameter RteLdComUserEcucPartitionRef of that RteLdComUser

(RTE-13820)

Improved Mixed Port Interface Mappings when RIPS is enabled

In addition to the support added in issues RTE-15327 and RTE-15512, RTA-RTE now prevents error E902155, allowing mixed port interface mappings between structs and primitives with RIPS enabled.

(RTE-15623)

Stop generating Rte_Catalog.xml

The `Rte_Catalog.xml` file will no longer be generated.

(RTE-14430)

Support bi-directional COM over multiple ISignals for ComSignals

In addition to the support added in issue RTE-14339, RTA-RTE now supports configuring bi-directional COM for `ComSignal`s with specified directions via multiple `ISignal`s.

(RTE-15551)

Reverted behavior from RTE-15056

RTA-RTE no longer raises error 100140 when, in the context of bi-directional COM, an `ISignalToIPduMapping` is only referenced by one `ComSignal` or `ComSignalGroup` with a specified direction. (see issue RTE-15056).

(RTE-15644)

Improve reporting of resources for zero or one OsApplications

When an input configuration contains zero or one `OsApplication`s, RTA-RTE now generates an `OsResource` named `RTE_RESOURCE` in `osNeeds.arxml`.

This updates issue RTE-1150.

(RTE-15047)

Improved handling of PrimitiveDataType mappings when RIPS is enabled

RTA-RTE now supports mapping of `ApplicationPrimitiveDataType` or `ImplementationDataType` of `Category VALUE` to a specific sub-element of `ApplicationRecordDataType` or `ImplementationDataType` of `Category STRUCTURE` via `VariableAndParameterInterfaceMapping` for RIPS enabled inter-ecu `SenderReceiver` communication graph.

(RTE-15364)

Version 12.2.1pr8

Improved RecordDataType mappings involving LdCom when RIPS is enabled

RTA-RTE now supports the usage of `ApplicationRecordDataType` or `ImplementationDataType` of `Category STRUCTURE` with dissimilar shapes in `VariableAndParameterInterfaceMapping` for RIPS enabled inter-ecu `SenderReceiver` communication graphs involving LdCom.

(RTE-15512)

Increase maximum allowed XfrmBufferSizeInBytes

RTA-RTE now accepts a `XfrmBufferSizeInBytes` in the range `1` to `4294967295` (inclusive) and raises error 100143 when a value outside this range is provided (see issue RTE-14745). Previously the upper limit was 65535 and error 2969 was

raised.

(RTE-15595)

Version 12.2.1pr7

Improved handling of RecordDataType mappings when RIPS is enabled

RTA-RTE now supports the usage of `ApplicationRecordDataType` or `ImplementationDataType` of `Category STRUCTURE` with dissimilar shapes in `VariableAndParameterInterfaceMapping` for RIPS enabled inter-ecu `SenderReceiver` communication graphs.

(RTE-15327)

Update to the RTA-RTE package layout

The Windows and Linux archives have been renamed to Windows.zip and Ubuntu20.04.tar.gz respectively. When extracted they produce a singe directory (for example, `RTA-RTE_V12.2.1pr6`) matching the structure previously generated by the Windows installer.

Additionally, the remaining artifacts related to the `Installer.exe` have been removed (see issue RTE-14215).

(RTE-15419)

Partial support for RTE_RIPS_INSTANTIATION_BY_PLUGIN

RTA-RTE now supports `RTE_RIPS_INSTANTIATION_BY_PLUGIN` for unqueued explicit sender/receiver communication.

(RTE-8934)

Updated MISRA Compliance Statement

The MISRA Compliance Statement has been updated to document non-conformance for specific instances of MISRA violations.

(RTE-15105)

Use SduLength for calculating inverse transformer buffer sizes

As part of support for dynamic PDU lengths when using the `SOMEIP` or `E2E` transformers, when calling inverse transformer chains in explicit queued sender/receiver communication, RTA-RTE will now use the `SduLength` parameter of the `PduInfoPtr` provided by `LdCom` when calculating transformer buffer lengths instead of assuming that the length of the entire transformer buffer is required.

(RTE-15184)

Version 12.2.1pr6

Allow transformerError to be independently configured for 1:N

RTA-RTE now generates the `transformerError` related code independently for each of N `Rte_Read` APIs based on the `PortAPIOption.errorHandling` when, as part of an inter-ECU 1:N COM communication graph with transformation, a single `SystemSignal` fans to N consumers.

Note RTA-RTE does not support the general case of fanning from a single `SystemSignal` to multiple consumers with transformation.

(RTE-15459)

SenderReceiverToSignalGroupMapping dataElement and typeMapping

RTA-RTE now raises error 100141 when a `SenderReceiverToSignalGroupMapping` relating to a structure references a `dataElement` that is typed by an `ApplicationRecordDataType` and aggregates a `typeMapping` that does not reference an `applicationRecordElement` or when a `SenderReceiverToSignalGroupMapping` references a `dataElement` that is typed by an `ImplementationDataType` and aggregates a `typeMapping` that does not reference an `implementationRecordElement`.

Note RTA-RTE does not reject a configuration when a `SenderRecRecordElementMapping` references both an `applicationRecordElement` and an `implementationRecordElement`.

(RTE-15336)

Do not raise error 2468 for non-bi-directional COM

RTA-RTE no longer raises error 2468 for signals referenced by an `ISignalTriggering` when not involved in a bi-directional COM configuration. (see issue RTE-11835)

(RTE-15449)

Use SduLength for calculating inverse transformer buffer sizes

As part of support for dynamic PDU lengths when using the `SOMEIP` or `E2E` transformers, when calling inverse transformer chains in explicit unqueued sender/receiver communication, RTA-RTE will now use the `SduLength` parameter of the `PduInfoPtr` provided by `LdCom` when calculating transformer buffer lengths instead of assuming that the length of the entire transformer buffer is required.

(RTE-15181)

Generate core-specific Rte_TickCounters

RTA-RTE now generates one `OsCounter` per used core when the Ecuc parameter `RteGeneration.RteScheduleTablePerCore` is set to `true`. In this context, a used core is a core whose `EcucCoreDefinition` references an `OsApplication` which references an `OsTask` that maps a `TimingEvent`. Each `OsCounter` will be named based on the core it was generated for, using the format `Rte_TickCounter_Core_<CoreID>`.

(RTE-15101)

Generate core-specific schedule tables

RTA-RTE now generates one `OsScheduleTable` per used core when the Ecuc parameter `RteGeneration.RteScheduleTablePerCore` is set to `true`

(RTE-14376)

Version 12.2.1pr5

Invoke Com_SendDynSignal to transmit queued dynamic PDU length data

RTA-RTE invokes `Com_SendDynSignal` instead of `Com_SendSignal` when using transformers of protocols `SOMEIP` or `E2E` and `ComSignal` of type `UINT8_DYN` to transmit queued dynamic PDU length data.

(RTE-15154)

Initialize the global holding the length of a dynamic signal

RTA-RTE now initializes the global variable generated to store the length of the

data received for a dynamic PDU to be the maximum size of the global buffer allocated for that communication (see issue RTE-15182).

(RTE-15296)

Version 12.2.1pr4

Validate ComSignal(Group) direction in bidirectional COM

RTA-RTE now raises error 100139 when, in the context of bi-directional COM (see issue RTE-11835), an `ISignalToIPduMapping` is referenced by more than one `ComSignal` or `ComSignalGroup` with the same or no direction.

(RTE-15055)

Error for sole directional ComSignal(Group) in bidirectional COM

RTA-RTE now raises error 100140 when, in the context of bi-directional COM (see issue RTE-11835), an `ISignalToIPduMapping` is only referenced by one `ComSignal` or `ComSignalGroup` with a specified direction.

(RTE-15056)

Invoke Com_SendDynSignal to transmit unqueued dynamic PDU length data

RTA-RTE invokes `Com_SendDynSignal` instead of `Com_SendSignal` when using transformers of protocols `SOMEIP` or `E2E` and `ComSignal` of type `UINT8_DYN` to transmit unqueued dynamic PDU length data.

(RTE-15153)

Invoke Com_ReceiveDynSignal to receive queued dynamic PDU length data

RTA-RTE invokes `Com_ReceiveDynSignal` instead of `Com_ReceiveSignal` when using transformers of protocols `SOMEIP` or `E2E` and `ComSignal` of type `UINT8_DYN` to receive queued dynamic PDU length data.

(RTE-15157)

Update to the RTA-RTE package

RTA-RTE no longer provides an installer. The RTA-RTE package now contains a single set of documents, one archive for Windows and another for Ubuntu20.04. Both of these archives have the same structure as previously generated by the `Installer.exe`.

Additionally, RTA-RTE no longer includes a version of the ETAS License Manager (LiMa) or registers RTA-RTE with any existing ETAS License Manager installation.

Note that there is no change when RTA-RTE is installed as part of RTA-CAR.

(RTE-14215)

Validation for max ComSignal(Group)s in bidirectional COM

RTA-RTE now raises error 100138 when, in the context of bi-directional COM (see issue RTE-11835), an `ISignalToIPduMapping` is referenced by more than two `ComSignal`s or `ComSignalGroup`s.

(RTE-15054)

Consider ComIPduDirection for bidirectional COM

In addition to the support added in issue RTE-11835 RTA-RTE now takes the `ComIPduDirection` into account passing the correct `Signal(Group)Id` when transmitting or receiving data to/from COM.

(RTE-14339)

Version 12.2.1pr3

Invoke Com_ReceiveDynSignal to receive dynamic PDU length data

RTA-RTE invokes `Com_ReceiveDynSignal` instead of `Com_ReceiveSignal` when using transformers of protocols `SOMEIP` or `E2E` and `ComSignal` of type `UINT8_DYN` to receive dynamic PDU length data.

(RTE-15186)

Use the correct destination type when referencing an OsAlarmAction

RTA-RTE now uses `EcucChoiceContainerDef` as the `DEST` attribute of the `DestinationRef` when referring to an `OsAlarmAction` in the generated `osNeeds.arxml`.

(RTE-11134)

Process ARXML with UTF-8 encoding

RTA-RTE now supports ARXML files with `UTF-8` encoding. Previously RTA-RTE treated all ARXML files as `Windows-1252`. This addresses a known difference in behavior between Window and Linux where on Linux `UTF-8` characters would be transcoded to `Windows-1252`. The related Known Difference Between OS Versions in Chapter 3 has been removed (RTE-13892).

(RTE-4285)

Generate a global to hold the length of a dynamic signal

In preparation for supporting `Com_ReceiveDynSignal` with unqueued data transformation, when the relevant transformer chain is composed of a `TransformerTechnology` with `protocol SOMEIP` and optionally another `TransformerTechnology` with `protocol E2E`, and the associated `ComSignal` has `ComSignalType UINT8_DYN`, RTA-RTE now generates a global variable to store the length of the data received and uses that global to pass the length to the first re-transformer.

(RTE-15182)

Removed warning 2984 when RIPS is enabled

RTA-RTE no longer raises warning 2984 when `DataTransformation` is configured on the `ISignal` for a RIPS enabled inter-ECU Sender/Receiver communication graph.

(RTE-14869)

Reduce the length of the OsIocCommunication shortName to comply with the AUTOSAR schema

RTA-RTE reduces the length of `OsIocCommunication` container `shortName` in the generated `iocNeeds.arxml` by applying MD5 hash value if the RTA-RTE internal computed name exceeds 128 characters, thereby complying to AUTOSAR schema.

(RTE-14700)

Improved validation check for warning 2636 when RIPS is enabled

RTA-RTE no longer raises warning 2636 when runtime data conversion is configured using `VariableAndParameterInterfaceMapping` for RIPS enabled inter-ecu

Sender/Receiver communication graph.

Note: RIPS must handle the runtime data conversion in this case.

(RTE-14867)

Version 12.2.1pr2

Improved validation check for message 2628 when RIPS is enabled

RTA-RTE no longer raises message 2628 when runtime data conversion is configured using `VariableAndParameterInterfaceMapping` for RIPS enabled inter-ecu Sender/Receiver communication graph on the sender side.

Note: RIPS must handle the runtime data conversion in this case.

(RTE-14868)

Improved validation check for message 2628 when RIPS is enabled

RTA-RTE no longer raises message 2628 when runtime data conversion is configured for `ApplicationRecordDataType` using `VariableAndParameterInterfaceMapping` for RIPS enabled intra-ecu Sender/Receiver communication graph.

Note: RIPS must handle the runtime data conversion in this case.

(RTE-14898)

Stop core specific OsScheduleTables

When `RteScheduleTablePerCore` is enabled, RTA-RTE now generates switch blocks inside the `Rte_Stop` API to call `StopScheduleTable`, passing in the parameter `OsScheduleTable` for the core where the `Rte_Stop` API is running in.

(RTE-14378)

Improvements to the RTA-RTE Reference Manual

Chapter 10.2.2 of the RTA-RTE Reference Manual has been updated to describe the usage of the vendor specific Ecuc parameter `RteScheduleTablePerCore`.

(RTE-15123)

Start core specific OsScheduleTables

When `RteScheduleTablePerCore` is enabled, RTA-RTE now generates switch blocks in the `Rte_Start` API to call `StartScheduleTable`/`StartScheduleTableRel`, passing in the relevant `OsScheduleTable` for the core the `Rte_Start` API is running in.

(RTE-14377)

Version 12.2.1pr1

RTE now validates SLATT compumethods for upper/lower limits

RTA-RTE now only generates an error if the configuration contains a `CompuMethod` from category `SCALE_LINEAR_AND_TEXTTABLE` and one of the `CompuScale` limitations is not present , the `LOWER-LIMIT` of `CompuScale` limitations is larger than `UPPER-LIMIT`, or the `UPPER-LIMIT` or the `LOWER-LIMIT` of the CompuScale limitations is not within or equal to the limits defined by the size of the Implementation DataType, which in turn shall be within or equal to the range of the `BaseType`, then RTEGen rejects that configuration with errors 100135, 100136, 100137 respectively.

(RTE-14494)

Activate unmapped DRE RunnableEntitys in Rte_LdComCbkRxIndication

RTA-RTE now invokes `RunnableEntity`s referenced by unmapped `DataReceivedEvent`s that reference unqueued data elements via a direct function call in the related `Rte_LdComCbkRxIndication` APIs.

(RTE-14775)

Support implicit S/R access for runnables started by ModeSwitchEvents

RTA-RTE now supports activation of runnables with `VariableDataAccess` as part of a sender receiver communication graph where those runnables are started by `ModeSwitchEvent`s via a direct call from the associated `Rte_Switch`. The implicit shadow buffers generated for that S/R communication are filled and flushed in the task body associated with the `RunnableEntity` aggregating the `ModeSwitchPoint` leading to the generated of the particular `Rte_Switch` API. Previously RTA-RTE rejected the configuration with error 7724.

(RTE-13576)

4.12 Version 12.2.0

Version 12.2.0pr7

Update the branding of the End User License Agreement

The branding of the End User License Agreement has been updated.

(RTE-15140)

Make LdComUserIPdu.LdComUserCbkHandleId optional

The VSMD file shipped with RTA-RTE has been updated to accept a `LowerMulitplicity` of `0` for the Ecuc parameter `LdComUserCbkHandleId`.

(RTE-15107)

Version 12.2.0pr6

The name of the RTA-RTE excutable for Linux has been updated

The executable shipped with RTA-RTE for Linux has been updated to `RTEGen` (See RTE-15023).

(RTE-15022)

Improved initialization of struct and array members

RTA-RTE now appends ''`.0`'' to the initialization of members of structs and arrays that are floating-point type when no decimal value is provided in the input. Additionally `32-bit` floating-point data are now given a suffix ''`F`''.

(RTE-14806)

Server invocation extracted from task bodies

When the **new** command line option `--optimize-task-stack-size` is provided RTA-RTE generates separate functions for the invocation of each server runnable in an asynchronous C/S communication graph that would previously have been executed in the task body. These static functions are inside the same translation unit and are surrounded by memory allocation keywords with the format `RTE_<SWCT>_[START|STOP]_SEC_CODE` where `<SWCT>` is the `shortName` of the server's `SwComponentType`.

(RTE-13790)

The RTA-RTE User Guide has been updated

The RTA-RTE User Guide section regarding Memory Allocation Keywords for task bodies has been updated in line with the current behavior of RTA-RTE (see RTE-12968).

(RTE-14592)

Support bidirectional routing in COM with multiple senders and receivers

In addition to the support added in issue RTE-11835 RTA-RTE now supports configurations fanning from a single outer port to several inner ports, connected via `DelegationSwConnectors`.

(RTE-14777)

Don't duplicate definitions of constants in the Application Types Header File

RTA-RTE now only generates a single definition for constants in the *Application Types Header File*. Previously where `ImplementationDataTypes` or `ApplicationDataTypes` referenced the same `CompuMethod` the definitions were duplicated.

(RTE-14738)

RTA-RTE should not generate non-applicable C-Type instances for Async Inter-OsApp

When a configuration consists of a queued/un-queued asynchronus client server communication graph, then RTA-RTE will not generate instances of C-Types Rte_REContainerType or Rte_QREType

(RTE-14883)

Two part activation for cross core LdCom aliveTimeout DREEs

When the command line option `--deviate-ldcom-alivetimeout` is provided, RTA-RTE now activates a `DataReceiveErrorEvent` that references an LdCom data element with an `aliveTimeout`, and is mapped to an `OsApplication` with a different `EcucCoreDefinition` compared to the `Rte_MainFunction`, in two parts, first in `Rte_MainFunction` and second in the appropriate task body, using the variables `Rte_<SWCI>_<EVENT>_PartA_Flag` and `Rte_<SWCI>_<EVENT>_PartB_Flag` where `<SWCI>` is the internal name of the relevant `SwComponentPrototype` and `<EVENT>` is the `shortName` of the `DataReceiveErrorEvent`.

(RTE-14735)

Error if ——deviate-cbk-xfrm not used with LdCom IF Queued Receiver

RTA-RTE now only raises error 100126 for the **reception** side of non-RIPS configurations that use queued S/R communication with LdCom IF when the command line option `--deviate-cbk-xfrm` is not provided. Previously issue RTE-14026 did not limit this error to non-RIPS or to reception only.

(RTE-14616)

The RTA-RTE Toolchain Integration Guide has been removed

Substantial portions of the RTA-RTE Toolchain Integration Guide were out of date and it has now been removed. If you are an RTA-CAR customer then consult that documentation for further information.

(RTE-11278)

References to the RTA-RTE Toolchain Integration Guide have been removed

Any references to this guide have either been removed, where the information was outdated, or updated to reference the current relevant documentation.

(RTE-11279)

Update to the EHOOKS report for different but compatible interfaces

RTA-RTE now generates identical EHOOKS reports (`rte_ehooksinfo_conf.xml`) when different but compatible interfaces are used for inner and outer ports (connected via a `DelegationSwConnector`) as when the same interface is used.

(RTE-14774)

Update the branding of the RTA-RTE Release Notes

The branding of the RTA-RTE Release Notes (this document) has been updated.

(RTE-13974)

Two part flags for cross core LdCom aliveTimeout DataReceiveErrorEvents

When the command line option `--deviate-ldcom-alivetimeout` is provided, RTA-RTE now generates the variables `Rte_<SWCI>_<EVENT>_PartA_Flag` and `Rte_<SWCI>_<EVENT>_PartB_Flag` for `DataReceiveErrorEvent`s referencing an LdCom data element with an `aliveTimeout`, and mapped to an `OsApplication` with a different `EcucCoreDefinition` compared to the `Rte_MainFunction`, where `<SWCI>` is the internal name of the relevant `SwComponentPrototype` and `<EVENT>` is the `shortName` of the `DataReceiveErrorEvent`.

Note: This ticket only generates the new variables, the activation of the `DataReceiveErrorEvent`'s `RunnableEntity` using the new variables will be generated in a future ticket.

(RTE-14734)

Version 12.2.0pr5

Default behavior for RteUseComShadowSignalApi updated

The VSMD file shipped with RTA-RTE has been updated to contain the legacy parameter `RteUseComShadowSignalApi`. The default behavior for RTA-RTE when this parameter is not provided or when it is set to `false` is to generate the modern APIs `Com_InvalidateSignal`, `Com_ReceiveSignal` and `Com_SendSignal` instead of their legacy counterparts `Com_InvalidateShadowSignal`, `Com_ReceiveShadowSignal` and `Com_UpdateShadowSignal` respectively. The legacy APIs can still be generated by setting `RteUseComShadowSignalApi` to `true`.

(RTE-9076)

Error if configuration requires multiple different periods for Rte_MainFunction

RTA-RTE will now raise an error 100133, when an input configuration requires multiple different periods for the `Rte_MainFunction`. This scenario can occur when configuring LdCom `aliveTimeout`s and multiple Com `aliveTimeout`s.

(RTE-14677)

Update transformer function bufferLengths from uint16 to uint32

RTA-RTE now uses the `uint32` type for `bufferLength` and `inputBufferLength` in the interface with transformer functions, this brings RTA-RTE up to date with

AUTOSAR 4.3 and higher. Previously the `uint16` type was used.

(RTE-14745)

Error if Rte_MainFunction has different periods via MSI

RTA-RTE will now raise an error 100132, if a configuration contains a `MinimumStartInterval`, a `BSWTimingEvent` that is not 10ms and the command line option `--period` is not provided, in the context of LdCom with `aliveTimeout`s.

(RTE-14678)

Update the branding of the RTA-RTE User Guide

The branding of the RTA-RTE User Guide has been updated.

(RTE-13824)

Allow a FID to rename ParameterDataPrototype symbols in Rte_Type.h

RTA-RTE now renames the symbol of the struct member generated in `Rte_Type.h` representing a `ParameterDataPrototype` using the `shortName` of a `FlatInstanceDescriptor` when one is configured to reference that `ParameterDataPrototype`. This matches the name in the generated `Rte_McSupportData.arxml`.

(RTE-13016)

Version 12.2.0pr4

Support for inPlace data transformation

RTA-RTE now supports the `inPlace` data transformation for explicit sender/receiver communication for `Rte_Write` and `Rte_Send` APIs when `inPlace` is enabled for a `TransformationTechnology`.

(RTE-14214)

Process only Rte specific RteComUser containers

RTA-RTE now ignores `RteComUser` containers which are not part of the `Rte` module.

(RTE-14543)

Error if Rte_MainFunction has a period via command line different from input config

RTA-RTE will now raise an error 100134, if the command line option `--period` is provided with a different period than the one provided for the `Rte_MainFunction` in the input config.

(RTE-14554)

Improved concurrency protection for LdCom aliveTimeouts

RTA-RTE now uses concurrency protection in the `Rte_CheckLdComAliveTimeouts` function only around the checking and setting of the LdCom timeout counter(s) and `Rte_Status` variable(s). Any calls to `ActivateTask` generated for `DataReceiveErrorEvent`s are outside of this protected region. This updates issue RTE-14460.

(RTE-14490)

Do not queue the composite type for queued RIPS inter-ECU S/R comms

RTA-RTE now no longer uses the composite type, see issue RTE-9593, in the RTE

managed queue created for queued RIPS explicit inter-ECU sender/receiver communication with transformation when the command-line option `--deviate-cbk-xfrm` is provided, instead the appropriate `ImplementationDataType` is used, as when `--deviate-cbk-xfrm` is not provided.

(RTE-14733)

Improved error message for RteComUser container validation

RTA-RTE now raises error 100131 (replacing 100120, see issue RTE-13813), the error has been improved to provide the `shortName` of the relevant `RteComUser` container.

(RTE-14542)

Support for inPlace data transformation

RTA-RTE now supports the `inPlace` data transformation for explicit sender/receiver communication for `Rte_Read` and `Rte_Receive` APIs when `inPlace` is enabled for a `TransformationTechnology`.

The command line option `--xfrm-ignore-inplace` has been removed, as the RTA-RTE now supports `inPlace` data transformation.

(RTE-14217)

Error if multiple RIPS plugins INDEX values are not unique or missing

RTA-RTE will now raise a separate error 100129, for each RIPS plugin whose INDEX values are either duplicated or missing.

(RTE-8981)

RteExclusiveAreaImplMechanisms for split SWCI/BSW modules

RTA-RTE now accepts configurations where split SWCI/BSW modules (requires `--deviate-split-swci-support`) are configured with an `RteExclusiveAreaImplMechanism` of `OS_SPINLOCK` when using `ExclusiveArea`s in the `canEnter` role. The code generated to handle the `ExclusiveArea` is the same as if the SWCI/BSW module was not split.

As part of this work error 1356 has been updated to 100128 and is now only raised when a split SWCI/BSW module is configured to use Exclusive Areas in the `runsInside` role.

Additionally RTA-RTE now raises error 100130 when a split SWCI/BSW module is configured to an `RteExclusiveAreaImplMechanism` other than `OS_SPINLOCK` when using `ExclusiveArea`s in the `canEnter` role.

(RTE-13449)

Version 12.2.0pr3

Data conversion where only one Unit is configured

RTA-RTE now accepts configurations where data conversion is taking place and one side of the data conversion does not reference a `Unit`. RTA-RTE behaves as if the missing `Unit` has default values for `factorSiToUnit` equal to one and `offsetSiToUnit` equal to zero.

(RTE-13429)

Do not emit Rte_LdComAliveTimeoutTickCount if there are no LdCom timeouts

RTA-RTE now only emits the `Rte_LdComAliveTimeoutTickCount` variable (see issue RTE-14035) if the command line option `--deviate-ldcom-alivetimeout` is provided **and** an LdCom communication graph contains a port with the attribute `aliveTimeout` set to a non zero value.

(RTE-14404)

Expanded support for enableTakeAddress

RTA-RTE now generates C functions, not function-like preprocessor macros, when a `PortPrototype` is referenced by `PortAPIOption` with `enableTakeAddress` set to `true`.

This updates known issue 454627.

(RTE-13785)

Update the EHOOKS report to include buffers that are written but not read

RTA-RTE now includes entries in the EHOOKS report (`rte_ehooksinfo_conf.xml`) for implicitly written `VariableDataPrototype`s that are not also implicitly read.

(RTE-14252)

Activate unmapped DRE RunnableEntitys via direct function call in Rte_-COMCbk

RTA-RTE now invokes `RunnableEntity`s referenced by unmapped `DataReceivedEvent`s via direct function call in related `Rte_COMCbk` APIs. If the `DataReceivedEvent`'s parent `SwComponentPrototype` is not mapped to the same `EcucPartition` as `COM`, then the direct function call instead occurs in the `Rte_ReceiverPullCB` API.

This is currently only supported for `DataReceivedEvent`s referencing unqueued data elements.

(RTE-14245)

Do not generate COM variables for LdCom with aliveTimeouts

RTA-RTE no longer generates unused `Rte_TOut` variables when LdCom has been configured with an `aliveTimeout`. Note that RTEGen only accepts LdCom with `aliveTimeout`s when the command line option `--deviate-ldcom-alivetimeout` is provided.

This is in addition to issue RTE-14264 that removed invalid `Rte_TOut` variables.

(RTE-14441)

Do not duplicate task activations in Rte_COMCbk

RTA-RTE no longer duplicates task activations inside the `read_ok` conditional in `Rte_COMCbk` when a `SystemSignal` referenced by multiple `ISignals` is mapped to multiple inner `RPortPrototype`s and the `variableDataPrototype`s defined in context of those `RPortPrototype`s are read by different `RunnableEntity`s that are activated by `DataReceivedEvent`s that are, themselves, mapped to the same task.

(RTE-10211)

Update to the EHOOKS report for Inter-ECU IOC reception

RTA-RTE now correctly adds the requirer / `READ` entry of the EHOOKS report (`rte_ehooksinfo_conf.xml`) and uses the `FlatInstanceDescriptor` from the requirer side in the `VAR-FID-REF` section for inter-ECU inter-`OsApplication`

sender/receiver implicit communications.

(RTE-14251)

The RTA-RTE User Guide has been updated

A note in the RTA-RTE User Guide section 5.3.4 has been updated to clarify that, while RTA-RTE does not natively support dynamic-sized arrays, they are supported through the `RTE-RIPS_SRCom` module.

(RTE-14258)

Version 12.2.0pr2

Generate modern Rte_COMCbk APIs

RTA-RTE now generates AUTOSAR R21-11 compliant Rte_<Partition>_COMCbk APIs when the input configuration contains an RteComUser aggregating a ComUserCallback with ComUserCallbackType COM_RX_ACK, where <Partition> is the EcucPartition referneced by an RteComUserEcucPartitionRef in that RteComUser.

(RTE-13815)

INDEX value is now optional when only a single RIPS plugin exists

RTA-RTE now accepts a RIPS plugin without an INDEX value when the configured RteRips container only includes a single RteRipsPluginConfigurationRef. RTEGen now behaves as if the INDEX value is '0'. Previously RTEGen silently ignored a RIPS plugin if it had no INDEX value.

(RTE-8980)

Don't use concurrency protection when calling NvM_SetRamBlockStatus

RTA-RTE now calls NvM_SetRamBlockStatus outside of any concurrency protection that may be generated for the associate data assignment.

(RTE-12841)

Version 12.2.0pr1

Data conversion where only one side references a CompuMethod

RTA-RTE now accepts configurations where linear data conversion is taking place and one side references a CompuMethod but the other side does not. RTA-RTE behaves as if the missing CompuMethod is of category IDENTICAL.

(RTE-13427)

Support for mayBeUnconnected

RTA-RTE now considers the attribute mayBeUnconnected when raising messages 7210, 7220, 6114 and 6238 for unconnected RPortPrototypes.

Note that the presence and severity of these messages is also influenced by the command line option strict-unconnected-rport-check. See the RTA-RTE Reference Manual for more information.

(RTE-12374)

Cross core concurrency protection with inter-ecu-no-ioc

RTA-RTE now uses Rte_GetResource and Rte_ReleaseResource in task bodies for implicit data received inter-ECU when concurrency protection is required around the shadow buffer fill and the command line option inter-ecu-no-ioc is

enabled. Previously RTA-RTE would use Rte_GetSpinlock and Rte_ReleaseSpinlock if the communication was across cores.

(RTE-13914)

Precedence of DataConstrs in MCSD

RTA-RTE now reports the DataConstr configured on the ApplicationDataType in the generated Rte_McSupportData.arxml when DataConstrs are configured on both the ImplementationDataType and ApplicationDataType for a reported data instance.

(RTE-13312)

Rte_IReadRef also for RIPS data elements

When RIPS is enabled, RTA-RTE now generates the Rte_IReadRef API, redirecting to Rte_Rips_IRBufferRef, for a VariableDataPrototype when a RunnableEntity has a VariableAccess in the dataReadAccess role referring to that VariableDataPrototype and the VariableDataPrototype is referenced by a RIPS FlatInstanceDescriptor.

(RTE-6909)

Error if deviate-cbk-xfrm not used with LdCom IF Queued S/R

RTA-RTE will now raise error 100126 for configurations which use queued S/R communication with LdCom IF when the command line option deviate-cbk-xfrm is not provided.

(RTE-14026)

Do not emit additional Rte_MainFunction BSWMD entries

RTA-RTE does not emit the entries related to the Rte_MainFunction in the generated Rte_BSWMD.arxml when a BswModuleEntry with shortName Rte_MainFunction is configured in the input to RTEGen.

(RTE-14477)

Do not raise error 7648 for non-reentrant RIPS servers

RTA-RTE no longer raises error 7648 when the runnable triggering this error is a non-reentrant RIPS server runnable. Previously when the command line option force-basic-tasks was enabled and a task contained a RIPS server runnable with canBeInvokedConcurrently set to false RTA-RTE would reject the configuration with error 7648.

(RTE-14034)

Updated naming for trusted function code related to LdCom aliveTimeouts

The code generated by RTA-RTE to support trusted function calls used in the context of LdCom with aliveTimeout (requires deviate-ldcom-alivetimeout) has been updated to use the SwComponentPrototype shortName (or referencing FlatInstanceDescriptor) instead of the SwComponentType shortName.

The new name is of the form <SwCP>_<RE> where <SwCP> is the SwComponentPrototype shortName (or referencing FlatInstanceDescriptor) and <RE> is the shortName of the RunnableEntity being activated by the relevant DataReceiveErrorEvent.

Note: Care should be taken not to choose a symbol that would clash with this generated code. For example, if the symbol for the RunnableEntity in the above

context were formed by prefixing the shortName of the RunnableEntity with the shortName of the SwComponentPrototype (or referencing FlatInstanceDescriptor) then the generated code would contain duplicate symbols leading to compiler issues.

(RTE-14432)

4.13 Version 12.1.2

Version 12.1.2pr6

DataSendCompletedEvent triggering in RIPS graphs

Support was added in RTE-13897 for RIPS redirection of Rte_Feedback APIs when the associated SenderComSpec specified the use of the transmissionAcknowledgement attribute.

RTA-RTE will now trigger any runnables referenced by a DataSendCompletedEvent on a RIPS communication graph using the transmissionAcknowledgement attribute.

(RTE-14538)

Version 12.1.2pr5

Limited support for SCALE_LINEAR_AND_TEXTTABLE initValues

RTA-RTE now supports initValues for ApplicationPrimitiveDataTypes referring to CompuMethods of category SCALE_LINEAR_AND_TEXTTABLE where the specified initValue falls within the LINEAR portion of the relevant CompuMethod.

Notes:

- RTA-RTE allows more than one CompuScale to be configured.
- The first matching CompuScale encountered, where the value being converted falls within the lowerLimit and the upperLimit, is used to generate the initValue.
- When a CompuScale has no defined limits it will always be applied if encountered first, even if subsequent CompuScales contain limits that match the value for conversion.
- RTA-RTE does not apply any validation to the CompuScale limits, e.g., there is no validation that lowerLimit < = upperLimit or that there are no overlapping CompuScale limits provided.
- When the initValue provided falls outside of any limits no conversion occurs and RTEGen does not generate any notification.

(RTE-13745)

Use concurrency protection when setting Rte_Status for LdCom aliveTimeouts

RTA-RTE now uses concurrency protection in the Rte_CheckLdComAliveTimeouts function when checking the LdCom timeout counter(s) and setting the Rte_Status variable(s).

Note that this concurrency protection is always generated and while some code for DataReceiveErrorEvents is also generated DataReceiveErrorEvents should

not be used at this time as it will lead to the generation of calls to ActivateTask within a protected region.

(RTE-14460)

Only activate DataReceiveErrorEvents for LdCom aliveTimeouts

RTA-RTE now only sets activation flags and calls ActivateTask for DataReceiveErrorEvents referencing a data element whose LdCom aliveTimeout expired.

Additionally, when setting activation flags via trusted functions, the first argument to CallTrustedFunction now omits the prefix `TRUSTED_`. Note that CallTrustedFunction is used when the DataReceiveErrorEvent is mapped to a different OsApplication than Rte_MainFunction (see RTE-14085).

(RTE-14447)

Rte_MainFunction rate when using deviate-ldcom-alivetimeout

When the command line option deviate-ldcom-alivetimeout is enabled RTA-RTE now determines the rate of the Rte_MainFunction from the period of the mapped BswTimingEvent (or default rate of 10ms). This updates the behavior described in RTE-14085.

Note: No validation is done to check for inconsistent configuration of the command line option period and the configured BswTimingEvent period.

(RTE-14444)

Use concurrency protection when setting LdCom timeout counters

RTA-RTE now uses concurrency protection in the Rte_LdComCbkRxIndication API when setting the aliveTimeout counter generated to support LdCom reception with an aliveTimeout configured (requires deviate-ldcom-alivetimeout).

(RTE-14406)

Do not make duplicate calls to IocWrite in Rte_COMCbk

RTA-RTE no longer generates duplicate calls to IocWrite (i.e., for the same channel) in Rte_COMCbk.

(RTE-14387)

Rename timeout variables for LdCom with aliveTimeouts

RTA-RTE now uses the SwComponentPrototype's name instead of the SwComponentType's for the individual aliveTimeout counters generated to support LdCom aliveTimeouts.

Note that RTEGen only accepts LdCom with aliveTimeouts when the command line option deviate-ldcom-alivetimeout is provided.

(RTE-14405)

Version 12.1.2pr4

Set status and activate tasks for LdCom with aliveTimeouts

RTA-RTE now sets the status to RTE_E_MAX_AGE_EXCEEDED, sets any associated activation flags, and activates the relevant OsTask when the individual tick counter (see RTE-14087) equals the global tick counter (see RTE-14035) for a particular data element with an LdCom aliveTimeout configured.

When the activated runnable is in another OsApplication compared to the Rte_-

MainFunctionthen CallTrustedFunction is used to set the activation flag.

Note: The integrator is required to activate the Rte_MainFunction at the rate specified by the period (or default rate of 10ms). This activation is done by configuring a BswSchedulableEntity referencing a BswModuleEntry (via ImplementedEntryRef) named Rte_MainFunction, the partition of the Rte_MainFunction is determined through the event mapping. Note further for this BswSchedulableEntity's BswImplementation, the vendorId and vendorApiInfix must be omitted.

(RTE-14085)

Use ImplementationDataTypes for RIPS global copy type definitions

When a communication graph is referenced by a RIPS FlatInstanceDescriptor, in the generated Rte_DataHandleType.h file, the typedef defining Rte_Rips_GlobalCopy_<FID>_Type to be the type of the global buffer associated with that communication graph, its global type is the ultimate ImplementationDataType of the RTE managed global buffer for that data element. <FID> is the shortName of the referencing RIPS FlatInstanceDescriptor

(RTE-9728)

Do not optimize 1:N IOC over a single channel

RTA-RTE now suppresses the optimization of 1:N IOC channels and instead uses multiple channels when the command line option no-optimized-ioc-channels is provided (see RTE-11324).

(RTE-14289)

Do not raise error 100072 when DataTypePolicy is set to TransformingISignal

RTA-RTE no longer raises error 100072 when an ISignal with DataTypePolicy set to TransformingISignal aggregates a NetworkRepresentationProps and generates as if that NetworkRepresentationProps was not present.

(RTE-13163)

Refactor inter-ECU communication graphs

As part of a refactor of the generator code the order that inter-ECU communication graphs are processed and hence the sequence number of RTE managed buffers associated with them has changed.

(RTE-12017)

Do not generate COM variables for LdCom with aliveTimeouts

RTA-RTE no longer generates Rte_TOut_<undefined> when LdCom has been configured with an aliveTimeout. Note that RTEGen only accepts LdCom with aliveTimeouts when the command line option deviate-ldcom-alivetimeout is provided.

(RTE-14264)

RTA-RTE now supports additional PRPorts scenarios

RTA-RTE now accepts additional PRPorts configurations including M:N communication graphs when the command line option extended-prports-support is enabled. See the RTA-RTE Reference Manual for limitations.

(RTE-13880)

Update the branding of RTA-RTE Error Description

The branding of the RTA-RTE Error Description has been updated.

(RTE-13973)

Version 12.1.2pr3

Global tick counter to support LdCom aliveTimeouts

When the command line option deviate-ldcom-alivetimeout is enabled RTA-RTE now generates a global variable named Rte_LdComAliveTimeoutTickCount. That variable counts ticks relative to the Rte_MainFunction function and is used to support LdCom with aliveTimeouts.

(RTE-14035)

Set AliveTimeout counters for LdCom

RTA-RTE now sets the individual aliveTimeout counters in the Rte_LdCom-CbkRxIndication APIs to the calculated tick timeout, when the command line option deviate-ldcom-alivetimeout is provided.

(RTE-14087)

RTA-RTE now implements Rte_IReadRef

RTA-RTE now generates the Rte_IReadRef API for a VariableDataPrototype when a RunnableEntity has a VariableAccess in the dataReadAccess role referring to that VariableDataPrototype.

(RTE-6908)

Update the branding of RTA-RTE Reference Manual

The branding of the RTA-RTE Reference Manual has been updated.

(RTE-13791)

Error when aliveTimeout with LdCom is smaller than Rte_MainFunction period

RTA-RTE now raises error 100122 when the aliveTimeout is smaller than the Rte_-MainFunction period.

Note: Support for LdCom with aliveTimeout requires command line option deviate-ldcom-alivetimeout to be enabled.

(RTE-14135)

Applying float suffix to RTE buffers with floating point initial values

In order to avoid compiler warnings and additional memory usage, initial values of 32-bit floating-point data are now given a suffix F.

(RTE-6701)

Error when aliveTimeout with LdCom is greater than :math:`2^{32}`

RTA-RTE now raises error 100123 when the aliveTimeout, converted to ticks based on the period of the Rte_MainFunction, is greater than 2^{32} (the maximum value of the LdCom tick counter).

Note: Support for LdCom with aliveTimeout requires command line option deviate-ldcom-alivetimeout to be enabled.

(RTE-14136)

AliveTimeout variables for LdCom

RTA-RTE now generates aliveTimeout variables for an LdCom communication graph with an aliveTimeout when the command line option deviate-ldcom-alive-timeout is provided.

(RTE-14079)

Fix support for legacy Rte_NvMNotifyJobFinished signatures in modern NvM

When generating a modern Rte_NvMNotifyJobFinished callback function and a RoleBasedPortAssignment whose effective Operation's ArgumentDataPrototypes' shortNames match the legacy Rte_NvMNotifyJobFinished callback signature, RTA-RTE now passes JobResult instead of the erroneous JobRequest as the second argument when calling the effective Operation.

(RTE-14188)

Support Rte_Feedback for communication graphs with RIPS

RTA-RTE now redirects calls to Rte_Feedback_<Port>_<VDP> to Rte_Rips_←PlugIn>_Feedback_<SwcBswI>_<CGI> and returns the return value, when transmission acknowledgement is enabled for a RIPS-relevant data element that is transmitted via LdCom/Com as part of an explicit sender/receiver communication graph. Where <PlugIn> is the shortName of the RIPS plugin, <SwcBswI> is the shortName of the relevant SwComponentPrototype, <CGI> is the shortName of the ARRIPS FlatInstanceDescriptor, <Port> is the shortName of the target PPortPrototype and <VDP> is the shortName of the target VariableDataPrototype.

Note that the RTE Implementation Plug-In author must provide the Rte_Rips_Feedback API and handle the feedback variable for that data element.

(RTE-13897)

Validation of the RteComUser container

RTA-RTE now raises error 100120 when a RteComUser container references an ISignalToIPduMapping, via a ComUserSystemTemplateSystemSignalRef or ComUserSystemTemplateSignalGroupRef, and that ISignalToIPduMapping is not also referenced by either a ComSignal or ComSignalGroup respectively.

Note support for partition specific callbacks will be added in future tickets.

(RTE-13813)

Version 12.1.2pr2

Status variable and Rte_IStatus API for LdCom with aliveTimeout

RTA-RTE now generates a global status variable for an LdCom communication graph with an aliveTimeout when the command line option deviate-ldcom-alive-timeout is provided. An Rte_IStatus API is also generated if the related VariableAccess is a DataReadAccess.

(RTE-14086)

Deprecated option com-symbolic-sigs

The command line option com-symbolic-sigs has been deprecated, as well as its abbreviation cy. Note that RTEGen now always behaves as if this options were enabled.

(RTE-13955)

AUTOSAR 22-11 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace http://autosar.org/schema/r4.0) against the AUTOSAR 22-11 schema.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to 22-11 will validate against the 22-11 schema.

(RTE-13923)

Error for LdCom with aliveTimeouts without deviate-ldcom-alivetimeout

RTA-RTE now raises error 100121 when a configuration contains aliveTimeouts with LdCom communication unless the command line option deviate-ldcom-alivetimeout is provided.

(RTE-14033)

Updated the branding of the RTA-RTE StarterKit

The branding of the RTA-RTE StarterKit has been updated.

(RTE-13972)

RTA-A2L distribution has moved

The RTA-A2L tool has moved from the RTA-RTE installation to RTA-CAR.

(RTE-13538)

Error messages 2692 wording correction

RTA-RTE now uses the correct operator (`+`, `*`, `/`) in error message 2692 when reporting that there is no sufficiently wide platform type available to hold the possible intermediate values calculated for a particular data conversion. Previously the operator `+` was always reported.

(RTE-12058)

Version 12.1.2pr1

Improved Compliance with MISRA Rule 8.2

RTA-RTE now names all the parameters of the function declaration of Rte_MSITest generated in Rte_Intl.h. These names are the same as the parameter names of the corresponding function definition of Rte_MSITest generated in Rte_Lib.c. This addresses violations of MISRA C:2012 Rule 8.2.

(RTE-6471)

Allow aliveTimeouts with LdCom using deviate-ldcom-alivetimeout

RTA-RTE no longer raises an error when the command line option deviate-ldcom-alivetimeout is provided and a configuration contains aliveTimeouts with LdCom communication.

Note: At this time RTA-RTE accepts the configuration but does not generate any additional code to support the aliveTimeout.

(RTE-14032)

4.14 Version 12.1.1

Version 12.1.1pr3

NvM callbacks now wrapped in partition specific memory allocation keywords

RTA-RTE now wraps the Rte_SetMirror, Rte_GetMirror, Rte_NvMNotifyInitBlock and Rte_NvMNotifyJobFinished NvM callback functions in memory allocation keywords of the form RTE_<PARTITION>_START_SEC_CODE and RTE_<PARTITION>__STOP_SEC_CODE, where <PARTITION> is EcucPartition mapped to the relevant NvMBlockDescriptor via an NvMBlockEcucPartitionRef. When no NvMBlockEcucPartitionRef is configured then <PARTITION> is the EcucPartition with EcucPartitionBswModuleExecution set to TRUE.

(RTE-12974)

Lower/UpperLimit defines for mapped ApplicationPrimitiveDataTypes

RTA-RTE now defines the macros <prefix><DataType>_LowerLimit and <prefix><DataType>_UpperLimit in the appropriate *Application Types Header File* with the effective internal range for each ApplicationPrimitiveDataType mapped to an ImplementationDataType via a DataTypeMappingSet referenced by a SwComponentType. Where <prefix> is the optional IncludedDataTypeSet.literalPrefix if the ApplicationPrimitiveDataType is referenced from one of the SwComponentType's IncludedDataTypeSets and <DataType> is the shortName of the ApplicationPrimitiveDataType.

The effective internal range is first determined from the ApplicationPrimitiveDataType's DataConstr. If a Unit is not defined on either the ApplicationPrimitiveDataType or the DataConstr, they are assumed to be the same. If the CompuMethod is not defined, it is assumed to be of category IDENTICAL. As a fallback, the effective internal range is then determined from the ImplementationDataType's DataConstr.

(RTE-13708)

Version 12.1.1pr2

Memory Allocation Keywords for activation flags

RTA-RTE now declares the start and stop Memory Allocation Keywords for activation flags, that are initialized, with the <SwAddrMethod> VAR_INIT. This updates the work done in issue RTE-12911 where the <SwAddrMethod> was VAR_CLEARED.

(RTE-13709)

Improved Compliance with MISRA Rule 21.1

RTA-RTE now omits the leading underscore from the #define generated in Rte_Const.h that maps the internal name of each SoftwareComponent instance to a SwComponentPrototype based on the full shortName of that SwComponentPrototype with forward slashes converted to underscores. This addresses violations of MISRA C:2012 Rule 21.1

(RTE-6470)

LdCom can be called from any partition when ——inter-ecu-no-ioc is enabled

RTA-RTE now assumes that IOC is not required for queued communication between the LdCom EcucPartition and the EcucPartition(s) of the receiving runnable(s) when the command line option --inter-ecu-no-ioc is enabled.

Specifically, RTA-RTE now uses non-IOC code to communicate between the Rte_LdComCbkRxIndication and the Rte_Receive API when the reading runnable is in a different EcucPartition to LdCom.

Note: For queued LdCom the command line option `--deviate-cbk-xfrm` must also be enabled.

(RTE-13795)

Memory Allocation Keywords for implicit N:1 communication buffers

RTA-RTE now declares the start and stop Memory Allocation Keywords for the RTE generated buffer for implicit N:1 communication in the format `RTE_<INFIX>_START_SEC_CODE` and `RTE_<INFIX>_STOP_SEC_CODE` respectively, where `<INFIX>` is the `ShortName` of the `EcucPartition` of the `OsApplication` that has exclusive access to the buffer.

(RTE-13803)

RTA-RTE is now available for Ubuntu 20.04

RTA-RTE is now available for Ubuntu 20.04. It is packaged alongside the existing Windows executable and documentation in `RTA-RTE_<Version>_Ubuntu20.04.tar.gz`, where `<Version>` refers to the version of RTA-RTE in that release.

To use RTEGen on Ubuntu 20.04:

> tar -xzf RTA-RTE_<Version>_Ubuntu20.04.tar.gz LC_CTYPE=.cp1252//-TRANSLIT ./RTEGen.exe –version

Known differences between OS versions are documented in Chapter 3.

(RTE-12332)

Updated ETAS License Manager

RTA-RTE now ships with ETAS License Manager (LiMa) version 1.8.9.37. Note: This is the minimum version tested for RTA-RTE on Linux.

(RTE-13750)

Updated RTA-RTE build type metadata

When RTEGen is invoked with the command line option `--version` the `Build Type` is now set to `Release`. This corrects a change made in issue RTE-9349 where the `Build Type` was set to `Debug`, which is used for internal testing only.

(RTE-13967)

Include Comstack_Types.h in Rte.h when (Ld)Com is used

RTA-RTE now includes `ComStack_Types.h` in the generated `Rte.h` file when a configuration contains an `EcucModuleConfigurationValues` element referencing an `EcuModuleDef` with `ShortName` 'LdCom' or 'Com'. It also includes `Rte_Type.h` in the generated `Rte_Cbk.h` file before any `LdCom` or `Com` callback function is declared. Previously RTA-RTE included `Comstack_Types.h` in `Rte_Cbk.h`, see issue RTE-4377.

(RTE-12798)

Memory Allocation Keywords for write-if-dirty code sections

RTA-RTE now declares the start and stop Memory Allocation Keywords for the RTE generated `RunnableEntity` that writes back to the `ramBlock` when the `DirtyFlag` is `TRUE` (write-if-dirty) in the format `RTE_<INFIX>_START_SEC_CODE` and `RTE_<INFIX>_STOP_SEC_CODE` respectively, where `<INFIX>` is the `shortName` of the `EcucPartition` that references the `SwComponentPrototype` typed by the `NvBlockSwComponentType` that aggregates the `NvBlockDescriptor` that aggregates

the ramBlock.

(RTE-13535)

Version 12.1.1pr1

Updated Windows runtime libraries and utilities

The Windows runtime libraries and utilities shipped in the RTA-RTE bin directory have been updated.

(RTE-10802)

Memory Allocation Keywords for BswExecutableEntity activation flags

RTA-RTE now declares the start and stop Memory Allocation Keywords for flags guarding the activation of BswExecutableEntitys, whether activated by RTEEvents or BswEvents, in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>_<ALIGNMENT> and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>_<ALIGNMENT> respectively, where <INFIX> is the shortName of the EcucPartition that owns the Event that activates the BswExecutableEntity, <SwAddrMethod> is VAR_CLEARED and <ALIGNMENT> is 8.

(RTE-13471)

Memory Allocation Keywords for client/server data structures

RTA-RTE now includes the shortName of the relevant EcucPartition in the memory allocation keywords for data structures used in asynchronous intra-OsApplication client/server communication graphs. This extends the work done in issue RTE-9241.

(RTE-9607)

Memory Allocation Keywords for IIRVs without EcucPartitions

RTA-RTE now uses the OsApplication associated with an ImplicitInterRunnableVariable in the Memory Allocation Keyword <INFIX> when no EcucPartition is configured (see issue RTE-12298).

(RTE-13356)

Memory Allocation Keywords for NvBlockDescriptor's DirtyFlag

RTA-RTE now declares the start and stop Memory Allocation Keywords for DirtyFlags in the format RTE_<INFIX>_START_SEC_INIT_8 and RTE_<INFIX>_STOP_SEC_INIT_8 respectively, where <INFIX> is the shortName of the EcucPartition that references the SwComponentPrototype typed by the NvBlockSwComponentType that aggregates the NvBlockDescriptor that aggregates the ramBlock.

(RTE-13200)

Improved compliance with MISRA Rule 9.1

RTA-RTE now initialises variables with automatic storage duration created to support DataTransformation. This addresses violations of MISRA C:2012 Rule 9.1 solely in the scope of DataTransformation.

(RTE-13409)

Update to RTA-RTE interaction with the ETAS License Manager

The error messages related to RTA-RTE interaction with the ETAS License Manager have been updated to remove OS specific wording.

(RTE-9348)

Update the branding of lRTA-RTEgettingstartedl

The branding of the RTA-RTE Getting Started Guide has been updated.

(RTE-13738)

Updated RTA-RTE metadata to include the operating system

In preparation for releasing a Linux version of RTA-RTE all C source and header files, ARXML files, and the output to `stdout` when RTEGen is invoked (with or without the command line option `--version`) now includes the operating system for which that executable was built. The output with the command line option `--version` has also been supplemented with additional information visible in the Windows file properties for the RTEGen executable.

(RTE-9349)

4.15 Version 12.1.0

Version 12.1.0pr10

Map subElement byte arrays to primitive signals with compatible interfaces

Support for mapping a `subElement` byte array to a single `SystemSignal`, introduced in issue RTE-13729, has been extended to also accept configurations where the `SenderReceiverToSignalGroupMapping` points to a `PortPrototype` referenced as an `outerPort` in a `DelegationSwConnector` and the referenced `innerPort` `PortPrototype`s are typed by different but compatible `SenderReceiverInterface`s.

(RTE-13856)

Version 12.1.0pr9

COM can be called from any partition when ——inter-ecu-no-ioc is enabled

RTA-RTE now assumes that IOC is not required when communicating between the COM `EcucPartition` and the `EcucPartition`(s) of the receiving runnable(s) when the command line option `--inter-ecu-no-ioc` is enabled.

Specifically, RTA-RTE now uses non-IOC code to communicate between the `COMCbk` and the global buffers or `Rte_Read` or `Rte_Receive` (dependent on communication semantics) when the reading runnable is in a different `EcucPartition` to COM.

(RTE-13852)

Additional license feature for RTA-RTE

RTA-RTE now accepts the license feature `RTA_RTE` in addition to the existing license feature `RTA-RTE`.

(RTE-13747)

Support data mapping of subElement byte array to single SystemSignal

RTA-RTE now accepts configurations where a `subElement` byte array is mapped to a single `SystemSignal` via the respective `SenderRecArrayElementMapping` / `SenderRecRecordElementMapping`'s `systemSignalRef`. Previously RTA-RTE rejected the configuration with error 1694.

Note that in this case a byte array is defined as an array whose elements have a `SwBaseType` with `baseTypeEncoding` `NONE` and `baseTypeSize` `8`.

(RTE-13729)

RIPS specific license feature removed

RTA-RTE no longer requires the users to have access to a specific license feature, `RTA-RIPS`, to enable the use of RIPS mode.

As a result of this change, RTA-RTE accepts any valid license feature, such as `RTA-RTE`, to generate an RTE that requires the use of RIPS mode.

(RTE-13612)

Update to the EHOOKS report for split-swc ImplicitInterRunnableVariables

The RTA-RTE EHOOKS report (`rte_ehooksinfo_conf.xml`) now only reports the implicit shadow buffers generated for runnables accessing split-swc `ImplicitInterRunnableVariable`s in the `MEASUREMENT` section generated for the globals used to fill that shadow buffer.

(RTE-13338)

Version 12.1.0pr8

Allow client/server cross cluster communication using compatible interfaces

RTA-RTE now accepts configurations that define a cross cluster RIPS `FlatInstanceDescriptor` referencing a `ClientServerOperation` in a delegation `PortPrototype` which is connected to an inner `PortPrototype` and both `PortPrototype`s are typed by different but compatible `ClientServerInterface`s. The generated output is the same as it would be if the `PortPrototype`s were typed by the same `ClientServerInterface`.

(RTE-13697)

Raise an error for synchronous cross cluster client/server communication

RTA-RTE now raises error 100118 for an input configuration with a `SynchronousServerCallPoint` whose related operation instance is part of a cross cluster communication graph.

(RTE-11701)

Mixing RIPS local and cluster plugins for client/server

When a RIPS local plugin is configured in combination with a RIPS cluster plugin as part of a client/server communication graph RTA-RTE behaves as if there was only the local plugin which is responsible for implementing all functionality.

Note that RTA-RTE continues to call the appropriate start/stop function for the RIPS cluster plugin (see issue RTE-12129).

(RTE-11700)

Validate cross cluster ASCREs/OIEs mapped to periodic OsTasks

RTA-RTE now raises error 100117 for an input configuration with an `AsynchronousServerCallReturnsEvent` or `OperationInvokedEvent` mapped to an `OsTask` that does not contain any `TimingEvent`s or `BswTimingEvent`s and the related operation instance is part of a cross cluster communication graph.

(RTE-12176)

Reduced validation of cluster configurations

RTA-RTE no longer raises error 100097 when generating for a cluster and a data element referenced by a `PortElementToCommunicationResourceMapping` is not

referenced by a cross-cluster RIPS `FlatInstanceDescriptor`. See issue RTE-10528. This error rejected valid configurations such as the use of `PortElementToCommunicationResourceMappings` for non cluster communication from an application cluster.

(RTE-13450)

Inclusion of SWC-BSW instance header for inter-cluster clients

For RIPS relevant inter-cluster client/server communication where a `SwComponentPrototype` acts as the client (aggregates an `AsynchronousServerCallResultPoint`), the generated application header file of the corresponding `SwComponentType` now includes the applicable `RIPS SWC-BSW-Instance Header File`.

(RTE-12165)

Add clarifications where documentation is specific to Windows

Various minor updates to RTA-RTE documentation in preparation for releasing a Linux version of RTEGen.

(RTE-13498)

Version 12.1.0pr7

Rte_Result for cross cluster asynchronous client/server communication

For cluster-relevant client/server communication where the client aggregates an `AsynchronousServerCallPoint`, the `Rte_Result` API generated for that server call point will now calls the `Rte_Rips_ReturnResult` API and return the result as the return of the `Rte_Result` API.

(RTE-11696)

Rte_Call for cross cluster asynchronous client/server communication

For cluster-relevant client/server communication where the client aggregates an `AsynchronousServerCallPoint`, the `Rte_Call` API generated for that server call point will now calls the `Rte_Rips_Invoke` API and return the result as the return of the `Rte_Call` API.

(RTE-11715)

Support RteRipsInvocationHandlerFncs for cross-cluster clients

RTA-RTE now calls the relevant `RteEventToTaskMapping`'s `RteRipsInvocationHandlerFnc`'s `RteRipsInvocationHandlerFncSymbol` in place of the standard `RunnableEntity` activation code for an `AsynchronousServerCallReturnsEvent` whose `ClientServerOperation` instance is part of a cross-cluster communication graph.

(RTE-12175)

Support RteRipsInvocationHandlerFncs for cross-cluster servers

RTA-RTE now calls the relevant `RteEventToTaskMapping`'s `RteRipsInvocationHandlerFnc`'s `RteRipsInvocationHandlerFncSymbol` in place of the standard `RunnableEntity` activation code for an `OperationInvokedEvent` whose `ClientServerOperation` instance is part of a cross-cluster communication graph.

(RTE-11698)

Update to the EHOOKS report for IOC with only a provider FID

RTA-RTE now uses the FlatInstanceDescriptor from the provider side of an inter-OsApplication sender/receiver implicit communication in the VAR-FID-REF section in the requirer / READ entry of the EHOOKS report (rte_ehooksinfo_conf.xml) when no FlatInstanceDescriptor is present on the requirer side.

(RTE-13419)

Improved Compliance with MISRA Rule 8.4

The template files shipped with RTA-RTE have been updated with the addition of a new macro RTE_LOCAL. This macro instantiates the storage-class specifier of the variable Rte_ResourceCount as static, however this can be changed by defining RTE_LOCAL in Rte_UserCfg.h. This addresses violations of MISRA C:2012 Rule 8.4.

Additionally the type-specifier suffix U has been added to the initialization value used when defining the Rte_ResourceCount variables.

(RTE-13167)

Version 12.1.0pr6

Memory Allocation Keywords for S/R ReceiverPullCBs

RTA-RTE now declares the start and stop Memory Allocation Keywords for Rte_ReceiverPullCB APIs, generated for inter-OsApplication sender-receiver with DataReceivedEvents, in the format OS_START_SEC_<SwAddrMethod> and OS_STOP_SEC_<SwAddrMethod> respectively, where <SwAddrMethod> is the SwAddrMethod . shortName if configured in the OsMemoryMappingCodeLocationRef of the receiving OsApplication. If OsMemoryMappingCodeLocationRef is not configured then OsApplication . shortName is used.

Note in either case RTA-RTE converts the full MAKW to upper case.

(RTE-9613)

Start Task Hooks for specific OsTasks

RTA-RTE now generates a call to user provided Start Task Hook API(s) as the first executable line of code inside the OsTask body of each OsTask that is referenced by an (RTA-RTE specific) EcucContainerValue with shortName RteFillFlushHookEnabledOSTasks. The Start Task Hook(s) have the format <Task>_TaskStartHook(), where <Task> is the shortName of the OsTask. The user must declare each Start Task Hook in Rte_UserCfg.h.

Note that the Start Task Hook(s) are not output when the command line option --os-task-as-function is enabled.

(RTE-12586)

End Task Hooks for specific OsTasks

RTA-RTE now generates a call to user provided End Task Hook API(s) directly before the call to TerminateTask() inside the OsTask body of each OsTask that is referenced by an (RTA-RTE specific) EcucContainerValue with shortName RteFillFlushHookEnabledOSTasks. The End Task Hook(s) have the format <Task>_TaskEndHook(), where <Task> is the shortName of the OsTask. The user must declare each End Task Hook in Rte_UserCfg.h.

Note that the End Task Hook(s) are not output when the command line option --os-task-as-function is enabled.

(RTE-12900)

Improved Compliance with MISRA Rule 2.4

Removed unused tags declarations from type definitions in the template files shipped with RTA-RTE. This addresses violations of MISRA C:2012 Rule 2.4.

(RTE-13170)

Changed Memory Allocation Keywords for OsTasks

RTA-RTE now declares the start and stop Memory Allocation Keywords for OsTasks in the format OS_START_SEC_<sadm> and OS_STOP_SEC_<sadm> respectively, where <sadm> is either the shortName of the SwAddrMethod referenced by the OsTask converted to upper case, or CODE_<Taskname> if no SwAddrMethod is referenced, where <Taskname> is the shortName of the corresponding OsTask converted to upper case. The Memory Allocation Keywords are followed by an inclusion of Os_MemMap.h.

(RTE-12968)

Memory Allocation Keywords for RomBlock

RTA-RTE now declares the start and stop Memory Allocation Keywords for RomBlocks in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>[_ALIGNMENT] and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>[_ALIGNMENT] respectively, where <INFIX> is the shortName of the EcucPartition that references the SwComponentPrototype typed by the NvBlockSwComponentType that aggregates the RomBlock, <SwAddrMethod> is the SwAddrMethod . shortName and [_ALIGNMENT] is the width of the corresponding datum and is only generated if the associated SwAddrMethod has memoryAllocationKeywordPolicy of value addrMethodShortNameAndAlignment.

(RTE-12976)

Version 12.1.0pr5

Map NvMBlockDescriptors to EcucPartitions and validate RteNvRamAllocations

RTA-RTE now raises info message 100110 or 100112 per NvMBlockDescriptor for an input configuration with EcucPartitions configured, depending on the success of mapping the NvMBlockDescriptor to an EcucPartition via its NvMBlockEcucPartitionRef.

RTA-RTE now raises error 100113 if an input configuration contains multiple RteNvRamAllocations that reference the same NvBlockDescriptor.

(RTE-12981)

Memory Allocation Keywords for Asynchronous C/S ReceiverPullCBs

RTA-RTE now declares the start and stop Memory Allocation Keywords for inter-OsApplication asynchronous client-server ReceiverPullCBs in the format OS_START_SEC_<SwAddrMethod> and OS_STOP_SEC_<SwAddrMethod> respectively, where <SwAddrMethod> is the SwAddrMethod . shortName if configured in the OsMemoryMappingCodeLocationRef of the receiving OsApplication. If OsMemoryMappingCodeLocationRef is not configured then OsApplication . shortName is used.

Note in either case RTA-RTE converts the full MAKW to upper case.

(RTE-9554)

Updated VSMD with new parameter for OsTask hooks

The VSMD file shipped with RTA-RTE has been updated with a new container to allow pre-fill and post-flush hook functions to be placed on a per task basis. This is achieved by referencing the OsTasks in which hooks are desired in an EcucContainerValue with shortName RteFillFlushHookEnabledOSTasks. Note that this is only the enabling mechanism and no output code is associated with this ticket.

(RTE-12903)

Only generate the RTE specific OsApplication when required

RTA-RTE now only generates the RTE specific OsApplication ('RTE_DEFAULT_OS_APP') when one or more of Rte_TOutCounter, Rte_TickCounter and Rte_ScheduleTable are not mapped to an OsApplication in the provided input configuration. This is in addition to the existing check that the input uses OsApplications and only mapping the constructs that are used in the generated C code.

(RTE-12717)

Do not optimize functions when VFB tracing is enabled

RTA-RTE now implements Rte_CData, Rte_Prm and Schm_CData as functions in the generated Rte.c rather than as comma expressions in the relevant header files when VFB tracing is enabled.

(RTE-13222)

Memory Allocation Keywords for RamBlock

RTA-RTE now declares the start and stop Memory Allocation Keywords for RamBlocks in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>[_ALIGNMENT] and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>[_ALIGNMENT] respectively, where <INFIX> is the shortName of the EcucPartition that references the SwComponentPrototype typed by the NvBlockSwComponentType that aggregates the RamBlock, <SwAddrMethod> is the SwAddrMethod . shortName and [_ALIGNMENT] is the width of the corresponding datum and is only generated if the associated SwAddrMethod has memoryAllocationKeywordPolicy of value addrMethodShortNameAndAlignment.

(RTE-12975)

Internal error 900445 replaced with informative error message

RTA-RTE now raises error 100114 instead of internal error 900445 when two connected structure data types are incompatible. This behavior is the result of a configuration which uses a combination of PortInterfaceMappings, InitialValues and CompuMethods.

(RTE-13237)

MAKWs for ParameterDataPrototypes in ParameterInterfaces

RTA-RTE now defines the start and stop Memory Allocation Keywords for ParameterDataPrototypes used in ParameterInterfaces, in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>_<ALIGNMENT> and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>_<ALIGNMENT> respectively, where <INFIX> is the shortName of the EcucPartition that accesses the ParameterDataPrototypes, <SwAddrMethod> is the SwAddrMethod (or CALPRM if not

set) and <ALIGNMENT> is the width of the corresponding datum.

(RTE-12304)

Improved Compliance with MISRA Directive 4.6

Template files in RTA-RTE, specifically Rte_h.template, were modified such that typedef RTE_ALARMREFTYPE is defined by an unsigned 16 bit integer uint16 as opposed to the previous unsigned int to have the appropriate size and signedness. This addresses violations of MISRA C:2012 Directive 4.6.

(RTE-13146)

Version 12.1.0pr4

Improved compliance with MISRA Rule 10.4

The template files shipped with RTA-RTE have been updated by adding the type-specifier suffix U to constants used in arithmetic operations involving other unsigned integers. This addresses violations of MISRA C:2012 Rule 10.4.

(RTE-13147)

Memory Allocation Keywords for VariableDataPrototypes

RTA-RTE now declares the start and stop Memory Allocation Keywords for VariableDataPrototypes in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>[_ALIGNMENT] and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>[_ALIGNMENT] respectively, where <INFIX> is the shortName of the EcucPartition that accesses the VariableDataPrototype, <SwAddrMethod> is the SwAddrMethod . shortName and [_ALIGNMENT] is the width of the corresponding datum and is only generated if the associated SwAddrMethod has memoryAllocationKeywordPolicy of value addrMethodShortNameAndAlignment.

(RTE-12613)

Remove misleading information from the RTA-RTE User Guide

The RTA-RTE User Guide previously had misleading information describing the legacy requirement to map service components at the root composition level which is no longer supported. That section has now been removed.

(RTE-10156)

Implicit shadow buffer naming fix

RTA-RTE now correctly names implicit shadow buffers rbuf / wbuf when an OsTask contains RTEEvents that trigger RunnableEntitys with implicit read/write access to matching VariableDataPrototypes in PortPrototypes (whose related SwcInternalBehaviour has supportsMultipleInstantiation enabled) that are connected in the context of different SwComponentPrototoypes but not the SwComponentPrototypes that the RTEEvent mappings belong to. Previously RTA-RTE incorrectly named the implicit shadow buffers sbuf which is used for buffers that are both read and written.

(RTE-13192)

Version 12.1.0pr3

Preparatory work for releasing a Linux version of RTA-RTE

In preparation for releasing a Linux version of RTA-RTE there have been several updates to the RTEGen codebase. The changes to the Windows version of RTA-RTE are cosmetic and primarily increase the consistency in outputs between

runs and when compared to the Linux version of RTEGen. The changes include:

- More consistent ordering of Fatal, Error, Warning and Info messages
- Minor changes to the wording of those messages
- Consistent ordering of elements within the generated `Rte_BSWMD.arxml`
- Consistent naming of asynchronous client-server data structures within the generated `Rte.c`

Note: For inquiries about obtaining a Linux version of RTA-RTE contact ETAS.

(RTE-12222)

Memory Allocation Keywords for activation flags

RTA-RTE now declares the start and stop Memory Allocation Keywords for activation flags in the format `RTE_<INFIX>_START_SEC_<SwAddrMethod>_<ALIGNMENT>` and `RTE_<INFIX>_STOP_SEC_<SwAddrMethod>_<ALIGNMENT>` respectively, where `<INFIX>` is the `shortName` of the `EcucPartition` that owns the `RTEEvent` which activates the runnable the `EcucPartition` is guarding, `<SwAddrMethod>` is `VAR_CLEARED` and `<ALIGNMENT>` is `8`.

RTA-RTE now also declares the start and stop Memory Allocation Keywords for server activation flags in the format `RTE_<INFIX>_START_SEC_<SwAddrMethod>_<ALIGNMENT>` and `RTE_<INFIX>_STOP_SEC_<SwAddrMethod>_<ALIGNMENT>` respectively, where `<INFIX>` is the `shortName` of the `EcucPartition` that owns the `RTEEvent` which activates the runnable the `EcucPartition` is guarding, `<SwAddrMethod>` is `VAR_INIT` and `<ALIGNMENT>` is `8`.

(RTE-12911)

Memory Allocation Keywords for SharedParameters

RTA-RTE now defines the start and stop Memory Allocation Keywords for `SharedParameter`s in the format `RTE_<INFIX>_START_SEC_<SwAddrMethod>_<ALIGNMENT>` and `RTE_<INFIX>_STOP_SEC_<SwAddrMethod>_<ALIGNMENT>` respectively, where `<INFIX>` is the `shortName` of the `EcucPartition` that accesses the `SharedParameter`, `<SwAddrMethod>` is the `SwAddrMethod` (or `CALPRM` if not set) and `<ALIGNMENT>` is the width of the corresponding datum.

(RTE-12301)

Update to RTA-RTE handling of OperationInSystemInstanceRef elements

The rules for resolving references to `ClientServerOperation`s as part of `OperationInSystemInstanceRef` elements have been updated. Previously, RTA-RTE would allow references with invalid context compositions, components and ports based on the rules of the AUTOSAR XML schema and MetaModel. In this update, RTA-RTE will now raise an error if any provided `OperationInSystemInstanceRef` elements do not match the rules outlined in the MetaModel, *including the explanatory text in the AUTOSAR System Template section B.7 (21-11)* which describes the exceptions to the MetaModel rules governing usage of this reference type across Systems of category `SYSTEM_EXTRACT` and `ECU_EXTRACT`.

Note that as part of this change RTA-RTE will now validate the correctness of DEST attributes used inside any PROVIDED-INTERFACE-TREF or REQUIRED-INTERFACE-TREF elements used within PPortPrototypes or RPortPrototypes which are referenced as part of a context chain within an OperationInSystemInstanceRef. Previously RTA-RTE did not validate that the DEST attribute stated the correct target interface type but RTA-RTE will now raise an error if this is incorrect.

(RTE-12594)

Version 12.1.0pr2

Support for different aliveTimeouts

RTA-RTE now supports configurations fanning from a single outer port to several inner ports, connected via DelegationSwConnectors, where the inner ports aggregate NonqueuedReceiverComSpecs with aliveTimeouts having different values.

Note RTA-RTE requires the ComTimeout for the ComSignal / ComSignalGroup related to the communication graph described above to be equal to the shortest non-zero aliveTimeout but this is not currently validated. Additionally RTA-RTE does not support having multiple ComSignals / ComSignalGroups fan in to the same outer port where the connected inner ports aggregate different non-zero aliveTimeouts and accepts the configuration generating one channel and one status variable shared by all of the receivers.

(RTE-12756)

Memory Allocation Keywords for PerInstanceParameters

RTA-RTE now defines the start and stop Memory Allocation Keywords for PerInstanceParameters in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>_<ALIGNMENT> and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>_<ALIGNMENT> respectively, where <INFIX> is the shortName of the EcucPartition that accesses the PerInstanceParameter, <SwAddrMethod> is the SwAddrMethod (or CALPRM if not set) and <ALIGNMENT> is the width of the corresponding datum.

(RTE-12300)

Memory Allocation Keywords for arTypedPerInstanceMemory

RTA-RTE now defines the start and stop Memory Allocation Keywords for arTypedPerInstanceMemory in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>_<ALIGNMENT> and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>_<ALIGNMENT> respectively, where <INFIX> is the shortName of the EcucPartition that accesses the arTypedPerInstanceMemory, <SwAddrMethod> is the SwAddrMethod and <ALIGNMENT> is the width of the corresponding datum.

(RTE-12303)

Version 12.1.0pr1

Partial support for different aliveTimeouts

RTA-RTE now accepts a configuration fanning from a single outer port to several inner ports, connected via DelegationSwConnectors, where the inner ports aggregate NonqueuedReceiverComSpecs with aliveTimeouts having different values.

RTA-RTE generates different buffers for each aliveTimeout as well as status variables for each distinct aliveTimeout greater than zero. Handling the status variables in Rte_COMCbk will be the subject of future work.

(RTE-12757)

Memory Allocation Keywords for ExplicitInterRunnableVariable

RTA-RTE now declares the start and stop Memory Allocation Keywords for ExplicitInterRunnableVariables in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>[_ALIGNMENT] and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>[_ALIGNMENT] respectively, where <INFIX> is the shortName of the EcucPartition that accesses the ExplicitInterRunnableVariable, <SwAddrMethod> is the SwAddrMethod . shortName and [_ALIGNMENT] is the width of the corresponding datum and is only generated if the associated SwAddrMethod has memoryAllocationKeywordPolicy of value addrMethodShortNameAndAlignment.

(RTE-12299)

Memory Allocation Keywords for ImplicitInterRunnableVariable

RTA-RTE now declares the start and stop Memory Allocation Keywords for ImplicitInterRunnableVariables in the format RTE_<INFIX>_START_SEC_<SwAddrMethod>[_ALIGNMENT] and RTE_<INFIX>_STOP_SEC_<SwAddrMethod>[_ALIGNMENT] respectively, where <INFIX> is the shortName of the EcucPartition that accesses the ImplicitInterRunnableVariable, <SwAddrMethod> is the SwAddrMethod . shortName and [_ALIGNMENT] is the width of the corresponding datum and is only generated if the associated SwAddrMethod has memoryAllocationKeywordPolicy of value addrMethodShortNameAndAlignment.

(RTE-12298)

Support legacy RoleBasedPortAssignments in modern NvM callbacks

When generating a modern Rte_NvMNotifyJobFinished callback function and a RoleBasedPortAssignment whose effective Operation's ArgumentDataPrototypes' shortNames match the legacy Rte_NvMNotifyJobFinished callback signature, RTA-RTE now converts BlockRequest to a ServiceId before calling the effective Operation.

When generating a modern Rte_NvMNotifyInitBlock callback function and a RoleBasedPortAssignment whose effective Operation does not aggregate any ArgumentDataPrototypes, matching the legacy Rte_NvMNotifyInitBlock callback signature, RTA-RTE does not pass any arguments when calling the effective Operation.

Note the effective Operation for a RoleBasedPortAssignment is the first Operation in the ClientServerInterface that types the RPortPrototype referenced by the RoleBasedPortAssignment.

(RTE-12560)

Don't error for missing DefinitionRef when merging ECU containers

RTA-RTE no longer raises internal errors 904056 and 904057 when two or more EcucModuleConfigurationValues or EcucContainerValues have the same path and shortName and at least one of them does not contain a DefinitionRef.

(RTE-13050)

Error when merging similar ECU containers with different definitions

RTA-RTE now raises error 2012 when two or more `EcucModuleConfigurationValues` or `EcucContainerValue`s have the same path and `shortName` but contain a different `DefinitionRef`.

(RTE-12028)

4.16 Version 12.0.0

Version 12.0.0pr6

Improvements to the RTA-RTE Reference Manual

The RTA-RTE Reference Manual has been updated to strengthen warnings about the potential non-AUTOSAR behavior when the command line option `--client-server-global-optimization` is enabled. Specifically the use of `--client-server-global-optimization` may result in generation of direct function calls between clients and servers that are mapped to different `EcucPartition`s.

(RTE-13025)

Version 12.0.0pr5

Update the RTA-RTE License feature number

The RTA-RTE License version has been updated to 12.0.0 to conform with the other components of the `RTA-CAR` solution.

As a result of this change it is not possible to use RTA-RTE with an old license. Please contact ETAS for a new license.

(RTE-12193)

Deterministically generate symbols when a FID references a delegation port

RTA-RTE now deterministically names symbols when a `FlatInstanceDescriptor` references the `outerPort` from a `DelegationSwConnector` and does not reference a specific data element in that port and that `outerPort` is connected to multiple `innerPort`s aggregated by multiple `SwComponentType`s.

(RTE-12866)

Generate the modern Rte_NvMNotifyInitBlock function body

RTA-RTE now generates the API body for `Rte_NvMNotifyInitBlock` complying with AUTOSAR 4.4 and higher that returns `RTE_E_RAM_UNCHANGED` if the `NvBlockDescriptor`'s RAM block was not changed, otherwise it returns `E_OK`.

Note: The behaviour has not changed for the `Rte_NvMNotifyInitBlock API` that complies with AUTOSAR 4.3 and earlier.

(RTE-12057)

Version 12.0.0pr4

RIPS FlatInstanceDescriptor RteRipsPluginCommunicationScope validation

RTA-RTE now raises error 100102 when a `FlatInstanceDescriptor` has an `associatedCrossSwClusterComRtePluginRef` that references a RIPS plugin that has the parameter `RteRipsPluginCommunicationScope` that is not set to `RTE_RIPS_CROSS_SW_CLUSTER_COM`.

RTA-RTE now raises error 100103 when a `FlatInstanceDescriptor` has an `associatedRtePluginRef` that references a RIPS plugin that has the parameter `RteRipsPluginCommunicationScope` that is specified and not set to `RTE_RIPS_LOCAL_SW_CLUSTER_COM`.

(RTE-10531)

Generate machine readable release notes

RTA-RTE now generates an XML version of the RTA-RTEreleasenote as part of each product release. The XML and associated XSD are available alongside the RTA-RTEreleasenote.

(RTE-12507)

Generate both legacy and modern Rte_NvMNotifyInitBlock APIs

RTA-RTE now generates the `Rte_NvMNotifyInitBlock` callback function with both legacy (prior to AUTOSAR 4.4) and modern (AUTOSAR 4.4 and higher) signatures, each prototype and definition is guarded by a macro checking whether or not `RTE_LEGACY_NVM_SUPPORT` is defined as appropriate.

Note that the arguments passed to any calls to runnables from the generated `Rte_NvMNotifyInitBlock` APIs will be based on the input configuration, specifically the `ClientServerOperation` aggregated by the `ClientServerInterface` indicated through the `RoleBasedPortAssignment`, therefore code generated for a legacy input will not work with a modern NvM and vice versa.

(RTE-12040)

Generate both legacy and modern Rte_NvMNotifyJobFinished APIs

RTA-RTE now generates the `Rte_NvMNotifyJobFinished` callback function with both legacy (prior to AUTOSAR 4.4) and modern (AUTOSAR 4.4 and higher) signatures, each prototype and definition is guarded by a macro checking whether or not `RTE_LEGACY_NVM_SUPPORT` is defined as appropriate.

Note that the arguments passed to any calls to runnables from the generated `Rte_NvMNotifyJobFinished` APIs will be based on the input configuration, specifically the `ClientServerOperation` aggregated by the `ClientServerInterface` indicated through the `RoleBasedPortAssignment`, therefore code generated for a legacy input will not work with a modern NvM and vice versa.

(RTE-12039)

Update the simple example input model shipped with RTA-RTE

The example input model shipped with RTA-RTE no longer generates warning 100088 (see issue RTE-12141). See RTA-RTE Getting Started Guide section 3 for more information on the example configuration.

(RTE-12418)

Error if an application cluster contains an incomplete NvBlockDescriptor

RTA-RTE now raises error 100107 when generating in application cluster mode and the input configuration contains an `NvBlockDescriptor` that does not aggregate a `romBlock`.

(RTE-10527)

Improved OperationInAtomicSwcInstanceRef validation

RTA-RTE validation of the input configuration for `OperationInAtomicSwcInstanceRef`s has been improved leading to some erroneous configurations raising a more specific error, for example, omission of the context when referencing an instance of a `ClientServerOperation` in an `OperationInvokedEvent` now raises 2416 where previously it raised 2403.

(RTE-12851)

Define RTE_LEGACY_NVM_SUPPORT

RTA-RTE now defines the macro `RTE_LEGACY_NVM_SUPPORT` in `Rte.c` if NvM API version 4.3 or older is detected, and the macro `RTE_NVM_VERSION_OVERRIDE` does not have a non-zero value defined in the user-supplied configuration file `Rte_Cfg.h`.

Note there is no change to the RTA-RTE NvM support.

(RTE-12478)

Version 12.0.0pr3

Internal error 903541 replaced with informative error message

RTA-RTE now raises error 100109 instead of internal error 903541 when a runnable is not mapped to an `OsTask` referenced by an `OsApplication` and is accessing an `InterRunnableVariable` belonging to a split SWC. This improves the validation related to known issue RTE-1312.

Note: split `InterRunnableVariable`s are only possible when the command line option `--deviate-split-swci-support` is enabled.

(RTE-6437)

Error for unused ISignalToIPduMappings

RTA-RTE now raises error 2468 when an `ISignal` or `ISignalGroup` is ultimately mapped to a data element with a `VariableAccess`, and its `ISignalToIPduMapping` is not referenced by either a `ComSignal`, `ComSignalGroup`, `ComGroupSignal` or `LdComIPdu`.

(RTE-12531)

No support for multiple cross-cluster plugins

RTA-RTE now raises error 100094 when generating for a cluster and a data element is referenced by more than one cross-cluster RIPS `FlatInstanceDescriptor`.

(RTE-10530)

Don't export intra-cluster global buffers for cross-cluster plugins

RTA-RTE no longer exports the global buffer for a data element to the generated `Rte_Buffers.h` when that data element is being transmitted cross-cluster and also received intra-cluster and only a cross-cluster RIPS plugin is used.

(RTE-12842)

Version 12.0.0pr2

Error if [SWS_Rte_CONSTR_80200] is violated

RTA-RTE now raises error 100105 when an input configuration violates AUTOSAR R21-11 constraint `[SWS_Rte_CONSTR_80200]`.

(RTE-10529)

Don't generate measurement buffer for cross-cluster data elements

RTA-RTE no longer generates a measurement buffer for a measureable data element that is being transmitted cross-cluster, but the sending `PPortPrototype` is also connected intra-cluster to an `RPortPrototype` that is aggregated by an `ApplicationSwComponentType` that does not have read access to the data element.

(RTE-12704)

Do not use IOC for intra-OsApp SR comms on split-swci direct call servers

RTA-RTE no longer uses IOC to transfer data to and from an unmapped server on a split-swci if all of its calling clients and `RunnableEntity`s participating in the sender-receiver communication are mapped to the same `OsApplication`.

(RTE-12070)

Do not generate unqueued buffers for RIPS-activated queued data elements

RTA-RTE no longer generates unqueued global buffers for RIPS-activated queued data elements.

(RTE-12694)

Error for cross-cluster with inconsistent transformerErrorHandling

RTA-RTE now raises error 100106 when some but not all of the ports involved in a cross-cluster communication graph are referenced by a `PortAPIOption` with `errorHandling` set to `transformerErrorHandling`. Info 1001 is also raised for each port in the communication graph.

(RTE-12082)

Error if ALL_INTERRUPT_BLOCKING is used in application cluster mode

RTA-RTE now raises error 100108 when generating in application cluster mode if a `RteExclusiveAreaImplMechanism` is specified and has a value of `ALL_INTERRUPT_BLOCKING`.

(RTE-10532)

Don't error attempting to define a SwAddrMethod to the same value twice

RTA-RTE no longer raises internal error 900591 when attempting to define the same `SwAddrMethod` twice for the same data element. Previously this error occurred in some fan-in communication graphs where one connection uses `COM`.

(RTE-12727)

Improvements to the lRTA-RTErefguidel Rte_MemMap.h section

The RTA-RTE Reference Manual has been updated to clarify that RTA-RTE does not generate a complete `Rte_MemMap.h`.

(RTE-7772)

Version 12.0.0pr1

Remove redundant assignment to rtn in explicit sender/receiver APIs

RTA-RTE no longer assigns `RTE_E_OK` to the local variable `rtn` in the `Rte_Send` and `Rte_Write` APIs generated for inter-cluster transmission. Previously the assignment of `RTE_E_OK` to `rtn` was immediately overwritten by assigning the return

value of the RIPS cluster plugin.

(RTE-12584)

RTA-RTE no longer ships with a standalone ComXf2Xf tool

RTA-RTE no longer ships the standalone `ComXf2Xf` tool used to rewrite parts of an input configuration so that the COM-based transformation appears in the same way to RTEGen as a regular data transformation. Where COM-based transformation is configured in an RTA-CAR project, the necessary conversion is handled automatically when the RTE is generated through the RTA Code Generator dialog. There is no change to the behavior of RTEGen.

(RTE-12409)

Support Rte_Feedback for explicit with cross-cluster only transmission

RTA-RTE now redirects calls to `Rte_FbVar_<SWCI>_<Port>_<VDP>` to `Rte_Rips_<PlugIn>_Feedback_<SwcBswI>_<CGI>` and returns the return value, when transmission acknowledgement is enabled for a cluster relevant data element that is only involved in cross-cluster transmission as part of an explicit sender/receiver communication graph. Additionally, RTA-RTE does not generate the associated feedback variable `Rte_FbVar_<SWCI>_<Port>_<VDP>`. Where `<PlugIn>` is the `shortName` of the RIPS cluster plugin, `<SwcBswI>` is the `shortName` of the relevant `SwComponentPrototype`, `<CGI>` is the `shortName` of the cross-cluster RIPS `FlatInstanceDescriptor`, `<SWCI>` is the internal name of the relevant `SwComponentPrototype`, `<Port>` is the `shortName` of the target `PPortPrototype` and `<VDP>` is the `shortName` of the target `VariableDataPrototype`.

(RTE-12469)

Error for cross-cluster RIPS FIDs when not generating for a cluster

RTA-RTE now raises error 100104 when the `System` being generated does not reference a `CpSoftwareCluster` in an input configuration and cross-cluster RIPS `FlatInstanceDescriptors` are present in the `System`'s `RootSwCompositionPrototype`'s `FlatMap`.

(RTE-12331)

Define RTE_CLUSTER_APPLICATION_MODE_ENABLED in Rte_Const.h

RTA-RTE now defines `RTE_CLUSTER_APPLICATION_MODE_ENABLED` in `Rte_Const.h` to 1 if it is generating in application cluster mode and to 0 if not.

(RTE-12159)

No Rte_Ioc buffers for RIPS enabled SR IOC

RTA-RTE no longer generates `Rte_Ioc` buffers for RIPS enabled systems in the case of inter-`OsApplication` Sender/Receiver communication where the receiving `RunnableEntity` is activated by a `DataReceivedEvent`.

(RTE-12306)

No request for IOC Channel for RIPS enabled SR IOC

RTA-RTE no longer generates an IOC Channel for RIPS enabled systems in the case of inter-`OsApplication` Sender/Receiver communication where the receiving `RunnableEntity` is activated by a `DataReceivedEvent`.

(RTE-12305)

Cluster FlatInstanceDescriptors must reference delegation ports

RTA-RTE now raises error 100092 when a cross-cluster RIPS FlatInstanceDescriptor references an inner port.

(RTE-10617)

No ReceiverPullCB functions for RIPS enabled SR IOC

RTA-RTE no longer generates ReceiverPullCB functions for RIPS enabled systems in the case of inter-OsApplication Sender/Receiver communication where the receiving RunnableEntity is activated by a DataReceivedEvent.

(RTE-12244)

Support Rte_Feedback for implicit with cross-cluster only transmission

RTA-RTE now calls Rte_Rips_<PlugIn>_Feedback_<Partition>_<CGI> and assigns its return value to the feedback variable Rte_FbVar_<SWCI>_<Port>_<VDP> after Rte_Rips_<PlugIn>_Write_<Partition>_<CGI> is called when a DataWriteAccess whose target data element has transmission acknowledgement enabled and is only involved in cross-cluster transmission, where <PlugIn> is the shortName of the RIPS cluster plugin, <Partition> is the shortName of the relevant EcucPartition, <CGI> is the shortName of the cross-cluster RIPS FlatInstanceDescriptor, <SWCI> is the internal name of the relevant SwComponentPrototype, <Port> is the shortName of the target PPortPrototype and <VDP> is the shortName of the target VariableDataPrototype.

(RTE-12475)

4.17 Version 7.6.0

Version 7.6.0pr9

Validation of cluster configurations

RTA-RTE now raises errors 100095, 100096 and 100097 respectively when generating for a cluster and the SystemMapping does not aggregate a PortElementToCommunicationResourceMapping, that PortElementToCommunicationResourceMapping does not reference a CpSoftwareClusterCommunicationResource, or a data element referenced by a PortElementToCommunicationResourceMapping is not referenced by a cross-cluster RIPS FlatInstanceDescriptor.

(RTE-10528)

Support for Rte_IsUpdated for cluster relevant data

When a runnable explicitly reads a cluster-relevant data element that has the enableUpdate attribute enabled, RTA-RTE implements the Rte_IsUpdated API using the RIPS cluster plugin Rte_Rips_<PlugIn>_DataIsUpdated_<SwcBswI>_<CGI> where <PlugIn> is the shortName of the RIPS cluster plugin, <SwcBswI> is the shortName of the relevant SwComponentPrototype, and <CGI> is the shortName of the cross-cluster RIPS FlatInstanceDescriptor.

(RTE-12462)

Allow cross-cluster communication using compatible interfaces

RTA-RTE now accepts configurations that define a cross-cluster RIPS FlatInstanceDescriptor referencing a VariableDataPrototype in a delegation PortPrototype which is connected to an inner PortPrototype and both

PortProtoypes are typed by different but compatible SenderReceiverInterfaces that share the same shortName. The generated output is the same as it would be if the PortPrototypes were typed by the same SenderReceiverInterface.

(RTE-12456)

Do not include the application header file for NvBlockSwcComponentTypes

In vendor mode, RTA-RTE no longer includes the application header files for NvBlockSwcComponentTypes corresponding to relevant RTEEvents in the files generated for each task body. Effectively this change removes NvBlockSwcComponentTypes from the scope of the changes introduced in issue RTE-12162.

(RTE-12470)

Version 7.6.0pr8

No support for WaitPoints in application clusters

RTA-RTE now raises error 100093 when generating for an application cluster and WaitPoints are configured.

(RTE-10525)

No global buffer for explicit queued cross cluster only data elements

RTA-RTE does not generate RTE managed queue structures for data elements which are only relevant to queued explicit cross cluster communication, i.e., they are not related to a local plugin, or accessed inter-ECU or intra-OsApplication.

(RTE-11557)

VSMD file for RTA-RTE updated

The VSMD file shipped with RTA-RTE, VSMD_RTA-Rte_4-7-0.arxml, now references the AUTOSAR schema definition AUTOSAR_4-2-2.xsd. This is in line with other output generated by RTA-RTE and allows for compatibility with other tools that use the RTA-RTE VSMD but don't currently ship with AUTOSAR_00050.xsd.

(RTE-12414)

Stop generating global buffers for cross-cluster data elements

RTA-RTE does not generate global buffers for data elements that are only relevant to explicit unqueued or implicit cross-cluster communication regardless of whether that data element is measurable or calibratable. This extends issue RTE-11556 and issue RTE-11563.

(RTE-12408)

Raise an error if multiple CpSoftwareClusters are referenced by the System

RTA-RTE now raises error 100090 when the System for which RTEGen is generating references more than one CpSoftwareCluster.

(RTE-10522)

DataReceivedEvents for cross cluster configurations

RTA-RTE now conditionally activates the RunnableEntity associated with a cross cluster DataReceivedEvent in the appropriate task body if a call to Rte_Rips_<PlugIn>_DataIsUpdatedEventActivation__<SwcBswI>_<DR>_<CGI>() returns TRUE, where <PlugIn> is the shortName of the RIPS cluster plugin, <SwcBswI> is the shortName of the relevant SwComponentPrototype, DR is is the shortName of the DataReceivedEvent and <CGI> is the shortName of the

cross-cluster RIPS `FlatInstanceDescriptor`.

Note that RTA-RTE does not currently validate that a cross cluster plug-in mapped to a DataReceivedEvent is periodically activated.

(RTE-11975)

Raise error if cross cluster composition contains any nested compositions

RTA-RTE does not support nested compositions in application cluster mode or host cluster mode and rejects such configurations with error 100089.

(RTE-10523)

Include BSW-Synchronized application header files in task body source files

In vendor mode, RTA-RTE now includes the application header files for `SwComponentTypes` whose `RunnableEntitys` are synchronized with `BswSchedulableEntitys` corresponding to relevant `BswEvents` in the files generated for each task body.

(RTE-12213)

Support mixed implicit and explicit unqueued cross cluster communication

RTA-RTE now supports configurations that mix implicit and explicit access to unqueued cluster-relevant data elements.

(RTE-11948)

Queued send for inter-cluster mixed with RTE managed communication

RTA-RTE now calls `Rte_Rips_<PlugIn>_Write_<SwcI>_<CGI>` (see issue RTE-11546) to write across clusters when a `RunnableEntity` aggregates a `VariableAccess` in the `DataSendPoint` role to a data element involved in queued inter-cluster communication in combination with other RTE managed queued communication after writing intra and inter-`OsApplication` (via IOC) and before writing inter-ECU (via COM) where appropriate.

(RTE-11960)

Mixing RIPS local and cluster plugins for explicit queued

When a RIPS local plugin is configured in combination with a RIPS cluster plugin as part of an explicit queued sender/receiver communication graph RTA-RTE behaves as if there was only the local plugin which is responsible implementing all functionality extending the work done for unqueued (see issue RTE-11543).

(RTE-11547)

Include the RIPS SWC-BSW Instance header file in the SWC application header

For queued inter-cluster sender/receiver communication the generated application header file of the `SwComponentType` that types the participating `SwComponentPrototype` now includes the applicable `RIPS SWC-BSW-Instance Header File`. Previously the `RIPS SWC-BSW-Instance Header File` related to cross-cluster plugins was not being included in the relevant application header files.

(RTE-12164)

Only initialize the used OsApp Resources when in application cluster mode

When generating for an application cluster and there is more than one

OsApplication in the input configuration, the file Rte_Lib.c will contain initialization only for the Resources mapped to OsApplications that are *used*.

Note that an OsApplication is *used* when it references an OsTask that is referenced by either an RteEventToTaskMapping or an RteBswEventToTaskMapping.

(RTE-12248)

Version 7.6.0pr7

Depricate support for other system categories than ECU_EXTRACT

When running in generation phase and a system is configured with category other than ECU_EXTRACT, RTA-RTE warns that support for the provided system category is deprecated and will be removed in a future version.

Note that RTA-RTE continues with the generation as before.

(RTE-12141)

Raise error if a negative value is supplied with ——period or ——p

RTA-RTE exits with error 1117 when the command line option –period or –p is used and it is supplied with an incorrect (negative) value.

(RTE-12274)

Explicit write for inter-cluster mixed with RTE managed communication

RTA-RTE now calls Rte_Rips_<PlugIn>_Write_<Partition>_<CGI> (see issue RTE-11541) to write across clusters when a RunnableEntity aggregates a VariableAccess in the DataSendPoint role to a data element involved in inter-cluster communication in combination with other RTE managed communication after writing intra and inter-OsApplication (via IOC) and before writing inter-ECU (via COM) where appropriate.

(RTE-11953)

Support for Rte_DRead for cluster relevant data

When a runnable explicitly reads a cluster-relevant data element by value, RTA-RTE implements the Rte_DRead API using the RIPS cluster plugin Rte_Rips_<PlugIn>_DRead_<Partition>_<CGI> where <PlugIn> is the shortName of the RIPS cluster plugin, <Partition> is the shortName of the relevant EcucPartition, and <CGI> is the shortName of the cross-cluster RIPS FlatInstanceDescriptor.

(RTE-11540)

Support for Rte_Send for cluster relevant data

When a runnable explicitly writes to a queued cluster-relevant data element, RTA-RTE implements the Rte_Send API using the RIPS cluster plugin Rte_Rips_<PlugIn>_Write_<SwcI>_<CGI> where <PlugIn> is the shortName of the RIPS cluster plugin, <SwcI> is the shortName of the relevant SwComponentPrototype, and <CGI> is the shortName of the cross-cluster RIPS FlatInstanceDescriptor.

(RTE-11546)

Return Std_ReturnType from cross cluster Rte_Read and Rte_Receive

RTA-RTE now generates Rte_Read and Rte_Receive APIs with a return type of Std_ReturnType when these APIs are generated in relation to a cross-cluster communication graph. Previously RTA-RTE was incorrectly generating these APIs

with a return type of `void`.

(RTE-12227)

Include the RIPS SWC-BSW Instance header file in the SWC application header

For inter-cluster snder/receiver communication the generated application header file of the `SwComponentType` that types the participating `SwComponentPrototype` now includes the applicable `RIPS SWC-BSW-Instance Header File`. Previously the `RIPS SWC-BSW-Instance Header File` related to cross-cluster plugins was not being included in the relevant application header files.

(RTE-12160)

Include the application header file in task body source files

In vendor mode, RTA-RTE now includes the application header files for `SwComponentTypes` corresponding to relevant `RTEEvents` in the files generated for each task body.

(RTE-12162)

Version 7.6.0pr6

Mixing RIPS local and cluster plugins

When a RIPS local plugin is configured in combination with a RIPS cluster plugin as part of an explicit unqueued sender/receiver communication graph RTA-RTE behaves as if there was only the local plugin which is responsible implementing all functionality.

Note that RTA-RTE continues to call the appropriate start/stop function for the RIPS cluster plugin (see issue RTE-12129).

(RTE-11543)

Support for Rte_Read for cluster relevant data

When a runnable explicitly reads a cluster-relevant data element, RTA-RTE implements the Rte_Read API using the RIPS cluster plugin `Rte_Rips_<PlugIn>_Read_<Partition>_<CGI>` where `<PlugIn>` is the `shortName` of the RIPS cluster plugin, `<Partition>` is the `shortName` of the relevant `EcucPartition`, and `<CGI>` is the `shortName` of the cross-cluster RIPS `FlatInstanceDescriptor`.

(RTE-11539)

Generate and include Rte_Buffers.h for RIPS cluster plugins

RTA-RTE now creates the file Rte_Buffers.h in the generated output when RIPS mode is enabled for configurations containing RIPS cluster plugins. Rte_Buffers.h `#include`s the RIPS Buffer Declaration File, `Rte_DataHandleType.h` and is itself `#include`d in Rte.c. `Rte_DataHandleType.h` `#include`s `Rte_Type.h`.

(RTE-12155)

Support for Rte_Write for cluster relevant data

When a runnable explicitly writes to a cluster-relevant data element, RTA-RTE implements the Rte_Write API using the RIPS cluster plugin `Rte_Rips_<PlugIn>_Write_<Partition>_<CGI>` where `<PlugIn>` is the `shortName` of the RIPS cluster plugin, `<Partition>` is the `shortName` of the relevant `EcucPartition`, and `<CGI>` is the `shortName` of the cross-cluster RIPS

`FlatInstanceDescriptor`.

(RTE-11541)

Include all Cross Cluster Plug-In Header Files in Rte.h

When RIPS mode is enabled, RTA-RTE now `#include`s the Cross Software Cluster Communication Plug-In Header File for each configured cluster plugin in the generated Rte.h file.

(RTE-12112)

No global buffer for explicit unqueued cross cluster only data elements

RTA-RTE does not generate a global buffer for data elements which are only cross cluster relevant during explicit unqueued communication, i.e., they are not related to any other local plugin, and are not measurable or calibratable. This extends issue RTE-11563.

(RTE-11556)

Support for RIPS cluster plugin Start and Stop

When RIPS is enabled and the configuration contains RIPS cluster plugins, RTA-RTE now generates a call to each plugin's `Rte_Rips_<plugin>_Rte_Start` / `Rte_Rips_<plugin>_Rte_Stop` function in the `Rte_Start` / `Rte_Stop` APIs respectively.

When the configuration contains multiple RIPS cluster plugins, the order of calls is determined by the `index` values provided in the `RteRipsPluginConfigurationRef`.

(RTE-12129)

Implicit flush for inter-cluster mixed with RTE managed communication

RTA-RTE now calls `Rte_Rips_<PlugIn>_Write_<Partition>_<CGI>` (see issue RTE-11555) to flush the RTE managed shadow buffer created when a `RunnableEntity` has `dataWriteAccess` to a data element involved in inter-cluster communication in combination with other RTE managed communication after flushing intra and inter-`OsApplication` (via IOC) and before flushing inter-ECU (via COM) where appropriate.

(RTE-11959)

Rte_IWriteRef not redirected for inter-cluster only communication

When a data communication graph with only cross cluster communication is configured, the Rte_IWriteRef API is not redirected to `Rte_Rips_<plugin>_IWriteRef` (compared to non-cluster RIPS, see issue RTE-5927).

(RTE-11568)

Version 7.6.0pr5

Rte_IRead not redirected for inter-cluster only communication

When a data communication graph with only cross cluster communication is configured, the Rte_IRead API is not redirected to `Rte_Rips_<plugin>_IRead` (compared to non-cluster RIPS, see issue RTE-5926).

(RTE-11564)

Rte_IWrite not redirected for inter-cluster only communication

When a data communication graph with only cross cluster communication is

configured, the Rte_IWrite API is not redirected to `Rte_Rips_<plugin>_IWrite` (compared to non-cluster RIPS, see issue RTE-5927).

(RTE-11565)

AUTOSAR 21-11 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace `http://autosar.org/schema/r4.0`) against the AUTOSAR 21-11 schema.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to 21-11 will validate against the 21-11 schema.

(RTE-11148)

Memory Allocation Keywords for Rte_WaitableDatum

For variables of type Rte_WaitableDatum generated for client/server communication graphs RTA-RTE now uses the memory allocation keyword `RTE_WD_<SWC-PROTO>_<EVENT>_VAR_CLEARED_<ALIGNMENT>`, where: `<SWC-PROTO>` is the `shortName` of the relevant `SwComponentPrototype`, `<EVENT>` is the `shortName` of the relevant `RTEEVent` (both in upper case), and `<ALIGNMENT>` is an appropriate `Alignment` for the variable.

(RTE-11602)

Rte_Receive for inter-cluster only communication

When a runnable explicitly reads by argument a queued data element that is involved in inter-cluster only communication, RTA-RTE implements the Rte_Receive API by calling `Rte_Rips_<PlugIn>_Read_<SwcBswI>_<CGI>` where `<PlugIn>` is the `shortName` of the RIPS cluster plugin, `<SwcBswI>` is the `shortName` of the relevant `SwComponentPrototype`, and `<CGI>` is the `shortName` of the cross-cluster RIPS `FlatInstanceDescriptor`.

(RTE-11545)

Implicit fill for inter-cluster only transmission

RTA-RTE now fills the RTE managed shadow buffer created when a `RunnableEntity` has **dataWriteAccess** to a data element involved in inter-cluster only communication by calling `Rte_Rips_<PlugIn>_Read_<Partition>_<CGI>` where `<PlugIn>` is the `shortName` of the RIPS cluster plugin, `<Partition>` is the `shortName` of the relevant `EcucPartition`, and `<CGI>` is the `shortName` of the cross-cluster RIPS `FlatInstanceDescriptor`.

(RTE-11551)

Reduce code complexity of Rte_WriteQueue.

The generated Rte_WriteQueue function has been refactored to reduce the code complexity.

(RTE-12091)

Version 7.6.0pr4

Implicit flush for inter-cluster only communication

RTA-RTE now flushes the RTE managed shadow buffer created when a `RunnableEntity` has `dataWriteAccess` to a data element involved in inter-cluster only communication by calling `Rte_Rips_<PlugIn>_Write_<Partition>_<CGI>`

where <PlugIn> is the shortName of the RIPS cluster plugin, <Partition> is the shortName of the relevant EcucPartition, and <CGI> is the shortName of the cross-cluster RIPS FlatInstanceDescriptor.

(RTE-11555)

The command line option ——buffers-use-data-name has been updated

When the command line option –buffers-use-data-name is set, RTA-RTE now names the RTE managed buffer generated on the reception side of inter-ECU communication (via COM) using the shortName of the relevant ComSignal or ComSignalGroup as appropriate. See the RTA-RTE Reference Manual for more details.

(RTE-11579)

Move event id definitions from Rte.c to Rte_Intl.h

RTA-RTE now generates the macro definitions of numeric identifiers used for events that are queued in Rte_Intl.h so they are available to be used in the task body when generating in Vendor mode.

(RTE-11518)

No global buffer for cross cluster only data elements

RTA-RTE does not generate a global buffer for data elements which are only cross cluster relevant, i.e., they are not related to any other local plugin, and are not measurable or calibratable.

(RTE-11563)

Implicit fill for inter-cluster only communication

RTA-RTE now fills the RTE managed shadow buffer created when a RunnableEntity has dataReadAccess to a data element involved in inter-cluster only communication by calling Rte_Rips_<PlugIn>_Read_<Partition>_<CGI> where <PlugIn> is the shortName of the RIPS cluster plugin, <Partition> is the shortName of the relevant EcucPartition, and <CGI> is the shortName of the cross-cluster RIPS FlatInstanceDescriptor.

(RTE-11550)

Do not call OS Scheduling APIs in application cluster mode

When generating for an application cluster RTA-RTE no longer calls StartScheduleTable, StartScheduleTableRel or StopScheduleTable in the Rte_Start and Rte_Stop functions output in Rte_Lib.c.

(RTE-10520)

Accept signal fanning in combination with port fanning

RTA-RTE now accepts configurations where a SystemSignal (or SystemSignalGroup) is referenced by multiple ISignals (or ISignalGroups) and that same signal is also mapped to multiple inner PortPrototypes. Previously RTA-RTE identified this as an M:N connection and rejected the configuration with error 001798.

(RTE-11693)

No support of mixed reception in a cluster without a RIPS plugin

RTA-RTE does not support inter-cluster reception in combination with reception managed by RTE and rejects such configurations with error 100087.

(RTE-11570)

Version 7.6.0pr3

Bidirectional routing in COM

RTA-RTE now supports explicit unqueued sending and receiving of data via COM using a single `ISignal` (or `ISignalGroup`) referenced by an `ISignalTriggering` when a single `SystemSignal` (or `SystemSignalGroup`) is referenced by a `SenderReceiverToSignalMapping` (or `SenderReceiverToSignalGroupMapping`) in the context of both a `PPortPrototype` and an `RPortPrototype`.

(RTE-11835)

Raise error when RIPS mode is enabled and no cross-cluster RIPS FIDs

RTA-RTE exits with error 100005 when the input configuration has RIPS mode enabled but does not contain any RIPS `FlatInstanceDescriptor`s, this now includes cross-cluster RIPS `FlatInstanceDescriptor`s, extending the work done in issue RTE-5884.

(RTE-11548)

Validate CpSoftwareCluster is the same as the RootSwCompositionPrototype

When a CpSoftwareCluster is present in the input configuration RTA-RTE now raises error messages 100079 or 100080 if the CpSoftwareCluster refers to a `CompositionSwComponentType` different from the RootSwCompositionPrototype's `CompositionSwComponentType` or contains a `SwComponentPrototypeAssignment` that refers to a `SwComponentPrototype` that is not inside the RootSwCompositionPrototype respectively.

(RTE-10511)

Remove volatile from Rte_Prm, Rte_CData and SchM_CData APIs

RTA-RTE no longer uses the memory modifier volatile in the declarations and definitions of Rte_Prm, Rte_CData and SchM_CData API functions, in the generated Rte.c and `Rte_Type.h` files.

(RTE-11771)

Refactor Vendor mode OsTask extern declarations

As part of a refactor of the generator code around Vendor mode OsTask extern declarations, RTA-RTE no longer outputs duplicate new lines for these extern declarations.

(RTE-11585)

Version 7.6.0pr2

Identify when RTA-RTE is generating for a CpSoftwareCluster

When a CpSoftwareCluster is referenced by the System in an input configuration RTA-RTE will generate for that `HOST_SOFTWARE_CLUSTER` or `APPLICATION_SOFTWARE_CLUSTER` as determined by the `category` (with default `APPLICATION_SOFTWARE_CLUSTER`) and raise info 100078. Additionally error 100086 is raised when an unsupported `category` is provided.

This replaces the command line switches `--appl-cluster` and `--host-cluster` added in issue RTE-10510 which have now been removed.

(RTE-11857)

Raise an error for incorrect event to SWC mapping

RTA-RTE now raises error 100084 when an event is mapped to a `SwComponentPrototype` and that event is also mapped to an `RteSwComponentInstance` that references a different `SwComponentPrototype`.

(RTE-11209)

Error when data conversion uses a floating point NetworkRepresentation

RTA-RTE now raises error 100085 when inter-ECU communication is configured which requires data conversion and uses an `ISignal` with a NetworkRepresentation referencing a `SwBaseType` with the attribute `baseTypeEncoding` set to `IEEE754`.

(RTE-11772)

Version 7.6.0pr1

Raise an info message when the RTE specific OsApp is mapped to a core

When cores are present in the input configuration RTA-RTE now raises info message 100083 informing the user which `EcucCoreDefinition` the RTE specific `OsApplication` (`'RTE_DEFAULT_OS_APP'`) has been mapped to.

(RTE-9529)

4.18 Version 7.5.4

Version 7.5.4pr3

Do not surround calls to locRead with resource protection

Extending the work done in issue RTE-11707 for communication that is both cross core and cross `OsApplication` and involves runnables mapped to `OsTasks` that are not the same priority, RTA-RTE does not apply any resource protection around calls to locRead as part of 1:N implicit sender/receiver communication that has been optimized to use a single IOC channel.

(RTE-11797)

Version 7.5.4pr2

Do not output the volatile memory modifier in module interlink header files

RTA-RTE no longer uses the memory modifier volatile in addition to the AUTOSAR compiler abstraction when outputting `extern`s and `#define`s in module interlink header files in the context of direct access to calibration parameters (rather than accessing through `Schm_CData`). This is supported by `[SWS_Rte_03787]` (AUTOSAR R21-11).

(RTE-11741)

Asynchronous client/server output and bi-directional parameters use P2VAR

This ticket extends the work done in issue RTE-3728, RTA-RTE now generates `Rte_Call` APIs for Asynchronous client/server using the P2VAR compiler abstraction for output and bi-directional parameters of type `ARRAY` and `RECORD`.

(RTE-11346)

Do not surround calls to locRead with resource protection

RTA-RTE does not apply any resource protection around calls to locRead as part

of 1:N implicit sender/receiver communication that has been optimized to use a single IOC channel.

(RTE-11707)

Ignore unused RIPS FlatInstanceDescriptors

RTA-RTE now accepts configurations containing RIPS FIDs that are not aggregated by the `Flatmap` referenced by the `RootSWCompositionPrototype` which is aggregated by the `System` being generated. Previously, RTA-RTE would raise internal error 903586 for a RIPS client-server configuration containing such a FlatInstanceDescriptor.

(RTE-11562)

Version 7.5.4pr1

Optimize unqueued sender/receiver to use a single IOC channel

RTA-RTE now uses a single IOC channel where possible for both implicit and explicit unqueued 1:N inter-OsApplication sender/receiver communication.

When considering merging IOC channels RTA-RTE validates the compatibility of the receiver `DataTypes` and generates one IOC channel per distinct receiver `DataType`. Each IOC channel is then typed by the relevant `ImplementationDataType`. Note that when comparing receiver `DataTypes` RTA-RTE does not does not consider `invalidValues` or `DataConstrs`.

When a single IOC channel is used RTA-RTE generates one IOC channel in `iocNeeds.arxml` for each 1:N inter-`OsApplication` sender-receiver communication graph where previously one channel was generated per receiving `OsApplication`. That IOC channel contains as many `OsIocReceiverProperties` as there are receiving `OsApplications` and each `OsIocReceiverProperties` contains a unique `OsIocReceiverId`.

The explicit APIs `Rte_Read`, `Rte_DRead` and `Rte_Write` have been updated to read and write, via IocRead and `IocWrite`, using a single IOC channel.

The implicit fills and flushes, via IocRead / `IocWrite` or `OsIocReadGroup` / `OsIocWriteGroup` have been updated to use the single IOC channel.

For both implicit and explicit RTA-RTE appends the `OsIocReceiverId` to the name of the `Rte_ReceiverPullCB` to provide a unique name allowing a single IOC channel to be used in conjunction with runnables activated by `DataReceivedEvents`.

RTA-RTE now appends the provider name to the IOC channel when the command line option `--append-name-to-buffer` is used. See the RTA-RTE Reference Manual for more details.

Note that these changes require RTA-OS 6.2.101 or later.

(RTE-11324)

Memory Allocation Keywords for Mode Machine Instance variables

RTA-RTE now correctly uses the `SwAddrMethod` of the FlatInstanceDescriptor in the Memory Allocation Keyword for Mode Machine Instance variables as described in issue RTE-5080. Previously in some circumstances the `SwAddrMethod` was omitted.

(RTE-11603)

Do not output the volatile memory modifier in application header files

RTA-RTE no longer uses the memory modifier volatile in addition to the AUTOSAR compiler abstraction when outputting `extern`s and `#define`s in application header files in the context of direct access to calibration data and parameters (rather than accessing through `Rte_Prm` and `Rte_CData` APIs). This is supported by `[SWS_Rte_03787]` (AUTOSAR R21-11).

(RTE-11368)

Consistently report implicit buffer copies in the EHOOKS report

When the EHOOKS report is generated RTA-RTE now reports implicit buffer copies, `VAR-COPY`, under the `MEASUREMENT` of the global buffer, `VAR-GLOBAL-BUFFER`, from which it is filled.

(RTE-11485)

The VSMD file for RTA-RTE has been updated

The VSMD file has been updated to remove the `EcucDestinationUriDefSets` container.

(RTE-11488)

Data conversion where only one Unit references a PhysicalDimension

RTA-RTE now accepts configurations where data conversion is taking place and one Unit references a PhysicalDimension but the connected one does not. RTA-RTE behaves as if the Units on both sides of the data conversion referenced the same PhysicalDimension.

(RTE-11208)

Raise an error for non-standard Rte_Cs_TransactionHandleType

RTA-RTE now raises error 100082 when an input configuration defines an `ImplementationDataType` with the `shortName` (or `symbol`) Rte_Cs_TransactionHandleType that does not match the AUTOSAR definition `[SWS_Rte_06712]` (AUTOSAR R21-11). If a matching definition is present in the input it is accepted without error (see issue RTE-11408).

(RTE-11479)

4.19 Version 7.5.3

Version 7.5.3pr1

Fixed an error introduced in RTA-RTE 7.5.1pr1

All versions of RTA-RTE 7.5.1 contain an error introduced by an internal refactor that leads to unpredictable output. The refactor has been reverted and the behavior in this specific area is now the same as 7.5.0.

(RTE-11637)

4.20 Version 7.5.1

Version 7.5.1pr6

Avoid duplicate definition of Rte_Cs_TransactionHandleType

RTA-RTE no longer defines Rte_Cs_TransactionHandleType in `Rte_Type.h` when it is included in the input configuration passed to RTEGen in accordance with

AUTOSAR requirement [SWS_Rte_06712] (AUTOSAR R21-11).

Note that since issue RTE-10737 Rte_Cs_TransactionHandleType is defined in Rte.h that is itself included in Rte_Type.h.

(RTE-11408)

Version 7.5.1pr5

VSMD file for RTA-RTE updated

The VSMD file has been updated to include the default RteCodeVendorId for **ETAS** (11).

(RTE-9880)

COMPONENT-REF now correctly validates COMPOSITION-SW-COMPONENT-TYPE

RTA-RTE now does not raise error 1022 if a COMPONENT-REF element has destination COMPOSITION-SW-COMPONENT-TYPE.

(RTE-10720)

Check compatibility of SwBaseTypes

RTA-RTE now conforms with AUTOSAR R20-11 [constr_1220] when performing compatibility checks for the SwBaseType of AutosarDataPrototypes.

This extends the work done in issue RTE-9942 to include ApplicationDataTypes and NvDataInterfaces and resolves the known RTE-10904, which has been removed.

(RTE-10964)

Update to the Rte_memcpy declaration guard

RTA-RTE checks for the definition of RTE_USER_MEMCPY to guard the declaration of the RTA-RTE implementation of Rte_memcpy, previously it was RTE_LIBC_MEMCPY. Additionally the guarded declaration of the RTA-RTE implementation of Rte_memcpy is no longer preceded by *undefining* Rte_memcpy.

(RTE-11333)

Updates to RTA-RTE concurrency control customization

The RTA-RTE concurrency control must now by selected by defining the appropriate values in Rte_UserCfg.h. The following customizations must be selected:

- Define RTE_RESOURCE_IMPL to select the implementation of Rte_GetResource() and Rte_ReleaseResource(). To keep the previous behaviour use the value RTE_RESOURCE_IMPL_RESOURCE.
- Define RTE_SUSPEND_RESUME_OS_INTERRUPTS_IMPL to select the implementation of Rte_SuspendOSInterrupts() and Rte_ResumeOSInterrupts(). To keep the previous behaviour use the value RTE_SUSPEND_RESUME_OS_INTERRUPTS_IMPL_INTLOCK.
- Define RTE_SUSPEND_RESUME_ALL_INTERRUPTS_IMPL to select the implementation of Rte_SuspendAllInterrupts() and Rte_ResumeAllInterrupts(). To keep the previous behaviour use the value RTE_SUSPEND_RESUME_ALL_INTERRUPTS_IMPL_INTLOCK.

See Rte_Intl_h.template for alternative values.

Additionally `Rte_Lib_c.template` is updated to selectively make use of the appropriate resources locking mechanisms when writing to an RTE managed queue based on the specific notification type (such as a task activation).

(RTE-11301)

Propagate SwDataDefProps for ARRAY DataPrototypes from input to MCSD

RTA-RTE now propagates the `SwDataDefProps.dataConstr`, `SwDataDefProps.displayFormat` and `SwDataDefProps.stepSize` for DataPrototypes of category ARRAY from the input configuration to the sub-elements of the arrays in the generated `Rte_McSupportData.arxml`. The reported values of the array sub-elements are inherited from the array, see `constr_1718`, `constr_1719` and `constr_1720` respectively.

Note that other than the change to `Rte_McSupportData.arxml` there is no change to the behaviour of RTEGen.

(RTE-8243)

Implicit fill/flush for EcucPartition separated shadow buffers

RTA-RTE now generates implicit fills and flushes for shadow buffers using the EcucPartition separated shadow buffer when EcucPartitions are configured.

(RTE-11077)

Limit the precision of linear data conversion

RTA-RTE now checks if the overall scale and offset that would be applied as part of a linear data conversion is within a small value (`epsilon`) of zero, if it is then no scaling or offset are applied. `Epsilon` is defined by RTA-RTE to be a base 2 value of 2 to the power of -24.

Previously, attempting higher precision calculations could result in RTA-RTE being unable to find a suitable platform type for the intermediate values and rejecting the configuration with error 2692.

Note: For intra-ECU communication graphs data conversion only applies when there is a `PortInterfaceMapping`. For inter-ECU communication graphs data conversion only applies when the relevant `ISignal` has a `dataTypePolicy` of `OVERRIDE` or `LEGACY`.

(RTE-10913, RTE-11151)

Version 7.5.1pr4

Memory section for implicit shadow buffers

RTA-RTE now generates the `struct` containing the implicit shadow buffers pertaining to a EcucPartition in a data memory section that is owned by that particular EcucPartition.

(RTE-10989)

Initialization of CDS handles for EcucPartition separated shadow buffers

RTA-RTE now initializes the CDS data handles associated with implicit data access using the implicit shadow buffers generated per EcucPartition (see issue RTE-10975).

(RTE-11073)

Update implicit APIs for separate EcucPartition shadow buffers

RTA-RTE now outputs the appropriate EcucPartition shadow buffers added in issue RTE-11259 when generating `Rte_IWrite`, `Rte_IWriteRef` and `Rte_IRead` APIs.

(RTE-11072)

Separate implicit shadow buffers per EcucPartition

RTA-RTE now generates separate `struct`s containing the implicit shadow buffers pertaining to each individual EcucPartition. Previously, RTA-RTE generated a single `struct` containing all implicit shadow buffers from all EcucPartitions.

(RTE-10975,RTE-11259)

Do not include Com.h in Rte.c when COM is not used

RTA-RTE now only defines `RTE_REQUIRES_COM` and hence only includes Com.h in Rte.c when COM is required for the system. Previously, the macro would be defined in some cases where LDCOM was used and COM was not.

(RTE-10422)

Memory Allocation Keywords for split InterRunnableVariables

The memory allocation keywords generated for split InterRunnableVariables now honor the `SwAddrMethod` configured in the input following the precedence rules defined in `[constr_1015]`.

Note split InterRunnableVariables are only possible when the command line option (`--deviate-split-swci-support`) is enabled.

(RTE-11157)

Version 7.5.1pr3

RTA-A2L now able to patch complex data

RTA-A2L now patches symbols that contain array indexes and/or structure member accesses where a symbol address is provided by the input symbol table.

(RTE-11191)

RTA-RTE now ships with rtadwarfextractor

RTA-RTE now ships with the rtadwarfextractor for gathering symbol data from built images. See the RTA-A2L User Guide for more information on usage.

(RTE-11190)

All output and bi-directional parameters shall use the P2VAR macro

RTA-RTE now generates APIs that make use of the macro P2VAR for output and bi-directional parameters, as specified by AUTOSAR R20-11 `[SWS_Rte_06806]`.

Previously `CONSTP2VAR` was used, which caused compiler warnings when used in conjunction with tools that already conform to `[SWS_Rte_06806]`.

(RTE-3728)

Deterministic output of messages related to LDCOM_TP

RTA-RTE now outputs messages 100037, 100070 and 3030 related to LDCOM_TP in a deterministic order.

(RTE-11131)

Version 7.5.1pr2

Fixed Rte_LdComTp_Cbk.h inclusion for RIPSified client-server LDCOM_TP

RTA-RTE now includes the file Rte_LdComTp_Cbk.h in `Rte_Cbk.h` deterministically for both clients and servers that are involved in communication graphs with a RIPS-relevant `LdComIPdu` having `LdComApiType` set to LDCOM_TP. This extends the work done in issue RTE-10697 resolving the non-deterministic treatments of client and adding the include for servers.

Note that LDCOM_TP is still not supported natively thus the Rte_LdComTp_Cbk.h header file must be provided by the system integrator and warning 100037 is still raised.

(RTE-11128)

Do not duplicate the input Rte_TickCounter in osNeeds.arxml

RTA-RTE no longer duplicates parameters of the RTE default `OsCounter`, Rte_TickCounter, that are present in the input configuration in the generated osNeeds.arxml, when the input values match the RTE defaults and the command line option `----os-output-param=changed` is provided. When the entire Rte_TickCounter is provided in the input with values matching the RTE defaults then no Rte_TickCounter is output in osNeeds.arxml.

(RTE-10577)

Fix validation of TARGET-P-PORT-REF destinationRef for PRPortPrototypes

RTA-RTE no longer raises error 1022 when a TARGET-P-PORT-REF has its destinationRef set to a PRPortPrototype.

Note: RTA-RTE will continue to raise the error 3006 when a PRPortPrototype is referenced as an `innerPort` or `outerPort` from a `DelegationSwConnector`. See issue RTE-3571.

(RTE-10909)

Introducing command line switches ——appl-cluster and ——host-cluster

RTA-RTE now accepts Application and Host Cluster mode command line switches –appl-cluster and –host-cluster along with the phase generation option `--rte`.

Note: Usage of `--rte` and `--contract` stays unchanged.

(RTE-10510)

Fix data conversion for complex ADT members transmitted over COM

RTA-RTE now applies data conversion for `ApplicationRecordElement`s and `ApplicationArrayElement`s typed by `ApplicationPrimitiveDataType`s when they are sent or received over COM via `ComGroupSignal`s. The behaviour is the same as if each `ApplicationPrimitiveDataType` was individually sent or received over COM via a `ComSignal`.

Previously, RTA-RTE would ignore the `ApplicationPrimitiveDataType`'s `CompuMethod`.

(RTE-5899)

Version 7.5.1pr1

Define Rte_Cs_TransactionHandleType in the RTE Header File

RTA-RTE now defines the type Rte_Cs_TransactionHandleType of the transaction handle for inter-ECU Client-Server communication as specified by AUTOSAR R20-11 `[SWS_Rte_08732]` in `Rte.h`.

Note that Rte_Cs_TransactionHandleType is always defined regardless of the communication paradigm in the input configuration.

(RTE-10737)

Define RTE_E_COM_BUSY in the RTE Header File

RTA-RTE now defines the standard error RTE_E_COM_BUSY in `Rte.h` as specified by AUTOSAR R20-11 `[SWS_Rte_01389]`.

(RTE-10362)

4.21 Version 7.5.0

Version 7.5.0pr9

Fixed intra-ECU behaviour when ––deviate-cbk-xfrm is enabled

RTA-RTE no longer raises an internal error when the command line option –deviate-cbk-xfrm is set and a configuration contains queued intra-ECU communication.

(RTE-11041)

Version 7.5.0pr8

Generate OsEvents for synchronised BswModuleEntitys in osNeeds.arxml

When an input configuration contains a BswModuleEntity mapped to a category 2 `RunnableEntity` via a `SwcBswMapping` and the BswModuleEntity is started by a `BswEvent` while the `RunnableEntity` is not started by any `RTEEvent`s, RTA-RTE now outputs OsEvents in the generated osNeeds.arxml as though the `RunnableEntity` is started by an `RTEEvent` directly.

(RTE-10012)

Propagate OsCounterType to osNeeds.arxml

RTA-RTE now propagates the OsCounterType to the generated osNeeds.arxml for `OsCounter`s with `shortName` `Rte_TickCounter` or `Rte_TOutCounter`.

In the event that the input configuration does not contain an `OsCounter` with `shortName` `Rte_TickCounter` and one is required, RTA-RTE will output one with OsCounterType `SOFTWARE` to the generated osNeeds.arxml.

(RTE-10539)

Only raise error 100056 if multiple ASCREs for the same call point are split

RTA-RTE now only raises error 100056, and raises it deterministically, when `AsynchronousServerCallReturnsEvent`s mapped to a `SWC` instance are mapped to different OsApplications as part of an inter-OsApplication asynchronous client-server communication graph. This error was created in issue RTE-9179 which incorrectly raised the error if any event in an inter-OsApplication asynchronous client-server communication graph that contained at least one `AsynchronousServerCallReturnsEvent` was mapped to a different OsApplication than the `AsynchronousServerCallReturnsEvent`.

(RTE-10314)

Optimization of buffers for IOC initial values

RTA-RTE now generates a single `const` buffer per DataType and initial value and records the symbol of that buffer in the `OsIocInitValueSymbol` of the generated

`iocNeeds.arxml` for each IOC channel typed by that DataType and with that initial value that is part of a sender/receiver communication graph. Previously RTA-RTE generated one buffer per channel.

(RTE-10583)

Improved error message 2692 for inter-ECU linear data conversion

RTA-RTE now includes the relevant CompuMethod, DataType and ISignal when error 2692 is raised for inter-ECU linear data conversion requiring an intermediate value that could overflow the largest available platform data type.

(RTE-10590)

Allow the RTE specific OsApplication to access all OsTasks

RTA-RTE now includes an `OsTaskAccessingApplication` referencing the RTE specific OsApplication (`'RTE_DEFAULT_OS_APP'`) for each OsTask reported in the generated osNeeds.arxml.

(RTE-8130)

Include Rte_LdComTp_Cbk.h in Rte_Cbk.h for RIPSified client-server LDCOM_TP

RTA-RTE now includes the file Rte_LdComTp_Cbk.h in Rte_Cbk.h when an input configuration contains a **client-server** communication graph with a RIPS-relevant LdComIPdu having `LdComApiType` set to LDCOM_TP. This extends issue RTE-9979 which included the header file for sender-receiver communication via LDCOM_TP.

Note that LDCOM_TP is still not supported natively thus the Rte_LdComTp_Cbk.h header file must be provided by the system integrator and warning 100037 is still raised.

(RTE-10697)

Check compatibility of SwBaseTypes for connected ImplementationDataTypes

RTA-RTE now conforms with AUTOSAR R20-11 `[constr_1220]` when performing compatibility checks for `AutosarDataPrototype`s, in `ClientServerInterface`s or `SenderReceiverInterface`s, that are typed directly by ImplementationDataTypes.

See issue RTE-10904 for more detailed information about RTA-RTE validation of the compatibility of SwBaseTypes.

(RTE-9942)

No optimization for implicit access to data elements with status

When the command line option `--implicit-use-global-buffers` is enabled, RTA-RTE no longer optimizes implicit accesses using the global buffer for any `data element with status`.

(RTE-10199)

Version 7.5.0pr7

Map Os constructs to the RTE specific OsApplication in osNeeds.arxml

When the RTE specific OsApplication is generated in the output osNeeds.arxml, RTA-RTE now maps the `Rte_ScheduleTable`, `Rte_TickCounter`, and `Rte_TOutCounter` to that OsApplication, if they are used in the generated C code.

(RTE-8129)

Add shadow buffer fills for write only runnables accessing IIRVs

When a runnable writes to an `ImplicitInterRunnableVariable` but does not read from it, RTA-RTE now fills the shadow buffer generated for that `ImplicitInterRunnableVariable` before calling the writing runnable. This is in accordance with AUTOSAR requirement `[SWS_Rte_08418]`.

(RTE-10687)

Generate an RTE specific OsApplication in osNeeds.arxml

RTA-RTE now outputs an RTE specific OsApplication in the generated osNeeds.arxml with the `shortName 'RTE_DEFAULT_OS_APP'` if OsApplications are present in the input configuration. Additionally RTA-RTE raises error 100057 if an OsApplication in the input configuration has the same `shortName` as the RTE specific OsApplication.

(RTE-8127)

Accept connection of BOOLEAN ApplicationDataTypes to VALUE DataTypes

RTA-RTE now accepts configurations that connect an ApplicationDataType of category BOOLEAN to either an ApplicationDataType of category VALUE or an ImplementationDataType of category VALUE. Previously such configurations would raise error 2650.

(RTE-8687)

Version 7.5.0pr6

Include write buffers for compatible interfaces in the EHOOKS report

RTA-RTE now includes write buffers for implicitly accessed `VariableDataPrototype`s when the sender and receiver use different but compatible interfaces.

Note: If a configuration contains `FlatInstanceDescriptor`s that reference both the provider side `VariableDataPrototype` and the requirer side `VariableDataPrototype` within a communication graph, then RTEGen will only report one of the `FlatInstanceDescriptor`s.

(RTE-10405)

Fix split-swci IIRV names in EHOOKS report

RTA-RTE now does not prepend the prefix `Rte_IirvD_` to the `VAR-NAME` element in the EHOOKS report for reported `ImplicitInterRunnableVariable`s that are accessed from multiple OsApplications.

Note: this is only possible when the command line option `--deviate-split-swci-support` is enabled.

(RTE-10412)

EHOOKS report has VAR-FID-REF for all split SWC IIRV MEASUREMENT sections

RTA-RTE now generates the same VAR-FID-REF for all MEASUREMENT sections reported in the EHOOKS report for an `ImplicitInterRunnableVariable` that is accessed from multiple OsApplications.

Note: this is only possible when the command line option

`--deviate-split-swci-support` is enabled.

(RTE-10367)

Composite type for inter-ECU S/R communication with transformation

RTA-RTE now declares the composite type generated for queued explicit inter-ECU S/R communication with transformation in `Rte_Type.h`. Previously this was declared in the receiver application header file (see issue RTE-9593).

(RTE-10621)

Fixed an issue with copying structs in Rte_Result

RTA-RTE now correctly uses `Rte_memcpy` to copy structs from the result of the `IocRead` API when generating the Rte_Result API for inter-OsApplication client/server communication graphs in which the client and server `ClientServerOperations` use different but compatible struct `ArgumentDataPrototypes`.

(RTE-10600)

Update to RTA-A2L error reporting

The error reporting in the RTA-A2L tool shipped with RTA-RTE has been updated with minor changes to the message wording and severity. No functional changes have been made.

(RTE-10628)

Fix CompuMethods for ARDTs w optional elements w/o availabilityBitfields

RTA-RTE no longer throws error 7062 when mapping an `ApplicationRecordDataType` with optional elements but without an availabilityBitfield to an ImplementationDataType that has `isStructWithOptionalElement` set to `TRUE` and that `ApplicationRecordDataType` contains an ApplicationRecordElement which references the same `TEXTTABLE` CompuMethod as the corresponding ImplementationDataTypeElement.

An *optional element* in this context is an ApplicationRecordElement with `isOptional` set to `true`.

(RTE-10461)

Accept BswInterruptEntitys when OsApplications used

RTA-RTE now accepts configurations that contain both BswInterruptEntitys and OsApplications. Previously such configurations would raise internal error 900832.

(RTE-5539)

Version 7.5.0pr5

Error when COM is required and is not configured

RTA-RTE now raises error 100069 when the merged `EcucValueCollection` does not contain an `EcucModuleConfigurationValues` referencing an `EcuModuleDef` with `shortName Com` and COM is required for the system.

(RTE-10252)

Include Rte_LdComTp_Cbk.h in Rte_Cbk.h for RIPSified LDCOM_TP

RTA-RTE now includes the file Rte_LdComTp_Cbk.h in Rte_Cbk.h when an input configuration contains a RIPS-relevant LdComIPdu having `LdComApiType` set to LDCOM_TP.

Note that LDCOM_TP is still not supported natively thus the Rte_LdComTp_Cbk.h header file must be provided by the system integrator and warning 100037 is still raised.

(RTE-9979)

Error if a transformed ISignal has NetworkRepresentationProps

RTA-RTE now raises error 100072 when an ISignal that aggregates a NetworkRepresentationProps also references a `DataTransformation`.

(RTE-8808)

Include struct member name in all relevant elements of the EHOOKS report

RTA-RTE now includes the struct member name in the elements `VAR-COPY-NAME` and VAR-FID-REF of the generated EHOOKS report.

(RTE-10296)

Allow ApplicationRecordElements to specify SwCalibrationAccess

RTA-RTE no longer throws error 2883 when a SwCalibrationAccess is specified for an ApplicationRecordElement that has `category` VALUE. The behaviour for this SwCalibrationAccess is the same as though the SwCalibrationAccess was instead specified for the corresponding ImplementationDataTypeElement.

(RTE-10309)

Allow RIPS AsynchronousServerCallPoints with timeouts

RTA-RTE now accepts a configuration where a `SoftwareComponent` contains multiple AsynchronousServerCallPoints and the AsynchronousServerCallPoints that specify a timeout attribute are RIPS relevant, and any of the AsynchronousServerCallPoints that do not specify a timeout attribute are part of a different, non-RIPS, inter-OsApplication asynchronous client-server communication graph. Previously RTA-RTE did not differentiate between RIPS relevant AsynchronousServerCallPoints where timeouts are allowed and non-RIPS inter-OsApplication AsynchronousServerCallPoints where timeouts are not allowed and erroneously rejected the configuration with error 100049.

(RTE-9974)

Allow ImplementationDataTypes to specify a SwCalibrationAccess

RTA-RTE now accepts configurations that define a SwCalibrationAccess directly in an ImplementationDataType that has `category TYPE_REFERENCE` and is referenced by a `FlatInstanceDescriptor`. The generated output is the same as if the SwCalibrationAccess was specified on the referencing `FlatInstanceDescriptor`.

(RTE-10310)

Allow ImplementationDataTypeElements to specify a SwCalibrationAccess

RTA-RTE now accepts configurations that define a SwCalibrationAccess directly in an ImplementationDataTypeElement that has `category TYPE_REFERENCE` and generates the same output as though the SwCalibrationAccess was specified on the referenced ImplementationDataType.

(RTE-10308)

Minor update to RTA-RTE application usage information

RTA-RTE application usage information, the output from `RTEGen` when called with

the command line option `--help`, has been revised. There are no functional changes to the behaviour of the other command line options.

(RTE-10142)

Version 7.5.0pr4

Error for invalid InstantiationTimingEventProps to TimingEvent reference

RTA-RTE now raises error 100067 when an InstantiationTimingEventProps references a TimingEvent that is contained in a different `ApplicationSwComponentType` than the one that types the `SwComponentPrototype` referenced by the InstantiationTimingEventProps.

(RTE-8219)

RTA-RTE now supports LdComTxTriggerTransmit

RTA-RTE now generates the `Rte_LdComCbkTriggerTransmit_<sn>` API where `<sn>` is the `shortName` of an LdComIPdu with LdComTxTriggerTransmit configured.

In the RTA-RTE implementation of LdComTxTriggerTransmit the untransformed data is stored in a global variable (see issue RTE-8998) by the transmitting API, either Rte_Write or Rte_Send depending on data semantics (see issue RTE-8999 and issue RTE-9678 respectively). Additionally, the transmitting API sets a flag marking that data as available (see issue RTE-8997).

When `Rte_LdComCbkTriggerTransmit_<sn>` is called, data is available, transformation is successful (or a transformer returns a soft error), and the length of the transformed data is less than or equal to the input value of `PduInfoPtr->SduLength` (from the `inout` parameter `PduInfoType* PduInfoPtr`), then that `PduInfoPtr` is updated with the data and its length based on the output of the last transformer and the API returns `E_OK`. Otherwise `Rte_LdComCbkTriggerTransmit_<sn>` returns `E_NOT_OK`

Note that support is limited to inter-ECU explicit sender-receiver communication via `LDCOM_IF` with transformation configured.

(RTE-9000)

Support for queued transformation with RIPS

RTA-RTE now supports reception of queued explicit inter-ECU communication (via LDCOM_TP) where LdCom is in a different OsApplication to the receiving SWC. Previously for this specific case RTA-RTE raised internal errors 901181 and 901192.

(RTE-10335)

Allow configuration of alternative Implementation Data Type symbols

RTA-RTE now uses the `SymbolProps.symbol` attribute of an ImplementationDataType as the *Implementation Data Type symbol* when it is present in the configuration.

(RTE-9578)

Copy untransformed data in Rte_Send for LdComTxTriggerTransmit

RTA-RTE now stores a global copy of the untransformed data from within the Rte_Send API that is generated for a transformed queued data element that is mapped to a `SystemSignal`, which is mapped to an LdComIPdu with LdComTxTriggerTransmit configured. In addition, the generated Rte_Send API will also set the

boolean flag that indicates that data is available for that `SystemSignal`. See issue RTE-8997 for more information about this flag.

(RTE-9678)

Version 7.5.0pr3

Generate a global variable to support LdComTxTriggerTransmit

RTA-RTE now generates a global variable, `Rte_Rx_<SystemSignal>`, for every `SystemSignal` mapped to an LdComIPdu with LdComTxTriggerTransmit configured. See issue RTE-8998 where that global variable will be used to store the untransformed data that will be transmitted inter-ECU.

(RTE-8998)

Allow ApplicationArrayDataType mapped "new-world" variable-size arrays

RTA-RTE now treats "new-world" variable-size arrays defined by means of an ApplicationArrayDataType as if they were defined by configuring the attributes of the relevant ImplementationDataType when said data elements are either RIPS-relevant or RTE implemented data transformation is requested.

The presence of a "new-world" variable-size array anywhere within the hierarchy of an ApplicationDataType (including as a sub-element of an `ApplicationRecordDataType`) will cause RTA-RTE to behave as if the ImplementationDataType was mapped directly.

Note that the underlying behaviour of RTA-RTE generated code with respect to variable-size arrays remains unchanged.

(RTE-9525)

Copy untransformed data in Rte_Write for LdComTxTriggerTransmit

RTA-RTE now stores a global copy of the untransformed data from within the Rte_Write API that is generated for a transformed data element that is mapped to a `SystemSignal`, which is mapped to an LdComIPdu with LdComTxTriggerTransmit configured. In addition, the generated Rte_Write API will also set the boolean flag that indicates that data is available for that `SystemSignal`. See issue RTE-8997 for more information about this flag.

(RTE-8999)

Warning, not error, for unused LDCOM_TP LdComIPdus

RTA-RTE now raises error 3030 for an LdComIPdu that has `LdComApiType` set to LDCOM_TP only when it is used (it references an ISignal which references a `SystemSignal` that has a data element mapped to it).

In addition, when an input configuration contains an LdComIPdu that has `LdComApiType` set to LDCOM_TP and that LdComIPdu is not used, RTA-RTE now accepts the configuration and raises warning 100070.

Previously, RTA-RTE would raise error 3030 for all LdComIPdus that had `LdComApiType` set to LDCOM_TP.

(RTE-8863)

Error on LdComTriggerTransmit Transmit and Callback naming mismatch

RTA-RTE now rejects the input configuration with Error Message 100068 if the LdComTxTriggerTransmit is configured for a data element and its name is not

Rte_LdComCbkTriggerTransmit_<sn>, where <sn> is the shortName of the LdComIPdu mapped to the ISignal transporting that data element.

Note: As the AUTOSAR spec is currently ambiguous on the LdComTxTrigger-Transmit naming, we are forcing RTA-RTE's callback naming convention on it as well to avoid any potential name clashes.

(RTE-9191)

Map ARDTs w/o availabilityBitfields to optional ImplementationDataTypes

RTA-RTE now allows the mapping of an ApplicationRecordDataType with optional elements but without an availabilityBitfield to an ImplementationDataType that has isStructWithOptionalElement set to TRUE as if the ApplicationRecordDataType aggregates a compatible ApplicationRecordElement to the availabilityBitfield. Previously RTEGen would raise error 2669 for such cases.

RTA-RTE now raises error 2669 if an ApplicationRecordElement and the corresponding ImplementationDataTypeElement do not have the same value for isOptional when mapping an ApplicationRecordDataType to an Implementation-DataType with category STRUCTURE.

An *optional element* in this context is an ApplicationRecordElement with isOptional set to true.

(RTE-9557)

Version 7.5.0pr2

Generate a boolean flag to support LdComTxTriggerTransmit

RTA-RTE now generates a boolean flag, Rte_Rx_<SystemSignal>_Avaialble, for every SystemSignal mapped to an LdComIPdu with LdComTxTriggerTransmit configured. That flag will be used to track when data is available to be returned by a call to an Rte_LdComCbkTriggerTransmit API that will be generated for the LdComIPdu mapped to that SystemSignal.

(RTE-8997)

Fix MAKW for 1:N sender/receiver involving mixed inter/intra-OsApplication

RTA-RTE now includes the partition name in the memory allocation keywords surrounding an RTE managed buffer generated for a 1:N mixed inter/intra-OsApplication sender/receiver communication graph when the command line option --use-partition-sections is enabled. Previously when measurement was enabled for this configuration, specifically by referencing the PPortPrototype with a FlatInstanceDescriptor, then the partition name was omitted.

(RTE-10149)

The RTA-RTEcustomfeaturesguide now includes the EHOOKS report

The EHOOKS report is now documented in section 2.8 of the RTA-RTEcustomfeaturesguide.

(RTE-9919)

VSMD file for RTA-RTE updated

The VSMD file has been updated so that the metadata now references the version of RTA-RTE with which it is being shipped.

Note that this does not imply the implementation of any new AUTOSAR features. New features will have their own release notes.

(RTE-9555)

Error for ASW/BSW mismatch between EventToTaskMappings and Events

RTA-RTE now raises error 100065 when an `RteEventToTaskMapping` references a `BswEvent` and error 100066 when an `RteBswEventToTaskMapping` references an `RteEvent`.

(RTE-9717)

Version 7.5.0pr1

Precedence of SwCalibrationAccess

RTA-RTE now conforms with `[TPS_SWCT_01505]@R20-11` when determining the precedence of SwCalibrationAccess along structure hierarchies in complex types and uses the determined SwCalibrationAccess when generating the `Rte_McSupportData.arxml`.

Notes: RTA-RTE does not accept configurations that define a SwCalibrationAccess directly in the ImplementationDataTypeElement or ApplicationRecordElement. In order to configure the SwCalibrationAccess for an element that element must reference an ImplementationDataType or an ApplicationDataType which itself contains the SwCalibrationAccess.

Additionally, when a `FlatInstanceDescriptor` with a SwCalibrationAccess references a sub-element then that SwCalibrationAccess is applied to the root element.

Finally, for client/server configurations the `shortName` of `McDataInstances` of sub-elements are taken from the ApplicationDataTypes not the ImplementationDataTypes which is the case in sender/receiver configurations.

(RTE-9097)

Rte_ReceiverPullCB for queued inter-ECU inter-OsApp S/R

RTA-RTE now generates an Rte_ReceiverPullCB API for queued explicit inter-ECU inter-OsApplication sender/receiver communications with transformation when the command line option –deviate-cbk-xfrm is set. Within that Rte_ReceiverPullCB API the activation flag for the relevant `DataReceivedEvent` is set and the mapped OsTask is activated (unless it is a task that has been forced to be basic).

(RTE-9779)

Extend the EHOOKS report for IOC Channels

RTA-RTE now includes IOC Channels in the generated EHOOKS report (`rte_ehooksinfo_conf.xml`). When an IOC Channel is reported the name of the `VAR-GLOBAL-BUFFER` is prefixed with `Os_Ioc_`. An implicit buffer copy, `VAR-COPY-NAME`, is reported under the `VAR-GLOBAL-BUFFER` from which it is filled regardless of whether that is an IOC Channel or an RTE managed buffer.

Note: the report is generated when the command line option `--report` is provided to RTA-RTE with the parameter `[null]EHOOKS`.

(RTE-9149)

Declare data types in IOC Channels for queued transformed inter-ECU

comms

RTA-RTE now declares the underlying data types used in IOC Channels in the generated `iocNeeds.arxml` for receivers in queued explicit inter-ECU inter-OsApplication sender/receiver communications with transformation when the command line option –deviate-cbk-xfrm is set.

(RTE-9912)

Rte_Receive for queued inter-ECU inter-OsApp S/R with transformation

RTA-RTE now receives retransformed data across partitions using `IocReceive` in the Rte_Receive API generated for queued explicit inter-ECU inter-OsApplication sender/receiver communication with transformation when the command-line option –deviate-cbk-xfrm is provided.

(RTE-9777)

Rte_LdComCbk for queued inter-ECU inter-OsApp S/R with transformation

RTA-RTE now transfers data across partitions using `IocSend` after the retransformation has taken place in the Rte_LdComCbk generated for queued explicit inter-ECU inter-OsApplication sender/receiver communications with transformation when the command-line option –deviate-cbk-xfrm is set.

(RTE-9776)

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Version 7.4.3pr5

The IRTA-RTEsafetymanuall has been updated

Section `5.3.16` has been added to the RTA-RTE Safety Manual relating to issue `RTE-10197` which describes the RTA-RTE behaviour with respect to the compatibility of `SwBaseType`s and the required validation of generated code.

(RTE-10198)

Extend the EHOOKS report to include ImplicitInterRunnableVariables

RTA-RTE now includes ImplicitInterRunnableVariables in the EHOOKS report (`rte_ehooksinfo_conf.xml`) which is generated when the command line option `--report` is provided to RTA-RTE with the parameter `[null]EHOOKS`.

(RTE-9148)

Extend the EHOOKS report to include structs

RTA-RTE now includes implicitly accessed `VariableDataPrototype`s of `category STRUCT` in the generated EHOOKS report (`rte_ehooksinfo_conf.xml`). For each struct individual members with `category VALUE` or `ARRAY` are reported and the member (or sub-member in case of nested structs) is included in the `VAR-GLOBAL-BUFFER` and corresponding `VAR-COPY-NAME`(s).

Note: the report is generated when the command line option `--report` is provided to RTA-RTE with the parameter `[null]EHOOKS`.

(RTE-9147)

Add concurrency protection for shadow buffer fills of RamBlock subElements

RTA-RTE now provides resource protection for shadow buffer fills using `Rte_GetResource` and `Rte_ReleaseResource` in the context of data elements in an

RPort or PRPort typed by an `NvDataInterface` where that data element is a subElement of the RamBlock and is implicitly written but not implicitly read. When RTA-RTE determines that a fill cannot be interrupted (e.g. when the fill is being done in the highest priority task) the resource protection is omitted.

Note: This addresses the missing protection noted in issue RTE-8635.

(RTE-10077)

Fix a bug with Rte_Receive not returning Std_ReturnType

RTA-RTE now returns from Rte_Receive using Std_ReturnType for queued explicit inter-ECU sender/receiver communication with transformation. Previously Rte_Receive returned `void`.

Note: Queued inter-ECU explicit sender/receiver communication with transformation is only supported when the command-line option `--deviate-cbk-xfrm` is provided.

(RTE-9761)

Generate fills and flushes for synchronous C/S inter-OsApp direct calls

RTA-RTE now generates fills and flushes in the task body of the calling client for ImplicitInterRunnableVariables accessed from more than one `OsApplication` as part of a synchronous inter-`OsApplication` client/server communication graph where the server is called directly, the client runnable is synchronized with a `BswExecutableEntity` and is activated by a `BswEvent`.

(RTE-10042)

Extend the EHOOKS report to include arrays

RTA-RTE now includes implicitly accessed `VariableDataPrototype`s of `category ARRAY` in the generated EHOOKS report (`rte_ehooksinfo_conf.xml`).

Note: the report is generated when the command line option `--report` is provided to RTA-RTE with the parameter `[null]EHOOKS`.

(RTE-9146)

Version 7.4.3pr4

Extend the EHOOKS report to include detailed information

RTA-RTE now includes detailed information in the EHOOKS report (`rte_ehooksinfo_conf.xml`) for the individual implicit buffers generated for each implicitly accessed primitive data element in an input configuration. This includes the `shortName` and `symbol` of each `RunnableEntity` with implicit access, the direction of access (whether reading, writing or both), the mapped OsTask and the referenced `OsApplication`.

Note the report is generated when the command line option `--report` is provided to RTA-RTE with the parameter `[null]EHOOKS`.

(RTE-9144)

Missing fill for RamBlock subElements that are implicitly written but not read

RTA-RTE now fills the shadow buffer generated for a data element in an RPort or PRPort typed by an `NvDataInterface` where that data element is a subElement of the RamBlock and is implicitly written but not implicitly read.

Note: No concurrency protection is generated around the shadow buffer fills, so

data corruption is possible if a higher priority task writes to the RamBlock while a non-atomic fill operation is taking place.

(RTE-8635)

Extend the EHOOKS report to include FlatInstanceDescriptors

RTA-RTE will now include an XML element in the EHOOKS report `rte_ehooksinfo_conf.xml` containing the FlatInstanceDescriptor for each implicitly accessed `VariableDataPrototype` for which a FlatInstanceDescriptor has been configured.

Note the report is generated when the command line option `--report` is provided to RTA-RTE with the parameter `[null]EHOOKS`.

(RTE-9145)

Support more than 256 OsTasks in a configuration

RTA-RTE now supports up to 4,294,967,296 OsTasks (subject to support from the underlying operating system), previously support was limited to 256 OsTasks.

(RTE-9940)

Error when an InstantiationRTEEventProps references a non-TimingEvent

RTA-RTE now raises error 100063 when an InstantiationRTEEventProps references an `RTEEvent` that is not a TimingEvent.

(RTE-8217)

Version 7.4.3pr3

Identify the COM partition based on event mappings

When the option `--deviate-bsw-any-partition` is in use **and** the **new** option `--deviate-event-based-com-mapping` is present RTA-RTE will now determine the execution partition for the COM module (and hence if IOC is required between an Rte_COMCbk and the receiving API) through event mappings.

To decide which `EcucPartition` COM should execute in, RTA-RTE will attempt to find a `BSWModuleDescription` with the `shortName Com`. If this `BSWModuleDescription` contains any `BSWEvents`, RTA-RTE will then try to find the mapped `EcucPartition` through the path of `RteBswEventToTaskMapping`s which map the event(s) to an OsTask. RTA-RTE will then accept the `EcucPartition` which is implemented by the `OsApplication` the relevant OsTask is mapped to as the `EcucPartition` that COM is executing in.

Note: When any link in the mapping chain is missing then RTA-RTE determines the COM partition to be the `EcucPartition` with `EcucPartitionBswModuleExecution` set to `true` (see issue RTE-6185).

(RTE-9871)

Version 7.4.3pr2

Activation flags for inter-partition queued explicit inter-ECU S/R comms

RTA-RTE now generates a runnable activation flag for queued explicit inter-ECU sender/receiver communication with transformation when COM or LdCom and the receiving SWC are in different partitions and the runnable is started by a `DataReceivedEvent`.

Note: Queued inter-ECU explicit sender/receiver communication with transfor-

mationis only supported when the command-line option `--deviate-cbk-xfrm` is provided.

(RTE-9780)

Rte_COMCbk for queued inter-ECU inter-OsApp S/R comms with transformation

RTA-RTE now transfers data across partitions using `IocSend` after the retransformation has taken place in the Rte_COMCbk generated for queued explicit inter-ECU inter-`OsApplication` sender/receiver communications with transformation when the command-line option `--deviate-cbk-xfrm` is set.

(RTE-9775)

Fix a bug with Rte_Receive return codes for queued explicit comms

RTA-RTE now returns `RTE_E_NO_DATA` from Rte_Receive when no data is available in the queue generated as part of a queued explicit inter-ECU sender/receiver communication with transformation. Previously if no data was available in the queue when Rte_Receive was called the API could potentially return `RTE_E_HARD_TRANSFORMER_ERROR`.

Note that queued explicit inter-ECU sender/receiver communication with transformation is only supported when the command-line option `--deviate-cbk-xfrm` is provided.

(RTE-9789)

Create IOC Channels for queued transformed inter-ECU inter-OsApp receivers

RTA-RTE now generates IOC Channels and declares them in the generated `iocNeeds.arxml` for receivers in queued explicit inter-ECU inter-`OsApplication` sender/receiver communications with transformation when the command line option `--deviate-cbk-xfrm` is set.

(RTE-9778)

Generate a new report of implicitly accessed data elements

RTA-RTE now generates a report named `rte_ehooksinfo_conf.xml` when the option `--report` is provided on the command line with the parameter `[null]EHOOKS`. This report currently provides minimal information for each implicitly accessed primitive data element from an input configuration. It will be extended to include complex data elements, details of implicit copies and details of the accessing runnables.

(RTE-9143)

Validate OsTask activation queue lengths for asynchronous client/server

RTA-RTE now raises warning 100052 when an OsTask in an inter-`OsApplication` asynchronous client/server communication graph contains an activation queue length that is less than the total of the number of `AsynchronousServerCallReturnsEvent`s mapped to the OsTask plus the number of `AsynchronousServerCallPoint`s referencing `ClientServerOperation`s that are referenced by `OperationInvokedEvent`s mapped to the OsTask. This changes earlier behavior as described in issue RTE-8798 which raised an error when there was an inconsistency in the queue length.

(RTE-9600)

4.23 Version 7.4.2

Version 7.4.2pr1

Use correct Rte_memcpy parameters when copying STRUCT subelements

A bug was fixed which caused incorrect sizes to be passed to Rte_memcpy when copying complex types. Previously, in some circumstances, when generating code which uses Rte_memcpy to copy subelements of STRUCT types, RTA-RTE would use an incorrect size calculation based on the type of the outer STRUCT rather than the type of the subelement being copied.

(RTE-9705)

4.24 Version 7.4.1

Version 7.4.1pr1

Fix bug with ——output command line option

A bug was fixed whereby uppercase characters in patterns given to the –output option in order to override specific file output locations were erroneously matching their lowercase equivalent instead.

(RTE-9862)

4.25 Version 7.4.0

Version 7.4.0pr5

Fix bug causing error 2125 when using ——deviate-bsw-any-partition

A bug was introduced in issue RTE-9699 which caused RTA-RTE to ignore the `EcucPartitionBswModuleExecution` ECUC option when the command line option –deviate-bsw-any-partition was in use causing RTA-RTE to spuriously raise error 2125. This has now been fixed.

(RTE-9790)

Rte_Send for queued explicit inter-ECU S/R comms with transformation

RTA-RTE now generates Rte_Send for queued explicit inter-ECU sender/receiver communication with transformation. Note that on the transmission side the transformation is done in the API and the option –deviate-cbk-xfrm has no effect.

(RTE-9548)

IOC channel names with ——buffers-use-provider-name

RTA-RTE now uses the providing `SwComponentPrototype`, `PortPrototype`, and `VariableDataPrototype` to determine the name of IOC channels when the command line option –buffers-use-provider-name is set. Previously RTA-RTE used the requirer (see issue RTE-8845). Moreover, when multiple `SwComponentPrototype`s share a single IOC channel to write the same data, the IOC channel name is formed using the `SwComponentPrototype` that sorts first alphabetically. See the RTA-RTE Reference Manual for more details.

(RTE-9577)

Version 7.4.0pr4

Ignore ——deviate-cbk-xfrm for unqueued explicit inter-ECU S/R comms

When the option –deviate-cbk-xfrm is provided, RTA-RTE now generates `Rte_Read`, Rte_COMCbk and `Rte_LdComCbkRxIndication` for unqueued explicit inter-ECU sender/receiver communication with transformation as though the option was not provided.

(RTE-9734)

Version 7.4.0pr3

Rte_LdComCbk for queued inter-ECU S/R comms with transformation

RTA-RTE now performs retransformation in the generated `Rte_LdComCbkRxIndication` for queued explicit inter-ECU sender/receiver communication with transformation when the command-line option –deviate-cbk-xfrm is provided.

Note: Queued inter-ECU explicit sender/receiver communication with transformation is only supported when the command-line option –deviate-cbk-xfrm is provided.

(RTE-9704)

Determine the owning OsApp for BSW components based on event mappings

RTA-RTE now determines which OsApplication a BSW component is mapped to based on which `OsTasks` that components events are mapped to when the option –deviate-bsw-any-partition is present. Previously RTA-RTE would always assume that the component was mapped to the OsApplication which implements the `EcucPartition` with `EcucPartitionBswModuleExecution` set to 1. RTA-RTE will continue to make this assumption for any BSW components which do not map events to tasks or where its events are mapped to an OsApplication which does not implement any partition.

(RTE-9699)

Rte_Receive for queued explicit inter-ECU S/R comms with transformation

RTA-RTE now generates Rte_Receive in support of queued explicit inter-ECU sender/receiver communication with transformation when the command-line option –deviate-cbk-xfrm is provided.

Note: When the option –deviate-cbk-xfrm is provided the transformation is performed in the callback, see the RTA-RTE Reference Manual for more information. Currently support is limited to COM.

(RTE-9547)

Rte_COMCbk for queued inter-ECU S/R comms with transformation

RTA-RTE now performs retransformation in the generated Rte_COMCbk for queued explicit inter-ECU sender/receiver communication with transformation when the command-line option –deviate-cbk-xfrm is provided.

Note: Queued inter-ECU explicit sender/receiver communication with transformation is only supported when the command-line option –deviate-cbk-xfrm is provided.

(RTE-9545)

Queue the composite type for inter-ECU S/R comms with transformation

RTA-RTE now uses the new composite type, see issue RTE-9593, in the RTE

managed queue created for queued explicit inter-ECU sender/receiver communication with transformation when the command-line option –deviate-cbk-xfrm is provided.

(RTE-9592)

Fixed an issue with Rte_memcpy for arrays in Rte_Result

RTA-RTE now correctly determines the underlying type of an `ArgumentDataPrototype`, which references an `ImplementationDataType` of category `TYPE_REFERENCE` and ultimately references an array, when generating the Rte_Result API for inter-OsApplication client/server communication graphs. This leads to the correct generation of Rte_memcpy which had previously behaved as if the argument was a primitive.

(RTE-9632)

Composite type for inter-ECU S/R communication with transformation

When the command-line option –deviate-cbk-xfrm is provided with a queued explicit inter-ECU S/R communication for which transformation is configured, RTA-RTE now generates a new composite type in the receiver application header file comprised of two fields: a `data` field typed by the ultimate `ImplementationDataType` of the data being received in the S/R interface and a field typed by `Rte_TransformerError` which will contain the corresponding `transformerError` data.

This is the first piece of the functionality, enabled by –deviate-cbk-xfrm, which will move the transformation from the API into the callback and will be required when using queued explicit inter-ECU sender/receiver communication with transformation.

(RTE-9593)

Raise an error for ——buffers-use-provider-name with multiple IOC channels

RTA-RTE now raises error 100060 when the command line option –buffers-use-provider-name is set and the input configuration contains an inter-OsApplication sender-receiver communication graph which generates multiple IOC channels.

(RTE-9575)

Raise an info for extended tasks

RTA-RTE now raises info 100058 for each task that has been extended due to a mix of sporadic and periodic events, or a need to wait for a runnable, or a background event triggered runnable. See RTA-RTE User Guide sections `24.4` and `24.5` for more information.

(RTE-8855)

Version 7.4.0pr2

Do not use IOC for explicit sender-receiver inter-ECU COM transmission

For any sending runnable on the sending side of an explicit sender-receiver inter-ECU (via COM) configuration involving transformation, RTA-RTE does not use IOC to transfer data between the partition containing that runnable and the BSW partition. The command line option –inter-ecu-no-ioc is not needed for this behaviour.

(RTE-8364)

Do not use IOC for explicit sender-receiver inter-ECU LdCom reception

When the command line option –inter-ecu-no-ioc is set, for any receiving runnable on the receiving side of an explicit sender-receiver inter-ECU (via LdCom) configuration involving transformation, RTA-RTE does not use IOC to transfer data between the BSW partition and the partition containing that runnable.

(RTE-8367)

Mixed inter/intra-OsApplication sender-receiver communication with PRPorts

RTA-RTE now updates the global buffer created for local read and write access to a data element in a PRPort in the generated `Rte_Write` API making it available to local readers in addition to calling `IocWrite` when there is read access from a connected RPort in a SWC in a different partition.

(RTE-9108)

Accept equivalent LINEAR CompuMethods with opposite directions

RTA-RTE now accepts configurations where two data elements in ports connected using a `PortInterfaceMapping` have LINEAR CompuMethods that yield the same conversion, regardless of the direction of the conversion, i.e., `compuInternalToPhys` or `compuPhysToInternal`.

(RTE-9302)

Do not use IOC for explicit sender-receiver inter-ECU LdCom transmission

For any sending runnable on the sending side of an explicit sender-receiver inter-ECU (via LdCom) configuration involving transformation, RTA-RTE does not use IOC to transfer data between the partition containing that runnable and the BSW partition. The command line option –inter-ecu-no-ioc is not needed for this behaviour.

(RTE-8366)

New command line option ——inter-ecu-no-ioc

RTA-RTE now provides the command line option –inter-ecu-no-ioc. When this option is set, for any receiving runnable on the receiving side of an explicit sender-receiver inter-ECU (via COM) configuration involving transformation, RTA-RTE does not use IOC to transfer data between the BSW partition and the partition containing that runnable.

(RTE-8365)

Generate compatible CDS in contract and generation phases with split SWCs

RTA-RTE now generates a compatible component data structure, comparing the output of contract and generation phase, when a configuration contains `ImplicitInterRunnableVariable`s that are accessed from different OsApplications (using the option `--deviate-split-swci-support=1`) and the configuration additionally contains implicit read access to a data element in a port. Previously in generation phase with this configuration RTA-RTE was merging the data handles section and the inter runnable variable handles section leading to incorrect and incompatible output.

Note: Runnables with both read and write accesses to an `ImplicitInterRunnableVariable` which is accessed from multiple OsApplications (through use of `--deviate-allow-split-swc=1`) are not supported and may result in either an incompatible component data structure between contract phase and generation phase or uncompilable RTE IIRV API code. See issue RTE-9573 for more information.

(RTE-9462)

Raise an error for ASCREs in a SWC mapped to multiple OsApplications

RTA-RTE now raises error 100056 when `AsynchronousServerCallReturnsEvent`s mapped to a SWC instance are mapped to different OsApplications as part of an inter-OsApplication asynchronous client-server communication graph (even when the option `--deviate-split-swc-support` is enabled). The error is followed by an info message 1001 for each `AsynchronousServerCallReturnsEvent` in the SWC instance.

(RTE-9179)

Version 7.4.0pr1

Don't attempt to output duplicate typedefs

RTA-RTE now correctly outputs one typedef in `Rte_Type.h` when given `ImplementationDataType`s that ultimately map to the same type. Previously some configurations caused RTA-RTE to output duplicate typedefs triggering Error 7061.

(RTE-8611)

VSMD file for RTA-RTE updated to AUTOSAR 4.6.0

The VSMD file shipped with RTA-RTE has been updated to AUTOSAR revision 4.6.0.

Note that this does not imply the implementation of any new AUTOSAR features added between 4.5.0 and 4.6.0. New features will have their own release notes.

(RTE-9217)

Use Rte_memcpy to copy binary compatible structures in 1:N explicit S/R

RTA-RTE now uses Rte_memcpy to copy structures that are typed by **binary compatible** `ImplementationDataType`s with different `shortName`s in `Rte_Write` APIs generated for 1:N explicit sender-receiver communication graphs where the global buffer is typed by a different but compatible data type to the `VariableDataPrototype` in the `PPortPrototype`.

(RTE-7531)

4.26 Version 7.3.1

Version 7.3.1pr11

Inter-OsApplication asynchronous client/server has been overhauled

RTA-RTE no longer raises warning 100047 when inter-OsApplication asynchronous client/server communication is configured. Please see issue RTE-9197 for known issues not included in the scope of this improvement.

(RTE-9267)

Fixed a bug in Rte_ReceiverPullCB APIs for mixed intra/inter-OsApp async c/s

An issue has been fixed that, in the context of mixed intra-OsApplication/inter-OsApplication asynchronous client-server, caused RTA-RTE not to generate the server side Rte_ReceiverPullCB function bodies and not declare the Rte_ReceiverPullCBs in the generated iocNeeds.arxml for intra-OsApplication clients using IOC for communication due to being part of a mixed communication graph. RTA-RTE now generates intra-OsApplication Rte_ReceiverPullCB function bodies and declares them in the generated iocNeeds.arxml.

(RTE-9353)

Fix a bug for compatible ClientServerInterfaces in inter-OsApp async c/s

RTA-RTE now emits the expected server side Rte_ReceiverPullCB APIs and corresponding entries in iocNeeds.arxml when an inter-OsApplication asynchronous client/server configuration is supplied where a client and a server in that configuration use different but compatible ClientServerInterfaces.

(RTE-9416)

Memory allocation keywords for client/server

RTA-RTE now includes the `shortName` of the relevant `EcucPartition` in the memory allocation keywords for data structures frequently used in asynchronous intra-OsApplication client/server communication graphs.

(RTE-9241)

Fixed a bug in Rte_ReceiverPullCB APIs for inter-OsApp async client/server

An issue has been fixed that, in the context of an inter-OsApplication asynchronous client-server, caused RTA-RTE to omit the iocNeeds.arxml entries for the `OsIocReceiverPullCB` parameter of the `OsIocCommunication` subcontainer depending on the lexicographical ordering of the client and server `SoftwareComponent`s in the input configuration.

(RTE-9328)

Human-readable names for IOC channels with ——buffers-use-provider-name

When the command line option –buffers-use-provider-name is set, RTA-RTE now generates IOC channels for inter-OsApplication data communication graphs with names of the form `Rte_<SWC>_<PORT>_<DATA>` where `<SWC>` is the `shortName` of the `SwComponentPrototype` that reads the data element, `<PORT>` is the `shortName` of the requiring `PortPrototype`, and `<DATA>` is the `shortName` of the `VariableDataPrototype`. Moreover, when multiple `SwComponentPrototype`s share a single IOC channel to read the same data, the IOC channel name is formed using the `SwComponentPrototype` that sorts first alphabetically.

(RTE-8845)

Fixed a bug in Rte_ReceiverPullCB APIs for inter-OsApp async client/server

An issue has been fixed that, in the context of inter-OsApplication asynchronous client-server, caused RTA-RTE to set activation flags for all servers in an OsTask in each Rte_ReceiverPullCB that correctly sets an activation flag for any server in that OsTask. RTA-RTE now only sets the flag for the server related to the specific IOC channel for which the Rte_ReceiverPullCB is generated.

(RTE-9355)

Fixed server activation flag bug for inter-OsApp asynchronous client/server

An issue has been fixed that caused RTA-RTE not to generate the server activation flag definitions if the activation flag would have been used inside an extended OsTask that has been forced to be a basic task through the use of `--force-basic-tasks`. RTA-RTE now generates the activation flag definitions regardless of the type of OsTask.

(RTE-9310)

Fix a bug causing compatible ClientServerInterfaces to be rejected

A bug has been fixed that caused RTA-RTE to reject valid inter-OsApplication asynchronous client/server configurations with error 1335 when a client and server were using different but compatible ClientServerInterfaces in which the `ClientServerOperation`s have arguments that use different but compatible `ApplicationDataType`s.

(RTE-9298)

Support compatible ClientServerInterfaces for inter-OsApp async c/s

RTA-RTE now accepts configurations where compatible ClientServerInterfaces are connected via an `AssemblySwConnector` within an inter-OsApplication **asynchronous** client/server communication graph. Compatibility is determined according to AUTOSAR R20-11 `[constr_1078]` and `[constr_1079]`, noting that `[constr_1079 part b]` is not currently supported.

(RTE-9259)

Align the IocReceive API naming with iocNeeds.arxml for N:1 client-server

RTA-RTE now generates calls to IocReceive APIs to match the shortname of the `OsIocCommunication` entry in the generated iocNeeds.arxml file when an IocReceive API is required as part of an N:1 synchronous inter-OsApplication client-server communication graph.

(RTE-9070)

Support server runnables with transformerError in inter-OsApp async c/s

RTA-RTE now supports the extension of the server runnable API with the optional in parameter transformerError in inter-OsApplication asynchronous client/server communication graphs.

As RTA-RTE does not currently support transformation for client/server communications, the generated RTE code always indicates to the server runnable that no transformer error occurred during the decoding of its input parameters, by passing `0x00` as `transformerError.errorCode` and `RTE_TRANSFORMER_UNSPECIFIED` as `transformerError.transformerClass`.

(RTE-9261)

Support application errors from server runnables for inter-OsApp async c/s

RTA-RTE now passes the return value from a server runnable into the appropriate member of an IOC result channel generated for a `ClientServerOperation` that references possible application errors and is part of an inter-OsApplication asynchronous client/server communication graph.

(RTE-9308)

No support for ClientServerInterfaceMappings in inter-OsApp async c/s

RTA-RTE does not support the use of ClientServerInterfaceMappings in the context of inter-OsApplication asynchronous client-server communication.

(RTE-9131)

Invalid queueLength value in Inter-OsApplication asynchronous client-server

RTA-RTE now raises error 100054 when the `ServerComSpec` belonging to a `PPort` defines a queueLength greater than one as part of an inter-OsApplication asynchronous client-server communication graph. Note that at this time RTA-RTE does not support the parameter `RteServerQueueLength`.

(RTE-9037)

Version 7.3.1pr10

Support compatible ClientServerInterfaces for inter-OsApp synchronous c/s

RTA-RTE now accepts configurations where compatible ClientServerInterfaces are connected via an `AssemblySwConnector` within an inter-OsApplication **synchronous** client/server communication graph. Compatibility is determined according to AUTOSAR R20-11 `[constr_1078]` and `[constr_1079]`, noting that `[constr_1079 part b]` is not currently supported.

(RTE-9276)

Support application errors in Rte_Result for inter-OsApp asynchronous c/s

RTA-RTE now returns the server application error as the return value of an Rte_Result API generated for a `ClientServerOperation` that references possible application errors and is part of an inter-OsApplication asynchronous client/server communication graph. Note that if application errors are not configured then Rte_Result will return `RTE_E_OK` and in either case if no result is available then Rte_Result will return `RTE_E_NO_DATA`.

(RTE-9043)

Fixed client activation flag bug for inter-OsApp asynchronous client/server

An issue has been fixed that caused RTA-RTE to set multiple activation flags in the Rte_ReceiverPullCB APIs generated for the response from server to client as part of inter-OsApplication asynchronous client/server. RTA-RTE now only sets the activation flag belonging to the client for which the Rte_ReceiverPullCB API was generated.

(RTE-9189)

Fix task activation flag bug for inter-OsApplication asynchronous c/s

A bug has been fixed where an activation flag guarding a client runnable was tested and reset multiple times in a task body, this lead to a nested condition where the client runnable could not be activated. RTA-RTE now only tests and resets each unique activation flag once within a task body.

(RTE-9129)

Raise error for ASCRE mapped to inter-OsApp async c/s extended OsTask

RTA-RTE now raises error 100051 when an `AsynchronousServerCallReturnsEvent` in an inter-OsApplication asynchronous client/server communication graph is mapped to an extended OsTask.

(RTE-9056)

Support Rte_Result with transformerError in inter-OsApp asynchronous c/s

RTA-RTE now supports the extension of the Rte_Result API with the optional out parameter transformerError in inter-OsApplication asynchronous client/server communication graphs.

As RTA-RTE does not currently support transformation for client/server communications, the generated RTE code always indicates to the client runnable that no transformer error occurred during the decoding of its input parameters, by passing `0x00` as `transformerError.errorCode` and `RTE_TRANSFORMER_UNSPECIFIED` as `transformerError.transformerClass`.

(RTE-9133)

Support Rte_Call with transformerError in inter-OsApp asynchronous c/s

RTA-RTE now supports the extension of the Rte_Call API with the optional out parameter transformerError in inter-OsApplication asynchronous client/server communication graphs.

As RTA-RTE does not currently support transformation for client/server communications, the generated RTE code always indicates to the server runnable that no transformer error occurred during the decoding of its input parameters, by passing `0x00` as `transformerError.errorCode` and `RTE_TRANSFORMER_UNSPECIFIED` as `transformerError.transformerClass`.

(RTE-9132)

Removal of the command line option ——experimental-interosapp-async-cs

The command line option –experimental-interosapp-async-cs which was required to generate the overhauled inter-OsApplication asynchronous client/server for Rte_Call (issue RTE-8734) and server task bodies (issue RTE-8733) has been removed. In both cases the overhauled code is now generated without the option and the legacy code is omitted.

(RTE-9010)

Rte_Result for inter-OsApplication asynchronous client/server

RTA-RTE now generates the overhauled Rte_Result API for inter-OsApplication asynchronous client/server communication graphs.

(RTE-8735)

Validate OsTask activation queue lengths for asynchronous client/server

RTA-RTE now raises error 100052 when an OsTask in an inter-OsApplication asynchronous client/server communication graph contains an activation queue length that is less than the total of the number of `AsynchronousServerCallReturnsEvent`s mapped to the OsTask plus the number of AsynchronousServerCallPoints referencing `ClientServerOperation`s that are referenced by `OperationInvokedEvent`s mapped to the OsTask. This updates issue RTE-8797 to also account for server side cases and to use a new error number.

(RTE-8798)

Version 7.3.1pr9

VSMD File for updated to AUTOSAR 4.5.0

The VSMD file shipped with RTA-RTE has been updated to AUTOSAR revision

4.5.0.

Note that this does not imply the implementation of any new AUTOSAR features added between 4.3.0 and 4.5.0. New features will have their own release notes.

(RTE-9102)

Server task bodies for inter-OsApplication asynchronous client/server

RTA-RTE now generates the overhauled inter-OsApplication asynchronous client/server code for server task bodies when the option –experimental-interosapp-async-cs is provided on the command line. Without this option both the old and new code will be generated. This option won't be required once the overhaul is complete and will be removed as part of a future release.

In the overhauled task bodies generated for OsTasks mapped to `OperationInvokedEvent`s that start server runnables RTA-RTE now reads the incoming call arguments via IOC, executes the server runnable, and returns the result via IOC. If multiple runnables are called in the same OsTask then the activation is guarded by an activation flag.

(RTE-8733)

Include server errors in IOC result channels for inter-OsApp asynchronous c/s

RTA-RTE now includes the server return value typed by `Std_ReturnType` in the IOC result channel generated for inter-OsApplication asynchronous client/server communication when the interface description of that client/server communication lists potential application errors.

(RTE-9041)

Rte_Call for inter-OsApplication asynchronous client/server

RTA-RTE now generates the overhauled inter-OsApplication asynchronous client/server Rte_Call API when the option –experimental-interosapp-async-cs is provided on the command line. Without the option the old code is generated. This option won't be required once the overhaul is complete and will be removed as part of a future release.

(RTE-8734)

Call Rte_ReceiverPullCB API for inter-OsApplication asynchronous c/s

RTA-RTE now generates an Rte_ReceiverPullCB API when a client call is written via IOC in an inter-OsApplication asynchronous client/server communication graph. Within that Rte_ReceiverPullCB API the activation flag for the relevant `OperationInvokedEvent` is set and the mapped server OsTask is activated.

(RTE-8730)

Server side global variables for inter-OsApplication asynchronous c/s

RTA-RTE now generates a flag that will be used to guard the activation of each server runnable for each client that calls it when they are part of an inter-OsApplication asynchronous client/server communication graph and there is at least one other runnable started in the OsTask mapped to the `OperationInvokedEvent` referencing the server.

(RTE-8728)

Validate unique Operation and RPort per AsynchronousServerCallPoint

RTA-RTE now raises error 100050 when a `SoftwareComponent` instance in an

inter-OsApplication asynchronous client/server communication graph contains multiple AsynchronousServerCallPoints referencing the same `ClientServerOperation` in the same `RPortPrototype`.

(RTE-9040)

AUTOSAR 20-11 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace `http://autosar.org/schema/r4.0`) against the AUTOSAR 20-11 schema.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to 20-11 will validate against the 20-11 schema.

(RTE-8560)

Version 7.3.1pr8

Result Rte_ReceiverPullCB API for inter-OsApplication asynchronous c/s

RTA-RTE now generates an Rte_ReceiverPullCB API when the result of an operation is written via IOC by a server in an inter-OsApplication asynchronous client/server communication graph. Within that Rte_ReceiverPullCB API RTA-RTE sets a flag to make the result available to a client runnable polling the relevant Rte_Result API. Additionally, if the client is referenced by an `AsynchronousServerCallReturnsEvent`, the relevant activation flag is set and the mapped OsTask is activated.

(RTE-8731)

Update flags for data elements implicitly written across partitions

RTA-RTE no longer incorrectly sets the update flag in the task body of the writing runnable for a data element that is implicitly written from one partition and explicitly read from another. The update flag is already correctly set in the generated Rte_ReceiverPullCB API in the same partition as the reading runnable, see issue RTE-8722 for more details.

(RTE-9072)

Client runnable activation for inter-OsApplication asynchronous c/s

RTA-RTE now activates a client runnable in an inter-OsApplication asynchronous client/server communication graph when the OsTask mapped to the `AsynchronousServerCallReturnsEvent` that references the client runnable is activated and the activation flag for that event is set.

(RTE-8732)

Validate OsTask activation queue lengths for client side asynchronous c/s

RTA-RTE now raises error 6930 when an OsTask in an inter-OsApplication asynchronous client/server communication graph contains an activation queue length that is less than the number of `AsynchronousServerCallReturnsEvent`s mapped to the OsTask.

(RTE-8797)

Use one global buffer for inter-OsApplication implicit access

RTA-RTE now generates a single global buffer for all OsTasks with `RunnableEntity`s which have `dataReadAccess`s or `dataWriteAccess`s to a data

element when no `RunnableEntity` that is mapped to the OsTasks has both `dataReadAccess` and `dataWriteAccess` to that data element. This fix addresses the outstanding case from issue RTE-8916 where there is no `RunnableEntity` with `dataReadAccess` mapped to the same OsApplication to which the `RunnableEntity`s with `dataWriteAccess` are mapped.

Previously RTA-RTE was generating multiple global buffers in such a scenario, which was resulting in the possibility of garbage data being read by some reading `RunnableEntity`s.

(RTE-8994)

Version 7.3.1pr7

IOC channels and data types for inter-OsApplication asynchronous c/s

RTA-RTE now generates an IOC call and result channel for each client in an inter-OsApplication asynchronous client/server communication graph. The appropriate data types are generated for each channel such that the IOC channel used when calling a server contains all `in` and `inout` `ArgumentDataPrototype`s and the channel used to pass the result from the server to the client contains all `inout` and `out` `ArgumentDataPrototype`s for a particular `ClientServerOperation`.

(RTE-8729)

Use one global buffer for mixed inter- and intra-OsApplication implicit access

RTA-RTE now generates a single global buffer for all OsTasks with `RunnableEntity`s which have `dataReadAccess`s or `dataWriteAccess`s to a data element when no `RunnableEntity` that is mapped to the OsTasks has both `dataReadAccess` and `dataWriteAccess` to that data element. This is currently limited to the case where all `RunnableEntity`s with `dataWriteAccess` and at least one `RunnableEntity` with `dataReadAccess` are mapped to the same OsApplication and one `RunnableEntity` with `dataWriteAccess` is mapped to a different OsApplication.

Previously RTA-RTE was generating multiple global buffers in such a scenario, which was resulting in the possibility of garbage data being read by some reading `RunnableEntity`s.

(RTE-8916)

Client side global variables for inter-OsApplication asynchronous c/s

RTA-RTE now generates two flags for each client in an inter-OsApplication asynchronous client/server communication graph. One will be used in the Rte_Call API to guard against subsequent calls to a server when an operation is in flight, the other is to guard against a client reading stale data in the Rte_Result API.

Additionally RTA-RTE generates a flag to guard the activation of the client runnable if that runnable is started by an `AsynchronousServerCallReturnsEvent` and there is at least one other runnable started in the same OsTask.

(RTE-8727)

Use one shadow buffer for implicit read access in a task

RTA-RTE now generates a single shadow buffer in the preemption area of an OsTask for all `dataReadAccess`s to a data element when no `RunnableEntity` that is mapped to that OsTask has both a `dataReadAccess` and a `dataWriteAccess` to that

data element. Previously, some cases were identified where RTA-RTE would generate a shadow buffer for each `dataReadAccess` to the data element.

(RTE-8807)

Version 7.3.1pr6

Inter-partition client/server with BSW synchronization

RTA-RTE now generates the appropriate IOC calls when a client runnable is synchronized with a `BswExecutableEntity` and is activated by a `BswEvent` that is ultimately mapped to a different partition than the server runnable. Previously RTA-RTE generated code as if the client and server were in the same partition.

(RTE-8800)

Explicit reading of NV data through RPorts connected to NV PRPorts

RTA-RTE now supports the use of RPorts for explicit sender-receiver communication of NV data elements with data semantics, specifically a configuration where an `ApplicationSwComponentType` with an `RPortPrototype` typed by an `NvDataInterface` is connected, via an `AssemblySwConnector`, to an `NvBlockSwComponentType` with a `PRPortPrototype` typed by an `NvDataInterface`. The `NvBlockSwComponentType` maps the NV data element to a RAM block using an `NvBlockDataMapping` that references the NV data element in the role `writtenReadNvData`.

(RTE-8457)

Raise errors for unsupported inter-OsApplication asynchronous c/s scenarios

RTA-RTE now raises error 100048 when a client runnable that aggregates a `WaitPoint` and is started by an `AsynchronousServerCallReturnsEvent` is involved in an inter-OsApplication asynchronous client/server communication graph.

RTA-RTE now raises error 100049 when an AsynchronousServerCallPoint in an inter-OsApplication asynchronous client/server communication graph contains a `timeout` attribute with a non-zero value.

(RTE-8726)

Inter-OsApplication asynchronous client/server is being overhauled

The RTA-RTE code generated for Inter-OsApplication asynchronous client/server is being overhauled and is not suitable for production use. Warning 100047 will be raised if the input configuration contains any relevant communication graphs.

(RTE-8725)

Accept equivalent LINEAR CompuMethods that are not textually identical

RTA-RTE now accepts configurations where two data elements in ports using a `PortInterfaceMapping` have LINEAR CompuMethods that yield the same conversion, regardless of textual differences in the CompuMethods, *e.g.*, when both the numerators and the denominator are doubled in one CompuMethod compared to the other. Previously RTA-RTE rejected configurations with textually different CompuMethods with internal error 902208.

(RTE-8244)

PRPorts internal improvements

The generator code related to `PRPortPrototype`s typed by `NvDataInterface`s has

been updated. Some configurations that previously required suppression of internal errors now generate without needing error suppression.

(RTE-8382)

Version 7.3.1pr5

Memory Allocation Keywords for update flags

RTA-RTE now places update flags, which are initialized to FALSE, in the `INIT` memory section `RTE_<INFIX>_[START|STOP]_SEC_VAR_INIT_8`. Previously they were in the `CLEARED` memory section.

(RTE-8817)

Set update flag for a data element read inter-ECU and inter-partition

RTA-RTE now sets the update flag to `TRUE` in the Rte_ReceiverPullCB API generated in the receiver partition when a data element that has the `enableUpdate` attribute enabled is communicated inter-ECU and inter-partition. Note that inter-partition communication is required when the receiving runnable is mapped to a non-BSW partition. See issue RTE-6185 for more information related to RTA-RTE specific support for BSW partitions.

(RTE-8724)

Version 7.3.1pr4

Set the update flag for a data element communicated inter-partition

RTA-RTE now sets the update flag to `TRUE` in the Rte_ReceiverPullCB API generated in the receiver partition when a data element that has the `enableUpdate` attribute enabled is written inter-partition (via IOC). Previously the update flag was set in the `Rte_Write` API using a global variable accessible to both sender and receiver partitions. As the update flag is now only accessed from one partition the memory allocation keyword associated with the update flag now only lists the OsApplication associated with the receiving runnable partition.

(RTE-8722)

Support LdCom for RIPS-relevant data elements without transformation

RTA-RTE now allows the transportation of RIPS-relevant data elements via LdCom without requiring transformation to be configured. Previously RTA-RTE rejected such configurations with error 3031.

(RTE-8708)

Reset update flag for a data element read inter-ECU

RTA-RTE now sets the update flag to `FALSE` in the `Rte_Read` or `Rte_DRead` API generated for a data element that has the `enableUpdate` attribute enabled and is read inter-ECU (via COM or LdCom).

(RTE-7803)

Support transportation of RIPS-relevant complex types via a single ISignal

RTA-RTE now accepts configurations where RIPS-relevant complex data elements are mapped to an ISignal that is not part of a COM signal group when that data element is involved in inter-ECU explicit sender-receiver communication. Previously RTA-RTE rejected such configurations with internal error 900714.

(RTE-8585)

Version 7.3.1pr3

Implement Com_InvalidateSignal instead of Com_InvalidateShadowSignal

RTA-RTE now emits Com_InvalidateSignal instead of Com_InvalidateShadowSignal when the input configuration contains the parameter RteUseComShadowSignalApi and it is set to false. The default behaviour for RTA-RTE is unchanged and will use Com_InvalidateShadowSignal to invalidate individual signals when they form part of a signal group.

(RTE-7695)

Implement Com_ReceiveSignal instead of Com_ReceiveShadowSignal

RTA-RTE now emits Com_ReceiveSignal instead of Com_ReceiveShadowSignal when the input configuration contains the parameter RteUseComShadowSignalApi and it is set to false. The default behaviour for RTA-RTE is unchanged and will use Com_ReceiveShadowSignal to receive individual signals when they form part of a signal group.

(RTE-7694)

Implement Com_SendSignal instead of Com_UpdateShadowSignal

RTA-RTE now emits Com_SendSignal instead of Com_UpdateShadowSignal when the input configuration contains the parameter RteUseComShadowSignalApi and it is set to false. The default behaviour for RTA-RTE is unchanged and will use Com_UpdateShadowSignal to update individual signals when they form part of a signal group.

(RTE-7693)

Reset update flag for a data element read via IOC

RTA-RTE now sets the update flag to `FALSE` in the `Rte_Read` or `Rte_DRead` API generated for a data element that has the `enableUpdate` attribute enabled and is read inter-partition.

(RTE-7769)

Memory Allocation Keywords for update flags

RTA-RTE now declares the start and stop Memory Allocation Keywords for the update flag associated with a data element in the format `RTE_<INFIX>_START_SEC_VAR_CLEARED_8` and `RTE_<INFIX>_STOP_SEC_VAR_CLEARED_8` respectively, where `<INFIX>` is an underscore separated, uppercase, alphabetical list of the unique `shortNames` of the OsApplications that write to the update flag.

(RTE-7765)

Version 7.3.1pr2

Fix internal error 901300 for NV PRPorts configurations

An issue has been fixed that caused RTA-RTE to reject configurations with internal error 901300 where NV data is accessed via a `PRPortPrototype` and a data element in that port is implicitly written and explicitly read but not implicitly read.

(RTE-8352)

Flush measurable data elements that are implicitly written but not read

When an implicitly-written `VariableDataPrototype` that is marked as measurable (by being referenced from a `FlatInstanceDescriptor` with `SwCalibrationAccess`

set to `READ-ONLY` or `READ-WRITE`) exists in a `PPortPrototype` that is connected to an `RPortPrototype`, but no runnable accesses the data element in that `RPortPrototype`, RTA-RTE now correctly flushes the contents of that data element's implicit shadow buffer to the corresponding measurement buffer.

(RTE-8480)

Read legacy configuration parameter RteUseComShadowSignalApi

RTA-RTE now raises info message 100046 when an input configuration contains the legacy parameter RteUseComShadowSignalApi. Future versions of RTA-RTE will take this parameter into account to determine if `ShadowSignal` APIs will be used.

(RTE-7692)

Only export UnitGroups that are used

RTA-RTE now only copies UnitGroups from the input configuration to the generated `Rte_McSupportData.arxml` when they reference a `Unit` that is used by RTA-RTE. Previously RTA-RTE copied all UnitGroups.

(RTE-8603)

Review of Boolean Options

Boolean options (that is, those that are enabled or disabled by means of their argument) have been rationalized.

They are now all enabled by an argument of `1` and disabled by `0`; all other parameter values are now deprecated and will raise warning 2977.

Formerly, validation failures for these options would give a variety of errors; now error 1114 is raised for all.

The following options are affected, along with their short versions where applicable;

- `--append-name-to-buffer`
- `--bsw-scope-limit-defns`
- `--client-server-global-optimization`
- `--deviate-allow-supportsmulti-sharedmemorys`
- `--deviate-allow-unmapped-swci-config`
- `--deviate-appl-impl-compu-method`
- `--deviate-appl-impl-display-format`
- `--deviate-bsw-any-partition`
- `--deviate-enum-cast`
- `--deviate-group-calibration-none`
- `--deviate-ignore-datatype-semantics`
- `--deviate-implicit-cat2-mdd`
- `--deviate-implicit-modify-for-loopbacks`
- `--deviate-memmap-decls`
- `--deviate-omit-implicit-cds`

- `--deviate-physical-dimension-compatibility`
- `--deviate-prefer-no-empty-executions`
- `--deviate-split-swci-support`
- `--deviate-trace-implicit-api`
- `--deviate-unconnected-pmode-behavior`
- `--have-64bit-int-types`
- `--implicit-read-return-const`
- `--incremental-build`
- `--mcore-spinlocks-always`
- `--os-fp`
- `--os-permit-extended-tasks`
- `--use-partition-sections`
- `--vfb-trace`
- `--warning-as-error`
- `--os-task-as-function`

(RTE-8149)

Version 7.3.1pr1

Fixed mode status updates for explicitly ordered ModeDeclarationGroups

An issue was fixed whereby a configuration that contained a `ModeSwitchPoint` to switch the mode from a `ModeDeclarationGroupPrototype`, but no `ModeSwitchEvent` referencing that same `ModeDeclarationGroupPrototype`, and where the relevant ModeDeclarationGroup has the category set to `EXPLICIT_ORDER` was resulting in the `Rte_Switch` API using incorrect values to update the relevant mode status variable.

RTA-RTE now generates the `Rte_Switch` API that updates the mode status variable using correct values.

(RTE-8535)

4.27 Version 7.3.0

Version 7.3.0pr7

Supplementary data for Units and UnitGroups in Rte_McSupportData.arxml

A new parameter, `copy-source-path`, has been added to the command line option `--mcsd-policy`. When this parameter is provided each Unit and UnitGroup in the generated Rte_McSupportData.arxml contains a `SpecialData` with the full package and `shortName` of that Unit or UnitGroup from the input configuration. See the RTA-RTE Reference Manual for more details.

(RTE-8472)

SensorActuatorSwComponentTypes now allowed to aggregate PRPortPrototypes

RTA-RTE now allows the aggregation of PRPortPrototypes by SensorActuator-

SwComponentTypes.SensorActuatorSwComponentTypes.

RTA-RTE handles the aggregation of a PRPort by an SensorActuatorSwComponentType as if it was an `ApplicationSwComponentType`.

(RTE-8373)

Fixed issue with RIPS sender-receiver with transformation and LdCom

An issue was fixed whereby configurations containing a delegation port on the `RootSwComposition` that is mapped to two or more inner ports via `DelegationSwConnector`s for inter-ECU sender-receiver communication graphs where both RIPS and transformation are enabled were being rejected with internal error 903228 when transmission occurred using LdCom. RTEGen now accepts such configurations.

(RTE-8523)

Version 7.3.0pr6

Fixed issue with RIPS sender-receiver with transformation

An issue was fixed whereby configurations containing a delegation port on the `RootSwComposition` that is mapped to three or more inner ports via `DelegationSwConnector`s for inter-ECU sender-receiver communication graphs where both RIPS and transformation are enabled were being rejected with internal error 900931. RTEGen now accepts such configurations.

Note that the fix only works for configurations where signals are transported using `Com`. Configurations with signals transported using LdCom will be handled in a future update.

(RTE-8458)

RIPS 1:N and N:1 explicit inter-ECU sender/receiver

RTA-RTE now correctly generates code when RIPS is used in 1:N and N:1 explicit inter-ECU sender/receiver scenarios.

This fixes a bug where not all connected ports were correctly identified as RIPS-relevant in a non-1:1 communication scenario when only one of the connected ports was referenced by a RIPS `FlatInstanceDescriptor`.

(RTE-8455)

Mixed inter and intra partition synchronous client/server

RTA-RTE now uses IOC between all clients/servers in the context of synchronous client/server communication where one or more clients are on the same partition as the server and one or more clients are on a different partition.

(RTE-8397)

Correct locNeeds.arxml for N:1 IOC communication

RTA-RTE now outputs the correct `OsIocCommunication.shortName` for each `IocSend` API in the generated locNeeds.arxml. Previously for N:1 communication the `<SenderId>` was omitted.

(RTE-7725)

Fixed internal error 900590

Fixed an issue that caused RTA-RTE to reject some fan-in sender/receiver mixed intra and inter-ECU (via COM) configurations where the receiver does not access

the data with internal error 900590 depending on the order that RTA-RTE treated the measurable `DataPrototypes` in the delegation and inner ports, ultimately dependant on the lexicographical order of the SWCs in the input arxml.

(RTE-8046)

EcuAbstractionSwComponentTypes now allowed to aggregate PRPortPrototypes

RTA-RTE now allows the aggregation of PRPortPrototypes by EcuAbstractionSwComponentTypes.

RTA-RTE handles the aggregation of a PRPort by an EcuAbstractionSwComponentType as if it was an `ApplicationSwComponentType`.

(RTE-8371)

Version 7.3.0pr5

Return the status variable value from Rte_Read when using IOC

RTA-RTE now returns the value of the status variable associated with a data element from the Rte_Read API when the communication is inter-partition and the data element has an `InvalidationPolicy` with `HandleInvalid` set to a value other than `dontInvalidate`.

(RTE-7630)

Copy all Units and UnitGroups to Rte_McSupportData.arxml

RTA-RTE now copies all Units and UnitGroups from the input configuration to the generated Rte_McSupportData.arxml. The `shortNames` in the generated Rte_McSupportData.arxml are updated to avoid potential duplicate entries (*e.g.* when there is more than one Unit with the same `shortName` in multiple packages in the input configuration). The format is: `<path>_<shortName>`, where the `shortName` is unchanged and the `<path>` is generated by concatenating the full ARXML path from the input configuration , with the leading forward slash removed and all remaining forward slashes replaced with underscores.

(RTE-8355)

Box comment removal from generated code

When the command line option `--disable-comments` is set, RTA-RTE now additionally omits from generated code comments of the form:

- `/* Box: <Comment string> begin */`
- `/* Box: <Comment string> end */`

Where `<Comment string>` is the text of the comment.

(RTE-8293)

Version 7.3.0pr4

Allow fan-in and fan-out for RIPS sender-receiver with transformation

RTA-RTE now supports mapping a delegation port on the `RootSwComposition` to multiple inner ports via `DelegationSwConnectors` for inter-ECU sender-receiver communication graphs where both RIPS and transformation are enabled. Previously this configuration was rejected with errors 2979 and 2980.

Note: In this configuration each data element involved in the communication

graph must be referenced by a RIPS `FlatInstanceDescriptor`. Note further that it is the responsibility of the RIPS plugin to handle the data transformation.

(RTE-8285)

Comment removal from generated code

RTA-RTE now omits some comments from generated code when the command line option `--disable-comments` is set.

(RTE-8292)

Error if data conversion for composite data element members is configured

RTA-RTE now raises error 100042 when floating point data conversion is configured for members of composite data element. Previously RTA-RTE raised internal errors 900005 and 900004 on the transmission and reception sides respectively.

(RTE-4030)

Extend support of floating point data conversion to 8-bit and 32-bit ISignals

RTA-RTE now supports the conversion of IEEE 754 floating point data elements to and from 8-bit and 32-bit unsigned integers in the context of inter-ECU COM transmission and reception of IEEE 754 floating point data elements via 8-bit or 32-bit unsigned integer ISignals. This is in addition to the existing support for 16-bit unsigned integer ISignals.

(RTE-8152)

Version 7.3.0pr3

Command line option to suppress comments

RTA-RTE now supports the command line option `--disable-comments`, this will suppress some comments in the generated C code. An info message (100043) is output when the option is enabled.

Note: Currently no comments are removed, this option will be expanded in the future.

(RTE-8291)

Version 7.3.0pr2

Activation flag and mode check optimization for BSW triggers

RTA-RTE's optimization to remove activation flags and move mode disabling dependency checks when all ExternalTriggerOccurredEvents in an `OsTask` originate from the same trigger source (where no other events are mapped to the `OsTask`) has been extended.

This optimization is now also performed when the source for the relevant event(s) is a `BswModuleEntry` and the sink(s) is either an `ApplicationSwComponentPrototype` or a mixture of `ApplicationSwComponentPrototype`s and `BswModuleEntry`s.

(RTE-8155)

Consistently apply CompuMethods for inter-ECU data conversion

RTA-RTE now consistently applies inter-ECU data conversion CompuMethods, previously if the data element was an IEEE 754 floating point type then the relevant CompuMethod was applied in the opposite direction when compared to a configuration where the converted data element is an integer type.

(RTE-5114)

Register with LiMa without installing it

The RTA-RTE installer now provides the ability to register RTA-RTE with the ETAS License Manager without installing the License Manager. This is to support the case where the ETAS License Manager is already installed, such as installing RTA-RTE alongside other RTA Solutions products.

The installation type "Typical" now selects this option to register RTA-RTE with the ETAS License Manager, and a new installation type of "Minimal" has been added to just install RTA-RTE and the supporting examples.

Note: Registering RTA-RTE with the ETAS License Manager requires Administrator Rights on the target computer.

(RTE-7808)

ModeDisablingDependency checks in Trigger API

When an `ExecutableEntity` is activated by a `TriggerOccurredEvent` that is not mapped to an `OsTask`, and that event has a `ModeDisablingDependency`, RTA-RTE guards the invocation of the `ExecutableEntity` in the trigger API with the correct MDD check.

This fixes an issue where the MDD checks would be incorrectly omitted if there was an `OsTask` where all of the `ExecutableEntity`s mapped to it were started by the same trigger, and that trigger also started `ExecutableEntity`s by direct function call.

(RTE-8148)

Extend the Rips_GlobalCopy for queued RIPS Explicit S/R to include 1:N

RTA-RTE now exports the RIPS global copy for queued explicit sender-receiver for 1:N communication graphs extending the work done in issue RTE-7968. Note that the name of the `typedef` defining the type of the global queue structure has been updated to `Rte_Rips_GlobalCopy_<SWC>_<PORT>_<DATA>_Queue_Type` where <SWC>, <PORT>, and <DATA> are as previously defined in issue RTE-7968.

(RTE-8207)

Version 7.3.0pr1

Rips_GlobalCopy for queued RIPS Explicit S/R

RTA-RTE now exports a global copy of the RTE queue structure created for queued explicit sender-receiver communication graphs that are referenced by a RIPS `FlatInstanceDescriptor`, this does not currently extend to 1:N communication graphs. The exported RIPS global copy is named `Rte_Rips_GlobalCopy_<SWC>_<PORT>_<DATA>_Queue` where `<SWC>` is the `shortName` of the `SwComponentPrototype` that accesses to the data element, `<PORT>` is the `shortName` of the `PortPrototype`, and `<DATA>` is the `shortName` of the `VariableDataPrototype`. Additionally the generated `Rte_Buffers.h` file now contains an extern declaration of the global queue and the generated `Rte_DataHandleType.h` file contains a `typedef` defining `Rte_Rips_GlobalCopy_<SWC>_<PORT>_<DATA>_Type` to be the type of the global queue structure.

Note: This behaviour is not specified by AUTOSAR.

(RTE-7968)

Support for InstantiationRTEEventProps

RTA-RTE now supports overriding the `period` of a `TimingEvent` at instance level by defining an InstantiationRTEEventProps that references the `TimingEvent` through a `RefinedEventIRef`. When an `InstantiationTimingEventProps` is defined the `InstantiationTimingEventProps.period` will be used in preference to the `TimingEvent.period` for that `RTEEvent` instance.

(RTE-7534)

Optimize activation flags and mode checks for BswSchedulableEntitys

When all of the schedulable entities in a task are started by the same BSW trigger source, RTA-RTE no longer creates activation flags for those schedulable entities. In addition, if the trigger events starting those schedulable entities are configured with mode disabling dependencies, the mode checks are now performed in the task body, rather than the trigger API.

(RTE-8156)

Error for ExternalTriggerOccurredEvent that does not reference a trigger

RTA-RTE now exits with error 100041 when an ExternalTriggerOccurredEvent either does not contain a `TriggerIRef` or the `TriggerIRef` itself does not contain at least a `ContextRPortRef` and a `TargetTriggerRef`.

Note: Whilst AUTOSAR R19-11 states that the multiplicity of the `TriggerIRef` for an ExternalTriggerOccurredEvent is 0..1, RTA-RTE requires it to be present.

(RTE-8151)

4.28 Version 7.2.0

Version 7.2.0pr12

Several improvements to PRPorts

When two connected `PRPortPrototype`s are involved in a communication graph that exhibits fan-in and fan-out RTA-RTE no longer incorrectly raises error 3010 and internal error 900621 which mention an M-N graph topology.

RTA-RTE no longer raises 6051 regarding a mix of OsApplications due to PRPorts.

RTA-RTE no longer raises internal error 900937 when a `PRPortPrototype` is not connected to another `PortPrototype` and is RIPS relevant. When a communication graph contains a RIPS-relevant FID that points at a `dataElement` in an NV PRPort that is mapped to that `subElement` then the generated global copy correctly points at the mapped `subElement` of the `RamBlock` instead of the entire `RamBlock`.

Finally, RTA-RTE now accepts configurations where a `dataElement` in a PRPort does not have any write access. Previously RTA-RTE rejected this configuration with error 3017.

(RTE-7826)

Restrict activation-flag optimizations to non-BSW cases

Issue RTE-7861 introduced an optimization of activation flags when all the events mapped to a task are started by the same source.

This is now restricted to `RteEvent`s. BSW events are no longer optimized.

(RTE-8154)

McSupportData symbols for measured RIPS

RTA-RTE now outputs the name of the global buffer associated with the `Rte_Rips_GlobalCopy_<FID>` as the value of the symbol for the `McDataInstance` with the `shortName` of that measurable RIPS `FlatInstanceDescriptor` (where `<FID>` is the `shortName` of the referencing RIPS `FlatInstanceDescriptor` with `SwCalibrationAccess` set to `READ-ONLY` or `READ-WRITE` marking it as measurable).

(RTE-8137)

Cache mode used for the mode disabling dependency check in the task body

RTA-RTE now caches the current mode at the start of a task body and uses the cached value to determine if it is appropriate to execute the `RunnableEntity` when implementing the mode disabling dependency check in the task body. See issue RTE-7485 for more details.

(RTE-7862)

Don't generate unused buffers for RIPS data elements

Fixed an issue whereby an implicitly accessed RIPS relevant data communication graph that involved a mix of inter-OsApplication and intra-OsApplication communication resulted in the generation of multiple global buffers for that communication graph. Now RTA-RTE only generates a single global buffer for each data communication graph.

(RTE-8136)

Version 7.2.0pr11

Optimization of activation flags

RTA-RTE no longer generates activation flags for `RunnableEntity`s when all of the `RunnableEntity`s in an OsTask are triggered by the same trigger source.

This partially addresses issue 341063

(RTE-7861)

Move mode disabling dependency check to the task body

RTA-RTE now implements the mode disabling dependency test for a `RunnableEntity` in the task body when all of the `RunnableEntity`s mapped to an OsTask are only mapped to trigger events, `ExternalTriggerOccurredEvent` or `InternalTriggerOccurredEvent`, and those `RTEEvent`s are associated with a single trigger emitter. Previously this check was implemented in the `Rte_Trigger`.

(RTE-7485)

Fixed a bug causing duplicate Rte_Rips_NotifyRxAck calls

When a signal is mapped multiple times to a RIPS-relevant client/server operation via the same inner port, RTA-RTE no longer outputs multiple calls to the same Rte_Rips_NotifyRxAck in the callback function.

(RTE-8052)

Definition of RTE_OSAPI_AUTOSAR_R40 in contract phase

RTA-RTE now emits a definition of RTE_OSAPI_AUTOSAR_R40 in the file `Rte_Const.h` when running in contract phase.

(RTE-7856)

Fixed a bug determining the ultimate data type in a multi-dimensional array

RTA-RTE now correctly generates the macro definitions for `TEXTTABLE CompuMethod`s with the ultimate type of a data element by recursing through the multi-dimensional array. Previously RTA-RTE always resolved the type of a multi-dimensional array to be signed which led to some valid configurations being rejected with error 7062.

(RTE-7468)

Version 7.2.0pr10

Implicit communication for RIPS AsynchronousServerCallReturnsEvents

When a runnable is started by a RIPS-relevant AsynchronousServerCallReturnsEvent that is configured with an RteRipsInvocationHandler, and that runnable has implicit access to a non-RIPS-relevant data element, RTA-RTE now emits the correct shadow buffer fills and flushes for that implicit access.

(RTE-8010)

LdCom callbacks for RIPS-relevant inter-ECU client/server communication

Where a client/server operation is referenced by a RIPS `FlatInstanceDescriptor` and is mapped to an `ISignal` that is mapped to an LdCom signal (indicating inter-ECU communication), RTA-RTE now generates the LdCom callback function, `Rte_LdComCbkRxIndication`, for the `LdComIPdu` mapped to that `ISignal`. Within the LdCom callback function RTA-RTE generates the appropriate calls to the RIPS-provided service Rte_Rips_NotifyRxAck with the input parameter `PduInfoType* PduInfoPtr`.

Note: Support is limited to configurations where data transformation is requested, the `ISignal` is referenced by a `DataTransformation`, and the `LdComApiType` and `LdComIPduDirection` referenced by the `LdComIPdu` are set to `LdCom_IF` and `LDCOM_RECEIVE` respectively.

(RTE-7948, RTE-7980)

RIPS implicit access from servers triggered by OperationInvokedEvents

RTA-RTE now accepts a configuration where a server runnable, that has implicit access exclusively to RIPS-relevant data, is triggered by an OperationInvokedEvent, which is not mapped to an OsTask, that is called by an intermediate runnable that is both a client in this communication graph and server in another. That intermediate runnable, also triggered by an OperationInvokedEvent, must be connected to a client that is mapped to an OsTask.

Previously the configuration was rejected with error 5111.

(RTE-7943)

RteRipsInvocationHandlers for AsynchronousServerCallReturnsEvents

When RIPS is enabled, `RteEventToTaskMapping`s that reference both an AsynchronousServerCallReturnsEvent and an RteRipsInvocationHandler will cause the RteRipsInvocationHandler's defined symbol to be called in place of the standard `RunnableEntity` activation code.

For client server RIPS to be configured correctly the `ClientServerOperation` referenced by the AsynchronousServerCallReturnsEvent must be RIPS relevant.

(RTE-7780)

Fixed issue reporting provider buffer for measurement

An issue was fixed whereby a 1:N communication graph, measured at the provider side, that contained both intra-OsApplication and inter-OsApplication communication was resulting in RTA-RTE failing to report a symbol for measuring the RTE-managed buffer in the MCSD report.

(RTE-7663)

Fixed issue with inter-ECU RIPS-relevant sender/receiver communication

An issue was fixed whereby RIPS-relevant inter-ECU sender/receiver communication was causing RTA-RTE to throw error 6708 when the communicating `SwComponentPrototype` and COM are mapped to different `EcucPartition`s.

(RTE-7949)

Allow data invalidation policies when ––buffers-use-provider-name is set

RTA-RTE now supports the use of the –buffers-use-provider-name command line option in configurations with `VariableDataPrototype`s that have `InvalidationPolicy`s. Previously some configurations resulted in internal error 900447.

(RTE-7900)

Support Rte_IrvIWriteRef for primitive RIPS implicitInterRunnableVariables

RTA-RTE now generates Rte_IrvIWriteRef when RIPS is enabled and a runnable has implicit write access to a primitive implicitInterRunnableVariable that is referenced by a RIPS `FlatInstanceDescriptor`. See issue RTE-5970 for further details.

(RTE-7745)

Version 7.2.0pr9

Rte_Write now fills the global buffer for a PRPort with no connections

When a data element in a PRPort, that is not connected to another `PortPrototype`, is both explicitly read and written, the explicit write API now populates the global buffer with the value passed to the write API. Previously the Rte_Write was incorrectly being optimized to a null implementation.

(RTE-7735)

Raise error for RIPS implicit inter-ECU sender/receiver communication

RTA-RTE now rejects the configuration and raises error 100040 when a RIPS-relevant data element is accessed implicitly through inter-ECU sender/receiver communication.

(RTE-7827)

Rte_COMCbk for RIPS 1:N and N:1 client/server communication

RTA-RTE now generates the Rte_COMCbk callback for each of the signals configured for RIPS relevant `ClientServerOperation`s. As in 1:1 scenarios, the callback is implemented by calling Rte_Rips_NotifyRxAck.

Previously, when multiple signals were configured for a single `ClientServerOperation`, *e.g.*, by mapping multiple delegation ports to an inner port, RTA-RTE raised internal error 903643.

(RTE-7652)

Validation of RteOsTaskChain containers

RTA-RTE now rejects the input configuration and raises error 100033 if an RteOsTaskChain does not contain exactly one `RtePredecessorOsTaskRef`.

Similarly, RTA-RTE rejects the configuration and raises error 100034 if an RteOsTaskChain does not contain exactly one `RteSuccessorOsTaskRef`.

Finally, RTA-RTE rejects the configuration and raises error 100035 if the same OsTask is referenced by both the `RtePredecessorOsTaskRef` and the `RteSuccessorOsTaskRef` in an RteOsTaskChain.

(RTE-7729)

Fix a bug with the return type of function prototypes for server runnables

RTA-RTE now declares the return type of the generated C-function prototype for a `RunnableEntity` that implements a server in client-server communication according to AUTOSAR `SWS_Rte_01130`. Briefly, where the interface description of that client server communication lists potential application errors the return type will be `Std_ReturnType` otherwise it will be `void`.

(RTE-7545)

Improve reporting of resources in osNeeds.arxml

RTA-RTE now declares an `OsResource` in the generated osNeeds.arxml for each OsApplication that is described in the input configuration. The `shortName` for the `OsResource` is of the form `RTE_RESOURCE_OS_APP_<OsAppName>`, where `<OsAppName>` is the `shortName` of the OsApplication for which the `OsResource` is being generated and the `OsResourceProperty` has the value `STANDARD`.

Additionally, where an OsTask is referenced by an OsApplication in the input configuration, the description of that OsTask in osNeeds.arxml will contain a reference to the `OsResource` that is defined in relation to the OsApplication that references the OsTask.

Previously, RTA-RTE would define a single `OsResource` named `RTE_RESOURCE` and all the OsTasks described in osNeeds.arxml would reference `RTE_RESOURCE`.

This updates issue RTE-1150.

(RTE-6297)

Improve validation of the allowable attributes of SwDataDefProps

RTA-RTE now allows an ApplicationPrimitiveDataType of category STRING to reference a unit from within SwDataDefProps. Previously RTA-RTE rejected this configuration with internal error 2883.

RTA-RTE now rejects configurations that have an `ApplicationRecordDataType` or `ApplicationArrayDataType` that reference a unit from within SwDataDefProps with internal error 2883.

(RTE-7875)

Fixed LdCom RIPS redirecting to NotifyTxAck instead of NotifyRxAck

Fixed a bug introduced in issue RTE-7678, RTA-RTE now properly redirects LdCom callbacks for RIPS to Rte_Rips_NotifyRxAck instead of `Rte_Rips_NotifyTxAck`.

(RTE-7866)

Support for RIPS Explicit S/R FIDs referencing delegation ports

RTA-RTE now allows a RIPS `FlatInstanceDescriptor` to reference an explicitly-accessed sender/receiver data element in the context of a delegation port on the configuration's `RootSwComposition`.

When a RIPS FID references an explicitly-accessed sender/receiver data element in the context of a delegation port, RTA-RTE generates the same code as it would if the RIPS FID referenced the same data element in the context of the connected inner port.

(RTE-7830)

Version 7.2.0pr8

Error if an OsTask is involved in multiple RteOsTaskChains

RTA-RTE now rejects the configuration and raises error 100032 if an OsTask is referenced by more than one `RtePredecessorOsTaskRef` followed by info 1001 for each of the relevant RteOsTaskChains. Similarly RTA-RTE rejects the configuration and raises error 100039 if an OsTask is referenced by more than one `RteSuccessorOsTaskRef` followed by info 1001 for each of the relevant RteOsTaskChains.

(RTE-5873)

Allow a unit for BOOLEAN ApplicationPrimitiveDataTypes

RTA-RTE now allows an ApplicationPrimitiveDataType of category BOOLEAN to reference a unit from within SwDataDefProps. Previously RTA-RTE rejected this configuration with internal error 2883.

(RTE-7228)

RTA-RTE does not support LdCom for implicit communication

RTA-RTE does not support the use of LdCom with implicit access and now rejects the configuration with error 100038, previously an internal error was raised.

(RTE-7407)

AUTOSAR R19-11 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace `http://autosar.org/schema/r4.0`) against the AUTOSAR R19-11 schema.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to R19-11 will validate against the R19-11 schema.

(RTE-7580)

Allow multiple Xfrm modules

RTA-RTE now accepts configurations with multiple Xfrm modules, optionally split over several files, RTA-RTE will implement all the `XfrmImplementationMapping`s for each Xfrm module listed in an `EcucValueCollection`. Previously RTA-RTE rejected this configuration with error 1210.

(RTE-4017)

Unit reference in ApplicationValueSpecification is optional

When an `ApplicationvalueSpecification` does not reference a `Unit` via the aggregated `SwValueCont` RTA-RTE now generates the RTE as if the `ApplicationvalueSpecification` references a `Unit` that has `FactorSiToUnit` set to `1.0` and `OffsetSiToUnit` set to 0.0, and has no reference to a `PhysicalDimension`.

Previously RTA-RTE would have rejected the configuration with error 1346.

(RTE-7523)

Allow LdCom TP API for RIPS

When an input configuration contains an `LdComIPdu` that has `LdComApiType` set to `LDCOM_TP` and that `LdComIPdu` is mapped to an `ISignal` that is itself mapped to a RIPS activated data element, RTA-RTE now accepts the configuration and raises warning 100037 to indicate that no code will be generated to support LdCom TP API.

(RTE-7676)

Deprecated option ——sws

The –sws option has been deprecated, as well as its abbreviation –sws. RTA-RTE now raises warning 2976 if –sws, or its abbreviated version, –sws, is used.

The –sws option should not be used for new projects, and where possible existing projects should be modified to no longer use it.

(RTE-7807)

Version 7.2.0pr7

Include fill and flush for implicit access from RIPS server runnable

When a runnable is started using an RteRipsInvocationHandler and that runnable has implicit access to a data element that is not RIPS-relevant, RTA-RTE generates the implicit buffer fill and flush either side of the RIPS invocation handler.

(RTE-7733)

Update to the command line option ——buffers-use-provider-name

The command line option –buffers-use-provider-name has been updated and now supports intra-ECU 1:1 and 1:N data communication graphs and also unconnected PPorts for which measurement is requested by marking the `SwDataDefProps\.SwCalibrationAccess` as `READ-ONLY` or `READ-WRITE`.

See the relevant section of the Reference Manual for more information.

(RTE-7793)

Fixed internal error for mixed RIPS C/S and implicit S/R communication

RTA-RTE now supports configurations where a RIPS-relevant server runnable in a client/server communication graph, that is mapped to an OperationInvokedEvent, has implicit access to a RIPS-relevant data element in a sender/receiver communication, and the `RteEventToTaskMapping` of the OperationInvokedEvent also configures a RIPS fill or flush routine.

Previously, RTA-RTE failed with internal error 1413.

(RTE-7486)

Fixed return type of Rte_DRead for RIPS-relevant data

When a runnable explicitly reads a RIPS-relevant data element by value, the

return value of the Rte_DRead API is now typed by the `ImplementationDataType` of the RIPS-relevant data element.

(RTE-7775)

LdCom callbacks for RIPS relevant inter-ECU communication

Where a data element is referenced by a RIPS `FlatInstanceDescriptor` and is mapped to an `ISignal` that is mapped to an LdCom signal (indicating inter-ECU communication), the LdCom callback functions generated for the `LdComIPdu` mapped to that `ISignal` are now implemented by calling the appropriate RIPS-provided service, Rte_Rips_NotifyRxAck, with the input parameter `PduInfoType* PduInfoPtr`.

(RTE-7678)

Include RIPS Header File in Application Header File for RIPS IIRVs

When a `SwComponentPrototype` has access to a RIPS-relevant ImplicitInter-RunnableVariable, the generated application header file of the corresponding `SwComponentType` now includes the applicable `RIPS SWC-BSW-Instance Header File`.

(RTE-7776)

ActivateTask now calls the first task in a chain

When an OsTask that is referenced by an RteOsTaskChain is activated, RTA-RTE now passes the first OsTask in the chain of RteOsTaskChains to the ActivateTask API.

This is currently supported for `DataReceivedEvent`s and OperationInvokedEvents.

(RTE-5872)

Version 7.2.0pr6

RIPS and forward-declaration of transformer functions

When a transformer is configured for a RIPS-relevant inter-ECU communication graph, RTA-RTE now does not forward declare the transformer functions.

(RTE-7677)

Fixed global copy type for RIPS-managed transformation

When RIPS is enabled on a communication graph for which data transformation takes place the declaration of the global buffer on the receiver side is now typed by the `ImplementationDataType`.

(RTE-7679)

Create RIPS global copy for explicit write access

When a `RunnableEntity` has explicit write access to a RIPS-activated data element, RTA-RTE now generates a global buffer corresponding to that data element.

(RTE-7680)

Error on circular task chains

When related RteOsTaskChain containers result in a circular chain of tasks, RTA-RTE now rejects the configuration and raises error 100031.

(RTE-5874)

Call ChainTask for tasks in chains

When two tasks are chained using an RteOsTaskChain, RTA-RTE now generates a call to ChainTask at the end of the task body of the predecessor task, in place of the usual `TerminateTask` call.

(RTE-5871)

Version 7.2.0pr5

Allow a RIPS-relevant runnable to be started by multiple tasks

When a `RunnableEntity` has implicit access to RIPS-relevant data, and no other implicit data access, RTA-RTE now allows the `RunnableEntity` to be started by events mapped to multiple `OsTasks`. Previously, RTA-RTE rejected several aspects of the configuration with the errors 1357, 2024 and 3038.

(RTE-6951)

Fixed a bug where the update flag is not set by implicit write

When a data element has the `enableUpdate` attribute enabled, is implicitly written and explicitly read, RTA-RTE now sets the update flag to `TRUE` after the implicit write.

Previously in this configuration the flag was not updated.

(RTE-7498)

Support for Rte_DRead for RIPS-relevant data

When a runnable explicitly reads a RIPS-relevant data element by value, RTA-RTE implements the Rte_DRead API using a RIPS plugin implementation, not currently defined by AUTOSAR, of the form `<return> Rte_Rips_<PlugIn>_DRead_<SwcBswI>_<CGI>()`, following the normal AUTOSAR conventions for each *‹variable›*.

Note: Rte_DRead does not currently support the optional OUT parameter `tranformerError` and by extension neither does the RIPS implementation.

(RTE-7655)

Fixed issue with inter-OsApplication RIPS ImplicitInterRunnableVariables

RTA-RTE now generates the APIs for RIPS-relevant ImplicitInterRunnableVariables that are accessed by `RunnableEntity`s spanning across multiple OsApplications in the same manner as if the `RunnableEntity`s are all activated within the same OsApplication. Previously, RTA-RTE also generated a mapping to an RTE-managed implementation.

Note: this behaviour is only valid when the command line option `--deviate-split-swci-support` is set to `1`.

(RTE-7657)

Raise an error if ——buffers-use-data-name leads to duplicated symbols

RTA-RTE now raises error 100030 when the command line option –buffers-use-data-name is set and the input configuration leads to duplicated internally named symbols.

(RTE-6747)

Inclusion of SWC-BSW instance header for RIPS relevant inter-ECU clients

For RIPS relevant inter-ECU client/server communication where a SwComponentPrototype acts as the client (aggregates an AsynchronousServerCallResultPoint or a SynchronousServerCallPoint), the generated application header file of the corresponding SwComponentType now includes the applicable RIPS SWC-BSW-Instance Header File.

This behavior only applies for ISignals - mapping a RIPS-relevant client server operation to an ISignalGroups is not currently supported.

(RTE-7594)

Version 7.2.0pr4

Template language section removed from Toolchain Integration Guide

The toolchain integration guide's section about the template language was out of date and has been removed.

(RTE-4871)

Option to replace buffer numbers with human-readable names

When the new command line option –buffers-use-provider-name is set and RTA-RTE generates exactly one global buffer for an intra-ECU 1:1 or 1:N communication graph, the name of the buffer takes the form Rte_<SWC>_<PORT>_<DATA> where <SWC> is the shortName of the SwComponentPrototype that writes to the data element, <PORT> is the shortName of the providing PortPrototype, and <DATA> is the shortName of the VariableDataPrototype.

(RTE-7656)

Toolchain Integration Guide refreshed

The Toolchain Integration Guide has been revised for readability, and obsolete material that referenced examples that are no longer shipped with RTA-RTE has been removed.

(RTE-4862)

Info message for RteOsTaskChains

RTA-RTE now raises info message 100029 for each RteOsTaskChain in the RteOsInteraction container.

(RTE-5870)

Do not generate implicit shadow buffers for RIPS-relevant IRVs

RTA-RTE no longer generates RTE-managed shadow buffers for implicitly-accessed implicitInterRunnableVariables that are handled by a RIPS plugin.

(RTE-5966)

Do not raise error 1360 for a RIPS relevant communication graph

RTA-RTE no longer raises error 1360 when a PortPrototype typed by a SenderReceiverInterface, that is read from or written to by tasks from different OsApplications, participates in a RIPS relevant communication graph and the option --deviate-split-swci-support is enabled.

(RTE-7264)

Omit fill and flush of RIPS-activated implicitInterRunnableVariables

RTA-RTE no longer generates code to fill and flush RTE-managed implicit shadow

buffers when access to RIPS-relevant implicitInterRunnableVariables is handled by a RIPS plugin.

(RTE-5968)

KEEP invalidation policy should not change the buffer value

When a `VariableDataPrototype` is referenced by an `InvalidationPolicy` with `HandleInvalid` set to `KEEP` and an explicit access exists for a `PPortPrototype` and the `VariableDataPrototype`, the generated buffer is no longer assigned an invalid value inside the `Rte_Invalidate` method generated for the `PPortPrototype`.

Previously the buffer was assigned the `InvalidationPolicy` configured for the `ImplementationDataType` that types the `VariableDataPrototype`.

(RTE-7313)

Version 7.2.0pr3

Fixed bug introduced in RTE-5965

Fixed a bug where the CDS handles for RIPS implicitInterRunnableVariables that were typed by complex data types were not initialized correctly.

(RTE-7636)

Support for Rte_Receive for RIPS-relevant data

When a runnable explicitly reads a RIPS-relevant queued data element by argument, RTA-RTE implements the Rte_Receive API using the RIPS plugin `Rte_Rips_<plugin>_Read`.

(RTE-7350)

Support for Rte_Send for RIPS-relevant data

For RIPS relevant explicit queued sender/receiver communication, where the sender aggregates a `VariableAccess` in the `DataSendPoint` role, the Rte_Send API generated for that communication will now call the `Rte_Rips_Write` API which will return the result of the Rte_Send API.

When RIPS is used, all aspects of the sender/receiver communication are expected to be handled by the RIPS plugin.

(RTE-7349)

Support for Rte_Write for RIPS-relevant data

When a runnable explicitly writes a RIPS-relevant data element, RTA-RTE implements the Rte_Write API using the RIPS plugin `Rte_Rips_<plugin>_Write`.

(RTE-7348)

Support for Rte_Read for RIPS-relevant data

When a runnable explicitly reads a RIPS-relevant data element by argument, RTA-RTE implements the Rte_Read API using the RIPS plugin `Rte_Rips_<plugin>_Read`. Note: This does not include the implementation of Rte_DRead for data elements read by value.

(RTE-7347)

Clearer documentation of issue RTE-7286

Section 10.11 of the User Guide and sections 6.4 and 10.4 of the Reference Manual have been updated to be clearer that RTA-RTE expects all tasks to run to comple-

tion,and that the consequences of exceptional termination may be significant. See issue RTE-7286.

(RTE-7576)

Support the value of NONE for RteExclusiveAreaImplMechanism

RTA-RTE now supports the value of NONE for RteExclusiveAreaImplMechanism, which results in the corresponding `Rte_Enter` and `Rte_Exit` APIs being optimized to a null implementation.

(RTE-7227)

Fixed internal error 901769 for NV PRPort configurations

Fixed an issue whereby a `SensorActuatorSwComponentType` that has write access to a `VariableDataPrototype` through a `PRPortPrototype` typed by an `NvDataInterface` was causing RTA-RTE to exit with internal error 901769 when the `AssemblySwConnector` referenced the `PRPortPrototype` aggregated by the `SensorActuatorSwComponentType` as its `requestor`.

(RTE-7467)

Inclusion of RIPS SWC Instance Header Files

When a `SwComponentPrototype` has explicit access to a RIPS-relevant data element, the generated application header file of the corresponding `SwComponentType` now includes the applicable `RIPS SWC-BSW-Instance Header File`.

Note, previously issue RTE-7263 limited this to `SwComponentPrototype`s with implicit access.

(RTE-7355)

Suppress the timestamp comment in Rte_Buffers.h with ——notimestamps

Previously RTA-RTE generated Rte_Buffers.h containing a timestamp comment when the –notimestamps option was present. This bug has now been fixed.

(RTE-7281)

Version 7.2.0pr2

Initialization of CDS handles for RIPS implicitInterRunnableVariables

When RIPS is enabled, and a SWC is delivered as object code and has access to an implicitInterRunnableVariable that is referenced by a RIPS `FlatInstanceDescriptor`, RTA-RTE initializes the CDS data handles associated with the inter-runnable variable using the RIPS API provided by the associated RIPS plugin.

CDS data handles associated with `readLocalVariable`s are initialized using `Rte_Rips_<plugin>_IRBufferRef`, while those associated with `writtenLocalVariable`s use `Rte_Rips_<plugin>_IWBufferRef`.

(RTE-5965)

Declare update flag in VAR memory section

When a data element has the `enableUpdate` attribute enabled and is explicitly read by a `RunnableEntity`, RTA-RTE now declares the associated update flag in the `VAR_CLEARED_8` memory section.

Previously some configurations could result in the update flag being declared in a

CONST memory section.

(RTE-7314)

Implicit write access to RIPS implicitInterRunnableVariables

When RIPS is enabled and a runnable has implicit write access to an implicitInterRunnableVariable that is referenced by a RIPS FlatInstanceDescriptor, RTA-RTE implements the Rte_IrvIWrite and Rte_IrvIWriteRef APIs that access that implicitInterRunnableVariable using the RIPS API provided by the associated RIPS plugin.

If the inter-runnable variable is primitive, Rte_IrvIWrite will be implemented using Rte_Rips_<plugin>_IWrite. For complex data elements Rte_Rips_<plugin>_IWBufferRef will be used.

The Rte_IrvIWriteRef API is implemented using Rte_Rips_<plugin>_IWBufferRef regardless of the data element type.

(RTE-5970)

RTA-CAR plugin

The installer for RTA-RTE no longer offers the option of installing an RTA-CAR plugin.

(RTE-7494)

Fixed bug using undefined Rte_ModeOld variable

Fixed an issue where BswModeSwitchEvents that are not mapped to an OsTask resulted in a generated RTE that cannot be compiled due to the use of an undefined Rte_ModeOld variable. RTA-RTE now uses the correctly defined Rte_ModeSchMOld variable instead.

(RTE-6354)

COM callbacks for RIPS relevant inter-ECU client/server communication

Where a client server operation is referenced by a RIPS FlatInstanceDescriptor and is mapped to an ISignal (indicating inter-ECU communication), the COM callback functions generated for that ISignal are now implemented by calling the appropriate RIPS-provided service.

In RIPS scenarios, the Rte_COMCbk callback is implemented by calling Rte_Rips_NotifyRxAck.

This behavior only applies for ISignals - mapping a RIPS-relevant client server operation to an ISignalGroups is not currently supported.

(RTE-7433)

Omit CDS handles for RIPS-relevant implicitInterRunnableVariables

When RIPS is enabled RTA-RTE does not generate a CDS handle associated with accesses to RIPS-relevant implicitInterRunnableVariables when the SWC is delivered as source code.

(RTE-5964)

Implicit read access to RIPS implicitInterRunnableVariables

When RIPS is enabled and a runnable has read access to an implicitInterRunnableVariable that is referenced by a RIPS FlatInstanceDescriptor, RTA-RTE implements the Rte_IrvIRead APIs that access that implicitInterRunnableVariable

using the RIPS API provided by the associated RIPS plugin.

If the inter-runnable variable is primitive, `Rte_IrvIRead` will be implemented using `Rte_Rips_<plugin>_IRead`. For complex data, `Rte_Rips_<plugin>_IRBufferRef` will be used.

(RTE-5969)

Version 7.2.0pr1

Allow unmapped events when cross-partition event mapping is enabled

RTA-RTE now allows the use of events not mapped to tasks in the input configuration when the command line option `--allow-cross-partition-event-mapping` is used.

Previously RTA-RTE would reject the configuration with internal error 903625.

(RTE-7344)

Fixed non-deterministic order of warning 3064 output by RTA-RTE

RTA-RTE now raises warning 3064 in a deterministic order.

(RTE-6934)

RTEGen exit status code

RTA-RTE's contract has been relaxed regarding exit status codes. Formerly, exit status 3 was used for internal errors and 1 was used for non-internal errors.

This has been simplified: exit status zero means success, non-zero means failure.

RTA-RTE's documentation has been updated to reflect this change.

(RTE-7436)

Rte_Result for RIPS relevant inter-ECU asynchronous client/server communication

For RIPS relevant inter-ECU client/server communication where the client aggregates an `AsynchronousServerCallResultPoint`, the Rte_Result API generated for that server call point now calls the appropriate `Rte_Rips_ReturnResult` API and returns the result as the return of the Rte_Result API.

When RIPS is used, all aspects of the client/server communication are expected to be handled by the RIPS plugin.

(RTE-7425)

Rte_Call for RIPS relevant inter-ECU synchronous client/server communication

For RIPS relevant inter-ECU client/server communication where the client aggregates an `SynchronousServerCallPoint`, the Rte_Call API generated for that server call point will now call the `Rte_Rips_Invoke` API and return the result as the return of the Rte_Call API.

When RIPS is used, all aspects of the client/server communication are expected to be handled by the RIPS plugin.

(RTE-7426)

Rte_Call for RIPS relevant inter-ECU asynchronous client/server communication

For RIPS relevant inter-ECU client/server communication where the client aggregates an `AsynchronousServerCallPoint`, the Rte_Call API generated for that server call point will now call the `Rte_Rips_Invoke` API and return the result as the return of the Rte_Call API.

When RIPS is used, all aspects of the client/server communication are expected to be handled by the RIPS plugin.

(RTE-7424)

4.29 Version 7.1.0

Version 7.1.0pr6

Omission of SWC-BSW instance header files for RIPS relevant servers

Application header files for SWCs that contain a server runnable for a RIPS relevant client-server communication graph no longer `#include` the RIPS SWC-BSW instance header file relating to that communication graph.

(RTE-7432)

Symbolic names for LdCom handles

Symbolic names used for LdCom handles in calls to LdCom APIs when the `--com-symbolic-sigs` option is used have been updated to conform to AUTOSAR 4.4.0 [TPS_ECUC_02108].

(RTE-7434)

Version 7.1.0pr5

Reference Manual error regarding ——vfb-trace option

The Reference Manual and help text now report correct information about the –vfb-trace option and `RteVfbTraceEnabled` parameter.

(RTE-7400)

Symbolic names for COM handles

Symbolic names used for COM handles in calls to COM APIs when the `--com-symbolic-sigs` option is used have been updated to conform to AUTOSAR 4.4.0 [TPS_ECUC_02108].

(RTE-6933)

User Guide updated

The User Guide has been updated to reflect that `SchM_Exit` and `SchM_Enter` APIs are supported.

(RTE-7283)

RTA-CAR registration

The installer for RTA-RTE can now integrate with ISOLAR-A/B as part of RTA-CAR: it will optionally install an RTA-CAR plugin and register the installation of RTA-RTE with it.

The RTA-CAR component is not installed by default.

(RTE-6923,RTE-7184)

New ——implicit-use-cds option

When the –implicit-use-cds option is provided and a `SwComponentType` is delivered

as source code, the implicit Sender/Receiver APIs (`Rte_IRead`, `Rte_IWrite` and `Rte_IWriteRef`) generated for that SWC will always access implicit shadow buffers using CDS handles.

RTA-RTE raises error 100028 when the –implicit-use-cds option is provided and the configuration contains a `PortProtoype` that is referenced by a `DataReadAccess` or `DataWriteAccess` and has `enableTakeAddress` set to `true`.

(RTE-6658)

Support for AUTOSAR 4.1 style Core References

RTA-RTE now supports the assignment of an `EcucPartition`'s core ID through the use of an `EcucCoreDefinition` referenced by the implementing `OsApplication`.

When an `OsApplication` references an `EcucCoreDefinition`, RTA-RTE generates the same code as it would if the `OsApplication` instead contained an `OsApplicationCoreAssignment` with the same value as the `EcucCoreId` of the `EcucCoreDefinition`.

Note: The use of an `OsApplicationCoreAssignment` for core assignment is deprecated.

RTA-RTE now raises warning 100026 for each `OsApplication` that contains an `OsApplicationCoreAssignment`.

RTA-RTE now raises error 100027 for each `OsApplication` that references an `EcucCoreDefinition` and contains an `OsApplicationCoreAssignment` with a different value than the `EcucCoreId` of the `EcucCoreDefinition`.

(RTE-5993)

Client/server RIPS SWC-BSW-Instance includes in application header files

When a `SwComponentPrototype` is a client of a RIPS-relevant client/server communication, the generated application header file of the corresponding `SwComponentType` now includes the applicable `RIPS SWC-BSW-Instance Header File`.

(RTE-7339)

Fixed RteRipsInvocationHandlerFncSymbol names for RIPS client server

An issue that caused RTA-RTE to fail to identify the function names for RteRipsInvocationHandlers has been fixed.

(RTE-7292)

Version 7.1.0pr4

COM callbacks for RIPS relevant inter-ECU sender/receiver communication

Where a data element is referenced by a RIPS `FlatInstanceDescriptor` and is mapped to an `ISignal` (indicating inter-ECU communication), the COM callback functions generated for that `ISignal` are now implemented by calling the appropriate RIPS-provided service.

In RIPS scenarios, the `Rte_COMCbk` callback is implemented by calling `Rte_Rips_NotifyRxAck`, `Rte_COMCbkTOut` by `Rte_Rips_NotifyRxTOut`, and `Rte_COMCbkTAck` by `Rte_Rips_NotifyTxAck`.

This behavior only applies for `ISignal`s - mapping RIPS-relevant data elements to `ISignalGroup`s is not currently supported.

(RTE-7282)

Raise error when first structure element is optional

RTA-RTE now raises error 100022 for each `ImplementationDataType` that has `isStructWithOptionalElement` set to `TRUE` when the first contained ImplementationDataTypeElement has `isOptional` set to `TRUE`. Such configurations violate `[constr_1638]` (AUTOSAR 4.4.0).

(RTE-7109)

Error when optional ImplementationDataTypeElement has category STRUCTURE

RTA-RTE now raises error 100021 for each ImplementationDataTypeElement that is of category STRUCTURE and has the attribute `isOptional` set to `TRUE`. Such configurations violate `[constr_1639]` (AUTOSAR 4.4.0).

(RTE-7112)

RIPS in mixed implicit and explicit communication scenarios

An issue has been fixed that caused RTA-RTE to fail to implement implicit APIs for a data element using RIPS, where the `FlatInstanceDescriptor` references the data element via a SWC in which it is explicitly accessed.

(RTE-7243)

Fixed internal error 902155 for NV PRPorts configurations

An issue has been fixed that caused RTA-RTE to reject configurations with internal error 902155 where NV data is accessed via a PRPort and a data element in that port is mapped onto a sub-element of a `RamBlock`.

(RTE-7266)

Inclusion of RIPS SWC Instance Header Files

RTA-RTE no longer includes `RIPS SWC-BSW-Instance Header Files` in the application header files of `SwComponentPrototypes` that contain only explicit accesses to a RIPS-relevant data element.

(RTE-7263)

Raise error if RIPS data element is not accessed

RTA-RTE now raises error 100024 when a RIPS `FlatInstanceDescriptor` references a data element via a `PortPrototype` in an instance of a `SwComponentType` which does not aggregate any `VariableAccess` referencing the data element.

(RTE-6956)

RIPS with mixed inter- and intra-ECU implicit communication

An issue has been fixed that previously caused RTA-RTE to raise fatal error 001413 when a RIPS-relevant data element is sent to a mix of inter-ECU and intra-ECU destinations via implicit communication.

(RTE-7036)

Raise error when non-optional implementation data type contains optional elements

RTA-RTE now raises error 100020 for each `ImplementationDataType` that does not have `isStructWithOptionalElement` set to `TRUE` but contains ImplementationDataTypeElements with `isOptional` set to `TRUE`. The error is accompanied by one

info message for each ImplementationDataTypeElement which has `isOptional` set to `TRUE`.

(RTE-7108)

FIDs for Client/Server RIPS in the context of delegation ports

RTA-RTE now allows a RIPS `FlatInstanceDescriptor` to reference a `ClientServerOperation` in the context of a delegation port on the configuration's `RootSwComposition`.

When a RIPS FID references an operation in the context of a delegation port, RTA-RTE generates the same code as it would if the RIPS FID referenced the same `ClientServerOperation` in the context of the connected inner port.

The equivalent sender/receiver scenario, where a RIPS FID references a data element in the context of a delegation port, is currently unsupported.

(RTE-6949)

Enforce [SWS_Rte_CONSTR_80005]

RTA-RTE now raises error 100019 when a RIPS `FlatInstanceDescriptor` references a `ClientServerOperation` in the context of an `RPortPrototype`, when that `RPortPrototype` participates in a client/server communication graph for which the configuration includes a `PPortPrototoype`.

Such configurations violate [SWS_Rte_CONSTR_80005], which stipulates that a RIPS `FlatInstanceDescriptor` must reference the `PPortPrototype` in a client/server communication graph when the configuration includes the server.

(RTE-7233)

Version 7.1.0pr3

Raise error when non-structure implementation data type supports optional elements

RTA-RTE now raises error 100018 for each `ImplementationDataType` that has `isStructWithOptionalElement` set to `TRUE` but does not have category STRUCTURE.

(RTE-7107)

RteRipsInvocationHandlers in RIPS activated client server

When RIPS is enabled, `RteEventToTaskMappings` that reference both an `OperationInvokedEvent` and an RteRipsInvocationHandler will cause the RteRipsInvocationHandler's defined symbol to be called in place of the standard `RunnableEntity` activation code.

For client server RIPS to be configured correctly the `ClientServerOperation` referenced by the `OperationInvokedEvent` must be RIPS relevant.

(RTE-6496, RTE-7273, RTE-7254)

Rte_Call for RIPS relevant intra-ECU synchronous client/server communication

For RIPS relevant intra-ECU client/server communication where the client aggregates a `SynchronousServerCallPoint`, the Rte_Call API generated for that server call point will now call the `Rte_Rips_Invoke` API and return the result as the return of the Rte_Call API.

When RIPS is used, all aspects of the client/server communication are expected to be handled by the RIPS plugin.

(RTE-6493)

AUTOSAR schema in generated iocNeeds.arxml

When IOC is in use, the generated `iocNeeds.arxml` file now refers to the AUTOSAR 4.2.2 schema.

(RTE-6618)

Version 7.1.0pr2

Memory Allocation Keywords for implicit buffers for cross-partition events

When a task contains cross-partition events with access to the task's implicit shadow buffer, the memory allocation keyword for the implicit buffer now includes the `shortName`s of partitions that access it.

The memory allocation keyword includes, in alphabetical order, the name of the partition that maps the task itself and the names of any partitions that map `SwComponentPrototype`s that own the relevant mapped events.

(RTE-7073)

Move flush of MST relevant implicit shadow buffers into trusted functions

When a runnable which participates in implicit communication is activated by a cross-partition event, RTA-RTE will now move any code used to perform flushes of the runnable's implicit shadow buffers into the trusted function generated to activate the runnable.

This change affects flushes which take place via a direct copy from the shadow buffer to the associated global and any flushes which take place within an `Rte_ImplWrite` function.

(RTE-7074)

Support for Rte_IsAvailable and Rte_SetAvailable APIs

RTA-RTE now generates Rte_IsAvailable and Rte_SetAvailable APIs for optional elements of complex `ImplementationDataType`s having `isStructWithOptionalElement` set to `true`.

An *optional element* in this context is an ImplementationDataTypeElement with `isOptional` set to `true`.

The generated APIs read and write the `availabilityBitfield`, which must be defined as the first element of the `ImplementationDataType` according to `[TPS_SWCT_01774]` (AUTOSAR 4.4.0).

The APIs are generated in `Rte_Type.h`, using the implementations described in `[SWS_Rte_03617]` and `[SWS_Rte_03618]` (AUTOSAR 4.4.0).

(RTE-7145, RTE-7146)

Allow PRPortPrototype aggregated by SensorActuatorSwComponentTypes

RTA-RTE now allows the aggregation of PRPortPrototypes by SensorActuator-SwComponentTypes.

RTA-RTE handles the aggregation of a PRPort by a SensorActuatorSwComponentType as if it was aggregated by an `ApplicationSwComponentType`.

(RTE-7149)

Version 7.1.0pr1

RTE-managed data structures relating to RIPS-relevant client/server communications

RTA-RTE no longer generates RTE-managed data structures for `ClientServerOperation`s that are handled by a RIPS plugin.

When a `ClientServerOperation` is handled by a RIPS plugin, any buffers and queues needed for transferring data to and from the server are managed by the RIPS plugin assigned to the operation.

(RTE-6497)

Do not raise internal errors 902199 and 902201

An issue that caused RTA-RTE to raise internal errors 902199 and 902201 for valid configurations that involved 1:N communication graphs where the requirer dataElements were typed by separate but compatible AutosarDataTypes has been fixed.

(RTE-7188)

Rte_Result for RIPS relevant intra-ECU asynchronous client/server communication

For RIPS relevant intra-ECU client/server communication where the client aggregates an `AsynchronousServerCallResultPoint`, the Rte_Result API generated for that server call point now calls the appropriate `Rte_Rips_ReturnResult` API and returns the result as the return of the Rte_Result API.

Although AUTOSAR only requires a RIPS plugin to provide the `Rte_Rips_ReturnResult` service for client/server communications that use data transformation (see `[SWS_Rte_70070]` in AUTOSAR 4.4.0), RTA-RTE allows RIPS to be used with any client/server communication. When RIPS is used, all aspects of the client/server communication are expected to be handled by the RIPS plugin.

The use of RIPS has not been tested in the case where the client software component is an instance of an `NvBlockSwComponentType`.

(RTE-6495)

Server runnable calls with transformer error handling

RTA-RTE now extends calls made to server runnables with a `transformerError` parameter when the associated server port is referenced by a `PortAPIOption` with `errorHandling` set to `transformerErrorHandling`.

As RTA-RTE does not currently support transformation for client/server communications, the generated RTE code always indicates to the server runnable that no transformer error occurred during the decoding of its input parameters, by passing `0x00` as `transformerError.errorCode` and `RTE_TRANSFORMER_UNSPECIFIED` as `transformerError.transformerClass`.

(RTE-6691)

Raise error when a SwComponentPrototype with cross partition events is targeted by multiple FlatInstanceDescriptors

RTA-RTE now raises error 100017 if a `SwComponentPrototype` that contains cross

partition events is a target of multiple `FlatInstanceDescriptors`. The error is accompanied by one info message for each `FlatInstanceDescriptor` which targets the `SwComponentPrototype`.

(RTE-6993)

Force use of task specific implicit buffer allocation for cross partition event activations

RTA-RTE now uses task specific implicit buffer allocation for tasks which participate in cross partition event activations.

(RTE-7072)

4.30 Version 7.0.2

Version 7.0.2pr2

VFB hooks for mixed-safety task RunnableEntitys

RTA-RTE now emits VFB hook calls before and after activation of RunnableEntitys in trusted functions generated for cross-partition event activation when VFB tracing is enabled. Declarations of any VFB hook calls are emitted in `Rte_Hook.h`.

(RTE-7076)

Fix incorrect OsTrustedFunctionNames in trustedFunctionNeeds.arxml

RTA-RTE previously emitted incorrect values for any OsTrustedFunctionNames generated in trustedFunctionNeeds.arxml, which has now been fixed. Any calls to `CallTrustedFunction` generated by RTA-RTE have also been fixed accordingly to use the correct value for their `FunctionIndex` parameter.

(RTE-7169)

Version 7.0.2pr1

Legacy RIPS ExclusiveArea macros in trusted functions

When a cross partition event activation invokes a RunnableEntity that runs inside an ExclusiveArea, if legacy RIPS is enabled, RTA-RTE now emits any calls to ExclusiveArea protection macros within the trusted function generated to support activation of the RunnableEntity.

(RTE-7075)

Fixed incorrect EcucModuleConfigurationValues shortName in trustedFunctionNeeds.arxml

An issue that caused RTA-RTE to always use the shortName "`Os`" for EcucModuleConfigurationValues in trustedFunctionNeeds.arxml has been fixed. RTA-RTE now uses the shortName of the EcucModuleConfigurationValues containing `Os` parameters from the input configuration.

(RTE-7134)

Fixed internal error 902155 for NV PRPorts configurations

An issue that caused RTA-RTE to reject configurations with internal error 902155 has been fixed.

Previously this internal error was raised for configurations where NV data is accessed via a PRPort and a data element in that port is mapped onto a `RamBlock` of a different type category.

(RTE-6624)

Declarations for trusted functions

RTA-RTE no longer emits declarations for generated trusted functions in `Rte.h`.

Declarations for these trusted functions are expected to be provided in `Os_Cfg.h`, which is produced during OS generation and is included from `Rte.c`.

(RTE-7116)

Fixed incorrect shortNames of EcucModuleConfigurationValues in trustedFunctionNeeds.arxml

An issue that resulted in RTA-RTE generating incorrect shortNames for the EcucModuleConfigurationValues in trustedFunctionNeeds.arxml has been fixed.

(RTE-7115)

4.31 Version 7.0.1

Version 7.0.1pr3

Global copy declarations and mapping macros for RIPS-relevant data

An issue that resulted in RTA-RTE failing to generate global copy declarations and mapping macros for some configurations of RIPS-relevant 1:N and N:1 communications has been fixed.

(RTE-7089)

Fixed internal error 902584 for RIPS configurations

An issue that caused RTA-RTE to reject some valid AUTOSAR RIPS configurations with internal error 902584 has been fixed.

(RTE-7100)

DataReceivedEvents referencing RIPS relevant data elements

Where an implicitly written RIPS-relevant data element is referenced by a DataReceivedEvent, RTA-RTE now sets the activation flag relating to that DataReceivedEvent and activates the `OsTask` that the DataReceivedEvent is mapped to.

This activation code is located at the end of the task body that the writing runnable.

(RTE-7077)

Version 7.0.1pr2

Do not use trusted functions for cross core events

If a cross partition event activation takes place cross-core, RTA-RTE now activates the runnable directly instead of generating a trusted function and invoking the Os API `CallTrustedFunction`.

(RTE-6910)

Message 100009 could reference an incorrect RteEventToTaskMapping

In some circumstances RTA-RTE could reference an incorrect `RteEventToTaskMapping` when outputting info message 100009 to report on the existence of cross partition event activations. This bug has now been fixed.

(RTE-7037)

CDS Handle status honoured when AUTOSAR RIPS mode is enabled

RTA-RTE no longer modifies the status of RIPS-relevant data elements that are used as CDS handle types.

Note: This fixes an issue introduced in issue RTE-5923 where RTA-RTE always generated data elements with status for CDS handle types.

(RTE-7032)

Initialization of CDS data handles for RIPS relevant data elements

RTA-RTE now correctly initializes the CDS data handles for RIPS relevant data elements when there is more that one RIPS relevant data element in the CDS. This fixes an issue where all of the data handles would be incorrectly initialized with the same RIPS macro.

(RTE-7034)

Version 7.0.1pr1

Trusted Function Needs

RTA-RTE now generates the file `trustedFunctionNeeds.arxml` containing Os configuration parameters for trusted functions which have been generated to support cross partition event activations.

(RTE-6266)

Cross partition trusted function naming updated

Trusted functions generated by RTA-RTE for activation of cross partition events now have the symbol `TRUSTED_<SwcP>_<Event>`, where `<Event>` is the `shortName` of the `RteEvent` which triggered their generation and `<SwcP>` is the `shortName` of the `SwComponentPrototype` which owns the cross partition event.

If the relevant `SwComponentPrototype` is referenced by a `FlatInstanceDescriptor`, the `<SwcP>` portion of the trusted function symbol will be replaced by the `shortName` of the `FlatInstanceDescriptor`. A `SwComponentPrototype` is considered to be referenced by a FID if the `SwComponentPrototype` is pointed to by the FID's `TARGET-REF`.

(RTE-6966)

4.32 Version 7.0.0

Version 7.0.0pr9

Removed historic release note entries

Release note entries for RTA-RTE versions older than V6.0.0 have been removed from this document.

If you need the release notes from any RTA-RTE 5.x version, please contact ETAS to obtain an earlier version of the release notes document.

(RTE-6611)

Implicit shadow buffers relating to RIPS-relevant data

RTA-RTE no longer generates RTE-managed shadow buffers for implicitly-accessed data elements that are handled by a RIPS plugin.

When implicitly-accessed data is handled by RIPS, any necessary shadow buffers must be created and managed by the RIPS plugin that is assigned to the commu-

nication.communication.

(RTE-5923)

Correction of initialization of CDS handles for accesses to a mixture of RIPS and non-RIPS relevent data elements

RTA-RTE now correctly initializes the CDS handles for `dataReadAccesses` and `dataWriteAccesses` when a `SwComponentPrototype` accesses both RIPS-relevent and non-RIPS-relevent data elements.

This fixes an issue introduced in issue RTE-5922 where RTA-RTE generated invalid CDS handle initialization code.

(RTE-6850)

Use of RIPS in 1:N and N:1 implicit sender/receiver scenarios

RTA-RTE now correctly generates code when RIPS is used in 1:N and N:1 implicit sender/receiver scenarios.

This fixes a bug where not all connected ports were correctly identified as RIPS-relevant in a non-1:1 communication scenario when one of the connected ports was referenced by a RIPS FlatInstanceDescriptor.

(RTE-6851,RTE-6942)

RIPS SWC-BSW-Instance includes in application header files

RTA-RTE now includes the appropriate `RIPS SWC-BSW-Instance Header File`s in the application header files relating to all `SwComponentPrototype`s that participate in RIPS-relevant communications.

This fixes an issue where RTA-RTE would not generate the include for all SWCs that participate in a communication graph.

(RTE-6835)

Isolation of activation flags

When RTA-RTE generates a runnable activation flag `Rte_ActCount_<SWC>_<EVENT>` that guards activation of a runnable, the runnable activation flag will be assigned to the memory section `RTE_<SWC-P>_<EVENT>_VAR_CLEARED_8`, where `<SWC-P>` is the `shortName` of the `SwComponentPrototype` that the activated runnable belongs to (in upper case) and `<EVENT>` is the `shortName` of the `RteEvent` which activates the runnable (in upper case).

(RTE-6580)

Documentation of ——buffers-use-data-name

The Reference Manual and issue RTE-6082 in this document now have the correct spelling of –buffers-use-data-name and the `--help` output now describes the new option.

(RTE-6883)

Fill and flush of implicit shadow buffers relating to RIPS-relevant data

RTA-RTE no longer generates code to fill and flush RTE-managed implicit shadow buffers when an implicitly-accessed data element is handled by a RIPS plugin. This addresses the potential for data corruption caused by the flushing back of unused shadow buffers.

When implicitly-accessed data is handled by RIPS, any necessary shadow buffers

will be managed by the RIPS plugin that is assigned to the communication. Filling and flushing of these plugin-managed buffers is performed by dedicated RIPS fill and flush routines, which must be configured separately (see issue RTE-5924).

Currently, RTA-RTE will continue to generate RTE-managed shadow buffers for RIPS-relevant data elements even though these will no longer be used.

(RTE-5925)

Version 7.0.0pr8

RIPS relevant application data instances

RTA-RTE now supports RIPS relevant sender/receiver data elements that are typed by complex `ApplicationDataTypes`.

This fixes an issue where RTA-RTE would exit with internal error 902584 in this situation.

(RTE-6833)

Rte_Call for RIPS relevant intra-ECU asynchronous client/server communication

For RIPS relevant intra-ECU client/server communication where the client aggregates an `AsynchronousServerCallPoint`, the Rte_Call API generated for that server call point will now call the `Rte_Rips_Invoke` API and return the result as the return of the Rte_Call API.

Although AUTOSAR only requires a RIPS plugin to provide the `Rte_Rips_Invoke` service for client/server communications that use data transformation (see `[SWS_Rte_70063]` in AUTOSAR 4.4.0), RTA-RTE allows RIPS to be used with any client/server communication. When RIPS is used, all aspects of the client/server communication are expected to be handled by the RIPS plugin.

The use of RIPS has not been tested in the case where the client software component is an instance of an `NvBlockSwComponentType`.

(RTE-6494)

Remove template references from generated files

Files generated from standard templates will no longer reference the template file that was used to generate them.

(RTE-6351)

Fix the duplication of errors, warnings and infos

RTA-RTE no longer raises duplicate Fatal, Error, Warning or Info messages.

Messages are considered duplicates when they have the same message ID, location and message text.

(RTE-6654)

Version 7.0.0pr7

Activation flags for cross partition events

RTA-RTE now resets activation flags associated with cross partition event activations in the trusted function that activates the associated runnable, and not in the body of the task that the event is mapped to.

This means that for cross-partition events, the associated activation flags are

written in the context of the activated runnable. If your system uses a memory protection scheme, you must ensure that the activation flags are allocated to a memory section that is writable from the appropriate context(s).

(RTE-6253)

Generate RIPS SWC Instance includes in the application header files

When a `SwComponentPrototype` has implicit access to a RIPS-relevant data element, the generated application header file of the corresponding `SwComponentType` now includes the applicable `RIPS SWC-BSW-Instance Header File`.

This change supports requirement `[SWS_Rte_80027]` in AUTOSAR 4.4.0.

(RTE-5906)

Fixed bug invoking RIPS Fill and Flush routines

RTA-RTE now correctly invokes RIPS Fill and Flush routines if they are configured in an event-to-task mapping.

This fixes a bug (introduced in RTE-5924) where RIPS Fill and Flush routines were not invoked for some valid AUTOSAR configurations.

(RTE-6719)

Fix internal error in queued 1:N communication scenarios

RTA-RTE no longer raises internal error 900634 when a queued data element participates in 1:N communication, and one of the destinations is not an RTE-managed buffer (such as a connection to `COM`).

(RTE-6728)

Client/server communication with RIPS

RTA-RTE now raises info message 100012 for each RIPS FlatInstanceDescriptor that references a client/server communication graph.

(RTE-6491)

Generate trusted functions for cross partition events

RTA-RTE now outputs a trusted function named `TRUSTED_<RUNNABLE>` for each cross partition event in the input config (where `<RUNNABLE>` is the `Symbol` of the activated `RunnableEntity`). These trusted functions contain a call to the `RunnableEntity` which is activated by the event.

Definitions for trusted functions are created in `Rte.c` and declarations are created in Rte.h. Both definitions and declarations of trusted functions have a MAKW `INFIX` based on the `shortName` of the `EcucPartition` that the event's parent SWC is mapped to.

`Rte_BSWMD.arxml` will also contain a `BswCalledEntity` and a `BswModuleEntry` (with the `BswCallType` of `CALLBACK`) for each generated trusted function.

(RTE-6265)

RIPS global copy type definitions

When a communication graph is referenced by a RIPS FlatInstanceDescriptor, the generated Rte_DataHandleType.h file now contains a `typedef` defining `Rte_Rips_GlobalCopy_<FID>_Type` to be the type of the global buffer associated with that communication graph (where `<FID>` is the `shortName` of the referencing

RIPS FlatInstanceDescriptor).

This change supports requirement [SWS_Rte_80079] in AUTOSAR 4.4.0.

(RTE-5904)

Support for Rte_Rips_<plugin>_Rte_Stop

When RIPS is enabled and the configuration contains RIPS plugins, RTA-RTE now generates a call to each plugin's Rte_Rips_<plugin>_Rte_Stop function in the Rte_Stop API.

When the configuration contains multiple RIPS plugins, the order of calls to the Rte_Rips_<plugin>_Rte_Stop functions is determined by the index values provided in the RteRipsPluginConfigurationRef.

(RTE-5894)

Support for Rte_Rips_<plugin>_Rte_Start

When RIPS is enabled and the configuration contains RIPS plugins, RTA-RTE now generates a call to each plugin's Rte_Rips_<plugin>_Rte_Start function in the Rte_Start API.

When the configuration contains multiple RIPS plugins, the order of calls to the Rte_Rips_<plugin>_Rte_Start functions is determined by the index values provided in the RteRipsPluginConfigurationRef.

(RTE-5893)

Version 7.0.0pr6

Return type of Rte_Feedback API declarations in contract phase headers

Declarations of Rte_Feedback APIs in contract phase header files are now generated with the correct return type (Std_ReturnType).

(RTE-6702)

Implicit write access to RIPS data elements

When RIPS is enabled and a runnable has implicit write access to a data element that is referenced by a RIPS FlatInstanceDescriptor, RTA-RTE implements the Rte_IWrite and Rte_IWriteRef APIs that access that data element using the RIPS API provided by the associated RIPS plugin.

If the data element is primitive, Rte_IWrite will be implemented using Rte_Rips_<plugin>_IWrite. For complex data elements Rte_Rips_<plugin>_IWBufferRef will be used.

The Rte_IWriteRef API is implemented using Rte_Rips_<plugin>_IWBufferRef regardless of the data element type.

(RTE-5927)

Initialization of CDS handles for RIPS data elements

When RIPS is enabled, and a SWC is delivered as object code and has implicit access to a data element that is referenced by a RIPS FlatInstanceDescriptor, RTA-RTE initializes the CDS data handles associated with that data element using the RIPS API provided by the associated RIPS plugin.

CDS data handles associated with dataReadAccesses are initialized using Rte_Rips_<plugin>_IRBufferRef, while those associated with dataWriteAccesses

use `Rte_Rips_<plugin>_IWBufferRef`.

(RTE-5922)

Implicit read access to RIPS data elements

When RIPS is enabled and a runnable has implicit read access to a data element that is referenced by a RIPS FlatInstanceDescriptor, RTA-RTE implements the `Rte_IRead` APIs that access that data element using the RIPS API provided by the associated RIPS plugin.

If the data element is primitive, `Rte_IRead` will be implemented using `Rte_Rips_<plugin>_IRead`. For complex data, `Rte_Rips_<plugin>_IRBufferRef` will be used.

(RTE-5926)

Invocation of RIPS fill and flush routines

RTA-RTE now supports the use of RIPS-provided fill and flush routines for `RteEvent` instances that are mapped to `OsTask`s.

RIPS fill routines, configured via an `RteRipsFillRoutineRef` from an `RteEventToTaskMapping`, will be invoked in the task body immediately before the `RunnableEntity` that is associated with the mapped `RteEvent`. RIPS flush routines, configured via an `RteRipsFlushRoutineRef`, will be invoked immediately after the `RunnableEntity`.

Currently, only one fill routine and one flush routine are supported for any given `RteEvent` instance.

(RTE-5924)

Conflicting initial values for NV data

RTA-RTE no longer checks the consistency of `initValue`s across the set of required NV data elements that are mapped to the same `NvRamBlock`.

Any `initValue` supplied for an NV data element is only relevant when its port is unconnected. When a data element is in a connected port (and so ultimately mapped to an `NvRamBlock`) its value will be read from NVRAM during the ECU startup. Any `initValue` supplied for a mapped NV data element will never be used, and therefore it does not matter whether it is consistent with any other `initValue` supplied for that NV data element.

Previously, RTA-RTE raised error 5030 when it detected conflicting initial values, as a result of this change this error is no longer raised for NV data elements.

(RTE-6639)

Error 7024 (Unconnected DSMQ RPort) is downgraded to an info by ——strict-unconnected-rport-check=off

RTA-RTE will now raise message 7024 (Unconnected RPort in a Distributed Shared Mode Queue) as an info instead of an error when –strict-unconnected-rport-check=off is set.

(RTE-6600)

RTA-RTE ships with RTA-A2L tool

RTA-A2L is now shipped with RTA-RTE and is installed into the folder `Tools/RTA-A2L`.

RTA-A2L converts AUTOSAR McSupportData into ASAM A2L data for use with measurement and calibration tools such as ETAS INCA. A User Guide is included that describes its features and operation in detail.

The use of RTA-A2L requires an additional license. Please contact ETAS for further information.

(RTE-6364)

Primitive data conversion for ParameterSwComponents

RTA-RTE now supports data conversion for ParameterAccess reads from ParameterSwComponents using primitive data types with CompuMethods of category LINEAR or TEXTTABLE.

(RTE-3440)

Version 7.0.0pr5

Declaration of resources in Rte_Intl.h

RTA-RTE no longer outputs OS resource declarations in the generated Rte_Intl.h file. This avoids conflicts with resource declarations that are contained in the OS header file.

(RTE-6630)

Use CallTrustedFunction to activate cross partition events

When RTA-RTE detects a cross partition event it will now output a call to the OS API CallTrustedFunction instead of directly calling the activated runnable.

This call references a trusted function named TRUSTED_<RUNNABLE>, where <RUNNABLE> will be the Symbol of the activated RunnableEntity.

Currently, RTA-RTE does not automatically generate the implementation of the referenced trusted function.

(RTE-6252)

Remove container ID and text from output messages

RTA-RTE now presents message identifiers in the format <S><NNNNNN>, where <S> is the severity of the message (F for fatal, E for error, W for warning and I for info), and <NNNNNN> is the message ID, zero-padded to six digits.

As messages can be identified using the message ID alone, message container IDs and descriptions are no longer presented when a message is raised. If your toolchain includes any processing of the messages produced by RTA-RTE, you may need to update this to accept the new message identifier format.

Command line options will accept message identifiers in both old and new formats (see issue RTE-4411), so existing command files do not need to be updated as a result of this change.

(RTE-3876)

Extension of Rte_Call and Rte_Result with transformerError parameter

RTA-RTE now extends Rte_Call and Rte_Result APIs with a transformerError parameter when the associated client port is referenced by a PortAPIOption with errorHandling set to transformerErrorHandling.

As RTA-RTE does not currently support transformation for client/server APIs, the

implementation of an extended Rte_Call or Rte_Result API always indicates that no transformer error occurred during the execution of that API, by setting `transformerError.errorCode` to `0x00` and `transformerError.transformerClass` to `RTE_TRANSFORMER_UNSPECIFIED`.

(RTE-6492)

Allow cross partition event activations

RTA-RTE now supports cross partition events if RIPS is enabled and the new command line option `--allow-cross-partition-event-mapping` is used.

A cross partition event is any `RTEEvent` mapped to an `OSTask` which is mapped to a different `EcucPartition` than the SWC instance which contains the event.

(RTE-6251)

Arrays of resources defined using OS_NUM_APPLICATIONS

RTA-RTE now generates the arrays `Rte_Resources` and `Rte_ResourceCount` with size `OS_NUM_APPLICATIONS + 1` using OS defined macro OS_NUM_APPLICATIONS.

This fixes a bug where RTA-RTE might declare the arrays with too few elements, then write outside the array bounds in `Rte_Start`.

(RTE-6450)

Version 7.0.0pr4

Messages ID formats for command line options

RTA-RTE now accepts more formats when providing message IDs to the `--disable-warning` and `--error-as-warning` command line options.

See the Reference Manual for further details on the available formats.

(RTE-4411)

Memory allocation of Rte_CalprmInitROM

RTA-RTE now generates the Rte_CalprmInitROM variable surrounded by the memory allocation key words `RTE_CALPRMINIT_START_SEC_CONST_32` and `RTE_CALPRMINIT_STOP_SEC_CONST_32`.

RTA-RTE now also outputs a `SectionNamePrefix` named `RTE_CALPRMINIT` and a `MemorySection` named `RTE_CALPRMINIT_CONST_32`.

(RTE-6250)

Option to replace buffer numbers with DataPrototype name

Now when –buffers-use-data-name is set RTA-RTE replaces the buffer number in buffer names with the `shortname` of the DataPrototype.

Note: this applies to buffer names, measurement variable names and IOC channel names.

(RTE-6082)

Remove all lint directives from the generated RTE code

RTA-RTE generated code no longer contains lint directives so that no MISRA violation is suppressed by default.

If you use `lint` as part of your static analysis then this change may result in an increase in static analysis findings. These findings should be managed as part of your MISRA case.

(RTE-5855)

Removed error 6122

RTA-RTE no longer raises error 6122 when calibration parameters generated due to RPorts typed by the same `ParameterInterface` are not referenced by a `FlatInstanceDesciptor`. Instead RTA-RTE generates a unique symbol name for each generated calibration parameter.

(RTE-5957)

RtePositionInTask can now be set to zero

RTA-RTE now allows `0` to be used as a value for RtePositionInTask.

As a result of this change, error 1296 (indicating a missing RtePositionInTask) is no longer raised when an RtePositionInTask with value `0` is supplied.

(RTE-4894)

Removed ––operating-system option

The –operating-system option and its short form `--os` have been removed.

If either the –operating-system or `--os` options are provided, RTA-RTE will exit with error 1103.

(RTE-5801)

Remove support for contract phase licenses

Licenses that only allow the use of RTA-RTE in contract phase are no longer supported.

As a result of this change, RTA-RTE requires the use of a full license in order to generate the contract for an RTE. Please contact ETAS for a new license.

(RTE-6066)

Version 7.0.0pr3

New license feature for RIPS mode

RTA-RTE now has a new license feature named `RTA-RIPS` to enable the use of RIPS mode.

As a result of this change, RTA-RTE requires an `RTA-RIPS` license to generate an RTE that requires the use of RIPS mode. Please contact ETAS for a new license.

(RTE-5907)

Declaration of RIPS-relevant global buffers in Rte_Buffers.h

When a communication graph is referenced by a RIPS FlatInstanceDescriptor, the generated Rte_Buffers.h file now contains an `extern` declaration of the global buffer associated with that communication graph.

This change supports requirement `[SWS_Rte_80005]` in AUTOSAR 4.4.0.

(RTE-5905)

Define AUTOSAR RIPS buffer mappings in Rte_Buffers.h

When a communication graph is referenced by a RIPS FlatInstanceDescriptor, the generated Rte_Buffers.h file now defines a mapping between the name of the global buffer associated with that communication graph and `Rte_Rips_GlobalCopy_<FID>` (where `<FID>` is the `shortName` of the referencing RIPS

FlatInstanceDescriptor).

This change supports requirement `[SWS_Rte_80006]` in AUTOSAR 4.4.0.

(RTE-5903)

Change to license feature name

RTA-RTE's license feature name has changed from `LD_RTA-RTE4.0` to `RTA-RTE`.

As a result of this change, existing electronic licenses for RTA-RTE will no longer work. Please contact ETAS for an updated license.

(RTE-5888)

Corrected floating-point literal with zero fractional part

Initial values of floating-point data are now given a trailing `.0` when the initial value is a whole number.

Formerly, RTA-RTE would truncate at the decimal point, resulting in a C integer literal, causing compilation issues.

(RTE-3659)

Remove support for BAS_RTA_RTE license feature

RTA-RTE no longer supports the BAS_RTA_RTE license feature, which was formerly used for the evaluation of RTA-RTE.

(RTE-6065)

Version 7.0.0pr2

Remove support for grace-mode licenses

RTA-RTE no longer supports licenses in grace mode.

(RTE-5129)

Initial values for global receive buffers with TEXTTABLE CompuMethods

Where a data element's initial value is attached to a `PPortPrototype`, RTA-RTE now converts the supplied value to the receiver's representation when initalizing the global receive buffer for that data element.

Previously the receive buffer was initialized using the provider's representation.

(RTE-6344)

Fixed bug introduced in RTE-5934

Fixed a bug where error 1356 was not raised if a `BswModuleDescription` used multiple exclusive areas and the option `--deviate-bsw-any-partition` was enabled. Error 1356 is now always raised if a `SwComponentType` has multiple `ExclusiveArea`s or a `BswModuleDescription` has multiple `ExclusiveArea`s and the option `--deviate-bsw-any-partition` is disabled.

(RTE-6401)

New warning 100007

RTA-RTE now raises warning message 100007 when the input configuration contains one or more RIPS FlatInstanceDescriptors but RIPS mode is not enabled.

(RTE-6378)

Data handles are not modified when AUTOSAR RIPS mode is enabled

RTA-RTE now does not modify data handle types for any 'data element without

status' in the 'Component Data Structure' (whether or not the data element is RIPS relevant) when AUTOSAR RIPS is enabled.

Note: This does not apply to RIPS relevant data elements on SWCs implemented with source delivery.

(RTE-5901)

New warning 100006

RTA-RTE now raises warning message 100006 when the input configuration contains one or more RTE Implementation Plug-Ins but RIPS mode is not enabled.

(RTE-6384)

Include all RIPS Plug-In Header Files in Rte.h.

When RIPS mode is enabled, RTA-RTE now `#include`s the RIPS Plug-In Header File for each configured RIPS plugin in the generated Rte.h file.

(RTE-5867)

Raise error when RIPS mode is enabled and no RIPS FlatInstanceDescriptors

RTA-RTE now exits with error 100005 when the input configuration has RIPS mode enabled but does not contain any RIPS FlatInstanceDescriptors.

(RTE-5884)

Include all RIPS Buffer Declaration Files in Rte_Buffers.h.

When RIPS mode is enabled, RTA-RTE now `#include`s the RIPS Buffer Declaration File for each configured RIPS plugin in the generated Rte_Buffers.h file.

(RTE-5866)

Raise error when configuration contains both AUTOSAR and legacy RIPS FlatInstanceDescriptors

RTA-RTE now exits with error 100004 when the input configuration contains both AUTOSAR RIPS FlatInstanceDescriptors and legacy RIPS FlatInstanceDescriptors.

(RTE-5883)

Version 7.0.0pr1

Include the Rte_Type.h file when RIPS mode is enabled.

RTA-RTE now `#include`s Rte_Type.h in the generated Rte_DataHandleType.h file when RIPS mode is enabled.

(RTE-5863)

Raise an info message for each RTE Implementation Plug-In.

RTA-RTE now raises info message 100003 for each RIPS plugin in the input configuration.

A RIPS plugin in configured with an `RteRipsPluginConfigurationRef` in the `RteRips` container.

(RTE-5865)

Configurations containing AUTOSAR-conformant RIPS FlatInstanceDescriptors

RTA-RTE now raises info message 100002 when the input configuration contains one or more RIPS FlatInstanceDescriptors.

In line with the RIPS specification introduced in AUTOSAR version 4.4.0, a `RIPS FlatInstanceDescriptor` is denoted by the aggregation of an `RtePluginProps` by a FlatInstanceDescriptor.

(RTE-5882)

Include the Rte_DataHandleType.h file when RIPS mode is enabled.

RTA-RTE now `#include`s Rte_DataHandleType.h in the generated Rte_Buffers.h file when RIPS mode is enabled.

(RTE-5862)

Include the Rte_Buffers.h file when RIPS mode is enabled.

RTA-RTE now `#include`s Rte_Buffers.h in the generated `Rte.c` file when RIPS mode is enabled.

(RTE-5861)

Preview Licensing

Preview releases now ship with a license valid for six weeks from the day the preview was built.

(RTE-6285)

Create the Rte_Buffers.h file when RIPS mode is enabled.

RTA-RTE now creates the file Rte_Buffers.h in the generated output when RIPS mode is enabled.

(RTE-5860)

Raise an info message when RIPS mode is enabled.

RTA-RTE now raises info message 100001 when RIPS mode is enabled.

RIPS mode is enabled when the configuration contains contains an EcuC module named `Rte`, which contains an `RteRips` container, which contains an `RteRipsSupport` parameter with the value `RTE_RIPS_ON`.

(RTE-5859)

4.33 Version 6.8.0

Version 6.8.0pr9

Removed duplication of ExclusiveArea protection for direct server calls

RTA-RTE no longer duplicates the calls to `Rte_SuspendOSInterrupts()` and `Rte_ResumeOSInterrupts()` when a server runs in multiple ExclusiveAreas and is called via a direct server call.

(RTE-6110)

Support for bi-directional AssemblySwConnectors with NV PRPorts

Further to issue RTE-5430 and issue RTE-5431, RTA-RTE now treats AssemblySwConnectors as bi-directional when they connect two `PRPortPrototype`s typed by `NvDataInterface`s.

As a result, RTA-RTE now generates the expected code when an AssemblySwConnector references a PRPort on an `NvBlockSwComponentType` as its `provider`.

(RTE-6246, RTE-6283)

Version 6.8.0pr8

Generation of McSupportData for ArTypedPerInstanceMemorys in BSW modules

RTA-RTE now supports the generation of McSupportData for ArTypedPerInstanceMemorys aggregated by BSW modules, when the per-instance memory is configured with a `SwCalibrationAccess` other than `NOT-ACCESSIBLE` and measurement is enabled.

(RTE-6262)

Explicit reading and writing of NV data from different runnables in an application SWC via a PRPort

RTA-RTE now supports the use of PRPorts for explicit reading and writing of NV data, in the case where one runnable has explicit write access to an NV data element via a `PRPortPrototype` typed by an `NvDataInterface`, and another runnable in the same SWC has explicit read access by arguments to the same NV data element via the same `PRPortPrototype`.

The `PRPortPrototype` is connected via an AssemblySwConnector to a `PRPortPrototype` aggregated by an `NvBlockSwComponentType`. This `NvBlockSwComponentType` maps the NV data element to a RAM block using an `NvBlockDataMapping` that references the NV data element in the role `writtenReadNvData`.

Multiply-instantiable SWCs are not currently supported with PRPorts (see issue RTE-3948). The use of multiple reading or writing runnables in the application SWC is untested at this time.

> **WARNING**
>
> At present, RTA-RTE does not fully support the bi-directionality of `AssemblySwConnectors` when two PRPorts are connected. The `PRPortPrototype` on the `ApplicationSwComponentType` must be referenced as the AssemblySwConnector . `provider`. Incomplete code will be generated if the AssemblySwConnector references the PRPort on an `NvBlockSwComponentType` as the `provider`.

(RTE-5431)

Do not error when multiple clients mapped to different OsApplications reference the same operation in port

RTA-RTE previously raised error 1361 when it found multiple RunnableEntitys executing in different OsApplications each with `SynchronousServerCallPoints` to the same `ClientServerOperation` in `Port`. This error is no longer raised if both `--client-server-global-optimization` and `--deviate-split-swci-support` are enabled.

(RTE-6284)

Error 3007 removed

RTA-RTE no longer raises error 3007 when a `PRPortPrototype` is referenced as

`requester` from an AssemblySwConnector.

Error 3063 will be raised instead, unless the `PRPortPrototype` is typed by a `NvDataInterface` (see issue RTE-5422).

(RTE-6212)

Warning instead of error in contract phase when RunnableEntity with implicit access has no activating event

In contract phase, RTA-RTE now raises warning 3066 instead of error 7223 when there is a RunnableEntity with implicit data access that is not activated by an RteEvent.

Raising a warning allows the RTA-RTE to generate contract phase headers when a RunnableEntity with implicit access is expected to be synchronized with a `BswExecutableEntity` and activated by a `BswEvent`.

(RTE-6092)

Error 7024 (Unconnected DSMQ R-Port) is downgraded to a warning by ––strict-unconnected-rport-check=warn

RTA-RTE will now raise message 7024 (Unconnected RPort in a Distributed Shared Mode Queue) as a warning instead of an error when –strict-unconnected-rport-check=warn is set.

(RTE-5572)

Fix to generated IRV-related API names

Previously RTA-RTE generated IRV-related APIs with an infix of `<undefined>` when the IRV was associated with a RunnableEntity that was synchronized with a `BswSchedulableEntity`.

Now, RTA-RTE will use the associated OsApplication.shortName of the synchronized `BswSchedulableEntity` in the infix of the generated API.

(RTE-6130)

Warning when an event instance is not referenced by an EventToTaskMapping

RTA-RTE now raises warning 3064 when an instance of an `AbstractEvent` is not referenced by an EventToTaskMapping.

The purpose of this warning is to indicate where EventToTaskMappings may be missing from the configuration. If an event is intentionally unmapped, the warning can be avoided by including an EventToTaskMapping for that event that does not reference an `OsTask`.

(RTE-6074)

Error when AssemblySwConnector.requester is a PRPort and a non-NV port is referenced as the AssemblySwConnector.provider

RTA-RTE now exits with error 3065 when the `PortPrototype` referenced as the `requester` by an AssemblySwConnector is a `PRPortPrototype` and the `PortPrototype` referenced as the `provider` is not typed by an `NvDataInterface`.

(RTE-5423)

Implicit reading and writing of NV data from a single runnable in an application SWC via a PRPort

RTA-RTE now supports the use of PRPorts for implicit reading and writing of NV data, in the case where a runnable has `DataReadAccess` and `DataWriteAccess` to an NV data element via a `PRPortPrototype` typed by an `NvDataInterface`.

The `PRPortPrototype` is connected via an AssemblySwConnector to a `PRPortPrototype` aggregated by an `NvBlockSwComponentType`. This `NvBlockSwComponentType` maps the NV data element to a RAM block using an `NvBlockDataMapping` that references the NV data element in the role `writtenReadNvData`

Multiply-instantiable SWCs are not currently supported with PRPorts (see issue RTE-3948). The use of multiple reading or writing runnables in the application SWC is untested at this time.

WARNING

At present, RTA-RTE does not fully support the bi-directionality of `AssemblySwConnectors` when two PRPorts are connected. The `PRPortPrototype` on the `ApplicationSwComponentType` must be referenced as the AssemblySwConnector . `provider`. Incomplete code will be generated if the AssemblySwConnector references the PRPort on an `NvBlockSwComponentType` as the `provider`.

(RTE-5430)

Error when non-NV PRPort referenced as AssemblySwConnector.requester

RTA-RTE now exits with error 3063 when an AssemblySwConnector references a `PRPortPrototype` in the role `requester`, and that `PRPortPrototype` is not typed by an `NvDataInterface`.

(RTE-5422)

Conversion of initial values used to initialize global receive buffers

Where a data element's initial value is attached to a `PPortPrototype`, RTA-RTE now converts the supplied value to the receiver's representation when initalizing the global receive buffer for that data element.

Previously RTA-RTE did not convert the value and the receive buffer was initialized incorrectly using the provider's representation.

Conversion will only be performed when the `compuMethods` involved in the conversion are `LINEAR`.

(RTE-5463)

Version 6.8.0pr7

Implicit reading and writing of NV data from different runnables in an application SWC via a PRPort

RTA-RTE now supports the use of PRPorts for implicit reading and writing of NV data, in the case where one runnable has `DataWriteAccess` to an NV data element via a `PRPortPrototype` typed by an `NvDataInterface`, and another runnable in the same SWC has `DataReadAccess` to the same NV data element via the same `PRPortPrototype`.

The PRPortPrototype is connected via an AssemblySwConnector to a PRPortPrototype aggregated by an NvBlockSwComponentType. This NvBlockSwComponentType maps the NV data element to a RAM block using an NvBlockDataMapping that references the NV data element in the role writtenReadNvData

Multiply-instantiable SWCs are not currently supported with PRPorts (see issue RTE-3948). The use of multiple reading or writing runnables in the application SWC is untested at this time.

(RTE-5429)

Support for arTypedPerInstanceMemorys aggregated by BswModuleDescriptions

RTA-RTE now supports the generation of variables and SchM_Pim APIs for arTypedPerInstanceMemorys aggregated by BswInternalBehaviors. These APIs are implemented as macros in the parent BSW modules' module interlink header files.

The variables for storing the arTypedPerInstanceMemorys are wrapped with memory allocation keywords of the form RTE_<BSWMD>_START_SEC_VAR_CLEARED_<ALIGNMENT> and RTE_<BSWMD>_STOP_SEC_VAR_CLEARED_<ALIGNMENT> where:

- <BSWMD> is the shortName of the BswModuleDescription aggregating the BswInternalBehavior
- <ALIGNMENT> is the alignment appropriate for the type of the VariableDataPrototype that types the arPerInstanceMemory

(RTE-5999)

canBeInvokedConcurrently is not checked if RIPS mode is enabled

When RIPS mode is enabled, RTA-RTE will not raise error 2024 for any runnable entities that are identified as being concurrently invokable, including runnables that do not have canBeInvokedConcurrently set to true.

(RTE-5677)

Inter Partition Client Server Direct Function Call

RTA-RTE now supports direct function calls of server runnables that are not mapped to tasks and where the invoking client is executing in a different OsApplication. This behavior is enabled when the --client-server-global-optimization option is enabled.

In the generated RTE, the server runnable will be called directly from the Rte_Call API, resulting in the server runnable being executed in the context of the invoking client. This may result in an MPU trap if memory protection is enabled.

(RTE-6077)

——terminate-background-tasks documented in Reference Manual

Documentation for the –terminate-background-tasks command-line option has been added to the RTA-RTE Reference Manual.

(RTE-5998)

——xfrm-ignore-inplace command-line option

The –xfrm-ignore-inplace command-line option allows RTA-RTE to ignore the requirement for in-place transformation.

This option does not provide full support for in-place transformation. When supplied, RTA-RTE will not raise error 2226 when a configuration contains in-place transformation, but the generated RTE code will continue to use out-of-place buffering and will call transformer functions using the out-of-place API format.

When this option is used, RTA-RTE will raise warning 3062 for each in-place transformer in the configuration.

(RTE-5880)

Version 6.8.0pr6

Rte_IrvIWrite, Rte_IrvIWriteRef, and Rte_IrvIRead APIs for RunnableEntities synchronized with BswSchedulableEntities

Further to issue RTE-5752, issue RTE-5754, issue RTE-5755, and issue RTE-5757, RTA-RTE supports the IRV implicit APIs Rte_IrvIWrite, Rte_IrvIWriteRef, and Rte_IrvIRead when a synchronized RunnableEntity requires them.

(RTE-5758)

Support of enableTakeAddress for Rte_IRead, Rte_IWrite and Rte_IWriteRef

RTA-RTE now supports generating Implicit Sender/Receiver APIs Rte_IRead, Rte_IWrite and Rte_IWriteRef as functions rather than macros when the enableTakeAddress attribute of their `PortAPIOption` is `true`.

(RTE-5354)

Definition of OSENV in contract phase

RTA-RTE now emits a definition of OSENV in the file `Rte_Const.h` when running in contract phase. The Os evironment that is defined in `Rte_Const.h` is controlled with the `--os-define-osenv` command line option.

Note: The two accepted values for `--os-define-osenv` are `RTAOS40` and `UNSUPPORTED`. If RTA-RTE is run with any other value for `--os-define-osenv`, it will exit with error 1136.

As part of this, RTA-RTE will no longer raise error 1112. Instead, RTA-RTE will raise error 1136 for any command line option that is provided with an invalid value.

(RTE-5796)

Version 6.8.0pr5

Disambiguation of SwBaseType shortNames in the generated Rte_BSWMD.arxml

RTA-RTE now disambiguates the shortNames of the SwBaseTypes emitted in the generated Rte_BSWMD.arxml file by using the full ARXML path of the corresponding type in the input configuration as the shortName of the SwBaseType, with the leading forward slash removed and all remaining forward slashes replaced with underscores.

(RTE-6028)

Disambiguation of SwBaseType shortNames in the generated Rte_McSupportData.arxml

RTA-RTE now disambiguates the shortNames of the SwBaseTypes emitted in the generated Rte_McSupportData.arxml file by using the full ARXML path of the corresponding type in the input configuration as the shortName of the SwBaseType, with the leading forward slash removed and all remaining forward slashes replaced with underscores.

(RTE-5614)

Rte_IFeedback API for RunnableEntities synchronized with BswSchedulableEntities

Further to issue RTE-5752, issue RTE-5754, and issue RTE-5755, RTA-RTE supports the implicit API Rte_IFeedback when a synchronized RunnableEntity requires it.

(RTE-5757)

Rte_IInvalidate and Rte_IStatus APIs for RunnableEntities synchronized with BswSchedulableEntities

Further to issue RTE-5752 and issue RTE-5754, RTA-RTE supports the implicit APIs Rte_IInvalidate and Rte_IStatus when a synchronized RunnableEntity requires them.

(RTE-5755)

RTA-RTE exits with error 3056 when an RteEvent references a RunnableEntity in a different SwcInternalBehavior

RTA-RTE now rejects input configurations with error 3056 when an RteEvent references a RunnableEntity that is aggregated by a SwcInternalBehavior which is not the same as the one which aggregates the RteEvent.

(RTE-5979)

Version 6.8.0pr4

BSW exclusive area access when ——deviate-bsw-any-partition enabled

RTA-RTE no longer raises error 1356 if a BSW module has access to an ExclusiveArea and BSW modules are mapped to multiple partitions using the –deviate-bsw-any-partition option.

(RTE-5934)

Internal error 901350 no longer raised in contract phase

RTA-RTE no longer raises internal error 901350 when the `--implicit-use-global-buffers` option is supplied in contract phase.

(RTE-5798)

Rte_IRead API for RunnableEntities synchronized with BswSchedulableEntities

In the case where a RunnableEntity with a `DataReadAccess` is synchronized with a `BswSchedulableEntity` via a `SwcBswRunnableMapping` and only the `BswSchedulableEntity` is referenced by an event, RTA-RTE supports implicit reads with the Rte_IRead API for the RunnableEntity.

(RTE-5754)

Fill implicit write buffers (inter-ECU cases)

RTA-RTE now fills implicit buffers for data elements that are only written (and not

read) from a task, when those data elements participate in inter-ECU communication.

In the case where the ECU contains an OsApplication an OsApplication-specific buffer is now created to store the written data element in the context of the transmitting OsApplication. The value stored in this buffer is then used to fill the implicit buffers for the data element at the start of a task body and before the execution of the writing runnable, in accordance with AUTOSAR requirement `[SWS_Rte_08418]`.

In the case where the ECU does not contain an OsApplication a global buffer is now created to store the written data element. The value stored in this buffer is then used to fill the implicit buffers for the data element at the start of a task body and before the execution of the writing runnable, in accordance with AUTOSAR requirement `[SWS_Rte_08418]`.

(RTE-5742)

Rte_IWrite and Rte_IWriteRef APIs for RunnableEntities synchronized with BswSchedulableEntities

In the case where a RunnableEntity with a `DataWriteAccess` is synchronized with a `BswSchedulableEntity` via a `SwcBswRunnableMapping` and only the `BswSchedulableEntity` is referenced by an event, RTA-RTE supports implicit writes with the Rte_IWrite and Rte_IWriteRef APIs for the RunnableEntity.

(RTE-5752)

Version 6.8.0pr3

Fill implicit write buffers (inter-OsApplication cases)

RTA-RTE now fills implicit buffers for data elements that are only written (and not read) from a task, when those data elements participate in inter-OsApplication communication.

An OsApplication-specific buffer is now created to store the written data element in the context of the transmitting OsApplication. The value stored in this buffer is then used to fill the implicit buffers for the data element at the start of a task body and before the execution of the writing runnable, in accordance with AUTOSAR requirement `[SWS_Rte_08418]`.

When the transmitted data element is written from multiple OsApplications, each has its own OsApplication-specific buffer, which is only updated as a result of writes from its owning OsApplication.

(RTE-5741)

Deprecated option ––operating-system

The –operating-system option has been deprecated, as well as its abbreviation `--os`. RTA-RTE now raises warning 2976 if –operating-system, or its abbreviated version, `--os`, is used.

The –operating-system option should not be used for new projects, and where possible existing projects should be modified to no longer use it.

(RTE-5792)

Removed warning message for warning 3054

Now that Known Issue RTE-3915 has been addressed by issue RTE-5286,

RTA-RTE no longer raises warning 3054 when the `--implicit-use-global-buffers` option is supplied.

(RTE-5795)

Removed all values for ——operating-system option other that autosar40

The –operating-system option and its short form `--os` no longer accept values other that autosar40.

If either the –operating-system or `--os` options are provided with values other than autosar40, RTA-RTE will exit with error 1136.

(RTE-5791)

Version 6.8.0pr2

Prevent optimisation of implicit accesses to direct global buffer accesses in the presence of explicit accesses

When the `--implicit-use-global-buffers` option is enabled, RTA-RTE no longer optimizes implicit accesses using the global buffer for any data element that is also accessed explicitly.

This fixes the known potential source of data corruption or data inconsistency previously described in Known Issue RTE-3915.

(RTE-5286)

Fill implicit buffers for data elements that are written from a task

RTA-RTE now fills implicit buffers for data elements that are written (and not read) from a task.

The implicit buffer for a data element is filled with the contents of that data element's global buffer at the start of a task body and before the execution of the writing runnable, in accordance with AUTOSAR requirement `[SWS_Rte_08418]`.

At present, the filling of implicit buffers for written data elements is not supported for inter-ECU or inter-OsApplication communications.

(RTE-5735)

Improved error message for error 5030

RTA-RTE now reports the name of the generated variable for which there are conflicting initial values when raising error 5030.

(RTE-4778)

4.34 Version 6.7.1

Version 6.7.1

Memory sections for runnable activation flags

When an input configuration contains a runnable started by a `DataReceivedEvent` that is mapped to the same `OsTask` as a runnable started by a `TimingEvent`, and the option `--force-basic-tasks` is enabled, the runnable activation flag for the runnable started by the `DataReceivedEvent` will be assigned to the memory section `RTE_<PARTITION>_VAR_CLEARED_8`. `<PARTITION>` is the `shortName` of the `EcucPartition` that the activated runnable is mapped to, in all uppercase.

(RTE-5730)

Reporting of OsResource and OsResource memory sections in the BSWMD

RTA-RTE now reports the memory section used to isolate the arrays of the `OsResources` and the `ResourceCounters`, and the memory sections used to isolate the `OsApplication` specific `ResourceCounters` in the BSWMD.

(RTE-5727)

Removed duplicate assignments to runnable activation counter in N:1 IOC

RTA-RTE no longer outputs duplicate runnable activation counter assignments when activating runnables via N:1 IOC callback functions.

See issue RTE-5663 for details on N:1 IOC runnable activation.

(RTE-5708)

4.35 Version 6.7.0

Version 6.7.0pr9

IOCReceiverPullCB for split SWC scenarios

Further to issue RTE-4770, activation of receivers by means of the IOCReceiver-PullCB is extended to scenarios where the communication receiver is on a split SWC.

(RTE-5664)

IOCReceiverPullCB for N to 1 cross-partition runnable activations

Further to issue RTE-4770, activation of receivers by means of the IOCReceiver-PullCB is extended to scenarios involving multiple senders and a single receiver where none of the senders are in the same OsApplication as the receiver.

(RTE-5663)

Warning if an ApplicationSwComponentType instance aggregates an NV PRPort that is not referenced by any AssemblySwConnector

RTA-RTE now raises warning 5207 if an ApplicationSwComponentType instance aggregates a PRPortPrototype that is typed by an NvDataInterface, but there is no AssemblySwConnector that references that instance of the PRPortPrototype.

As support for PRPorts for NV data is under development at this time, RTA-RTE continues to raise error 3008 for all configurations in which a PRPortPrototype is typed by an NvDataInterface

(RTE-5427)

ActivateRE calls no longer duplicated

An issue that caused RTA-RTE to incorrectly generate multiple ActivateRE calls for the same `RunnableEntity` has been fixed.

(RTE-5633)

Isolation of resource counters

RTA-RTE now allows the isolation of resource counters to improve support for freedom-from-interference.

Previously, RTA-RTE stored resource counters in the array `Rte_ResourceCount`. Now, the resource counter for a given OsApplication is a separate variable named `Rte_ResourceCount_<osAppName>`. This counter variable is wrapped in memory allo-

cationkeywords of the form RTE_<OSAPPNAME>_START_SEC_VAR_CLEARED_16 and RTE_<OSAPPNAME>_STOP_SEC_VAR_CLEARED_16, where <osAppName> is the shortName of the corresponding OsApplication and <OSAPPNAME> is that name converted to upper case.

Additionally, the infix of the memory allocation keywords wrapping the arrays Rte_Resources and Rte_ResourceCount has been changed to SYS. The Rte_ResourceCount array is populated with pointers to the resource counter variables at runtime by the function Rte_Start.

(RTE-4763)

Error if an AssemblySwConnector connects two NV PRPorts that are both aggregated by NvBlockSwComponentTypes

RTA-RTE now raises error 5206 if an AssemblySwConnector connects two PRPortPrototypes that are typed by an NvDataInterface, and both ports are aggregated by NvBlockSwComponentTypes.

As support for PRPorts for NV data is under development at this time, RTA-RTE continues to raise error 3008 for all configurations in which a PRPortPrototype is typed by an NvDataInterface.

(RTE-5425)

Error if a PRPortPrototype aggregated by an NvBlockSwComponentType is not typed by an NvDataInterface

RTA-RTE now raises error 3050 if a PRPortPrototype aggregated by an NvBlockSwComponentType is not typed by an NvDataInterface.

As support for PRPorts for NV data is under development at this time, RTA-RTE continues to raise error 3012 for all configurations in which a PRPortPrototype is aggregated by an NvBlockSwComponentType.

(RTE-5420)

The use of ——implicit-use-global-buffers now raises a warning

RTA-RTE will now raise warning 3054 when the option –implicit-use-global-buffers is used as the option may generate compilable but invalid code.

(RTE-5657)

Removed RTE_RESOURCE_CORE error from ——strict-config-check

From issue RTE-5476, RTA-RTE uses an OsApplication-based scheme for OsResources instead of a core-based scheme, making OsResources RTE_RESOURCE_CORE_x obsolete.

The Strict Config Check feature now no longer checks for the existence of the obsolete core-specific OsResources and no longer raises error 2902 if they are not supplied on the input.

This issue updates issue RTE-1150 and issue RTE-5541

(RTE-5611)

MODULE-REF references no longer validated

RTA-RTE no longer raises error 2411 for XML MODULE-REF references that cannot be resolved.

(RTE-5603)

Version 6.7.0pr8

Error if an AssemblySwConnector connects two NV PRPorts that are both aggregated by ApplicationSwComponentTypes

RTA-RTE now raises error 5205 if an AssemblySwConnector connects two PRPortPrototypes that are typed by an NvDataInterface, and both ports are aggregated by ApplicationSwComponentTypes.

As support for PRPorts for NV data is under development at this time, RTA-RTE continues to raise error 3008 for all configurations in which a PRPortPrototype is typed by an NvDataInterface.

(RTE-5424)

IOCReceiverPullCB for 1 to N cross-partition runnable activation when the sender and one receiver are in one OsApplication and another recieved is in a different OsApplication

Further to issue RTE-4770 activation of receivers by means of the `IOCReceivePullCB` is extended to the case where there are multiple receivers in different OsApplications where one reciever is in the same OsApplication as the sender.

(RTE-5546)

IOCReceiverPullCB for 1 to N cross-partition runnable activation when receivers are in different OsApplications

Further to issue RTE-4770 activation of receivers by means of the `IOCReceivePullCB` is extended to the case where there are multiple receivers in different OsApplications.

(RTE-5544)

IOCReceiverPullCB for 1 to N cross-partition runnable activation when receivers share OsApplication

Further to issue RTE-4770 activation of receivers by means of the `IOCReceivePullCB` is extended to the case where there are multiple receivers in the same OsApplication.

(RTE-5545)

Raise an error if multiple IncludedModeDeclarationGroupSets reference a single ModeDeclarationGroup

RTA-RTE will now raise an error 3052 for any ModeDeclarationGroup referenced by multiple IncludedModeDeclarationGroupSets in the composition.

(RTE-5135)

Warning if runtime range checking of ISignals requested

RTA-RTE will now raise a warning 3053 for each ISignal that aggregates an `ISignalProps` whose `handleOutOfRange` attribute is not equal to the only supported value, `NONE`.

(RTE-4029)

Version 6.7.0pr7

RTA-RTE no longer outputs warning 2634

RTA-RTE no longer outputs warning 2634 when an ISignal has a `DataTypePolicy` other than `PORT-INTERFACE-DEFINITION`.

(RTE-4524)

Do not output SpecReq comments.

RTA-RTE no longer outputs comments in the generated code relating to SpecReqs.

(RTE-4535)

Error when constr_2026 is violated

If the input configuration contains a `VariableAccess` in the role `ImplicitInterRunnableVariable` or `ExplicitInterRunnableVariable` that references an Inter-runnable variable in a different `SwcInternalBehavior` then RTA-RTE exits with error message 3049 for each invalid `VariableAccess`.

(RTE-4808)

ServiceSwComponentTypes, EcuAbstractionSwComponentTypes, and ComplexDeviceDriverSwComponentTypes are not multiply instantiable.

RTA-RTE no longer supports multiply instantiable ServiceSwComponentTypes, EcuAbstractionSwComponentTypes, or ComplexDeviceDriverSwComponent-Types

If an input configuration contains a ServiceSwComponentType, EcuAbstraction-SwComponentType, or a ComplexDeviceDriverSwComponentType which is configured to support multiple instantiation, RTA-RTE will exit with error 3047.

RTA-RTE no longer raises error 1305 when the input configuration contains multiple instances of an EcuAbstractionSwComponentType.

(RTE-5470)

RTA-RTE ignores ECUC-MODULE-DEF and LIFE-CYCLE-INFOS XML elements

RTA-RTE now skips any ECUC-MODULE-DEF or LIFE-CYCLE-INFOS XML elements appearing in the input.

RTA-RTE does not use these elements. Where possible you should not pass the Module Definitions File to the generator.

Previously, RTA-RTE sometimes gave validation errors about the contents of these elements, despite them not being relevant for RTE Generation.

(RTE-3318)

Error when LdComApiType is invalid

If the LdComApiType of an `LdComIPdu` is configured with a value that is invalid in the AUTOSAR metamodel (any value other than `LDCOM_IF` and `LDCOM_TP`) then RTA-RTE will exit with error message 3048 for each invalid LdComApiType.

(RTE-4476)

OsResource scheme based on OsApplications, not cores.

RTA-RTE has a new scheme for OsResource use.

For configurations having one OsApplication or no OsApplication at all, RTA-RTE uses `RTE_RESOURCE`. For configurations having more than one OsApplication, RTA-RTE requires one OsResource for each OsApplication, named

RTE_RESOURCE_OS_APP_ *OsApp.shortName*.

The required resources must be configured in the Operating System. The resources for the previous scheme, named RTE_RESOURCE_[CORE_*], are no longer used (but see issue RTE-5541).

(RTE-5476)

Version 6.7.0pr6

IOCReceiverPullCB for cross-partition runnable activation

Normally when activating a runnable via a DataReceivedEvent RTA-RTE generates a runnable activation flag Rte_ActCounter_<SWC>_<DRE> followed by a task activation activateTask(<TASK>) in the provider's write function.

Now, when activating a runnable via a DataReceivedEvent across partition boundaries in a 1:1 communication scenario, RTA-RTE instead generates an IOC receiver pull callback function in 'Rte.c' that sets the runnable flag and activate the task. Also, in the 'iocNeeds.arxml' the IOC channel that transfers the data will be configured to invoke this callback in the context of the receiving partition.

Note: If the communication receiver is on a split SWC then RTA-RTE will use the previous behavior of activation on writing.

Memory sections for runnable activation flags

When RTA-RTE generates a runnable activation flag Rte_ActCounter_<SWC>_<DRE> that starts a runnable mapped to an EcucPartition, the runnable activation flag will be assigned to the memory section RTE_<PARTITION>_VAR_CLEARED_8. <PARTITION> is the shortName of the EcucPartition that the activated runnable is mapped to, in all uppercase.

(RTE-4770)

Version 6.7.0pr5

AUTOSAR 4.4.0 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace http://autosar.org/schema/r4.0) against the AUTOSAR 4.4.0 schema.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to 4.4.0 will validate against the 4.4.0 schema.

(RTE-5399)

Binding of SYSC-REFs in CompuMethods

Previously, RTA-RTE did not support SYSC-REF in CompuMethods. If a CompuMethod contained a V element with a SYSC-REF then, in Rte_McSupportData.arxml, the V element would contain an XML comment stating that the SYSC-REF was not supported.

Now, RTA-RTE will bind the variability of the SYSC-REF and evaluate the value of the V element that contained it. This evaluated value is now output where the comment was previously output in Rte_McSupportData.arxml.

Note: This change only applies if the binding time of the SYSC-REF is CODE-GENERATION-TIME, or if the binding time is PRE-COMPILE-TIME and the option

`--variability-also-bind` is set to “PRE-COMPILE-TIME”.

(RTE-5003)

Version 6.7.0pr4

RTA-RTE no longer outputs information message 1564

Previously, RTEGen would output the informational message 1564 for each `RTEEvent` that was mapped to an OsTask but which could be optimized to a direct call. This message is no longer output.

(RTE-4528)

RTA-RTE now honors RunnableEntityArguments

When a `RunnableEntity` implements a server in client-server communication and the `RunnableEntity` has RunnableEntityArguments each RunnableEntityArgument `.symbol` is used as the parameter name for the `RunnableEntity`’s generated C-function prototype in the associated component’s header file.

When the number of RunnableEntityArguments does not equal the number of associated `ClientServerOperation.argumentDataPrototype`s plus the number of associated `PortApiOption.portDefinedArgumentValue`s error 3046 is raised.

(RTE-2651)

——mcore-spinlocks-always is now ignored on single core configurations

When RTA-RTE is generating for a single core configuration the option –mcore-spinlocks-always will be ignored as if it were not set.

Note: As of writing RTA-RTE only considers the system single core when there are no `OsApplications` mapped to cores greater than 0.

(RTE-4650)

RTA-RTE now honors SwAddrMethod SectionType VAR if present when generating symbols for unconnected data receive points

In the case of an unconnected sender-receiver `RPortPrototype` with an associated `SwAddrMethod` which has `SectionType` VAR, RTA-RTE now ensures that the compiler abstraction used for any symbol defined in a VAR memory section is also VAR.

(RTE-4582)

Version 6.7.0pr3

RTA-RTE no longer raises an error when a split-SWC Runnable without data access is started by multiple events

RTA-RTE previously raised the error E16-1357 if a `Runnable` in a split-SWC (with the `--deviate-split-swci-support` option enabled) was started by multiple `Events`. Now, this error is only raised for such a `Runnable` that also uses `VariableAccess` in any of the following roles: `dataSendPoint`, `dataReadAccess`, `dataWriteAccess`, `dataReceivePointByVal`, `dataReceivePointByArg`, `readLocalVariable` and `writtenLocalVariable`.

Additionally, the error E16-1357 is now raised for each `Runnable` that accesses data and is started by multiple events rather than just for one such `Runnable` per SWC. The error message has been corrected and now uses the `Runnable` for the reported location, and the corresponding section of the Reference Manual has

been updated accordingly.

(RTE-4581)

RTA-RTE no longer raises an error when a PRPort relates to a mode state machine participating in a DSMQ

RTA-RTE will no longer raise the error 901761 when a PRPort relates to a mode state machine participating in a DSMQ.

(RTE-4909)

Version 6.7.0pr2

New command-line option for controlling scheduling of Background OsTasks

By default, RTA-RTE re-schedules `background OsTasks` (OsTasks to which only background events are mapped) by ensuring that the final executable statement in the generated task body is a call to `ChainTask` with the OsTask given as an argument. RTA-RTE now provides a command-line option `--terminate-background-tasks` which, when enabled, suppresses this mechanism by replacing these calls to `ChainTask` with calls to `TerminateTask`.

This command-line option may be enabled by including `--terminate-background-tasks` (with no arguments) on the command-line.

(RTE-4896)

Extended support for Compound Primitive Types of order higher than MAP

RTA-RTE now permits the categories `CUBOID`, `CUBE_4` and `CUBE_5` to be applied to the `ApplicationPrimitiveDataType`, `ApplicationRecordElement`, `ApplicationArrayElement` and `ApplicationValueSpecificiation` types. This change only extends to the *acceptance* of the listed categories: the listed categories are otherwise treated exactly as if they were synonyms for MAP.

Additionally, if the input configuration contains a `ParameterDataPrototype` typed by an `ApplicationPrimitiveDataType` with a `category` equal to any of `CUBOID`, `CUBE_4` or `CUBE_5`, the resulting `McDataInstance` defined in the generated `Rte_McSupportData.arxml` will now have the same `category` as the originating `ApplicationPrimitiveDataType`.

(RTE-4804)

AUTOSAR standard types reference the standard path

Previously, when an object reported to BSWMD or IOC Needs used `Std_ReturnType`, RTA-RTE would reference a locally-generated, incomplete XML description of the type. Now, any type references to `Std_ReturnType` use the standard location: `/AUTOSAR_Std/ImplementationDataTypes/Std_ReturnType`.

(RTE-5207)

Version 6.7.0pr1

Code Section: Prescaler counter variables now wrapped in OsTask-specific memory allocation keywords

RTA-RTE now wraps internal Prescaler Counter Variables (PCV) `Rte_RECount` in data sections, with memory allocation keywords of the form `RTE_<TASKNAME>_START_SEC_<SW-ADDR-METHOD>[_<ALIGNMENT>]` and `RTE_<TASKNAME>_STOP_SEC_<SW-ADDR-METHOD>[_<ALIGNMENT>]`. The `<TASKNAME>` of the memory allocation keyword is the `ShortName` of the OsTask that uses the PCV

for scheduling runnables contained in that task. The <SWADDRMETHOD> of the memory allocation keyword is the appropriate SwAddrMethod associated with the PCV. The <ALIGNMENT> of the memory allocation keyword is the appropriate Alignment associated with the PCV.

In the generated Rte_BSWMD.arxml file, there now exists a MemorySection named RTE_<TASKNAME>_<SWADDRMETHOD>. It references SectionNamePrefix RTE_<TASKNAME> and SwAddrMethod <SWADDRMETHOD>, and has symbol <SWADDRMETHOD> and alignment <ALIGNMENT>.

(RTE-4239)

4.36 Version 6.6.0

Version 6.6.0

SwAddrMethod from FlatInstanceDescriptor for use in Mode Machine Instance variables' Memory Allocation Keywords.

MemorySections for Mode Machine Instance variables generated by RTA-RTE will now use the SwAddrMethod referenced by the associated FlatInstanceDescriptor with role MMI_SWADDRMETHOD, if one is available.

The associated FlatInstanceDescriptor is the single FlatInstanceDescriptor that references the ModeDeclarationGroupPrototype in the PPortPrototype of the ModeManager. The FlatInstanceDescriptor must have role MMI_SWADDRMETHOD for the SwAddrMethod it references to be used in the Mode Machine Instance variables' Memory Allocation Keywords.

The use of the SwAddrMethod from the associated FlatInstanceDescriptor, if one exists, takes precedence over any SwAddrMethods associated with the Mode Machine Instance via the ImplementationDataType that the instance is mapped to.

The Memory Allocation Keywords for Mode Machine Instance variables keep the same form as previous versions.

(RTE-5080)

SwAddrMethods and EcucPartition for use in Mode Machine Instance variables' Memory Allocation Keywords.

MemorySections for Mode Machine Instance variables generated by RTA-RTE will now use the associated SwAddrMethod and EcucPartition if mapped to a single EcucPartition.

The associated EcucPartition is the single partition to which the ModeManager and any ModeUsers are mapped to.

The associated SwAddrMethod is the first found SwAddrMethod that applies to the ImplementationDataType that is mapped to the ModeDeclarationGroup provided by the mode manager. This may include traversing any referenced ImplementationDataTypes. If no user defined SwAddrMethod is found, RTA-RTE applies a default SwAddrMethod as in previous versions.

As a result, the Memory Allocation Keywords for Mode Machine Instance variables will be of the form RTE_<ECUC>_START_SEC_<SwAddrMethod>[_ALIGNMENT] and RTE_<ECUC>_STOP_SEC_<SwAddrMethod>[_ALIGNMENT] where <ECUC> is the EcucPartition . \shortName, <SwAddrMethod> is the SwAddrMethod . \shortName and

[_ALIGNMENT] is the width of the corresponding datum and is only generated if the associated SwAddrMethod has memoryAllocationKeywordPolicy of value addrMethodShortNameAndAlignment.

Additionally, ModeStatus is now handled in the same way as the other MMI variables (ModeCurrent, ModeNext, ModeQueue, ModeQindex and ModeOld), so will be allocated to a memory section based on the EcucPartition and SwAddrMethod related to the Mode Machine Instance (when the Mode Machine Instance is accessed from a single EcucPartition).

(RTE-5079)

SwAddrMethods for task body memory allocation keywords

Task body MemorySections will now use the SwAddrMethod from the OsTask's OsMemoryMappingCodeLocationRef, when one is provided.

As a result, the memory allocation keywords for task bodies will now be of the form RTE_<TASK>_START_SEC_<SwAddrMethod> and RTE_<TASK>_STOP_SEC_<SwAddrMethod>, where <SwAddrMethod> is either the shortName of the task's referenced SwAddrMethod, or CODE if the OsTask does not supply an OsMemoryMappingCodeLocationRef.

(RTE-5191)

New MISRA Compliance statement

A MISRA Compliance Statement has been added to the installed product documents. This replaces the various references to MISRA throughout the RTA-RTE documentation.

(RTE-4537)

Version 6.6.0pr8

INF and -INF now supported as values in DataConstr limits

Previously, RTA-RTE only allowed infinite DataConstrs to be defined using Limits of intervalType INFINITE, and raised error 2951 if a Limit of a type other than INFINITE specified a value of INF or -INF.

RTA-RTE now accepts the values INF and -INF as the Limits for a DataConstr of IntervalType OPEN or CLOSED.

The legacy IntervalType INFINITE is still supported by RTA-RTE, although it is no longer considered a valid type in AUTOSAR versions from 4.2 onwards.

The use of INF or -INF as a value in any other circumstances is unsupported and may cause undefined behaviour.

(RTE-4328)

Updates to Reference Manual - code and data sections topics

Chapter 9 of the Reference Manual has been updated and expanded to cover the code and data section changes made since RTA-RTE V6.5.0 was released.

(RTE-4891)

AUTOSAR 4.3.1 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace http://autosar.org/schema/r4.0) against the AUTOSAR 4.3.1 schema.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to 4.3.1 will validate against the 4.3.1 schema.

(RTE-5190)

Version 6.6.0pr7

Code Sections: Rte_ReceiverPullCB functions and ALARMCALLBACK macros now wrapped in memory allocation keywords

RTA-RTE now wraps the Rte_ReceiverPullCB functions and ALARMCALLBACK macros in the memory allocation keywords `RTE_OSCBK_<NAME>_START_SEC_CODE` and `RTE_OSCBK_<NAME>_STOP_SEC_CODE`, where `<NAME>` is the name of the function for Rte_ReceiverPullCB functions or the name passed to the macro for ALARMCALL-BACK macros.

> The generated Rte_BSWMD.arxml file now outputs a MemorySection named `RTE_OSCBK_<NAME>_CODE` for each Rte_ReceiverPullCB function and ALARMCALLBACK macro. The MemorySections for Rte_ReceiverPullCB functions reference the `BswCalledEntity` for the function. The MemorySec-tions forALARMCALLBACK macros do not reference a `BswCalledEntity`.
>
> (RTE-5123)

Data Sections: Mode machine instance variables now wrapped in Parti-tion-specific memory allocation keywords

RTA-RTE now wraps mode machine instance (MMI) variables ModeCurrent, ModeNext, ModeQueue, ModeQindex and ModeOld in data sections, in memory allocation keywords of the form `RTE_<PARTITION>_START_SEC_<SW-ADDR-METHOD>[_<ALIGNMENT>]` and `RTE_<PARTITION>_STOP_SEC_<SW-ADDR-METHOD>[_<ALIGNMENT>]`. The `<PARTITION>` of the memory allocation keyword is the `ShortName` of the `EcucPartition` mapped to the `PPortPrototype` typed by a `ModeSwitchInterface` associated with the MMI. The `<SWADDRMETHOD>` of the memory allocation keyword is the appropriate `SwAddrMethod` associated with the MMI. The `<ALIGNMENT>` of the memory allocation keyword is the appropriate `Alignment` associated with the MMI.

In the generated Rte_BSWMD.arxml file, there now exists a MemorySection named `RTE_<PARTITION>_<SWADDRMETHOD>`. It references `SectionNamePrefix` `RTE_<PARTITION>` and `SwAddrMethod` `<SWADDRMETHOD>`, and has symbol `<SWADDRMETHOD>`.

(RTE-5078)

Code Sections: Rte_Tick_Timeouts and SchM_IModeInit functions now wrapped in memory allocation keywords

RTA-RTE now wraps the Rte_Tick_Timeouts and SchM_IModeInit functions in the memory allocation keywords `RTE_SYS_START_SEC_CODE` and `RTE_SYS_STOP_SEC_CODE`.

In the generated Rte_BSWMD.arxml file, the MemorySection named `RTE_SYS_CODE` now references the `BswCalledEntity`s for the functions Rte_Tick_Timeouts and SchM_IModeInit.

(RTE-5118, RTE-5120)

Correction to release note for RTE-4325 in 6.6.0pr1

The release note entry for RTE-4325 was incomplete, as it did not include the fix to floating point data conversion type casting.

This has now been corrected.

(RTE-5127)

Clarification of comments in generated output files

Comments in C source code and ARXML report files produced by RTA-RTE have been reworded to avoid using words derived from *elide*, as these words are uncommon in normal English usage.

The comments have been updated with simpler wording that has the same meaning.

(RTE-5137)

RTE no longer adds empty MemorySection CODE to Rte_BSWMD.arxml

As generated code objects are located in specific memory sections, RTA-RTE no longer generates a MemorySection named CODE in Rte_BSWMD.arxml.

(RTE-5124)

Code Sections: memory section start/stop sequences no longer generated for optimized function definition comments

RTA-RTE generates a comment when an API function body is optimized (such as when that API is implemented as a macro).

Previously, these comments were associated with the memory section that would have contained the optimized-away function, and the memory allocation keywords were generated even where no C symbols existed in the memory section.

These comments are now not considered to belong to any memory section, so they are no longer wrapped by memory allocation keywords in the generated output. As a consequence, memory allocation keywords that previously wrapped nothing but comments are no longer generated.

(RTE-5119)

Removed declaration of Rte_GetCurrentTaskAlarm from Rte_Intl.h

The external declaration of the obsolete Rte_GetCurrentTaskAlarm function has been removed from the template file from which Rte_Intl.h is generated.

(RTE-4920)

Version 6.6.0pr6

Efficient NV Runnables are now wrapped in NVRAM-specific memory allocation keywords

RTA-RTE generates runnable bodies to support models which use the `storeCyclic`, `storeImmediate` or `storeAtShutdown` flags set in an `NvBlockNeeds` element within an `NvBlockSwComponentType`. These runnables are now wrapped in memory allocation keywords of the form `RTE_<SWCT>_<NvBlockDescriptorName>_START_SEC_CODE` where `<SWCT>` is the name of the `NvBlockSwComponentType` related to the runnable and

<NvBlockDescriptorName> is the name of the NvBlockDescriptor related to the runnable.

In the generated Rte_BSWMD.arxml file, the MemorySection named RTE_<SWCT>_<NvBlockDescriptorName>_CODE now references the BswCalledEntitys for generated NV runnables.

(RTE-5034)

Declarations for queue variables removed from Rte_buffers.h when RIPS is enabled

When a configuration contains queued data and RIPS is enabled, RTA-RTE no longer declares global variables associated with the queued data in Rte_buffers.h.

(RTE-5088)

Code Sections: Explicit IRV API declarations in application header files now located in code sections

RTA-RTE now locates the declarations for RTE explicit IRV APIs in code sections in application header files. Previously, the declarations of these APIs in application header files were not located in any code section.

In line with RTA-RTE's approach for allocating code objects, the code sections for these APIs use memory allocation keywords of the form RTE_<SWCT>_START_SEC_CODE and RTE_<SWCT>_STOP_SEC_CODE, where <SWCT> is the shortName of the APIs associated ApplicationSwComponentType.

(RTE-4893)

Code Sections: SchM_Enter, SchM_Exit, SchM_Switch, SchM_Mode, SchM_SwitchAck, SchM_Trigger and SchM_ActMainFunction functions now wrapped in memory allocation keywords

RTA-RTE now wraps the SchM_Enter, SchM_Exit, SchM_Switch, SchM_Mode, SchM_SwitchAck, SchM_Trigger and SchM_ActMainFunction functions in the memory allocation keywords of the form RTE_<INFIX>_START_SEC_CODE and RTE_<INFIX>_STOP_SEC_CODE, where <INFIX> is the ShortName of the BswMD associated with the function.

In the generated Rte_BSWMD.arxml file, the MemorySection named RTE_<INFIX>_CODE references the BswCalledEntitys for the functions that are associated with that BswMD.

(RTE-4910)

Code Sections: SchM_Init, SchM_Deinit, SchM_GetVersionInfo and Rte_-MainFunction functions now wrapped in memory allocation keywords

RTA-RTE now wraps theSchM_Init, SchM_Deinit, SchM_GetVersionInfo and Rte_MainFunction functions in the memory allocation keywords RTE_MAIN_START_SEC_CODE and RTE_MAIN_STOP_SEC_CODE.

In the generated Rte_BSWMD.arxml file, the MemorySection named RTE_MAIN_CODE now references the BswCalledEntitys for the functions SchM_Init, SchM_Deinit, SchM_GetVersionInfo and Rte_MainFunction.

(RTE-4911)

Code Sections: Generated internal library functions now wrapped in catego-

ry-specificmemory allocation keywords

RTA-RTE now wraps several generated internal library functions in memory allocation keywords of the form RTE_<INFIX>_START_SEC_CODE and RTE_<INFIX>_STOP_SEC_CODE, where <INFIX> is EXT or SYS, depending on the category of the function.

The generated internal library functions Rte_MSACbk and Rte_ReplyCallback are now wrapped in memory allocation keywords with infix EXT.

The generated internal library function Rte_IModeInit is now wrapped in memory allocation keywords with infix SYS.

In the generated Rte_BSWMD.arxml file, the MemorySection named RTE_EXT_CODE now references the BswCalledEntitys for the generated internal library functions Rte_MSACbk and Rte_ReplyCallback, and the MemorySection named RTE_SYS_CODE now references the BswCalledEntity for the generated internal library function Rte_IModeInit.

(RTE-4901)

——use-partition-sections no longer affects task bodies

As task bodies are now located in their own task-specific MemorySections, the behavior of the –use-partition-sections option has been changed so that it no longer affects MAKWs relating to task bodies.

(RTE-4875)

Version 6.6.0pr5

Data Sections: Data associated with Intra-Partition Queued communication is now wrapped in Partition-specific memory allocation keywords

RTA-RTE now wraps all non-const data associated with Intra-Partition Queued communication in memory allocation keywords of the form RTE_<PARTITION>_START_SEC_<SW-ADDR-METHOD>[_<ALIGNMENT>] and RTE_<PARTITION>_STOP_SEC_<SW-ADDR-METHOD>[_<ALIGNMENT>], where <PARTITION>, <ALIGNMENT> and <SW-ADDR-METHOD> are the Partition's ShortName, Alignment and SwAddrMethod respectively associated with the Queue. This includes the QBuf array and QDyn struct.

The –use-partition-sections option no longer applies to this data.

RTA-RTE also outputs a SectionNamePrefix named RTE_<PARTITION> and a MemorySection named RTE_<PARTITION>_<SW-ADDR-METHOD>[_<ALIGNMENT>] in the generated Rte_BSWMD.arxml file when a Partition is associated with an Intra-Partition Queue.

(RTE-4915)

Code Sections: NvM callback functions now wrapped in NvBlockDescriptor-specific memory allocation keywords

RTA-RTE now wraps the Rte_SetMirror, Rte_NvMNotifyInitBlock and Rte_NvMNotifyJobFinished NvM callback functions in memory allocation keywords of the form RTE_<SWCT>_<NvBlockDescriptor>_START_SEC_CODE and RTE_<SWCT>_<NvBlockDescriptor>_STOP_SEC_CODE.

Similarly, RTA-RTE also now wraps the Rte_GetMirror NvM callback function in memory allocation keywords of the form

RTE_NVM_<SWCT>_<NvBlockDescriptor>_START_SEC_CODE and RTE_NVM_<SWCT>_<NvBlockDescriptor>_STOP_SEC_CODE.

In both cases, <SWCT> is the ShortName of the asscociated NvBlockSwComponentType and <NvBlockDescriptor> is the ShortName of the associated NvBlockDescriptor.

RTA-RTE also outputs SectionNamePrefixs named RTE_<SWCT>_<NvBlockDescriptor> and RTE_NVM_<SWCT>_<NvBlockDescriptor>, as well as MemorySections named RTE_<SWCT>_<NvBlockDescriptor>_CODE and RTE_NVM_<SWCT>_<NvBlockDescriptor>_CODE, in the generated Rte_BSWMD.arxml file. This happens for any NvBlockDescriptor that causes the generation of NvM callback functions. The MemorySections reference the BswCalledEntitys for the NvM callback functions they correspond to.

(RTE-4785)

Code Sections: Rte_ImplCall, Rte_ImplInvalidate and Rte_ImplWrite internal library functions now wrapped in SWCT-specific memory allocation keywords

RTA-RTE now wraps the Rte_ImplCall, Rte_ImplInvalidate and Rte_ImplWrite internal library functions in memory allocation keywords of the form RTE_<SWCT>_START_SEC_CODE and RTE_<SWCT>_STOP_SEC_CODE, where <SWCT> is the ShortName of the SWCT associated with that internal library function.

RTA-RTE also outputs a SectionNamePrefix named RTE_<SWCT> and a MemorySection named RTE_<SWCT>_CODE in the generated Rte_BSWMD.arxml file when a SWCT is associated with an internal library function. The MemorySection references the BswCalledEntitys for the internal library functions which that SWCT is associated with.

(RTE-4900)

Code Sections: Templated internal library functions now wrapped in category-specific memory allocation keywords

RTA-RTE now wraps several templated internal library functions in memory allocation keywords of the form RTE_<INFIX>_START_SEC_CODE and RTE_<INFIX>_STOP_SEC_CODE, where <INFIX> is LIB, SYS or EXT, depending on the category of the function.

The following templated internal library functions are wrapped in memory allocation keywords with infix LIB:

- Rte_GetCurrentTaskAlarmIndex
- Rte_GetCurrentTaskIndex
- Rte_MSITest
- Rte_memcpy

The following templated internal library functions are wrapped in memory allocation keywords with infix SYS:

- Rte_DecrementWaitingCount
- Rte_ReadQueue
- Rte_WaitWithTimeout

- Rte_OnStart

The following templated internal library functions are wrapped in memory allocation keywords with infix EXT:

- Rte_ActivateRE
- Rte_SetEvent
- Rte_WriteQueue

In the generated Rte_BSWMD.arxml file, RTA-RTE now outputs the following items for each of the three infix categories LIB, SYS and EXT:

- A SectionNamePrefix named RTE_<INFIX>, with symbol RTE_<INFIX>
- A MemorySection named RTE_<INFIX>_CODE, which has symbol CODE and references SectionNamePrefix RTE_<INFIX>, SwAddrMethod CODE and the relevant BswCalledEntitys for the internal library functions allocated to that section

(RTE-4898)

Code Sections: Rte_TTM, Rte_ConvertModeValueToIndex, Rte_DsmqEmptyForTBI and Rte_DsmqNextMode internal functions now wrapped in memory allocation keywords with LIB infix

RTA-RTE now wraps the Rte_TTM, Rte_ConvertModeValueToIndex, Rte_DsmqEmptyForTBI and Rte_DsmqNextMode internal library functions in memory allocation keywords of the form RTE_LIB_START_SEC_CODE and RTE_LIB_STOP_SEC_CODE.

(RTE-4899)

Code Sections: Call-backs for LdCom and COM now wrapped by signal-specific memory allocation keywords

RTA-RTE now wraps call-backs for communication over AUTOSAR Com and AUTOSAR LdCom in signal-specific memory allocation keywords. These memory allocation keywords are of the form RTE_SIG_<SN>_START_SEC_CODE and RTE_SIG_<SN>_STOP_SEC_CODE, where <SN> is the shortName of the ComSignal, ComSignalGroup or LdComIPdu related to the callback (whichever is appropriate).

RTA-RTE also outputs a SectionNamePrefix named RTE_SIG_<SN> and a MemorySection named RTE_SIG_<SN>_CODE in the generated Rte_BSWMD.arxml file when one of these call-backs exists. The MemorySection references the BswCalledEntitys for the call-backs corresponding to that ComSignal, ComSignalGroup or LdComIPdu.

(RTE-4761)

Fix to MemorySection.symbol in generated Rte_BSWMD.arxml

The symbol attribute of MemorySections associated with code objects in the generated Rte_BSWMD.arxml has been corrected to CODE. Previously, the value of symbol for these MemorySections was <INFIX>_CODE.

This fix avoids the duplication of the infix in MAKWs when they are generated by external tools following [SWS_MemMap_00022], as the infix is also included as a

component of the SectionNamePrefix.

(RTE-4806)

Version 6.6.0pr4

PortDefinedArgumentValues honoured when comparing ClientServerOperations triggering the same RunnableEntity

RTA-RTE now honors PortDefinedArgumentValues in addition to the ClientServerOperation . argumentDataPrototypes when determining the compatibility of ClientServerOperations that start the same RunnableEntity, in accordance with [constr_2000].

Previously, when multiple OperationInvokedEvents triggered the same RunnableEntity, RTA-RTE only considered the corresponding ClientServerOperation . argumentDataPrototypes when determining compatibility of the ClientServerOperations.

PortDefinedArgumentValues are always supplied to the called RunnableEntity at the start of the argument list, ahead of those specified as argumentDataPrototypes in the ClientServerOperation.

(RTE-4137)

Removed ——deviate-task-sections option

The –deviate-task-sections option and its short form --dv8ts have been removed.

If either the –deviate-task-sections or --dv8ts options are provided, RTA-RTE will exit with error 1103.

(RTE-4629)

Code Sections: Rte_Enter, Rte_Exit and Rte_Feedback APIs now wrapped in SWCT-specific memory allocation keywords

RTA-RTE now wraps the Rte_Enter, Rte_Exit and Rte_Feedback APIs in memory allocation keywords of the form RTE_<SWCT>_START_SEC_CODE and RTE_<SWCT>_STOP_SEC_CODE, where <SWCT> is the ShortName of the SWCT associated with that API.

RTA-RTE also outputs a SectionNamePrefix named RTE_<SWCT> and a MemorySection named RTE_<SWCT>_CODE in the generated Rte_BSWMD.arxml file when a SWCT has one of these APIs. The MemorySection references the BswCalledEntitys for the APIs corresponding to that SWCT.

(RTE-4675)

Code Sections: Trigger APIs now wrapped in SWCT-specific memory allocation keywords

RTA-RTE now wraps Trigger APIs in memory allocation keywords of the form RTE_<SWCT>_START_SEC_CODE and RTE_<SWCT>_STOP_SEC_CODE, where <SWCT> is the ShortName of the SWCT associated with that API.

RTA-RTE also outputs a SectionNamePrefix named RTE_<SWCT> and a MemorySection named RTE_<SWCT>_CODE in the generated Rte_BSWMD.arxml file when a SWCT has an associated Trigger API. The MemorySection references the BswCalledEntitys for the Trigger APIs corresponding to that SWCT.

(RTE-4674)

Code Sections: SchM_CData APIs now wrapped in BswModuleDescription-specific memory allocation keywords

RTA-RTE now wraps generated SchM_CData APIs in memory allocation keywords of the form `RTE_<BSW>_START_SEC_CODE` and `RTE_<BSW>_STOP_SEC_CODE`, where `<BSW>` is the `ShortName` of the BswModuleDescription associated with that API.

RTA-RTE also outputs a `SectionNamePrefix` named `RTE_<BSW>` and a MemorySection named `RTE_<BSW>_CODE` in the generated Rte_BSWMD.arxml file when a BswModuleDescription has an associated SchM_CData API. The MemorySection references the `BswCalledEntitys` for the SchM_CData APIs corresponding to that BswModuleDescription.

(RTE-4820)

Code Sections: Rte_CData and Rte_Prm APIs now wrapped in SWCT-specific memory allocation keywords

RTA-RTE now wraps generated Rte_CData and Rte_Prm APIs in memory allocation keywords of the form `RTE_<SWCT>_START_SEC_CODE` and `RTE_<SWCT>_STOP_SEC_CODE`, where `<SWCT>` is the `ShortName` of the `SWCT` associated with that API.

RTA-RTE also outputs a `SectionNamePrefix` named `RTE_<SWCT>` and a MemorySection named `RTE_<SWCT>_CODE` in the generated Rte_BSWMD.arxml file when a SWCT has an associated Rte_CData or Rte_Prm API. The MemorySection references the `BswCalledEntitys` for the parameter APIs corresponding to that SWCT.

(RTE-4673)

ERCOSek removed from documentation

As support for operating systems other than AUTOSAR4.x is now deprecated, ERCOSek has been expunged from the user documentation, with the exception of the Error Description document, which reflects the current state of the generator.

(RTE-4574)

Unique identifiers for internal assertion failures

Each internal assertion failure has been assigned a unique six-digit error number in the range `900000 - 999999`. This provides more clarity when internal assertion failures occur, as each internal assertion is separately identifiable. Previously, all internal assertion failures were reported as error 1002.

If your configuration causes RTA-RTE to raise an internal error, you should contact ETAS technical support for assistance.

(RTE-4609,RTE-4774,RTE-4775)

Version 6.6.0pr3

Inter-ECU transmission acknowledgement when System is of category ECU_EXTRACT

Previously, feedback APIs, variables and COM acknowledgement callback functions were not generated correctly in cases where transmission acknowledgement is enabled for an inter-ECU communication and the System has category ECU_EXTRACT. This has now been corrected.

(RTE-4781)

Code Sections: Client-Server APIs now wrapped in SWCT-specific memory allocation keywords

RTA-RTE now wraps Client-Server APIs in memory allocation keywords of the form `RTE_<SWCT>_START_SEC_CODE` and `RTE_<SWCT>_STOP_SEC_CODE`, where `<SWCT>` is the `ShortName` of the `SWCT` associated with that API.

RTA-RTE also outputs a `SectionNamePrefix` named `RTE_<SWCT>` and a MemorySection named `RTE_<SWCT>_CODE` in the generated Rte_BSWMD.arxml file when a SWCT has an associated Client-Server API. The MemorySection references the `BswCalledEntitys` for the mode management APIs corresponding to that SWCT.

(RTE-4672)

Code Sections: Mode Management APIs now wrapped in SWCT-specific memory allocation keywords

RTA-RTE now wraps mode management APIs in memory allocation keywords of the form `RTE_<SWCT>_START_SEC_CODE` and `RTE_<SWCT>_STOP_SEC_CODE`, where `<SWCT>` is the `ShortName` of the `SWCT` associated with that API.

RTA-RTE also outputs a `SectionNamePrefix` named `RTE_<SWCT>` and a MemorySection named `RTE_<SWCT>_CODE` in the generated Rte_BSWMD.arxml file when a SWCT has an associated mode management API. The MemorySection references the `BswCalledEntitys` for the mode management APIs corresponding to that SWCT.

(RTE-4671)

Code Sections: Change of infix position for memory allocation keywords

RTA-RTE now outputs memory allocation keywords in the form `RTE_<INFIX>_START_SEC_CODE` and `RTE_<INFIX>_STOP_SEC_CODE`.

Changes have also happened to the generated Rte_BSWMD.arxml file. RTA-RTE now outputs:

- A `SectionNamePrefix` named `RTE_<INFIX>`, with symbol `RTE_<INFIX>`
- A MemorySection named `RTE_<INFIX>_CODE`, which has symbol `<INFIX>_CODE` and references:
- `SectionNamePrefix` `RTE_<INFIX>`
- `SwAddrMethod` CODE
- The relevant `BswCalledEntity`s for the APIs allocated to that section

These changes are in response to AUTOSAR RfCs including 78213 and 79707.

(RTE-4754)

Version 6.6.0pr2

Code Sections: Explicit IRV APIs now wrapped in SWCT-specific memory allocation keywords

RTA-RTE now wraps explicit inter-runnable variable (IRV) APIs in memory allocation keywords of the form `RTE_START_SEC_<SWCT>_CODE` and `RTE_STOP_SEC_<SWCT>_CODE`, where `<SWCT>` is the `ShortName` of the `SWCT`.

RTA-RTE now outputs a MemorySection and a `SwAddrMethod` both named

<SWCT>_CODE in the generated Rte_BSWMD.arxml file when a SWCT has an associated explicit IRV API. The new MemorySection references the relevant BswCalledEntitys for any explicit IRV APIs corrosponding to that SWCT, and those BswCalledEntitys reference the <SWCT>_CODE SwAddrMethod.

(RTE-4670)

Code Sections: Explicit Sender/Receiver APIs now wrapped in memory allocation keywords

RTA-RTE now wraps Explicit Sender/Receiver APIs in memory allocation keywords of the form RTE_START_SEC_<SWCT>_CODE and RTE_STOP_SEC_<SWCT>_CODE, where <SWCT> is the ShortName of the SWCT.

RTA-RTE now outputs a MemorySection and a SwAddrMethod both named <SWCT>_CODE in the generated Rte_BSWMD.arxml file when a SWCT has an associated explicit Sender/Receiver API. The new MemorySection references the relevant BswCalledEntitys for any explicit Sender/Receiver APIs corrosponding to that SWCT, and those BswCalledEntitys reference the <SWCT>_CODE SwAddrMethod.

(RTE-4151)

Code Sections: Call-backs for NvM now wrapped in memory allocation keywords

RTA-RTE now wraps call-backs for communication over AUTOSAR NvM in memory allocation keywords of the form RTE_START_SEC_SYS_CODE and RTE_STOP_SEC_SYS_CODE.

RTA-RTE now outputs a MemorySection and a SwAddrMethod both named SYS_CODE in the generated Rte_BSWMD.arxml file when at least one NvM callback exists. The new MemorySection references the relevant BswCalledEntitys for any NvM callbacks, and those BswCalledEntitys reference the SYS_CODE SwAddrMethod.

(RTE-4600)

Version 6.6.0pr1

Code Sections: Call-backs for Com now wrapped in memory allocation keywords

RTA-RTE now wraps call-backs for communication over AUTOSAR Com in memory allocation keywords. These memory allocation keywords are of the form RTE_START_SEC_SYS_CODE and RTE_STOP_SEC_SYS_CODE.

RTA-RTE now outputs a MemorySection and a SwAddrMethod both named SYS_CODE in the generated Rte_BSWMD.arxml file when at least one Com callback exists. The new MemorySection references the relevant BswCalledEntitys for any Com callbacks, and those BswCalledEntitys reference the SYS_CODE SwAddrMethod.

(RTE-4598)

Code Sections: Call-backs for LdCom now wrapped in memory allocation keywords

RTA-RTE now wraps call-backs for communication over AUTOSAR LdCom in memory allocation keywords. These memory allocation keywords are of the form RTE_START_SEC_SYS_CODE and RTE_STOP_SEC_SYS_CODE.

RTA-RTE also now outputs a MemorySection and a `SwAddrMethod` both named `SYS_CODE` in the generated Rte_BSWMD.arxml file. The new MemorySection references the relevant `BswCalledEntity`s for any LdCom callbacks, and those `BswCalledEntity`s reference the `SYS_CODE SwAddrMethod`.

(RTE-4599)

Code Sections: Lifecycle APIs now wrapped in memory allocation keywords

RTA-RTE now wraps APIs `Rte_Start` and `Rte_Stop` in memory allocation keywords. These memory allocation keywords are of the form `RTE_START_SEC_MAIN_CODE` and `RTE_STOP_SEC_MAIN_CODE`.

RTA-RTE also now outputs a MemorySection and a `SwAddrMethod` both named `MAIN_CODE` in the generated Rte_BSWMD.arxml file. The new MemorySection references the relevant `BswCalledEntity`s for `Rte_Start` and `Rte_Stop`, and those `BswCalledEntity`s reference the `MAIN_CODE SwAddrMethod`.

(RTE-4597)

Code Sections: Task bodies now wrapped in task-specific memory allocation keywords

RTA-RTE now wraps all of the task bodies it generates with memory allocation keywords specific to the task. These memory allocation keywords are of the form `RTE_START_SEC_<TASK>_CODE` and `RTE_STOP_SEC_<TASK>_CODE`, where `<TASK>` is the `shortName` of the `OsTask` that a particular task body corresponds to.

(RTE-4315)

Floating point data conversion: Support compuMethods with Offsets & Fix type casting

Where a 32-bit floating point data is transmitted inter-ECU as a `uint16` value, RTA-RTE now allows the transporting `ISignal` to reference (via its `networkRepresentationProps`) a compuMethod of category LINEAR with an offset (i.e. where the first numerator `N0` has a value other than 0).

Additionally, floating point data conversion code now explicitly casts the integer data to a float before applying the conversion.

(RTE-4325)

Corrected guard symbol for iocNeeds.h

`iocNeeds.h` is now protected by guard symbol `RTE_IOCNEEDS_H` rather than `__iocNeeds_H__`, which violated MISRA C:2012 Rule-1.3 and MISRA C:2012 Rule-21.2.

(RTE-3905)

Support for merging of EcucValueCollections

RTA-RTE now handles EcucValueCollections as if they were `atpSplitable`, allowing an EcucValueCollection to be split across multiple AUTOSAR XML description files.

Although EcucValueCollection is not defined as splittable by AUTOSAR, this restores the behaviour of RTA-RTE that was observed in versions prior to 6.4.0, with some additional validation to ensure that conflicting references are detected and reported.

RTA-RTE now raises error 3043 when an EcucValueCollection (or, if split, the result of merging multiple EcucValueCollection fragments) references multiple Systems.

RTA-RTE continues to raise error 1301 when an EcucValueCollection (or, if split, the result of merging multiple EcucValueCollection fragments) does not reference a System. The text and prefix of this error has been revised.

(RTE-4383)

Floating point data conversion: Error when CompuMethod used is not of category IDENTICAL or LINEAR

RTA-RTE now raises error 3044 when a CompuMethod used for floating-point data conversion is not of category IDENTICAL or LINEAR.

(RTE-4324)

Implicitly written values flushed to measurement buffer when PPort is unconnected

When an implicitly-written `VariableDataPrototype` exists in an unconnected `PPort`, and is marked as measurable (by being referenced from a `FlatInstanceDescriptor` with `SwCalibrationAccess` set to `ReadOnly`), RTA-RTE now correctly flushes the contents of that data element's implicit shadow buffer to the corresponding measurement buffer at the end of any task in which the data element is written.

The measurement buffer is mapped to the symbol of the measured data element instance's `McDataInstance` in `Rte_McSupportData.arxml`.

Previously, a measurement buffer was declared but never updated because implicitly-written values were not flushed back to it.

(RTE-4519)

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Version 6.5.0pr8

Incorrect range error for IEEE 754 floating point types removed

RTA-RTE no longer raises error 2668 when a configuration contains an IEEE 754 floating-point data type with infinite data constraints.

(RTE-4566)

Fixed internal error occurrence relating to IRV validation

An occurrence of internal error 1002 relating to IRV validation was found and fixed.

(RTE-4561)

Internal error corrected in the case of implicit access to parameter data

Formerly, RTA-RTE failed with internal error 1002 when generating code for a configuration containing a `PPortPrototype` typed by a `ParameterInterface` connected to a `RPortPrototype` typed by a `SenderReceiverInterface` and referenced by a `DataReadAccess`. This has now been corrected.

(RTE-4532)

Internal error corrected in the case of data accessed via PRPorts in different OsTasks

Formerly, RTA-RTE failed with internal error 1002 when generating code for a configuration containing data accessed via PRPorts in different OsTasks. This has now been corrected.

(RTE-4530)

Floating point data conversion: Error when a CompuMethod results in a divide by zero and warn when scale is constant zero

RTA-RTE now raises error 3041 when a CompuMethod is used for floating point data conversion and that CompuMethod results in a division by zero when evaluated.

RTA-RTE now raises warning 3042 when a CompuMethod is used for floating point data conversion and that CompuMethod results in a scale of zero when evaluated.

(RTE-4540)

Removed ——diagnostic option

The `--diagnostic` option and its short form `--diag` have been removed.

If either the `--diagnostic` or `--diag` options are provided, RTA-RTE will exit with error 1103.

(RTE-4072)

Fatal errors no longer cause runtime abort

In previous versions, RTA-RTE called the runtime `abort()` function when it encountered a fatal error.

Depending on the system configuration, calling `abort()` may trigger the raising of an OS or debugger dialog, preventing control from being returned to the calling process in the event of a fatal error.

RTA-RTE now calls `exit()` in all handled error scenarios. Fatal errors will result in a return status of `3`.

The undocumented `--suppress-abort` option has been removed.

(RTE-4418)

Version 6.5.0pr7

Floating point data conversion: support for COM reception of floating-point data elements via integer-typed ISignals

RTA-RTE's support for floating point data conversion has been extended to allow inter-ECU `COM` reception of IEEE 754 floating point data elements via ISignals that are of 16-bit unsigned integer type.

On the reception side of inter-ECU communication, data conversion is performed in the relevant COM receive callback.

If the transporting ISignal references (via its `networkRepresentationProps`) a `compuMethod` of category `LINEAR` that resolves to a value other than `1`, the value received from COM is typecast to a floating point value, which is then scaled using a floating-point division operation before being stored in the appropriate reception buffer.

If the CompuMethod is of category `IDENTICAL`, or is of category `LINEAR` and

resolves to the value `1`, then the value received from COM is typecast to a floating point value that is then stored directly in the appropriate reception buffer.

Any referenced `LINEAR` `compuMethod` must not apply any constant offset - i.e. the value of its first numerator `N0` must be 0.

(RTE-4039)

Fixed spurious error 7062 relating to VAL_BLK

Formerly, if a `TEXTTABLE` CompuMethod was referenced by an `ApplicatonPrimitiveDataType` that was mapped to an `ARRAY` `ImplementationDataType`, RTEGen would treat the resulting enumeration constants as signed, and omit the `U` suffix, even if the array element type was unsigned. If an unsigned `ImplementationDataType` also referenced the CompuMethod then error 7062 would be raised.

This is now fixed: The enumeration preprocessor symbols are now defined with a `U` suffix in these cases.

(RTE-4117)

Floating point data conversion support extended to allow scaled fixed point transmission

RTA-RTE support for floating point data conversion has been extended to allow 32-bit floating point data to be transmitted inter-ECU as a scaled `uint16` value.

Scaling will be applied when the ISignal that transports the floating point data element references (via its `networkRepresentationProps`) a `compuMethod` of category `LINEAR`. The scaling uses a floating-point multiplication operation, the result of which is then typecast to a `uint16` value that is passed to COM.

The referenced `compuMethod` must not apply any constant offset - i.e. the value of the first numerator `N0` must be 0.

(RTE-4250)

Error when an OsTask has no OsTaskPriority

RTA-RTE now raises error 2052 when an OsTask has no OsTaskPriority.

(RTE-4516)

Version 6.5.0pr6

Floating point data conversion: support for COM transmission of floating-point data elements via integer-typed ISignals (no scaling)

RTA-RTE now supports floating point to integer data conversion, allowing inter-ECU `COM` transmission of IEEE 754 floating point data elements via ISignals that are of 16-bit unsigned integer type.

Floating point data conversion is only supported for data transmitted inter-ECU via explicit unqueued Sender/Receiver communication. Other types of communication are not supported. Fan-out and mixed inter-/intra-ECU scenarios are also not supported.

Support for floating point data conversion is restricted to primitive 32-bit floating point data elements (i.e. those where the underlying `SwBaseType` has `baseTypeSize` of `32` and `baseTypeEncoding` of `IEEE754`) and 16-bit unsigned integer ISignals (i.e. those that aggregate a `networkRepresentationProps` referencing a

SwBaseType with baseTypeSize of 16 and baseTypeEncoding of NONE). The use of floating point data conversion with any other types is unsupported.

Scaling of the data before transmission is not supported, so only networkRepresentationProps that aggregate a CompuMethod of category IDENTICAL are supported. The omission of a CompuMethod or the aggregation of a LINEAR CompuMethod whose normalized value results in 1 are considered IDENTICAL by RTA-RTE.

The generated RTE code casts the provided floating point value to a uint16 type, therefore the exact value that will be passed to COM depends on the target compiler's specific implementation of float to integer casting.

There is no support for runtime range checking or other validation in RTA-RTE. Conversion of floating point values that lie outside the range of uint16 will result in the transmission of incorrect or nonsense values.

(RTE-4248)

Error 3036 no longer raised for unused LdComIPdus

RTA-RTE no longer raises error 3036 against unused LdComIPdus (i.e. those that are not referenced from any ISignalToIPduMapping).

Previously, RTA-RTE raised error 3036 if a configuration contained more than one unused LdComIPdu, although error 3036 does not apply in this situation. This has been corrected.

(RTE-4503)

Error when a RunnableEntity with implicit access is started by multiple OsTasks

RTA-RTE now raises error 3038 when a RunnableEntity which aggregates a DataWriteAccess or DataReadAccess is started by multiple RteEvents which are mapped to different OsTasks. Info 1001 is also raised for each RteEventToTaskMapping mapping an RteEvent which starts that RunnableEntity.

Note: RTA-RTE used to raise error 7645 when a RunnableEntity which aggregates a DataWriteAccess or DataReadAccess is started by multiple RteEvents which are mapped to different OsTasks with different OsPriorities. This error has been replaced with error 3038, which covers the case for 7645 and the similar case where the involved OsTasks have the same OsPriority.

(RTE-4478)

Additional RIPS functionality

Additional RIPS functionality has been added. These changes do not affect configurations that do not use RIPS.

(RTE-4496)

Version 6.5.0pr5

Support AUTOSAR 4.2.x PduInfoType in LdCom transmit

RTA-RTE no longer initializes the MetaDataPtr member of the PduInfoType in RTE APIs that transmit data via LdCom.

This is to support the use of PduInfoType as defined in earlier AUTOSAR versions (up to and including 4.2.2) which do not include a MetaDataPtr member. In earlier

previews, a `nullptr` was assigned to the `MetaDataPtr` member which restricted compatibility with earlier AUTOSAR versions.

If the AUTOSAR 4.3.x version of `PduInfoType` is used, no value will be assigned to the `MetaDataPtr`. This may lead to runtime errors if a null or invalid pointer is passed to `LdCom_Transmit`.

(RTE-4460)

Version 6.5.0pr4

PDU-REF now correctly validates CONTAINER-I-PDU

RTA-RTE now does not raise error 1022 if a PDU-REF element has destination CONTAINER-I-PDU.

(RTE-4316)

PORT-REF now correctly validates PR-PORT-PROTOTYPE

RTA-RTE now does not raise error 1022 if a PORT-REF element has destination PR-PORT-PROTOTYPE.

(RTE-4344)

Implicit RIPS communication from different contexts no longer rejected

In the case where a runnable executed in different contexts has implicit RIPS communication and does not have any non-RIPS implicit communication, RTA-RTE now does not raise error 7645.

(RTE-4401)

Improved strict configuration check error relating to required OsAlarms

The detection of required `OsAlarm`s when `--strict-config-check` is enabled has been improved so that RTA-RTE does not raise error 2900 in cases where it previously did so incorrectly.

(RTE-4009)

Include ComStack_Types.h in Rte_cbk.h when LdCom is used

RTA-RTE now includes ComStack_Types.h in the generated Rte_cbk.h file when a configuration contains an `EcucModuleConfigurationValues` element referencing an `EcuModuleDef` with `ShortName` 'LdCom'.

(RTE-4377)

Compiler abstraction macro for LdCom callback parameter

RTA-RTE now types the `PduInfoType` parameter of the `Rte_LdComCbkRxIndication` callback function with the AUTOSAR `P2CONST` compiler abstraction macro, i.e. in the form `P2CONST(PduInfoType, AUTOMATIC, RTE_APPL_DATA) PduInfoPtr`.

In earlier previews, the parameter was declared without compiler abstraction (as `const PduInfoType* PduInfoPtr`).

(RTE-4360)

Symbolic names for LdCom signal handles

RTA-RTE now supports the use of the `--com-symbolic-sigs` command line option with LdCom. When this option is specified, the generated RTE uses the `shortName` of the LdComIPdu instead of the `LdComHandleId` when invoking LdCom API functions.

(RTE-4358)

Internal error when signal name is invalid

The detection of invalid signal names has been improved so that RTA-RTE now raises internal error 7418 when it is unable to find a valid signal name.

(RTE-4369)

LdCom callback DataReceivedEvents

The activation of a receiving runnable via a `DataReceivedEvent` raised in an LdCom callback is now supported.

(RTE-4106)

Version 6.5.0pr3

Reception of transformed signals with LdCom (without a DataReceivedEvent)

When a transformed receiving ISignal is mapped via an `ISignalToPduMapping` to an LdComIPdu, an `Rte_LdComCbkRxIndication` callback function is generated to copy the data received from LdCom into the RTE-managed global receive buffer.

The `Rte_Read` API reads the received data from the global buffer and applies the reverse transformation before supplying the original value to the receiving runnable.

At present, the activation of receiving runnables via a `DataReceivedEvent` raised in the LdCom callback is untested and should be considered unsupported.

(RTE-4105)

Internal error when signal name is invalid

The detection of invalid signal names has been improved so that RTA-RTE now raises internal error 7418 when it is unable to find a valid signal name.

(RTE-4326)

Error when a COMBased transformed signal is mapped to an LdComIPdu

RTA-RTE now raises error 3037 when an ISignal references a `DataTransformation` which has a `SERIALIZER TransformationTechnology` and `protocol` set to COMBased, and that ISignal is also mapped via an ISignalToIPduMapping to an LdComIPdu.

(RTE-4200)

Error when a ComSignal and an LdComIPdu reference the same ISignalToIPduMapping

RTA-RTE raises error 3036 when a ComSignal and an LdComIPdu reference the same ISignalToIPduMapping. The error message is accompanied one info message 1001 for each LdComIPdu or ComSignal referencing the ISignalToIPduMapping.

(RTE-4098)

Error when multiple LdComIPdus reference the same ISignalToIPduMapping

RTA-RTE now raises error 3036 when multiple LdComIPdus reference the same ISignalToIPduMapping. The error message is accompanied by one info message 1001 for each LdComIPdu referencing the ISignalToIPduMapping.

(RTE-4097)

Version 6.5.0pr2

Support for DataTypeMappingSets in NvBlockSwComponentTypes

RTA-RTE now honors the use of DataTypeMappingSets referenced in a `NvBlockDescriptor` for a particular NvBlockSwComponentType as well as DataTypeMappingSets referenced in a `SwcInternalBehavior`.

(RTE-4128)

Improved internal error

Configurations that raised error 1002 with a message relating to `getLocations().size()` in `getChildNode` now raise more helpful error message 3034, giving the location of the configuration object where the problem occurred.

(RTE-4253)

Additional RIPS functionality

Additional RIPS functionality has been added. These changes do not affect configurations that do not use RIPS.

(RTE-4247,RTE-4266)

Error when a non-transforming ISignal is mapped to an LdComIPdu

RTA-RTE now raises error 3031 when an ISignal that does not reference a `DataTransformation` is mapped via an `ISignalToPduMapping` to an LdComIPdu.

(RTE-4095)

Version 6.5.0pr1

Prototype support for transmission of transformed signals with LdCom

When a transforming ISignal is mapped via an `ISignalToPduMapping` to an LdComIPdu, the generated `Rte_Write` API calls `LdCom_Transmit` transmit the transformed byte array.

The generated code expects to get `PduInfoType` from `LdCom.h`, which is `#included` by `Rte.c`.

The generated code allocates and initializes a `PduInfoType` for the transmitted data but does not populate PDU metadata at this time. RTE initializes `MetaDataPtr` with a null pointer.

(RTE-4099)

Error when AUTOSAR [constr_3514] is violated

RTA-RTE now raises error 3032 when an ISignal is referenced by more than one ISignalToIPduMapping, in violation of AUTOSAR `[constr_3514]`. The error message is accompanied by one info message 1001 for each ISignalToIPduMapping referencing the ISignal.

(RTE-4096)

Error when an LdComIPdu uses the LdCom TP API

RTA-RTE now raises error 3030 when an LdComIPdu has `LdComApiType` set to `LDCOM_TP`.

(RTE-4094)

4.38 Version 6.4.0

Version 6.4.0

Fixed internal error occurrence for EcucValueCollections

An occurrence of internal error 1002 relating to EcucValueCollection objects was found and fixed.

(RTE-4197)

Version 6.4.0pr11

Support for DataReceivedEvent referencing a data item in a PRPortPrototype

RTA-RTE now allows a DataReceivedEvent to reference a data item in a PRPortPrototype. Previously, this scenario was rejected with error 2418.

(RTE-4189)

ModeSwitchedAckEvents with PRPorts

RTA-RTE now completes successfully when mode switch acknowledgment is requested for a PRPort, and a ModeSwitchedAckEvent is attached to the `ModeSwitchPoint` that references the corresponding `ModeDeclarationGroupPrototype`. Previously, error 1500 was raised when the ModeSwitchedAckEvent was present.

The code generated to implement the mode switch acknowledgment via a PRPortPrototype is identical to that which implements the mode switch acknowledgement via a `PPortPrototype`.

(RTE-4187)

Correction to spelling of library API

Since RTA-RTE 6.1.0rc1 there has been a chance that RTA-RTE would incorrectly capitalise the `OS` in `Rte_SuspendOSInterrupts` and `Rte_ResumeOSInterrupts`.

This is now corrected.

(RTE-4174)

Fixed regression of Implicit S/R API types

RTA-RTE Version 6.4.0pr9 introduced a regression which caused the return values of implicit S/R APIs to have incorrect types. This has now been fixed.

(RTE-4164)

Enhancements to RIPS functionality

Enhancements to RIPS functionality have been merged into the product line. These changes do not affect configurations that do not use RIPS.

(RTE-4169, RTE-4170, RTE-4191)

Version 6.4.0pr10

Configurations containing ParameterInterface connected to SenderReceiverInterface accepted

RTA-RTE now accepts configurations where a PPort typed by a ParameterInterface on a `ParameterSwComponentType` is connected to an RPort on an `ApplicationSwComponentType` typed by a SenderReceiverInterface.

This connection is permitted by AUTOSAR [constr_1036], as both SenderReceiverInterface and ParameterInterface are subclasses of `DataInterface`. Previously RTA-RTE rejected these configurations with error 2600.

(RTE-3748)

Fixed assertion in Optimize::createExterns

Certain configurations caused this internal error to be raised. This assertion was testing an incorrect assumption about the design. After code review the design issue is resolved and the assertion removed.

(RTE-4142)

DataReceiveEvents referencing RPorts connected to PRPorts now supported

RTA-RTE now supports the use of DataReceiveEvents referencing data elements in the context a `RPortPrototype`, where that `RPortPrototype` is connected to a PRPortPrototype via an `AssemblySwConnector`. The generated RTE code is the same as it would be if the `RPortPrototype` was connected to a `PPortPrototype`.

(RTE-4104)

ServiceSwComponentTypes now allowed to aggregate PRPorts

RTA-RTE now allows the use of PRPorts in ServiceSwComponentTypes. RTA-RTE handles the aggregation of a PRPort by a ServiceSwComponentType as if it was an `ApplicationSwComponentType`.

(RTE-4086)

ComplexDeviceDriverSwComponentTypes now allowed to aggregate PRPorts

RTA-RTE now allows the use of PRPorts in ComplexDeviceDriverSwComponentTypes. RTA-RTE handles the aggregation of a PRPort by a ComplexDeviceDriver as if it was an `ApplicationSwComponentType`.

(RTE-4085)

Version 6.4.0pr9

PRPorts: Support for 1:n explicit Sender-Receiver communication on a PRPort with explicit write and read by value in the same runnable

RTA-RTE now allows the use of PRPorts for explicit sender/receiver communication, in the case where one runnable has a `dataSendPoint` and a `dataReceivePointByValue` to a data item via the PRPort, and at least one other runnable has a `dataReceivePoint` to the same data item. The `dataReceivePoint`s in the other reading runnables may be any combination of `dataReceivePointByArgument` or `dataReceivePointByValue`.

Runnables in a connected SWC can also have a `DataReceivePointByArgument` or a `DataReceivePointByValue` to the same data element via the connected SWC's `RPortPrototype`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948). Support for `DataReceivePointByValue`s is limited to data elements of primitive type.

(RTE-3727)

PRPorts: Support for 1:n explicit Sender-Receiver communication on a PRPort

with explicit write and read by argument in the same runnable

RTA-RTE now allows the use of PRPorts for explicit sender/receiver communication, in the case where one runnable has a `dataSendPoint` and a `dataReceivePointByArgument` to a data item via the PRPort, and at least one other runnable has a `dataReceivePoint` to the same data item. The `dataReceivePoint`s in the other reading runnables may be any combination of `dataReceivePointByArgument` or `dataReceivePointByValue`.

Runnables in a connected SWC can also have a `DataReceivePointByArgument` or a `DataReceivePointByValue` to the same data element via the connected SWC's `RPortPrototype`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948). Support for `DataReceivePointByValue`s is limited to data elements of primitive type.

(RTE-3726)

PRPorts: Support for 1:n explicit Sender-Receiver communication on a PRPort in different runnables

RTA-RTE now allows the use of PRPorts for explicit sender/receiver communication, in the case where one runnable has a `dataSendPoint` to a data item via the PRPort, and at least two other runnables have a `dataReceivePoint` to the same data item. The `dataReceivePoint`s across all the reading runnables may be any combination of `dataReceivePointByArgument` or `dataReceivePointByValue`.

Runnables in a connected SWC can also have a `DataReceivePointByArgument` or a `DataReceivePointByValue` to the same data element via the connected SWC's `RPortPrototype`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948). Support for `DataReceivePointByValue`s is limited to data elements of primitive type.

(RTE-3725)

Enhancements to RIPS functionality

Enhancements to RIPS functionality have been merged into the product line. These changes do not affect configurations that do not use RIPS.

(RTE-4084)

PRPorts: Support for explicit reading by value and writing of the same data element by the same runnable in a SWC

RTA-RTE now allows the use of PRPorts for explicit sender/receiver communication, in the case where a single runnable has both a `DataSendPoint` and a `DataReceivePointByValue` to a `VariableDataPrototype` via a PRPortPrototype.

Runnables in a connected SWC can also have a `DataReceivePointByValue` to the same data element via the connected SWC's `RPortPrototype`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948). Support for `DataReceivePointByValue`s is limited to data elements of primitive type.

(RTE-3724)

Version 6.4.0pr7

Error when a data element in the context of a PRPortPrototype is never written

RTA-RTE now raises error 3017 when a `VariableDataPrototype` in the context of a PRPortPrototype is never referenced by a `DataWriteAccess` or `DataSendPoint`.

(RTE-3718)

PRPorts: Support for explicit reading by argument and writing of the same data element by the same runnable in a SWC

RTA-RTE now allows the use of PRPorts for explicit sender/receiver communication, in the case where a single runnable has both a `DataSendPoint` and a `DataReceivePointByArgument` to a `VariableDataPrototype` via a PRPortPrototype.

Runnables in a connected SWC can also have a `DataReceivePointByArgument` to the same data element via the connected SWC's `RPortPrototype`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948).

(RTE-3723)

PRPorts: Support for explicit reading by value and writing of the same data element by different runnables in a SWC

RTA-RTE now allows the use of PRPorts for explicit sender/receiver communication, in the case where one runnable has a `DataSendPoint` to a `VariableDataPrototype` via a PRPortPrototype, and another runnable in the same SWC has a `DataReceivePointByValue` to that `VariableDataPrototype` via the same PRPortPrototype.

Runnables in a connected SWC can also have a `DataReceivePointByValue` to the same data element via the connected SWC's `RPortPrototype`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948). Support for `DataReceivePointByValue`s is limited to data elements of primitive type.

(RTE-3722)

PRPorts: Support for explicit reading by argument and writing of the same data element by different runnables in a SWC

RTA-RTE now allows the use of PRPorts for explicit sender/receiver communication, in the case where one runnable has a `DataSendPoint` to a `VariableDataPrototype` via a PRPortPrototype, and another runnable in the same SWC has a `DataReceivePointByArgument` to that `VariableDataPrototype` via the same PRPortPrototype.

Runnables in a connected SWC can also have a `DataReceivePointByArgument` to the same data element via the connected SWC's `RPortPrototype`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948).

(RTE-3721)

RTEGen generate explicit write communication APIs for PRPorts

RTA-RTE now generates `Rte_Write` APIs for VariableAccesses in the `DataSendPoint` role, where the accessed data element exists in the context

of a PRPortPrototype. The APIs generated are the same as if a `PPortPrototype` was used in place of the PRPortPrototype.

(RTE-3720)

Version 6.4.0pr6

Support for COM-based transformer with workaround

RTA-RTE now allows you to generate code compatible with COM-based transformer.

When an `ISignal` is referenced by a `DataTransformation` which has a `SERIALIZER` `TransformationTechnology` and `protocol` set to `COMBased`, RTA-RTE invokes `Com_SendSignalGroupArray` or `Com_ReceiveSignalGroupArray` to send or receive data respectively.

RTA-RTE does not accept AUTOSAR-conformant configuration input for COM-based transformer; only the pattern decribed above.

(RTE-4015,RTE-4016)

CURVE, MAP and COM_AXIS in explicit port communication

When portdata typed by an `ApplicationPrimitiveDataType` of `category` `CURVE`, `MAP`, or `COM_AXIS` takes part in unqueued explicit sender-receiver comms over an `NvDataInterface` between an `ApplicationSwComponentType` instance and an `NvBlockSwComponentType` instance, RTA-RTE generates the APIs as if the port data had been typed by the mapped `ImplementationDataType`.

`DataReceivePointByValue` is not supported.

(RTE-3979)

PRPorts typed by a ModeSwitchInterface can now support a mode queue with queue size greater than one

RTA-RTE now supports a queue size greater than one for a mode in a PRPortPrototype typed by a ModeSwitchInterface.

(RTE-3814)

Error when data filtering is enabled in the context of a PRPortPrototype

RTA-RTE now raises error 3029 when a `NonqueuedReceiverComSpec` aggregated in a PRPortPrototype has a `dataFilter` attribute.

(RTE-3825)

Mode Switch task now only activates once with PRPorts

Formerly, when PRPorts were used and typed by a ModeSwitchInterface, the Mode Switch task would be activated twice.

This has now been fixed.

(RTE-4001)

Reverted behaviour from RTE-3591

RTA-RTE now no longer raises error 3011 when a `RunnableEntity` with a `ModeSwitchPoint` referencing a `ModeDeclarationGroupPrototype` in a PRPort is referenced by a `SwcModeSwitchEvent` that also references the same `ModeDeclarationGroupPrototype`.

(RTE-3980)

Version 6.4.0pr5

Reverted behaviour from RTE-3827

RTA-RTE now no longer raises error 3026 when more than one `DataWriteAccess` or `DataSendPoint` in the same `SwcInternalBehavior` references the same `VariableDataPrototype` in a PRPortPrototype.

(RTE-3982)

PRPorts: Support for implicit 1:N communication with the writer and at least one reader in the same runnable in the same SWC

RTA-RTE now supports the use of PRPorts for implicit sender/receiver communication, in the case where one runnable has a `DataWriteAccess` and a `DataReadAccess` to a `VariableDataPrototype` via a PRPortPrototype, and at least one other runnable in the same SWC has `DataReadAccess` to that `VariableDataPrototype` via the same PRPortPrototype.

Runnables in a connected SWC can also have `DataReadAccess` to the same data element via the connected SWC's `RPortPrototype`.

Note: issue RTE-3942 noted that the use of multiple reading runnables in the SWC containing the PRPort was untested. This is now no longer the case.

(RTE-3640)

COM_AXIS in port data

When portdata typed by an `ApplicationPrimitiveDataType` of `category COM_AXIS` takes part in implicit sender-receiver comms over an `NvDataInterface` between an `ApplicationSwComponentType` instance and an `NvBlockSwComponentType` instance, RTA-RTE generates the APIs as if the port data had been typed by the mapped `ImplementationDataType`.

(RTE-3962)

PRPorts: Support for implicit 1:N communication with the writer and readers in different runnables in the same SWC

RTA-RTE now supports the use of PRPorts for implicit sender/receiver communication, in the case where one runnable has `DataWriteAccess` to a `VariableDataPrototype` via a PRPortPrototype, and more than one other runnable in the same SWC has `DataReadAccess` to that `VariableDataPrototype` via the same PRPortPrototype.

Runnables in a connected SWC can also have `DataReadAccess` to the same data element via the connected SWC's `RPortPrototype`.

Note: issue RTE-3942 noted that the use of multiple reading runnables in the SWC containing the PRPort was untested. This is now no longer the case.

(RTE-3639)

Empty osNeeds.arxml fails schema validation

This applies when the `autosar40` OS is selected.

Formerly, if RTA-RTE did not find anything to report to the `osNeeds.arxml` file then it would write an empty file. This is not well-formed XML, so it fails schema validation.

Now, RTA-RTE writes a valid XML file containing an AUTOSAR tag containing only a comment.

(RTE-2590)

Validation of PRPorts configurations now covers PRPorts typed by SenderReceiverInterfaces

RTA-RTE now raises errors for invalid PRPorts configurations involving PRPortPrototypes typed by SenderReceiverInterfaces.

(RTE-3830)

Generation of McSupportData for data elements that exist in a PRPortPrototype

When measurement is enabled and a measurable data element exists in a PRPortPrototype, RTA-RTE now generates a `McDataInstance` for that data element in the `Rte_McSupportData.arxml` file.

(RTE-3985)

Crash when ApplicationErrors used with RIPS

Formerly, if ApplicationErrors were used in a client-server configuration with RIPS turned on, RTA-RTE would crash.

This has now been fixed.

(RTE-3729)

Version 6.4.0pr4

PRPorts: Support for implicit reading and writing of the same data element from a single runnable

RTA-RTE now allows the use of PRPorts for implicit sender/receiver communication, in the case where a single runnable has both `DataWriteAccess` and `DataReadAccess` to a `VariableDataPrototype` via a PRPortPrototype.

Within that single runnable, values written via the PRPort are made immediately available to subsequent reads, including subsequent reads with the same invocation of that runnable. This behaviour complies with AUTOSAR requirement `[SWS_Rte_08408]`.

Runnables in a connected SWC can also have `DataReadAccess` to the same data element via the connected SWC's `RPortPrototype`.

(RTE-3638)

Error when queued data used in the context of a PRPortPrototype

RTA-RTE now raises error 3027 when an instance of a `VariableDataPrototype` exists in the context of a PRPortPrototype, and also aggregates a `swImplPolicy` that is set to `queued`. The use of queued data elements with PRPorts violates AUTOSAR `[constr_1200]`.

(RTE-3822)

PRPorts: Support for implicit reading and writing of the same data element by different runnables in a SWC

RTA-RTE now allows the use of PRPorts for implicit sender/receiver communication, in the case where one runnable has `DataWriteAccess` to a

`VariableDataPrototype` via a PRPortPrototype, and another runnable in the same SWC has `DataReadAccess` to that `VariableDataPrototype` via the same PRPortPrototype.

Runnables in a connected SWC can also have `DataReadAccess` to the same data element via the connected SWC's `RPortPrototype`.

The runnables accessing the data via the PRPortPrototype can be scheduled in separate tasks. When a data element is written in a task via a PRPort, the shadow buffer for that data element will be filled at the start of the task body, in accordance with AUTOSAR requirement `[SWS_Rte_08418]`.

Support for PRPorts in sender/receiver communication is restricted to singly-instantiable SWCs (see issue RTE-3948), and RTA-RTE only supports one runnable with `dataWriteAccess` to a given `VariableDataPrototype` (see issue RTE-3827).

At present, the use of multiple reading runnables in the SWC containing the PRPort is untested.

(RTE-3942)

Error if a PRPort exists in a SWC which can have multiple instances

RTA-RTE now raises error 3028 when an `AtomicSwComponentType` with the attribute `supportsMultipleInstantiation` set to `TRUE` also aggregates a PRPortPrototype typed by a SenderReceiverInterface.

(RTE-3948)

Error if more than one VariableAccess writes to a PRPort

> **WARNING**
>
> This was reverted before release of 6.4.0. See issue RTE-3982.

RTA-RTE now raises error 3026 when more than one `DataWriteAccess` or `DataSendPoint` in the same `SwcInternalBehavior` references the same `VariableDataPrototype` in a PRPortPrototype. Info message 1001 is also raised for each `DataWriteAccess` or `DataSendPoint` referencing the `VariableDataPrototype`.

(RTE-3827)

Version 6.4.0pr3

CURVE and MAP in port data

When portdata typed by an `ApplicationPrimitiveDataType` of `category CURVE` or `MAP` takes part in implicit sender-receiver comms over an `NvDataInterface` between an `ApplicationSwComponentType` instance and an `NvBlockSwComponentType` instance, RTA-RTE generates the APIs as if the port data had been typed by the mapped `ImplementationDataType`.

(RTE-3941,RTE-3898)

Version 6.4.0pr2

Optimized copying of structure data types within explicit sender-receiver APIs

RTA-RTE now applies the optimization of using assignment instead of `Rte_memcpy`

for copying structure data, when possible, within generated `Rte_Read` APIs as well as generated `Rte_Write` APIs. Furthermore, in both APIs the optimization is now additionally performed when the data types are the same but configured using different definitions of the type, i.e. when identical `ImplementationDataTypes` with the same `shortName` but in different `Packages` are used.

Note that as a side-effect of this change the generation of `Rte_memcpy` calls within `Rte_Read` APIs, for arrays or when the aforementioned optimization cannot be applied, are now generated in a consistent manner to such calls in `Rte_Write` APIs; the changes are minor and do not change behaviour.

(RTE-2572)

Support for implicit write to data in a PRPortPrototype typed by a SenderReceiverInterface

RTA-RTE now supports a `RunnableEntity` having `DataWriteAccess` to a data element in a PRPortPrototype, where the PRPort is typed by a SenderReceiverInterface.

Reading from the PRPort in the same SWC is not yet supported, however if the PRPort is connected to an RPort on another SWC, the data written via the PRPort will be accessible from the connected SWC.

The generated RTE will contain `Rte_IWrite` and `Rte_IWriteRef` APIs corresponding to the runnable, port and data element. The code generated for these APIs is identical to that which would be generated if a PPort was used in place of the PRPort.

(RTE-3636)

Error when SenderReceiverInterfaces on the end of a connection using a PRPortPrototype are incompatible or the connection uses a PortInterfaceMapping

RTA-RTE now raises error 3021 when a PRPortPrototype typed by a SenderReceiverInterface is connected via an `AssemblySwConnector` to an `RPortPrototype` and the SenderReceiverInterfaces the ports aggregate are not compatible according to AUTOSAR `[constr_1069 part 1a]` and `[constr_1068 parts 1a and 1b]`.

RTA-RTE now raises error 3022 when a PRPortPrototype typed by a SenderReceiverInterface is connected via an `AssemblySwConnector` to an `RPortPrototype` and the `AssemblySwConnector` references a PortInterfaceMapping.

(RTE-3846)

Error when status is used with PRPortPrototypes or handleDataStatus is true or handleOutOfRange is not none for a PortPrototype

RTA-RTE now raises error 3020 when a PRPortPrototype aggregates a `NonqueuedReceiverComSpec` with `aliveTimeout` set to a value greater than zero.

RTA-RTE now raises error 3020 when a PRPortPrototype aggregates a `NonqueuedReceiverComSpec` with `handleNeverReceived` set to TRUE.

RTA-RTE now raises error 3020 when a PRPortPrototype aggregates a `NonqueuedReceiverComSpec` with `handleDataStatus` set to TRUE.

RTA-RTE now raises error 3020 when a PRPortPrototype aggregates a

`NonqueuedReceiverComSpec` with `handleOutOfRange` set to a value other than none.

RTA-RTE now raises error 3024 when a PortPrototype aggregates a `NonqueuedReceiverComSpec` with `handleDataStatus` set to TRUE.

RTA-RTE now raises error 3025 when a `RPortPrototype` aggregates a `ReceiverComSpec` with `handleOutOfRange` set to a value other than none.

RTA-RTE also now raises error 3025 when a `PPortPrototype` aggregates a `SenderComSpec` with `handleOutOfRange` set to a value other than none.

(RTE-3847)

Version 6.4.0pr1

Error when Acknowledgement or Invalidation is used with a PRPortPrototype or externalReplacement Invalidation is used with any PortPrototype

RTA-RTE now raises error 3018 when a PRPortPrototype typed by a SenderReceiverInterface aggregates a `SenderComSpec` that in turn aggregates a `TransmissionAcknowledgementRequest` as transmissionAcknowledge.

RTA-RTE now raises error 3019 when a PRPortPrototype is typed by a SenderReceiverInterface that aggregates an `InvalidationPolicy` with handleInvalid set to a value other than dontInvalidate.

RTA-RTE now raises error 3023 when any PortPrototype is typed by a SenderReceiverInterface that aggregates an `InvalidationPolicy` with handleInvalid set to externalReplacement.

(RTE-3845)

Error when PRPortPrototype data element is not implicitly written

RTA-RTE now raises error 3017 when a PRPortPrototype typed by a SenderReceiverInterface contains a data element that is not referenced by any `DataWriteAccess` from a `RunnableEntity`.

(RTE-3635)

RTEGen no longer raises error 3008 when a PRPort is typed by a SenderReceiverInterface

RTA-RTE no longer raises error 3008 when a PRPortPrototype is typed by a SenderReceiverInterface.

(RTE-3821)

4.39 Version 6.3.0

Version 6.3.0pr11

Internal error corrected in the case of complex inter-ECU sender-receiver transmission needing concurrency protection

Formerly, RTA-RTE failed with internal error 1002 when generating code for a configuration containing inter-ECU sender-receiver transmission of complex data where concurrency protection was required due to potential execution of the transmission from different execution contexts with possible pre-emption. This has now been corrected.

(RTE-3844)

Return status when internal errors are suppressed

When internal errors occur, RTA-RTE returns exit status 3.

Formerly, if the deprecated option `--error-as-warning` was used to change the internal error into a warning (not recommended) then the exit status 3 would still be returned.

This has now been corrected. An internal error subject to `--error-as-warning` is no longer regarded as an internal error, and does not cause the internal-error status code. If there are no other unsuppressed errors, the return status is 0.

(RTE-3862)

Enumeration Literals for SwSystemconst

In preview 10 of RTA-RTE 6.3.0, RTEGen would generate enumeration literals in the *Application Types Header File* to support the use of `CompuMethod`s in `SwSystemconst`s. (issue RTE-3333)

This behavior was found to be contrary to `[TPS_GST_00262]` and `[TPS_GST_00263]` and has been reverted. These requirements rule out using the symbolic literals in the C-code for the SW-C.

In case the symbolic literals are still required, `IncludedDataTypeSet`s can be used to cause RTEGen to generate the `#define`s.

(RTE-3840)

Correction of internal error in cases of global receive buffer sharing between implicit receivers with differing DataConstrs

Formerly, RTA-RTE failed with internal error 1002 when generating code for a configuration containing 1:N sender-receiver communication with implicit reception where the receivers were able to share a global receive buffer but had different DataConstrs, even when those DataConstrs were all compatible with those of the sender. This has now been corrected.

(RTE-3839)

RTEGen now raises error when mandatory EventSourceRef is missing

RTA-RTE now raises error 3016 when an event requires an EventSourceRef and that EventSourceRef is missing. The event types which this error applies to are `DataWriteCompletedEvent`, `DataSendCompletedEvent`, `AsynchronousServerCallReturnsEvent`, `ModeSwitchedAckEvent`, `InternalTriggerOccurredEvent` and `BswInternalTriggerOccurredEvent`.

(RTE-3785)

RTA-RTE now uses SwBaseType.shortName when SwBaseType.nativeDeclaration is missing

When a `NetworkRepresentation` applies to an `ISignal` that results in COM communication, RTA-RTE uses the `NetworkRepresentation.baseType.nativeDeclaration` to determine the C datatype of the conveyed data.

Previously where no `nativeDeclaration` was supplied RTA-RTE emitted an empty string as the type name, resulting in uncompilable code. Now the `NetworkRepresentation.baseType.shortName` is used, where this matches the name of a standard platform type.

RTA-RTE emits error 1796 in cases where no `nativeDeclaration` is supplied and

the `shortName` does not match a platform type.

(RTE-3787)

Version 6.3.0pr10

Comment and Info message now appear for VFB tracing on COM transmission of array elements

When inter-ECU transmission is present for an array, COM transmit hooks are enabled and VFB tracing is enabled, RTA-RTE now emits an info message 3015 for each `ISignalToIPduMapping` which maps to an array element to be transmitted. RTA-RTE also adds a comment to Rte.c in place of where the ComHook API for the array would be generated. Previously, RTA-RTE failed with an internal error in this case.

(RTE-3803)

Support use of PCT variation on ReadLocalVariable

RTA-RTE now supports basic PCT variation of a `VariableDataPrototype` in the role `implicitInterRunnableVariable`. When such a `VariableDataPrototype` is referenced by exactly one ReadLocalVariable, RTA-RTE supports the use of a `PreCompileTime` (PCT) `VariationPoint` on that ReadLocalVariable. PCT variation is not supported when a `VariableDataPrototype` is referenced by multiple ReadLocalVariables.

When a ReadLocalVariable is subject to PCT variation in this way, the `Rte_Cfg.h` file contains a condition value macro definition corresponding to the `VariationPoint`, and preprocessor guards testing that condition value are placed around the `ComponentDataStructure` elements, `ComponentDataStructure` initializers and `Rte_IrvIRead` API associated with the ReadLocalVariable.

(RTE-3530)

RTEGen now emits SwSystemconst referenced by VariationPointProxys

WARNING

This was reverted before release of 6.3.0. See issue RTE-3840.

When an `AtomicSoftwareComponent` aggregates a `VariationPointProxy`, the binding function of which references any `SwSystemconst` that aggregates a `compuMethod` with category `TEXTTABLE`, RTA-RTE now generates a C macro definition in the component type header file for each `CompuScale` with a point range that is aggregated by that `CompuMethod`.

Previously, no C macro definitions were generated in this situation, leading to uncompilable code.

(RTE-3333)

Version 6.3.0pr9

Fixed internal error when VFB tracing enabled and inter-ECU reception of array data exists

Previously, RTA-RTE failed with an internal error when VFB tracing was enabled for models containing inter-ECU reception of array data. This has now been

corrected.

(RTE-3666)

External/AUTOSAR40 folder renamed to External/AUTOSAR

An installation of RTA-RTE used to contain the folder External/AUTOSAR40. The folder has been renamed to External/AUTOSAR, with no changes to the folder contents.

(RTE-3528)

VSMD File for updated to AUTOSAR 4.3.0

The VSMD file shipped with RTA-RTE has been updated to AUTOSAR revision 4.3.0. Note that this does not imply implementation of any AUTOSAR feature. New features will have their own release notes.

(RTE-3759)

Error 2999 no longer erroneously appears alongside error 3013

RTA-RTE used to incorrectly raise error 2999 (for an InitEvent with no `RteEventToTaskMapping`) alongside error 3013 (for an InitEvent with an `RteEventToTaskMapping` but without an `RteMappedToTaskRef`). This has now been corrected so that when error 3013 is raised, it is raised without error 2999.

(RTE-3755)

Rte_SysCon definitions guarded in Application Header Files

When multiple `AtomicSwComponentTypes` define a `VariationPointProxy` with the same `shortName`, conflicting preprocessor definitions for the `SwSystemconst` are written to the application header files. If these conflicting preprocessor definitions are `#included` in the same file (for example, `Rte.c`) the result may be uncompilable.

As the SWC-specific definitions are not required when the application header files are included in `Rte.c`, the preprocessor definitions of the `SwSystemconsts` are now guarded by `RTE_CORE`, allowing `Rte.c` to compile:

Rte_swcA.h:

```
#ifndef RTE_CORE
#define Rte_SysCon_VPP0 Rte_SysCon_swcA_VPP0
#endif /* RTE_CORE */
```

Rte_swcB.h:

```
#ifndef RTE_CORE
#define Rte_SysCon_VPP0 Rte_SysCon_swcB_VPP0
#endif /* RTE_CORE */
```

(RTE-3398)

Make ——help output and Reference Manual documentation consistent

RTA-RTE The following options that are documented in the reference manual are now listed in the `--help` output: `--deviate-split-swci-support`, `--deviate-task-sections`, `--initial-value-rounding`, `--strict-initial-values-check`, `--use-partition-sections`.

(RTE-2981)

Version 6.3.0pr8

Error 1570 updated and raised when InitEvents mapped to the same OsTask have clashing position in task values

The text of error 1570 has been updated, and each occurrence of error 1570 is now accompanied by an info message 1001 for each relevant `RteEventToTaskMapping` or `RteBswEventToTaskMapping`.

Additionally, RTA-RTE now raises error 1570 when multiple InitEvents that are mapped to the same OsTask have clashing RtePositionInTask values.

(RTE-3687)

Floats generated with more precision

When writing non-integer values from doubles in 64-bit IEEE-754 representation, RTA-RTE now uses 17 significant figures (`limits<double>::max_digits10`) rather than 15 (`limits<double>::digits10`).

This has two main effects:

- The C compiler will have the same interpretation of the number as RTEGen. (Assuming it uses IEEE-754 representation internally).
- The loss of accuracy due to IEEE-754 will sometimes be explicit in the generated code.

Examples:

Expression:	0.14285714285714285
IEEE-754 64 bit to 30 sig figs:	0.142857142857142849212692681249
With double::digits10:	0.142857142857143
With double::max_digits10:	0.14285714285714285
Shortest Unique representation:	0.142857142857
Expression:	0.033333333333333333
IEEE-754 64 bit to 30 sig figs:	0.033333333333333333287074040640 62
With double::digits10:	0.0333333333333333
With double::max_digits10:	0.033333333333333333
Value to 30 sig figs:	0.033333333333333333287074040640 62
Shortest Unique representation:	0.0333333333333
Expression:	0.1
IEEE-754 64 bit to 30 sig figs:	0.100000000000000005551115123126
With double::digits10:	0.1
With double::max_digits10:	0.10000000000000001
Shortest Unique representation:	0.1

(RTE-3481)

Rte raises an error when an InitEvent is mapped to an OsTask which is does not exclusively contain InitEvents.

RTA-RTE raises error 3014 when an OsTask contains an InitEvent and that task

does not exclusively contain InitEvents.

(RTE-3674)

Rte raises an error when an InitEvent is not mapped to a OsTask

RTA-RTE no longer supports the use of InitEvent instances that are not mapped to an OsTask.

Error 3013 is now raised when a configuration contains an InitEvent instance that is referenced from an `RteEventToTaskMapping` that does not contain an `RteMappedToTaskRef`.

(RTE-3673)

InitEvents mapped to OsTasks are now supported

RTA-RTE now supports InitEvents that are mapped to an OsTask via an `RteEventToTaskMapping`. The associated RunnableEntity will be called from the body of the mapped task once that task becomes active.

When multiple InitEvents are mapped to the same OsTask, the order of the associated RunnableEntitys is determined by the values of RtePositionInTask in each `RteEventToTaskMapping`. Lower values of RtePositionInTask indicate earlier execution.

In order for the RunnableEntity to be executed on startup, the task must be scheduled to run once at system start with an appropriate priority. It is the responsibility of the ECU integrator to ensure that the tasks are scheduled appropriately.

(RTE-3675,RTE-3676,RTE-3677)

Rte raises an error when an InitEvent does not reference a RunnableEntity

RTA-RTE now raises error 3004 when an InitEvent does not reference a RunnableEntity in the role `startOnEvent`.

(RTE-3583)

Error 2994 no longer raised for clashing RtePositionInTask of unmapped SwcModeSwitchEvents

RTA-RTE no longer raises error 2994 in the case where two SwcModeSwitchEvents which are not mapped to any OsTask have clashing values of RtePositionInTask in their respective `RteEventToTaskMappings`.

(RTE-3669)

Rte raises errors for no position in task and out of bounds position in task

RTA-RTE now raises error 3002 when an InitEvent instance is referenced by an `RteEventToTaskMapping` which does not aggregate an RtePositionInTask.

RTA-RTE now raises error 3003 when an InitEvent instance is referenced by an `RteEventToTaskMapping` which aggregates an RtePositionInTask `. value` that is resolved to a numerical value larger than 65535.

(RTE-3582)

AUTOSAR 4.3.0 schema support

RTA-RTE now validates all input ARXML files (which must use the namespace `http://autosar.org/schema/r4.0`) against the AUTOSAR 4.3.0 schema, allowing configurations to contain elements defined by AUTOSAR versions later than

4.2.2. Note however that at present no AUTOSAR 4.3.0 features or behavioural changes are supported by RTA-RTE, so this is only useful for parts of the configuration that are not directly relevant to RTE generation.

Since the AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, any AUTOSAR XML declared as using that namespace and schema versions 4.0.1 up to 4.3.0 will validate against the 4.3.0 schema.

(RTE-3588)

Version 6.3.0pr7

Fix uncompilable code issue when data elements with binary-compatible RECORD ImplementationDataTypes share an implicit shadow buffer

RTA-RTE uses a common shadow buffer when RunnableEntitys have implicit access to RECORD data elements with different but binary compatible ImplementationDataTypes.

In previous versions, RTA-RTE used assignment operations in the generated fill and flush code for these shared buffers. The use of different types in these assignment operations caused compilation errors. This has now been corrected.

Where possible, RTA-RTE generates implicit shadow buffers with the same type as the shadowed global buffer or data channel, permitting the use of assignments in the fill and flush code. When a type mismatch cannot be avoided, RTA-RTE now generates fill and flush code using `Rte_memcpy` to avoid compilation errors.

> **WARNING**
>
> This fix only addresses the case where shadow buffers are shared between data elements with different *binary compatible* data types. Where data elements with *incompatible* types share a shadow buffer, or where data conversion or filtering apply, RTA-RTE will still generate incorrect code. For more details see issue EHI-496920.

(RTE-3507)

Version 6.3.0pr6

Assertion failure messages replaced with internal errors

Formerly, when RTA-RTE detected a violation of an internal assumption a message was printed but generation continued, with undefined results. Now internal error 1002 is raised, and hence generation fails, in such cases.

(RTE-3626)

Assertion failure messages during Flattening Root Composition

Formerly, RTA-RTE printed assertion failure messages during the 'Flattening Root Composition' step for some configurations. This has now been corrected.

(RTE-3404)

Support for enhanced mode API when used with PRPortProtype typed by ModeSwitchInterface

RTA-RTE now generates the enhanced `Rte_Mode` API when a runnable has

`ModeAccessPoint` to a `PRPortPrototype`, and the PRPort is typed by a `ModeSwitchInterface` that aggregates a `ModeSwitchSenderComSpec` with the `enhancedModeApi` attribute set to `true`.

(RTE-3433)

Support for ModeAccessPoint to PRPortPrototype typed by ModeSwitchInterface

RTA-RTE now supports `ModeAccessPoint` to a `PRPortPrototype` typed by a `ModeSwitchInterface`. Currently only the simple Rte_Mode API is supported.

The use of `PRPortPrototype` is restricted to SWCs used as mode managers, as referencing a `PRPortPrototype` as the `requester` from a `AssemblySwConnector` is not currently supported (see RTE-3572).

(RTE-3431)

Support for ModeSwitchPoint to PRPortPrototype typed by ModeSwitchInterface

RTA-RTE now supports `ModeSwitchPoint` to a `PRPortPrototype` typed by a `ModeSwitchInterface`. The triggering of any associated `ModeSwitchEvent` by the `Rte_Switch` API is not currently tested.

(RTE-3432)

Version 6.3.0pr5

Validation of accesses to sender-receiver port data in conjuction with inter-OsApplication direct server calls

This issue relates to accesses to port data from server RunnableEntitys that are unmapped for invocation by direct function call within the relevant `Rte_Call` APIs. When the `--client-server-global-optimization` option is enabled these direct function call invocations may occur across OsApplication boundaries.

In the case that a server RunnableEntity is invoked by direct function call from multiple OsApplications and accesses sender-receiver port data RTA-RTE now raises error 1360; previously invalid code was generated in this case.

Note that this validation does not apply to RIPS-relevant data, for which the RIPS macros are assumed to handle the buffering needs for all possible execution contexts.

(RTE-3188)

Improved validation of VariableAccesses and ParameterAccesses

RTA-RTE now raises error 2134, 2135, 2415 or 2422 when it encounters a VariableAccess where the referenced data element is not valid for the role of the VariableAccess. For example a VariableAccess in the role `DataSendPoint` that references a data element in a require port is now rejected with error 2134.

Furthermore, RTA-RTE now raises warning 2133 when it encounters a VariableAccess that contains no `accessedVariable` reference at all, or a ParameterAccess that contains no `accessedParameter` reference.

Previously the behaviour in such cases was undefined.

(RTE-3169)

RTEGen now raises an error when a PRPortPrototype is found on an unsup-

portedclass of SwComponentType in contract phase

If in `RTE Contract Phase` error 3012 will be raised if an `AtomicSwComponentType` is selected and aggregates one or more `PRPortPrototypes`.

(RTE-3621)

Removal of inter-ECU communication report

RTA-RTE no longer produces the inter-ECU communication OIL file `rta-com.oil`.

(RTE-3517)

RTEGen now raises an error when a PRPortPrototype is found on an unsupported class of SwComponentType in generation phase

When a `SwComponentPrototype` is instantiated on the selected ECU during `RTE Generation Phase` and is typed by a `ParameterSwComponentType`, `CompositionSwComponentType`, `EcuAbstractionSwComponentType`, `NvBlockSwComponentType`, `SensorActuatorSwComponentType`, `ComplexDeviceDriverSwComponentType`, `ServiceProxySwComponentType`, or `ServiceSwComponentType` and aggregates one or more `PRPortPrototypes` then error 3012 is raised.

(RTE-3592)

Errors raised with mixture of terms used for missing information

RTA-RTE previously raised errors with a mixture of terms (“undefined”, “<undefined>”, “null”, “<null>”) as placeholder output for missing parameters. RTA-RTE now raises these errors using “<undefined>” for missing parameters.

(RTE-3526)

RTEGen now raises an error for RunnableEntity triggering itself involving PRPort

WARNING

This was reverted before release of 6.4.0. See issue RTE-3980.

RTA-RTE now raises error 3011 when a RunnableEntity with a `ModeSwitchPoint` referencing a `ModeDeclarationGroupPrototype` in a PRPort is referenced by a SwcModeSwitchEvent that also references the same `ModeDeclarationGroupPrototype`. Info message 1001 is also raised for each SwcModeSwitchEvent involved.

(RTE-3591)

Version 6.3.0pr4

Raise error for fan-in communication involving PRPort

RTA-RTE now raises error 3010 when a `PRPortPrototype` is part of a fan-in communication graph.

(RTE-3590)

Raise error when PRPort is not typed by a ModeSwitchInterface

RTA-RTE now raises error 3008 when a `PRPortPrototype` is typed by a `PortInterface` that is not a `ModeSwitchInterface`.

(RTE-3573)

Raise error when AssemblyConnector.requester is PRPort

RTA-RTE now raises error 3007 when a `PRPortPrototype` is referenced as `requester` from an `AssemblySwConnector`.

(RTE-3572)

Raise error when DelegationSwConnector references PRPortPrototype

RTA-RTE now raises error 3006 when a `PRPortPrototype` is referenced as `innerPort` or `outerPort` from a `DelegationSwConnector`.

(RTE-3571)

Raise error when configuration is not ECU_EXTRACT and PRPortPrototype used

RTA-RTE now raises error 3005 when an input configuration describes an AUTOSAR `System` of a category other than `ECU_EXTRACT`, and it includes one or more `PRPortPrototype`s.

(RTE-3570)

Raise error when InitEvent is mapped zero or many times

If an InitEvent instance is not referenced by any `RteEventToTaskMapping`, RTA-RTE now rejects the configuration with error `2999`.

If an InitEvent instance is referenced by more than one `RteEventToTaskMapping`, RTA-RTE now rejects the configuration with error `3000` and uses info `3001` to list the `RteEventToTaskMappings` causing the error.

(RTE-3581)

Version 6.3.0pr3

Validation for when InitEvent is mapped to an OsTask

RTA-RTE now raises error 2998 where a configuration contains an InitEvent instance that is referenced from an `RteEventToTaskMapping` containing an `RteMappedToTaskRef`.

(RTE-3580)

Interleaving of InitEvents and InitMode SwcModeSwitchEvents now supported

RTA-RTE now supports the case where a configuration contains one or more InitEvents and one or more initial SwcModeSwitchEvents.

In the generated code, the RunnableEntitys referenced by the InitEvents and initial SwcModeSwitchEvents are called by `Rte_IModeInit`, which is in turn called by `Rte_Start`.

The order of the RunnableEntitys is determined by the RtePositionInTask values of the relevant `RteEventToTaskMapping`s. These values must be unique. RTA-RTE will raise error 2994 if any combination of InitEvents and initial SwcModeSwitchEvents are mapped to the same RtePositionInTask.

RTA-RTE only supports InitEvents interleaved with initial SwcModeSwitchEvents where the `RteEventToTaskMapping` for each does not reference an OsTask or `RteInitializationRunnableBatch`.

(RTE-3515)

Validation schema can be overridden in RTEGen.ini file

The XML schema used by RTA-RTE to validate AUTOSAR XML description files can now be overridden from the RTEGen.ini file.

The location of the schema file is supplied using the key `ValidateARXMLAgainstSchema=<path>` in the `[Options]` section:

```
[Options]
\* ValidateARXMLAgainstSchema\=c\:\\schemas\\schema_file.xsd
```

This can be supplied either as an absolute path, or as a relative path in which case the base location is the folder containing the `RTEGen.exe` executable.

> **WARNING**
>
> Overriding the validation schema is for development use only, and the generated code must not be used in production. RTA-RTE will raise warning 2996 whenever the validation schema is overridden.

(RTE-3491)

Restore file header in Task body source files

In `Vendor` mode, RTA-RTE generates seperate source files for each of the task bodies.

In release 6.1.0, the generation of these files was perturbed and the source files were emitted with the headers omitted. The correct file headers have now been restored.

(RTE-3544)

Renamed PlatformBaseTypes file

The sample PlatformBaseTypes file has been renamed to `AUTOSAR_MOD_PlatformBaseTypes_RTARTE.arxml`, to show that the file is customized for the sample application.

(RTE-3198)

Additional validation where InitEvent instances share RtePositionInTask across SWC instances

RTA-RTE now raises error 2994 in the cases where InitEvents across differing SWC instances or differing SWCs are mapped to the same RtePositionInTask.

(RTE-3535)

Deprecated option ——ioc-xml-namespace

The `--ioc-xml-namespace` option has been deprecated, as well as its abbreviation `--iocxmlns`. RTA-RTE now raises warning 2976 if `--ioc-xml-namespace`, or its abbreviated version, `--iocxmlns`, is used.

The `--ioc-xml-namespace` option should not be used for new projects, and where possible existing projects should be modified to no longer use it.

(RTE-3511)

Version 6.3.0pr2

Multiple InitEvents now supported

RTA-RTE now supports the case where a configuration contains multiple InitEvents and no initial `ModeSwitchEvent`s.

In the generated code, the RunnableEntitys referenced by the InitEvents are called by `Rte_IModeInit`, which is in turn called by `Rte_Start`.

The order of the RunnableEntitys is determined by the RtePositionInTask values of the relevant `RteEventToTaskMapping`s. These values must be unique. RTA-RTE will raise error 2994 if multiple InitEvents are mapped to the same RtePositionIn-Task.

RTA-RTE only supports InitEvents where the `RteEventToTaskMapping` does not reference an OsTask or `RteInitializationRunnableBatch`.

(RTE-3514)

RTEGen.ini: Schemas and URIs sections unrecognized

RTA-RTE no longer recognizes sections '[URIs]' and '[Schemas]'. RTA-RTE will raise error 2990 if unrecognized sections appear in the RTEGen.ini file.

(RTE-3520)

RTEGen uses internal schemas

Previously, RTA-RTE loaded schemas provided by the 'Schemas' section in the RTEGen.ini file. Now, the AUTOSAR 4.2.2 schema is built into the RTA-RTE executable and is loaded automatically. The 'Schemas' section in RTEGen.ini is now not used.

(RTE-3488)

Supplied platform types conform with AUTOSAR 4.2.2

The example platform types files shipped with RTA-RTE now conform with AUTOSAR 4.2.2. `uint64` and `sint64` types are included, and the package location for base types is now `AUTOSAR_Platform/BaseTypes`.

(RTE-2721)

Single InitEvent now supported

RTA-RTE now supports the case where a configuration contains a single InitEvent and no initial `ModeSwitchEvent`s. In this case, the RunnableEntity referenced by the InitEvent is called by `Rte_IModeInit` (which is in turn called by `Rte_Start`) in the generated code.

(RTE-3379)

Removed ––bsw-xml-namespace option

The –bsw-xml-namespace option and its short form `--bswxmlns` have been removed.

If either the –bsw-xml-namespace or `--bswxmlns` options are provided, RTA-RTE will exit with error 1103.

(RTE-3510)

Incorrect RTA-IOC write group generation when runnables in different tasks implicitly write the same data element

Formerly, when multiple RunnableEntitys within a software component had implicit write access to the same data element within the same `Port` but were mapped to different tasks, RTA-RTE could generate the RTA-IOC write groups for the implicit shadow buffer flushes within those tasks incorrectly. This led to RTA-IOC failing to generate the optimized APIs or to data being written to the wrong destinations. This has now been corrected.

(RTE-3521)

Removed ——xmlentity option

The –xmlentity option and its short form `--xe` have been removed.

If either the –xmlentity or `--xe` options are provided, RTA-RTE will exit with error 1103.

(RTE-3486)

Removed ——xmlschema option

The –xmlschema option and its short form `--xs` have been removed.

RTA-RTE now validates all input files with the default supported schema version. AUTOSAR XML schemas are designed to be backwards-compatible within the AUTOSAR 4.0 namespace, so any AUTOSAR XML description declaring the namespace `http://autosar.org/schema/r4.0` using schema versions 4.0.1 up to 4.2.2 will validate against the 4.2.2 schema.

If either the –xmlschema or `--xs` options are provided, RTA-RTE will exit with error 1103.

(RTE-3485)

Validation of AUTOSAR XML Description fragments

Validation of input files has been tightened. AUTOSAR XML description files provided to RTA-RTE must now declare `http://autosar.org/schema/r4.0` as the default namespace `[TR_APRXML_00052]`.

RTA-RTE validates all AUTOSAR XML Description fragments against http://autosar.org/schema/r4.0 using a pre-loaded copy of the AUTOSAR schema for AUTOSAR revision 4.2.2. Thus, RTA-RTE accepts input files created with tools conforming to any AUTOSAR revision from 4.0.1 to 4.2.2 inclusive.

The `schemaLocation` attribute is ignored and it is not supported to dynamically load schemas declared in the instance documents. (To support this change, RTA-RTE also pre-loads the XML schema.)

Non-conforming input files are rejected with error 2991.

(RTE-3487)

Rejection of initialization runnable batches as unsupported

Previously, RTA-RTE silently ignored `RteInitializationRunnableBatch` containers and any mappings of `ModeSwitchEvent`s to them. These are now detected and rejected as unsupported with error 2993.

(RTE-3513)

Version 6.3.0pr1

Invalid code generated for linear data conversion requiring intermediate

values overflowing 64-bits

Previously, RTA-RTE correctly raised error 2692 for a linear data conversion requiring intermediate values overflowing 32-bits but representable using 64-bits, when the `--have-64bit-int-types` command-line option was not enabled, but silently generated invalid code for a linear data conversion requiring intermediate values overflowing 64-bits.

This has now been corrected and error 2692 is raised whenever a linear data conversion requires intermediate values that overflow the largest available platform data type.

(RTE-3479)

Platform types path now uses AUTOSAR_Platform rather than AUTOSAR_-PlatformTypes

RTA-RTE now reads and writes references to the AUTOSAR platform types using the package path `AUTOSAR_Platform`. This is the correct path for all AUTOSAR versions from 4.0.3 onwards.

(RTE-3442)

Removed stale source code from Examples folder

Until recently, RTA-RTE shipped with a number of headers originally intended to help you build the example application for several targets. This is no longer the case.

(RTE-2969)

ErrorAsWarningOption disabled in provided RTEGen.ini file

RTA-RTE now has the deprecated option `--error-as-warning` disabled by default in the provided RTEGen.ini file. This means that if `--error-as-warning` is provided on the input and the provided RTEGen.ini file is used then error 1123 will be raised by RTA-RTE.

The use of the `--error-as-warning` option is discouraged because it causes undefined behavior. Any configurations requiring the `--error-as-warning` option should be updated to avoid the referenced error being raised.

(RTE-3204)

Error raised when ISignal does not reference SystemSignal

RTA-RTE now raises error 2989 if an `ISignal` in the input configuration does not reference a `systemSignal`.

(RTE-3028)

4.40 Version 6.2.0

Version 6.2.0

Fixed internal error when a SystemSignal or SystemSignalGroup mapped to port data elements with different data types

Previously, RTA-RTE raised internal error 1413 for configurations where the same SystemSignal or SystemSignalGroup was mapped to multiple port data elements with different data types.

This issue has been addressed so that these configurations now no longer raise

an internal error.

> **WARNING**
>
> Although these configurations no longer raise an error, they may cause RTA-RTE to generate incorrect code. For more details, refer to issue RTE-3464.

(RTE-3458)

Internal error when receiving invalidatable inter-ECU data

In PR5 RTA-RTE incorrectly raised error 7417 when recieving Inter-ECU data with invalidation enabled. This error is no longer incorrectly raised.

(RTE-3448)

Rte_SwitchAck in unmapped server runnable

This concerns the case where a `ModeSwitchPoint` is used with acknowledgement in a `RunnableEntity` for which RTA-RTE cannot determine the calling context (for example, a server runnable that is not mapped to an `OsTask`).

Formerly, RTA-RTE could generate the code with an unsafe level of concurrency protection or reject the configuration with error 6708.

Now, RTA-RTE generates the API and protects the assignment in the API body with `SuspendAllInterrupts`.

(RTE-3450)

Deprecated option ——os-xml-namespace

The `--os-xml-namespace` option has been deprecated, as well as its abbreviation `--osxmlns`. RTA-RTE now raises warning 2976 if `--os-xml-namespace`, or its abbreviated version, `--osxmlns`, is used.

Any configurations requiring the `--os-xml-namespace` option should be updated to avoid the referenced error being raised.

(RTE-3357)

Updated transformers User Guide documentation, removed KIs RTE-3068, RTE-3121 and RTE-3122

The documentation in the User Guide describing the use of data transformation in RTA-RTE has been updated to provide additional information on the support for `ECU_EXTRACT` and to outline constraints on the mapping of signals when transformation is in use.

Additionally, a number of known issues have been subsequently addressed and so their entries are no longer applicable. The entries for KIs RTE-3068, RTE-3021 and RTE-3022 have been removed from the document.

(RTE-3367)

Error when multiple ISignals reference the same SystemSignal and transformation in use

RTA-RTE now raises error 2986 when a configuration contains a SystemSignal that has multiple `ISignal`s mapped to it, and any of those `ISignal`s references a

`DataTransformation`.

(RTE-3378)

Error when multiple SystemSignals mapped to a data element and transformation in use

RTA-RTE now raises error 2979 when a configuration contains a data element that has multiple SystemSignals mapped to it, when any of those SystemSignals is referenced by an `ISignal` that references a `DataTransformation`.

(RTE-3214)

Deterministic order of error messages

Formerly, RTA-RTE visited SWCs in a non-deterministic order when creating Application Header Files. This could result in errors being written in a different order from one run to the next. This has now been fixed.

(RTE-3394)

Uncompilable code for implicit writes to unconnected or unread data elements when implicit shadow buffer optimization enabled

Previously, when the `--implicit-use-global-buffers` option was enabled, RTA-RTE generated uncompilable code for implicit writes to sender-receiver data elements in unconnected provide ports or in connected provide ports with no readers of the connected required data elements.

This has now been corrected.

(RTE-3291)

Deprecated option ——error-as-warning

The –error-as-warning option has been deprecated, as well as its abbreviation `--eaw`. RTA-RTE now raises warning 2976 if –error-as-warning, or its abbreviated version, `--eaw`, is used.

The use of this option causes undefined behavior. Any configurations requiring the –error-as-warning option should be updated to avoid the referenced error being raised.

(RTE-3364)

Version 6.2.0pr5

RTA-IOC integration in the case of implicit and explicit access to the same data element or connected data elements

RTA-RTE no longer generates code for using optimized RTA-IOC APIs, when available, for implicit shadow buffer fills and flushes for data elements that are also accessed via explicit sender-receiver APIs or that are connected to other senders / receivers in the same OsApplication that are accessed via explicit sender-receiver APIs.

Previously, uncompilable code was generated in these cases, because RTA-RTE does not yet support RTA-IOC integration within explicit sender-receiver APIs and RTA-IOC does not (and cannot) generate plain AUTOSAR IOC APIs for IOC channels for which the optimized RTA-IOC APIs are generated.

Now RTA-RTE uses the plain AUTOSAR IOC APIs for both the implicit and explict accesses for such data elements.

(RTE-3403)

Error when a SwcModeSwitchEvent has zero or more than two ModeDeclarations referenced

RTA-RTE now raises error 2987 where a `SwcModeSwitchEvent` references zero `ModeDeclarations` as `mode`. It also raises error 2987 in the case of a `SwcModeSwitchEvent` referencing more than two `ModeDeclarations` as `mode` when schema validation has been turned off.

(RTE-3356)

Location information corrected for several messages

Message 1309 was found to be logging an AUTOSAR path in place of its filename.

Error logging internals have been updated with the result that messages 1307, 1308, 1309, 1310, 1311, 1312, 1313, 1314, 1315, 6127 and 6146 now write correct location info.

(RTE-3362)

Error when data transformation is used with nested compositions

RTA-RTE now raises error 2985 where a transforming `ISignal` is mapped to a delegation port on the `RootSwComposition`, and that delegation port is connected via a `DelegationSwConnector` to a delegation port on a `CompositionSwComponent`.

(RTE-3186)

Error when data transformation is used with a delegation port that uses a PortInterfaceMapping

RTA-RTE now raises error 2984 where a transforming `ISignal` is mapped to a delegation port on the `RootSwComposition`, and the `DelegationSwConnector` mapping that delegation port to the inner port references a `PortInterfaceMapping`.

(RTE-3183)

Version 6.2.0pr4

Error when data transformation is used with a non-ECUExtract system

RTA-RTE now raises error 2983 if data transformation is used when `System.category` is not `ECU_EXTRACT`. Data transformation is defined as being used when an `XfrmImplementationMapping` exists in the ECU Value file.

(RTE-3185)

Data transformation support for signals that are mapped to delegation ports

RTA-RTE now supports data transformation where a transforming `ISignal` is mapped to a delegation port on the `RootSwComposition`, and the delegation port is typed by an interface containing interface elements that are directly compatible with those in the inner port (i.e. the interfaces are compatible without the use of a `PortInterfaceMapping`).

(RTE-3182)

Error when ManagedModeGroup references are not valid

RTA-RTE now raises error 2982 in the case where a `BswModuleEntity` references as a `ManagedModeGroup` a `ModeDeclarationGroupPrototype` that is not aggregated as a `ProvidedModeGroup` within the same `BswModuleDescription` as the

`BswModuleEntity`.

(RTE-2966)

Removed ——os-file and ——ioc-file options

The `--os-file` and `--ioc-file` options have been removed.

RTA-RTE now generates the OS configuration in the file `osNeeds.arxml`, and the IOC configuration in the file `iocNeeds.arxml`. If different filenames are required for subsequent toolchain steps, rename the generated files after generation is complete.

If either of the `--os-file` or `--ioc-file` options are provided, RTA-RTE will exit with error 1103.

(RTE-3307)

Rte.c does not compile when unconnected Rx ports are present in SC3 or SC4 configuration

In previous versions, RTA-RTE generated uncompilable code in the case of reading a data item from an unconnected require port in a configuration containing OsApplications. This has now been corrected.

(415750)

Version 6.2.0pr3

Removed option ——deviate-allow-unconnected-receiver

Command-line option –deviate-allow-unconnected-receiver is now removed.

Please consider using `--strict-unconnected-rport-check` instead.

(RTE-973)

Generation of Rte_IrvIWriteRef API for split SWC configurations

RTA-RTE now generates an `Rte_IrvIWriteRef` API in split SWC configurations, where the IRV is accessed across partition boundaries.

Previously, the `Rte_IrvIWriteRef` API was only generated for atomic SWCs.

(RTE-3215)

Spurious suffix on RTA-IOC write group parameter name

This issue relates to the names of RTA-IOC write group parameters for implicit writes to port data with fan-out.

When the connected requirers have different data representations, RTA-RTE creates write group parameters for each representation in the write group for the flush of the implicit shadow buffer. The names of such write group parameters are derived by appending a unique suffix to the implicit shadow buffer name.

Formerly, RTA-RTE added a suffix to such names even when there was only one inter-OsApplication connection with actual readers of the requirer in the fan-out, i.e., only one write to an IOC channel was used and one such write group parameter actually generated for the implicit shadow buffer.

Now, the suffix is added only when more than one write group parameter is actually generated for the flush of the implicit shadow buffer.

(RTE-3157)

Transmission side DataTransformation supported across OsApplications

RTA-RTE now supports DataTransformation for data transmission when COM and the `ApplicationSoftwareComponent` execute in different `OSApplications`.

(RTE-3235)

Version 6.2.0pr2

Spurious VariableAccess reference errors when different split SWCIs contain inter-OsApplication IRVs with the same name

Previously, RTA-RTE raised spurious errors in the case that different split SWCIs contained inter-OsApplication `InterRunnableVariables` with the same name. This has now been corrected.

(RTE-3216)

Error when IOC is used with receive side DataTransformation

RTA-RTE now raises error 2981 when a configuration requires transformed data to be transported via an IOC channel. An IOC channel is required for data reception via COM when COM and the `ApplicationSoftwareComponent` execute in different `OSApplications`.

(RTE-3208)

Corrected validation for atpSplitable merging

Previously RTA-RTE attempted to merge a `ServiceSwComponentType` and a `BswModuleDescription` if they had the same path. Now, it rejects this case with error 2012. In general RTA-RTE performs improved validation for atpSplitables with the same path.

(RTE-2915)

Error 6801 downgraded to a warning

RTA-RTE now raises warning 6801 and continues generating in the case where RTA-RTE calculates that an input configuration requires more than 10,000 expiry points. Previously, RTA-RTE aborted with error 6801 in this situation.

(RTE-2927)

Validation of execution contexts for implicit accesses to port data

This issue relates to any `RunnableEntity` that has implicit read or write access to port data and is not mapped to a task (i.e., its referencing `RTEEvents` are not mapped to a task).

Formerly, the behavior of RTA-RTE was undefined for such runnables, except where the runnable was a pure server and every client was mapped to some task.

Now, RTA-RTE rejects such runnables in cases where the implicit semantics cannot be implemented. This covers runnables in the following conditions:

- It can never execute because nothing triggers it and, if it is mapped to a `BswModuleEntity`, nothing triggers the `BswModuleEntity`.
- It can never execute because the referencing `RTEEvent` is not mapped to a task, and no API has been generated in which to write the fill and flush code for the runnable. For example, this is the case for an unmapped server for an operation in an unconnected provide port.
- It can never execute because the referencing `RTEEvent` is not mapped to a

task, and although an API has been generated, the `RunnableEntity` that calls the API cannot execute.

- It is synchronized with a `BswInterruptEntity` so executes in an interrupt context.
- It is directly invoked in an API that can be called in an interrupt context.

Furthermore, RTA-RTE now rejects such runnables where an implementation would be technically possible but is not yet supported. This comprises runnables started by the following means:

- Direct function call in an `Rte_Switch` API
- Direct function call in an `Rte_Trigger` API
- Direct function call in an `Rte_IrTrigger` API
- Synchronization with a `BswCalledEntity`.

Note that none of this validation applies to RIPS-relevant data, for which the RIPS macros are assumed to handle the buffering needs for all possible execution contexts.

(RTE-3172)

Fixed internal error when transformers used with ––append-name-to-buffer=1 option

In previous releases, the use of data transformation in combination with the `--append-name-to-buffer` option would cause an internal error if the appended VDP name contained an underscore character followed by a number. This has now been fixed and the VDP name is appended correctly.

(RTE-3202)

Error when multiple inner ports connected via delegation connectors to outer port involved in DataTransformation

RTA-RTE now raises error 2980 when a delegation port on the `RootSwComposition` is referenced by multiple `DelegationSwConnectors`, and the delegation port contains a data element that is subject to data transformation.

(RTE-3196)

Version 6.2.0pr1

Invalidation for a sender-receiver communication with no reader

This item applies to cases where unqueued sender-receiver communication is configured with invalidation enabled but the receiving software component does not have any read access to the port data.

Previously, RTA-RTE allocated a global receive buffer and status variable for such a connection and generated a write of the relevant invalid value to the buffer and set of the invalid bit in the status variable in the `Rte_Invalidate` API, even though the buffer and status variable were otherwise unused. This has now been corrected and the `Rte_Invalidate` API now only performs any necessary runnable activations (for `DataReceiveErrorEvents`) in such cases.

(RTE-3209)

Improved concurrency protection within implicit shadow buffer fill and flush code

RTA-RTE now generates implicit shadow buffer fill and flush code in tasks, and the neccessary concurrency protection code around it, in a way that is sometimes more optimal than previously and that allows more user control over the trade-off between runtime performance and jitter in higher priority tasks. In particular, RTA-RTE now groups together copies needing protection as much as possible so that fewer blocks of protection are required, subject to user preference regarding jitter, and copies within fill blocks are no longer grouped by class of data type.

The `--complex-dra-protection-strategy` `--dra-protection-mechanism` command-line options have been removed and replaced with more general and finer-grained options: `--protection-threshold-copy-bytes` and `--preferred-intra-core-protection-scheme`. At present these apply only to implicit shadow buffer fill and flush code within task bodies but ETAS intends to extend their applicable scope in the future.

Full details of the new options can be found in the Reference Manual.

As part of this work, improvements have also been made regarding the generation of concurrency protection around Autosar IOC API calls and RTA-IOC API calls, removing the protection in cases where it was previously generated but not needed. Furthermore, with the removal of the grouping of fills by data type, a single RTA-IOC grouped read is now created for each task fill block, containing all the IOC reads.

Note that whilst this change may result in a different order of fills and / or flushes compared to previous RTA-RTE versions no functional change results and the order remains stable for a given input configuration and set of command-line options.

Note also that the `RTE_ATOMIC<x>` macros are no longer used for protection within the implicit fill and flush blocks, since the `--atomic-assign` option already provides a finer-grained mechanism for specifying the data types for which simple copies are atomic operations on the target platform. ETAS intends to deprecate the use of these macros over time.

(RTE-3016)

RTA-IOC integration for implicit writes involving fan-out to intra- and inter-OsApplication receivers

RTA-RTE now generates RTA-IOC API calls as part of the implicit shadow buffer flush code for writes involving fan-out to a mixture of intra- and inter-OsApplication receivers, in addition to writes to only inter-OsApplication receivers. These calls and supporting declarations are made within a pre-processor conditional section and standard Autosar IOC API calls are made when the symbol `RTA_IOC_OPTIMIZED` is not defined.

Note that at present RTA-IOC integration still does not occur within explicit sender-receiver APIs, nor for implicit writes involving runnable activations, filtering or fan-out involving NV data receivers, nor for implicit communication involving feedback or status. Note however that a few minor, non-functional,

changes may be observed within explicit sender-receiver APIs, mostly to generated comments, in corner cases such as where there is a connection to a requirer but no actual reads from it. In particular, note issue RTE-3209.

To facilitate this extension of RTA-IOC integration, the way that concurrency protection is generated within implicit shadow buffer fill and flush code has been changed and the order of fills and flushes may change compared to previous versions of RTA-RTE; see issue RTE-3016.

(RTE-2813)

Data transformation now requires signals to be mapped to delegation ports

RTA-RTE now requires that signals conveying transformed data in an AUTOSAR `System` with category `ECU_EXTRACT` must be mapped to a delegation port on the `RootSwComposition`.

Error 2978 will be raised when a signal conveying transformed data is mapped to any port that is not a delegation port on the `RootSwComposition`. In earlier releases no error was raised in this situation.

(RTE-3181)

Data transformation when signals are mapped to delegation ports

RTA-RTE now supports data transformation for AUTOSAR `Systems` with category `ECU_EXTRACT`, where a transforming `ISignal` is mapped to a delegation port on the `RootSwComposition`. The delegation port must aggregate the same interface as the connected port on the inner SWC.

In earlier releases incorrect code was generated when transformation was requested for an `ISignal` mapped via a delegation port.

(RTE-3179)

——error-as-warning option deprecated

The reference manual has been updated to make it clearer that the behavior of RTA-RTE is undefined when –error-as-warning is used.

(RTE-3191)

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Version 6.1.0

Deprecated ——operating-system option

Support for operating systems other than an AUTOSAR 4.0 OS is now deprecated.

The only valid value for command-line option –operating-system is `autosar40`, in lower case.

See issue RTE-2456.

(RTE-3163)

Implicit write access to queued communication

AUTOSAR forbids implicit read access to queued communication but does not specify whether or not implicit write access is supported. RTA-RTE now supports implicit write access to queued communication, with the data sent once when the shadow buffer is flushed. Previously the behaviour was undefined.

(RTE-3161)

Support for data transformation

- Explicit unqueued sender/receiver communications: RTA-RTE supports data transformation for explicit sender/receiver communications using the `Rte_Read` and `Rte_Write` APIs.
- Serialization with SOME/IP plus optional end-to-end protection: RTA-RTE supports serialization of the input data using the SOME/IP transformer, optionally followed by end-to-end protection using the E2E transformer.
- Out-of-place buffering: RTA-RTE supports the use of the E2E transformer when it is configured for out-of-place buffering.

(RTE-3155)

Upgrade of warning 2173 to an error

Previously, RTA-RTE would issue a warning in the case of multiple references in a VariableAccess and continue processing. However, this would sometimes lead to an internal error.

RTA-RTE now issues message 2173 as an error.

(RTE-3134)

Multiple VariableDataAccesss on inter-partition implicit communication endpoint

This item applies when multiple VariableDataAccesss give rise to multiple generated APIs for the SWC to access the same port data, including at least one implicit API, and the connection crosses a partition (`OsApplication`) boundary.

Previously, RTA-RTE failed to generate such a configuration with an internal error. This has now been corrected.

(RTE-3125)

Intra-partition queued communication with no writer

This item applies to intra-partition (intra-`OsApplication`) cases where queued sender-receiver communication is configured but the sending software component does not have any write access to the port data.

Previously, RTA-RTE allocated a queue that was never written to. The generated `Rte_Receive` API performed a read on the queue, returning `RTE_E_NO_DATA` if non-blocking, returning `RTE_E_TIMEOUT` after expiry of the timeout if blocking. As well as an unnecessary allocation, this meant that the API provided no indication to the caller that data could never be received, potentially leading to needless repeated polling by caller.

Now, RTA-RTE no longer creates a queue in such cases and the `Rte_Receive` API returns `RTE_E_UNCONNECTED`.

See also issue RTE-3151.

(RTE-3153)

Inter-partition communication with no sender (queued)

This item applies to cases where queued sender-receiver communication crosses a partition (`OsApplication`) boundary but no `RunnableEntity` has write access, hence there is no sending API.

Formerly, RTA-RTE either crashed or generated invalid code that attempted to read from a non-existent IOC channel.

RTA-RTE now correctly generates code that always indicates that there is no data available, but see issue RTE-3151.

(RTE-3132)

Inter-partition communication with no writer (unqueued)

This item applies to cases where unqueued sender-receiver communication crosses a partition (`OsApplication`) boundary but no `RunnableEntity` has write access, hence there is no writing API, neither explicit nor implicit.

Formerly, RTA-RTE either crashed or generated invalid code that attempted to read from a non-existent IOC channel.

RTA-RTE now correctly generates code that reads from a constant buffer, but see issue RTE-3151.

(376571)

Fixed internal error when unconnected receive-side data is measured

RTA-RTE no longer raises an internal error for the case where an intra-ECU receive-side unconnected data element is measured.

(RTE-3133)

RTA-RTE now generates Rte_IrvIWriteRef API

RTA-RTE now generates an `Rte_IrvIWriteRef` API for each VariableAccess in role `writtenLocalVariable` to an `implicitInterRunnableVariable`, as per `[SWS_Rte 06208]`.

The `Rte_IrvIWriteRef` API provides its caller with a reference to the shadow buffer containing the IRV data. The calling SWC can then write data directly into the shadow buffer. The use of a reference allows the calling SWC to make a partial update of complex IRV datatypes without reading and writing every leaf element.

(RTE-2097)

RTA-RTE now handles OPAQUE packing byte order

RTA-RTE now allows `OPAQUE` to be used as an `ISignalToPduMapping . packingByteOrder` value. It will be treated as per `MOST-SIGNIFICANT-BYTE-FIRST`, i.e. big-endian.

(RTE-3123)

Internal assertion fixed

Formerly RTA-RTE reported a spurious 'Assertion failed' message when reading a configuration with fan-in or fan-out via `SystemSignals/SystemSignalGroups` or `ISignals/ISignalGroups` with the same name but in different packages (different full paths). This has been corrected.

(RTE-3003)

Improved validation of ParameterAccess and VariableAccess references

RTA-RTE now performs better validation of the ParameterAccess `. accessedParameter` and VariableAccess `. accessedVariable` references. Previously

some invalid references, for example those with references to data prototypes not in the interface of referenced port, were not detected and led to invalid code silently being generated or internal errors being raised without any location information. This has now been corrected and errors indicating the location and nature of the problem are now raised when such invalid references are found.

(RTE-3101)

ROM block symbol name changed

In previous releases, generated ROM block symbol names began with the prefix `Rte_RO_`. The prefix is now `Rte_RomBlk_`, for consistency with the RAM block naming scheme.

(RTE-3012)

Wrong values received by some receivers in a 1:N connection when they have compatible data types but different sub-element mappings or different text table mappings

Formerly, when receivers connected to the same provider had compatible data types, they were allocated a common receive buffer even when different sub-element mappings and/or different text table mappings applied, causing the wrong values to be received by some of them.

This has now been corrected and such receivers now only share buffers when the same sub-element and/or text table mappings apply.

(RTE-2690)

Warning about unsupported SwAddrMethod.sectionType

When an `SwAddrMethod` has a `sectionType` whose value is not recognised or supported (e.g. `CALIBRATION-ONLINE`), previous versions of RTA-RTE would silently treat the `SwAddrMethod` as if no `sectionType` were specified.

RTA-RTE still treats such a `SwAddrMethod` in the same way, but it will now emit a warning to explain this.

(RTE-3014)

DataConstraints with large upper limits

Previous versions of RTA-RTE incorrectly treated the upper limit of an unsigned, 64-bit `SwBaseType` to be $2^{63} - 1$, instead of $2^{64} - 1$. This caused RTA-RTE to raise an error when using a `DataContraint` with an upper limit of 2^{63} or more, stating that the underlying type had insufficient range.

The bug has been fixed, and it is now possible to use a `DataConstraint` with an upper limit up to and including $2^{64} - 1$.

(RTE-2935)

Additions to supported SwAddrMethod.sectionType values

Previous versions of RTA-RTE would ignore the `MemorySectionType` values `CALIBRATION-VARIABLES`, `CONFIG-DATA`, and `EXCLUDE-FROM-FLASH`. Any `SwAddrMethod` using one of these values for its `sectionType` would be treated as if no `sectionType` had been specified.

Now, RTA-RTE recognizes them as valid values for SwAddrMethod.sectionType, and will also correctly report them in Rte_BSWMD.arxml.

(RTE-3053)

Error when networkRepresentationProps is missing

Since RTA-RTE 5.11.0, RTA-RTE would crash if an `ISignal` had a `dataTypePolicy` of `OVERRIDE` or `LEGACY` but no networkRepresentationProps.

Now, RTA-RTE rejects the configuration according to AUTOSAR `[SWS_Rte 7842]` with error 2971.

(RTE-3043)

Partition-specific default memory sections for sender-receiver buffers for intra-partition parts of mixed intra- and inter-partition communications

The `--use-partition-sections` command-line option requests that RTA-RTE creates and uses partition-specific memory sections for communication buffers.

This fix addresses the case where 1:N communication contains a mix of inter- and intra-partition (intra-`OsApplication`) connections.

Previously, RTA-RTE failed to create partition-specific memory sections for the intra-partition buffers. This has now been corrected.

(RTE-2728)

Compiler warnings for IOC write API calls for implicit shadow buffer flushes for array data

RTA-RTE now generates a cast of the pointer to the implicit shadow buffer to the argument type of the locWrite API for flushes to IOC channels for array data types to avoid warnings with some compilers. This is necessary because by a strict interpretation of the C type compatibility rules neither an array variable not the address of an array variable are compatible with a const pointer to an array. Hence without a cast some compilers raise a warning whether the address of the variable is taken or not.

(RTE-3018)

Location information corrected in 2205

Formerly, RTA-RTE wrote an AUTOSAR path into the `File:` field of message 2205. This has been corrected.

(RTE-2951)

RTA-IOC API calls for implicit shadow buffer fills and flushes

In preparation for an upcoming feature RTA-RTE now generates a call to an `RtaIoc_Read` API function for every group of IOC reads within the same code section in the implicit shadow buffer fill code of a task body and similarly a call to an `RtaIoc_Write` API function for every group of IOC writes within the same code section in the implicit shadow buffer flush code of a task body. These calls, and supporting declarations, are made within a pre-processor conditional section and standard Autosar IOC API calls are made when the symbol `RTA_IOC_OPTIMIZED` is not defined.

Note that to facilitate this the order of fills / flushes may change compared to previous versions of RTA-RTE but no functional change results and the order remains stable.

(RTE-2986)

Filters now support values input as binary, octal or hexadecimal

In previous versions of RTA-RTE filters in `UnqueuedReceiverComSpecs` had to use decimal representation for the values of any `MASK`, `X`, `MIN`, or `MAX` attributes. Using binary, octal or hexadecimal values may have led to incorrectly generated code. This has now been corrected.

(RTE-2928)

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Unexpected attribute in RTEGen XML output

When an input file is validated against the AUTOSAR 4.2.2 schema, certain elements, for example the `VT` node of a `CompuMethod`, acquire attribute `space="preserve"` directly from the schema.

Formerly, RTA-RTE would write this attribute to report files without any namespace qualifier, causing downstream tools to reject the file.

Now, RTA-RTE does not write this attribute, nor any attribute declared in the namespace `http://www.w3.org/XML/1998/namespace`.

(RTE-2936)

TextValueSpecs provided as PortAPIOptions without CompuMethods: command-line option

A new option `--text-value-spec-policy` can take two values: `compumethod-resolution` or `symbolic-pdav-always`. The former is the default behavior of RTA-RTE and indicates that a `TextValueSpecification.value` in the role of a `PortDefinedArgumentValue` must match a `CompuScale.compuConst` belong to an associated `CompuMethod` of `category TEXTTABLE` or `BITFIELD_TEXTTABLE` so that the numerical value can be resolved, otherwise an error is raised. The latter specifies that the `value` can be put out verbatim into the generated C code without needing to reference a `CompuMethod`.

When `symbolic-pdav-always` is used, it is necessary to provide compile-time definitions of the symbols written to `Rte.c` relating to the `TextValueSpecification.value`(s) in roles of `PortDefinedArgumentValue`(s).

(RTE-2890,RTE-2797)

Parse error detection for DataConstraints with maximum limits improved

In previous versions of RTA-RTE a Parse error was incorrectly raised for the situation where a `VariableDataPrototype` is typed by an `ApplicationDataType` mapped to an `ImplementationDataType` which is either a signed or unsigned 64-bit integer, with a `DataConstraint` containing the maximum possible limits. In this scenario, the maximum possible limits for the `DataConstraint` are valid, but RTA-RTE did not recognize that. This has now been corrected.

(RTE-2663)

Warning 1313 detection improved

Warning 1313 about a `portDefinedArgumentValue` in an `NvBlockSwComponent`'s require-port was not raised if the `NvBlockSwComponent` did not also include some legitimate use of `portDefinedArgumentValue`, that is, with an `OperationInvokedEvent`.

This is now fixed; RTA-RTE now performs the check on the associated R-Port when seeking a `blockId` for an efficient NV-RAM writing strategy.

(RTE-2893)

Multiple contract-phase headers in one session

RTA-RTE can generate contract-phase headers for multiple `SoftwareComponentTypes` in one run: specify the `--contract` option multiple times on the command line.

Similarly, the `--local-mcsd` option can write the McSupportData for more than one `SoftwareComponentType` in one run.

The user documentation has been updated to reflect this (see Reference Manual sections for `--contract` and `--local-mcsd`).

(343523)

Incorrect warning regarding missing FlatInstanceDescriptor for measurable requirer connected to measurable provider also having inter-ECU connection

In RTA-RTE v6.0.0pr2 warning 6623 was incorrectly raised for a measurable sender-receiver data item in a require port when connected to a measurable data item in a provide port that was also connected to inter-ECU receivers. In this scenario the measurability at the provide port takes precedence and no `FlatInstanceDescriptor` is required for the data item in the require port, contrary to the incorrect warning. This has now been corrected.

(RTE-2871)

Efficient NV Writing Strategies

Previous releases of RTA-RTE provided limited support for the implementation of efficient NV writing strategies involving `DirtyFlag` support.

RTA-RTE previously only supported `storeCyclic` and `storeImmediate`. It now additionally supports `storeAtShutdown` which does not require the generation of a `RunnableEntity`, rather RTA-RTE calls `NvM_SetRamBlockStatus` in the generated write API.

RTA-RTE previously used `PortDefinedArgumentValues` to derive the `blockId` symbol for NVM API calls. This functionality has been removed in favor of using the EcucConfigurationValue collection. Any `PortDefinedArgumentValue` associated with a non-client-server port in an `NvBlockSwComponentType` will be ignored with warning 1313.

To configure a `blockId` in the Ecu Value Collection, aggregate an `NvRamAllocation` in the relevant `RteSoftwareComponentInstance` which aggregates a `EcucReferenceValue` which maps a `RteSwNvBlockDescriptor` to the corresponding `RteNvmBlock`. The `blockId` symbol is derived using the referenced path of the `RteNvmBlock`. In the event of ambiguous references or no reference at all, then RTA-RTE raises error 1314. If the error is suppressed, then the string `0xffff` is used for the `blockId` to allow compilation and linking.

RTA-RTE previously raised warning 2107 in associated with a subset of `constr_2015` if an `NvBlockSwComponentType` aggregated anything other than `OperationInvokedEvents`. The warning is now only raised when `NvBlockSwComponentType` aggregates anything other than `DataReceivedEvents`,

`TimingEvent`s or `OperationInvokedEvent`s since RTA-RTE now supports a more of `constr_2015`.

RTA-RTE now applies stricter validation regarding NvRam writing strategies: The `NvBlockDescriptor.nvBlockNeeds` and `NvBlockDescriptor.supportsDirtyFlag` are checked for consistency, the correct events must be present and must participate in the correct writing strategies, and the blockId must be uniquely identifiable via the Ecu Value Collection as above. Errors 1307–1315 are used for this purpose.

In particular, RTA-RTE now raises error 1312 when `constr_1404` is contravened: in short, `nvData` cannot be associated with more than one `NvBlockDescriptor`.

In version 5.11.0, RTA-RTE would generate duplicate writing code in the Runnable body if there were multiple `RTEEvent`s starting the runnable on behalf of the same `blockId`. This has been corrected.

If no valid code can be generated for a `RunnableEntity`, then RTA-RTE raises suitable errors and does not generate the Runnable body. If the errors are suppressed, then RTA-RTE generates an empty Runnable body to allow compilation and linking.

RTA-RTE now reports the generated `RunnableEntity` to the `Rte_BSWMD.arxml` report.

(442248)

Corrected command-line option ——deviate-trace-implicit-api

AUTOSAR does not support VFB tracing of Implicit APIs, but RTA-RTE supports tracing of `Rte_IWrite`, `Rte_IWriteRef` and `Rte_IRead` APIs if option `--deviate-trace-implicit-api` is enabled.

This option was found to be inoperative in RTA-RTE 5.11.0: the deviation was always in place, regardless of whether the option was specified.

Now, VFB trace calls are no longer added to Implicit API functions, unless `--deviate-trace-implicit-api` is enabled.

(RTE-2875)

Avoid validation of EOCEventRef

When encountering an `EOCEventRef` in the input, previous versions of RTA-RTE would attempt to resolve the reference (since its XML name ends in `-REF`). This could fail and cause an error.

Since it has no direct relevance to the RTE, `EOCEventRef` has now been added to the list of tags that are excluded from this reference validation.

(535436)

Grouping of IOC reads in implicit shadow buffer fill blocks

In preparation for an upcoming feature RTA-RTE now groups together IOC reads in implicit shadow buffer fill blocks in task bodies. In tasks with runnables with implicit read access to both intra-OsApplication and inter-OsApplication reception the order of the implicit shadow buffer fills will therefore change; no functional change results and the order remains stable. The grouping occurs within each concurrency protection section.

(RTE-2878)

Corrected multiplicity of ApplicationDataTypes to AUTOSAR 4.2.2 standard

Formerly RTA-RTE would reject an ApplicationType having no SwCalibrationAccess. Now RTA-RTE will accept ApplicationTypes of category: VALUE, VAL_BLK, STRUCTURE, ARRAY, STRING, and BOOLEAN having no SwCalibrationAcces. In this event it shall be assumed that the ApplicationType has a SwCalibrationAccess of NOT-ACCESSIBLE.

(RTE-2662)

Deterministic ordering for IRVs with split SWCs

When using the `--deviate-split-swci-support` option and reading an inter-runnable variable in multiple `OsApplications`, a bug in previous versions of RTA-RTE caused the ordering of the writes in the `ImplWrite` API to change sporadically. This has been fixed and now the ordering of these writes is deterministic.

(526288)

Fixed bug with exponents with leading zeros

Formerly, formula expressions in scientific notation where the exponent had a leading 0 were treated as octal. Further an invalid exponent was silently treated as 0.

Now RTA-RTE treats all exponents as decimal and will reject invalid exponents.

(532999)

Warning 1366 removed.

RTA-RTE no longer issues warning 1366 when AUTOSAR 4.2.2 schema detected.

Note this does not imply the implementation of any AUTOSAR 4.2.2 feature or model element. Any feature implemented to AUTOSAR later than 4.0.3 will be documented in a separate release note.

(RTE-2584)

Consistent naming scheme for sender-receiver buffers and queues

Previously RTA-RTE used a different naming scheme for RTE-managed queues compared to RTE-managed buffers and IOC channels (queued or unqueued) when generating sender-receiver communications. For consistency, and to help avoid name length issues, RTE-managed queues for sender-receiver communications are now named with the same naming scheme as RTE-managed buffers and IOC channels for sender-receiver communications.

Note that the order of assignment of the unique numbers for sender-receiver buffers and queues has changed compared to previous versions of RTA-RTE but the order remains stable.

(RTE-2846)

Correction to validation of presence of signal mappings when inter-ECU AssemblyConnector present

(This relates to a legacy model style where an `AssemblySwConnector` might be presented to RTA-RTE that connects a port on a SWC mapped to the ECU to a port on a SWC on another ECU.)

Previously, RTA-RTE failed to validate the presence of signal mappings for data

items in a require port with only inter-ECU connections, leading to invalid code generation for those data items where the signal mappings were missing.

This has now been corrected and an error is raised if signal mappings are not present for all data items in such a require port.

(RTE-2847)

Improved handling of measurement for connections writing to NV Data not referenced by any NvBlockDataMapping

Previously, RTA-RTE did not implement measurement for a connection to NV data item in a require port when no mapping of the data item to a `RamBlock` was present.

This has now been corrected in the case that measurement is requested for the provider and measurement is not also requested for the requirer or the provider has precedence; in this case RTA-RTE now allocates and uses a dedicated measurement buffer.

(RTE-2853)

Option ——rxbuf-use-data-name removed.

Option –rxbuf-use-data-name used the associated `DataPrototype.shortName` for the name of RTE-created receive buffers and measurement buffers. This led to name clashes and excessively constrained the input model, so the option has been removed.

Please use option `--append-name-to-buffer` if you require human-readable RTE buffer names. See the reference manual for details.

(RTE-2834)

Simplified version information

Previous releases of RTA-RTE reported both a product version and a separate generator/backend version. Now that there is no distinction between frontend and backend (since RTA-RTE V5.9), the version information has been simplified to match.

The version information reported in the header comments for generated source files, along with the output of the `--version` option, now contains only the product version.

(RTE-2759)

Allow ApplicationError 0 = E_OK

In previous versions of RTA-RTE, an `ApplicationError.errorCode` outside of the range [1..63] was rejected according to `constr_1108`.

RTA-RTE now supports the update to `constr_1108` in AUTOSAR 4.1, and will accept an `ApplicationError.errorCode` with value 0 if and only if the corresponding `shortName` is `E_OK`.

Similarly, RTA-RTE now supports the update to `constr_1078` in AUTOSAR 4.1. This means that, if a provided `ClientServerOperation` refers to a `possibleError` with `shortName E_OK` and `errorCode` 0, the required `ClientServerOperation` can still be considered compatible even if it does not refer to a matching error.

(478030)

Reports are now schema-conformant to AUTOSAR 4.2.2

Formerly, `Rte_BSWMD.arxml` and `Rte_McSupportData.arxml` were not conformant to the AUTOSAR schema. This has been fixed.

1) The Task Data Block in the support of implicit communication is changed from `_task` to `Rte_task`. This has affected the generated code.

2) The `_dummy` member of otherwise-empty `struct` types is now named `Rte_dummy` (`SWS_Rte_03724`). This has affected the generated code.

3) A bug where the `GetTask` OS API was being reported with the `shortName` "`( void )GetTask`" has been corrected.

Additionally the schema hint has been updated to `AUTOSAR-4_2_2.xsd`, since the output now passes a schema check against that revision. No changes have been made to support any new semantics however.

(362896)

5 ETAS Contact Addresses

5.1 Technical Support

Technical support is available to all RTA-RTE users with a valid support contract. If you do not have a valid support contract, please contact your regional sales office (see below).

The best way to get technical support is by email. Any problems or questions about the use of RTA-RTE should be sent to: rta.hotline@etas.com

If you prefer to discuss your problem with the technical support team, you can call the support hotline on: +44 (0)1904 562624. The hotline is available during normal office hours (0900-1730 GMT/BST).

In either case, when contacting technical support, it is helpful if you can provide as much of the following information (where relevant):

- Your support contract number.
- The versions of the ETAS tools you are using.
- The version of the compiler tool chain you are using.
- The command line (or reproduction of steps) that result in an error message.
- The error messages or return codes you received (if any).
- Your .xml and .arxml files.
- Your OS configuration files.
- The diagnostic.dmp file (if it was generated).

5.2 General Enquiries

5.2.1 ETAS Global Headquarters

ETAS GmbH
Borsigstrasse 24
70469 Stuttgart
Germany

Phone: +49 711 3423-0
Fax: +49 711 3423-2106
www: https://www.etas.com

5.2.2 ETAS Local Sales and Support Offices

Contact details for your local sales office and local technical support team (where available) can be found on the ETAS website:

ETAS subsidiaries: https://www.etas.com/en/contact.php
ETAS technical support: https://www.etas.com/en/hotlines.php